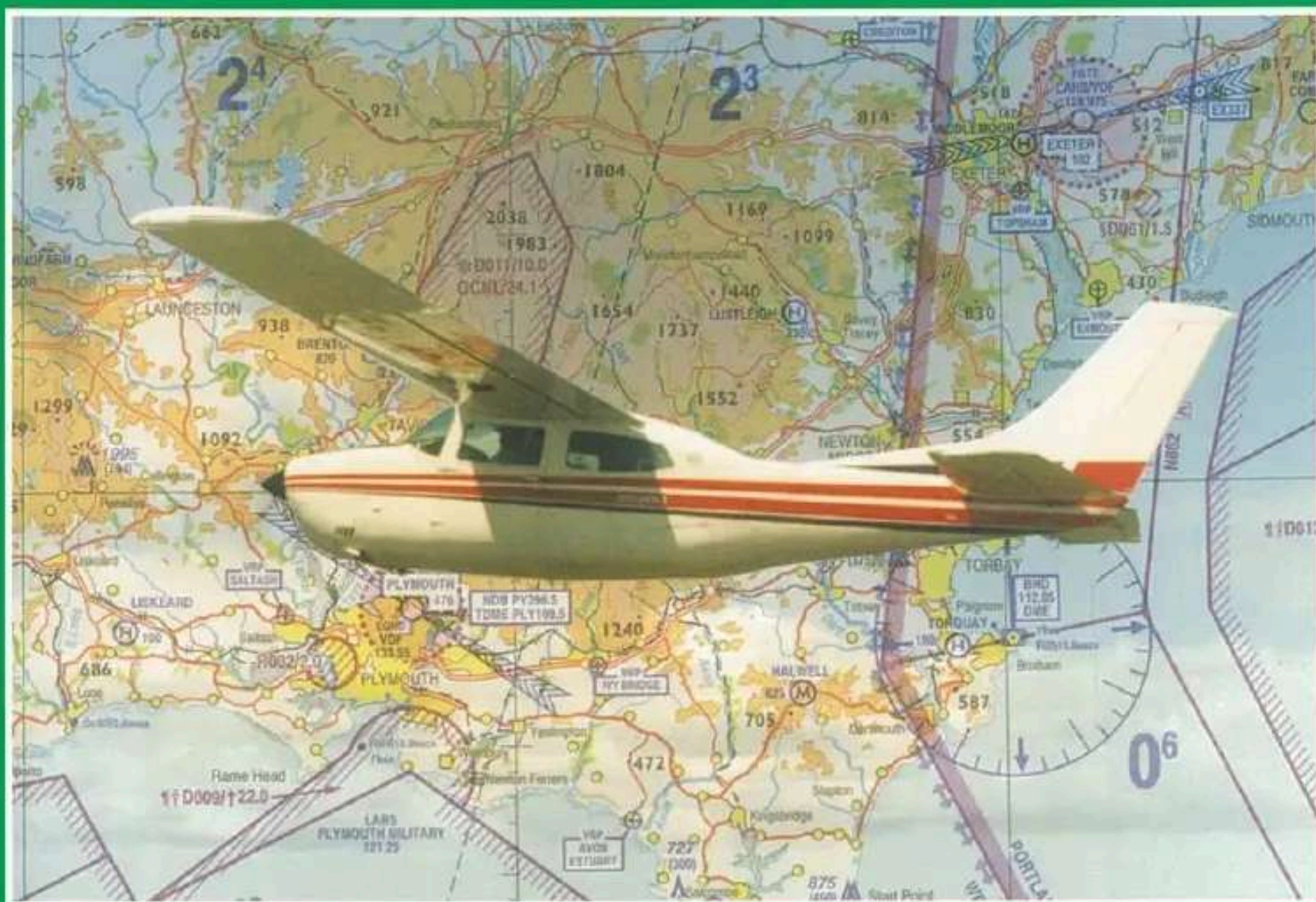


Air Pilot's Manual

3

Air Navigation



POOLEY'S
Air Pilot Publishing

CAA Recommended
JAR – FCL Syllabus & EASA Part-FCL Compliant
LAPL & National PPL Syllabus Compliant
Includes Question & Answer Section

The Air Pilot's **Manual**

Volume 3

Air Navigation

'Recommended reading'
Civil Aviation Authority



POOLEY'S
Air Pilot Publishing

Nothing in this manual supersedes any legislation, rules, regulations or procedures contained in any operational document issued by The Stationery Office, the Civil Aviation Authority, the manufacturers of aircraft, engines and systems, or by the operators of aircraft throughout the world. Note that as maps and charts are changed regularly, those extracts reproduced in this book must not be used for flight planning or flight operations.

Jeppesen charts in this manual have been reproduced with permission and are copyrighted by Jeppesen & Co GmbH.

Aerad charts in this manual have been reproduced with permission and are copyrighted by Thales Avionics.

Copyright © 2013 Pooleys-Air Pilot Publishing

ISBN 1 84336 067 5

First edition published 1987

Second revised edition 1987

Third revised edition published 1997

Fourth edition 1999

Fifth revised edition 2003

Reprinted with amendments 2004

Reprinted with amendments 2005

Reprinted with revisions and colour illustrations 2007

Reprinted with amendments 2008

Sixth Edition 2010

Reprinted with amendments 2011

Reprinted with revisions 2013

All rights reserved. No part of this book may be reproduced or transmitted in any form by any means, electronic or mechanical, including photocopying, recording or by any information storage and retrieval system, without permission from the publisher in writing.

Origination by Pooleys-Air Pilot Publishing Limited.

Printed in England by Portland Print, Kettering NN16 8UN.

Published by Pooleys-Air Pilot Publishing Ltd

Highdown House

Shoreham Airport

West Sussex BN43 5PB England

Tel: +44(0)208 207 3749

Web: www.pooleys.com

e-mail: sales@pooleys.com

The Air Pilot's Manual

Volume 3

Contents

Introduction

vii

Section One – Basic Navigation Theory

1. The Pilot/Navigator	3
2. Speed	15
3. Direction	31
4. Wind Side of the Navigation Computer	49
5. Calculator Side of the Navigation Computer	81
6. Vertical Navigation	103
7. Time	131
8. The Earth	149
9. Aeronautical Charts	163

Section Two – Flight Planning

10. Introduction to Flight Planning	197
11. Pre-Flight Briefing	203
12. Route Selection and Chart Preparation	215
13. Compiling a Flight Log	221
14. The Flight Plan Form	239

Section Three – En-Route Navigation

15. En-Route Navigation Techniques	245
16. Navigation In Remote Areas	291
17. Entry/Exit Lanes and Low-Level Routes	297

Section Four – En-Route Navigation with Radio Nav aids

18. Introduction to Radio Navigation Aids 305

19. Radar 307

20. DME 323

21. The NDB and the ADF 329

22. The Relative Bearing Indicator (RBI) 345

23. The Radio Magnetic Indicator (RMI) 361

24. The VOR 379

25. VHF Direction Finding (VDF) 411

26. Introduction to RNAV and GPS 423

*** Section Four covers the en route navigation requirements of the JAR-FCL syllabus. Refer to the notes on page 304. Section Four will not apply to the UK National PPL training syllabus nor to the LAPL syllabus.

Appendix 1

The Navigation Skill Test..... 429

Appendix 2

Planning the Climb 435

Exercises and Answers

Section One Exercises 445

Section Two Exercises 465

Section Three Exercises 469

Section Four Exercises 471

Section One Answers 479

Section Two Answers 486

Section Three Answers 493

Section Four Answers 495

Index 497

Editorial Team

Dorothy Saul-Pooley LLB(Hons) FRAeS

Dorothy holds an ATPL (A) and a CPL (H), and is both an instructor and examiner on aeroplanes and an instructor on helicopters. She is Head of Training for a school dedicated to running Flight Instructor courses at Shoreham. She is also a CAA Flight Instructor Examiner. In addition, having qualified as a solicitor in 1982, Dorothy acted for many years as a consultant specialising in aviation and insurance liability issues, and has lectured widely on air law and insurance issues. This highly unusual combination of qualifications led to her appointment as Honorary Solicitor to the Guild of Air Pilots and Navigators (GAPAN). Dorothy is a Fellow of the Royal Aeronautical Society, past Chairman of both the GAPAN Instructor Committee and the Education & Training Committee, as well as serving as a Warden on their Court. She is currently Master Elect and will be installed as Master in March 2014. In 2003 she was awarded the Jean Lennox Bird Trophy for her contribution to aviation and support of Women in Aviation and the BWPA (British Women Pilots Association). In 2013 Dorothy was awarded the prestigious Master Air Pilot Certificate by GAPAN. A regular contributor to seminars and conferences, Dorothy is the author and editor of a number of flying training books and has published articles in legal and insurance journals.

David Robson

David is a career aviator having been nurtured on balsa wood, dope and tissue paper. He won an ATC Flying Scholarship and made his first solo flight in a Chipmunk at Biggin Hill shortly after his seventeenth birthday. His first job was as a junior draughtsman at the Commonwealth Aircraft Corporation in Melbourne. He joined the RAAF in 1965 and served for 21 years as a fighter pilot and test pilot. He flew over 1,000 hours in Mirages and 500 on Sabres. He completed the Empire Test Pilot's course at Boscombe Down in 1972, flying everything from gliders, the magnificent Hunter, to Lightnings and Argosies. He completed a tour in Vietnam as a forward air controller in support of the First Australian Task Force. After retiring from the Air Force, he became a civilian instructor and lecturer. During 1986–88 he was editor of the *Aviation Safety Digest*, which won the Flight Safety Foundation's international award. He was awarded the Australian Aviation Safety Foundation's Certificate of Air Safety for 1998. He loves aeroplanes, aerobatics and instructing and still dreams of one day, flying a Spitfire.

Acknowledgements

The Civil Aviation Authority; ICAO; Cessna, Piper, and Gulfstream American for technical material; Daljeet Gill, Peter Godwin, Capt. R.W. K. Snell, Lotti Skeen and Dan Robertson; and the many other instructors and students whose comments have helped to improve this manual.

A Condensed History of the Air Pilot Manuals

For 25 years the Air Pilot Manuals have led the academic training of pilots in the United Kingdom and in many countries around the world.

I first met Trevor Thom, a professional pilot and natural teacher, in Melbourne during a visit to Australia in January 1985. He already had his series of PPL Manuals for the Australian market and I asked him to produce a series for the New Zealand market where we had a small aviation business. Having completed this task, Trevor immediately began writing the first of the Air Pilot Manuals for the United Kingdom market and this project began in earnest on 5th December 1985.

Both Trevor Thom and Robert Johnson commenced the task in my office at Feldon. By the end of the following year, all four volumes were complete and were published in February 1987. By the end of that year, we estimated that 95% of all the UK Flying Schools were using our manuals. Volumes 5, 6 and 7 followed, so completing the full series.

Unfortunately, Trevor Thom had a serious accident at home which prevented him from continuing with the editing of the manuals. His rights were eventually sold to David Robson, another experienced pilot and natural teacher, who progressively improved the drawings and brought colour into the manuals for the first time.

Over the years there have been many assistant editors, in particular Peter Godwin, whose help I first asked for in the very early days with Trevor Thom and which continued until quite recently. The rights in the Air Pilot Manuals are now vested with the Pooley family and they continue to be edited and published from our offices and the Flying Instructor School at Shoreham Airport.

The Air Pilot Manuals have an outstanding reputation for accuracy and are continuously updated. They are recommended CAA reading material and are referred to extensively in the CAA examination answer booklet.

Robert Pooley
CStJ FRIN FRAeS

Introduction

Volume 3 of *The Air Pilot's Manual – Air Navigation* – presents this important area of training for the Private Pilot's Licence in a logical sequence of theory, preparation and performance.

The Cockpit is a Difficult Environment in which to Learn

As with the other volumes of *The Air Pilot's Manual*, in *Air Navigation* we have avoided the presentation of 'facts only'. A thorough understanding of the principles will enable you to gain maximum benefit from your actual navigation exercise flights.

This approach will enable you to become a competent pilot/navigator and will also help to minimise your flight training hours. (It does, however, mean that our book is a little longer than it could be if the aim was only to cram in facts without a reasonable understanding.)

Understanding makes
for remembering.

In determining the order in which the information is presented, care has been taken to keep things as logical and practical as possible. Consequently, in the first section – *Basic Navigation Theory* – the simpler, more practical topics of *Speed, Direction* and *Using the Navigation Computer* come first, to give you the feel of practical operations, before some more involved subjects: *Vertical Navigation, The Earth* and *Aeronautical Charts*.

Operational Decisions

Navigation of an aeroplane consists mainly of making common sense operational decisions. These decisions are based on knowledge and experience. Very few are difficult to make – most being logical and simple – but occasionally there are difficult decisions (both on the ground and in flight) to be made. These are the ones for which we must prepare for.

We have adopted a professional approach right from the start, whether your ultimate aim is to be a private pilot or to go on and make aviation your career.

Operational decisions will often have to be taken well away from your home base, and to a large extent you will be on your own. They fall into two categories:

- those made on the ground during pre-flight planning; and
- in-flight operational decisions.

Many decisions are so simple and 'second nature' that you don't realise you are making them. Others require a calm, cool but quick assessment, followed by a decision and action. Proceeding into an area of poor visibility could fall into this category.

The aeroplane will not stand still while you decide what to do in difficult in-flight situations. You cannot just pull over to the side of the road and study your maps. Good pre-flight planning, with many operational decisions taken on the ground – and alternative courses of action considered in the event of in-flight problems occurring – takes a lot of pressure off the pilot/navigator.

The Navigation Computer

As a pilot/navigator you will become adept at estimating angles, distances, time intervals, fuel consumption, and so on. The art of estimating is an important skill to develop. It is also important that you can calculate these various quantities easily and accurately. To achieve this you will use a navigation computer. It is a simple device (looks complicated but isn't) that allows us to carry out almost every navigation calculation with speed and accuracy.

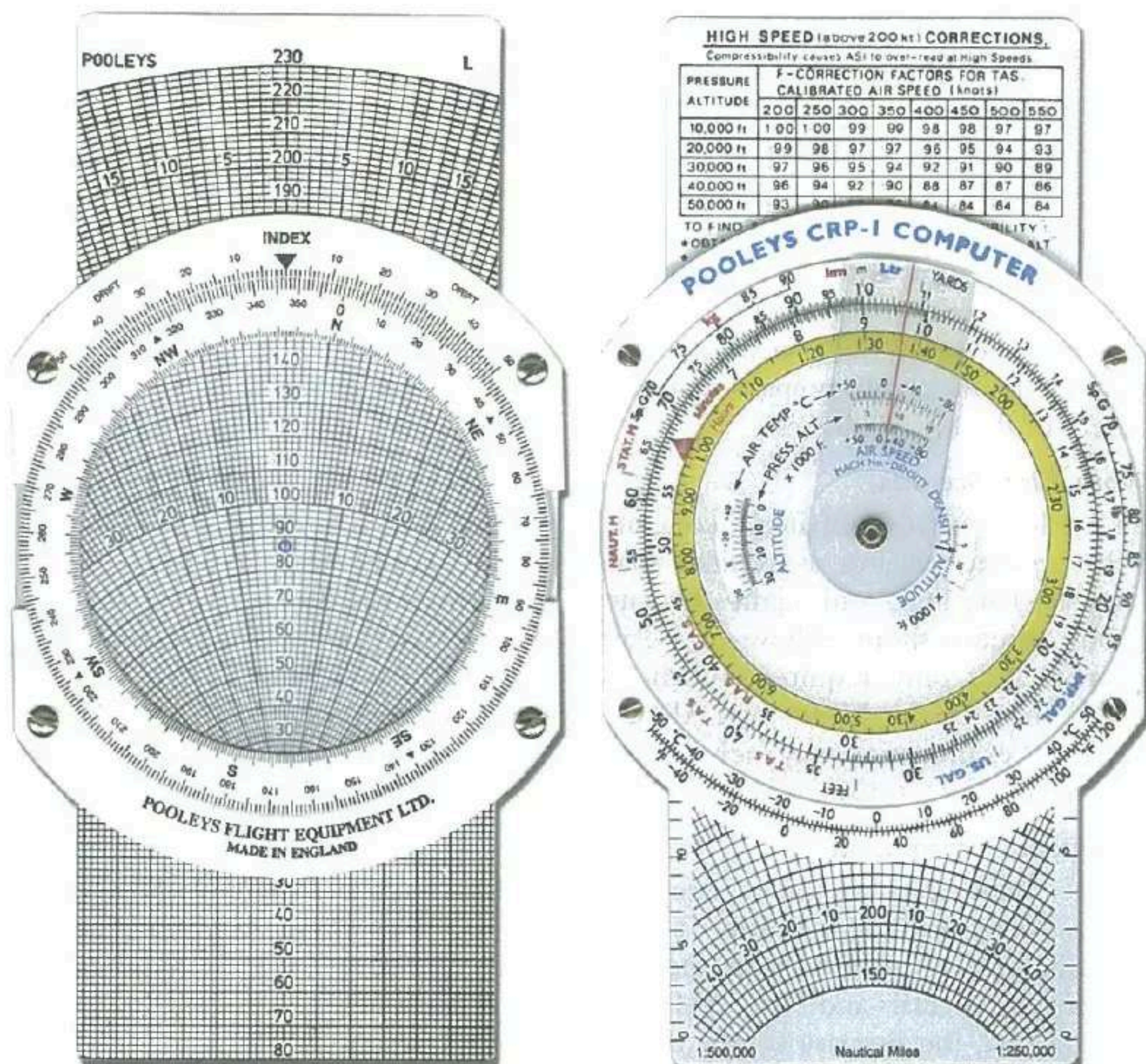
Electronic navigation computers are available, but we suggest you steer away from them, at least initially, because they do not encourage the pilot/navigator to visualise each situation – an important ability to develop. Once you are adept at the various computing problems involved in air navigation, you might decide to 'go electronic'. Beware of becoming over-reliant on electronic computers because you are not permitted to use them in the examination.

The basic concept of the slide navigation computer dates back to early navigation days. The modern version is an essential piece of equipment for a pilot/navigator.

The slide navigation computer has two sides:

- **a wind side**, which enables solution of *triangle of velocities* problems for flight-planning and en route navigation; and
- **a calculator side** (the main component of which is a circular slide-rule on the outer scales), used to perform the simple arithmetical calculations involved in flight operations, e.g. distance, speed and time; conversion of units; fuel quantities and consumption; true airspeed.

Two chapters in the first section describe using the navigation computer – one chapter for each side. Although it may appear a little complicated at first, working through the examples and illustrations we have set out will make using the computer logical and simple.



■ *The wind and calculator sides of a navigation computer*

The Theory Examination

Navigation is part of one of the theory examinations for the UK Private Pilot's Licence (PPL), which you will sit at your flying school. Prior to this you should be achieving considerable success in completing the **Exercises** at the back of the book. They are mentioned at the relevant places in each chapter, and in this volume some chapters have exercises interspersed through the text to give you practice on a particular aspect of the chapter before moving on.

The Exercises form an important part of the course and we recommend that you work through them carefully.

This manual is more than just a text to allow you to pass the examination, though this is one of its aims. It is designed to remain as a reference text on your shelf for as long as you fly.

In places we have included more information than is required for the Air Navigation examination section. For example,

Appendix 2 shows you how to plan a climb – something which, although not required of you in the PPL examination, will enable you to plan longer, higher altitude flights in the future.

The En route Navigation Section (PPL Skill Test)

This is the province of your flying instructor. The test is carried out at the completion of your flying training and is part of the PPL Skill Test (although with the agreement of your examiner it may be flown as a separate section.) It is designed to assess your ability as a pilot/navigator. This manual, and your navigation cross-country training, will prepare you fully for the Navigation element of the PPL Skill Test.

Private Pilot Licences

This edition covers the material contained in the JAR-FCL and NPPL training syllabus as well as that required by the new European Part-FCL and Light Aircraft Pilot Licence (LAPL) syllabus. Students should follow one syllabus only. Section Four of this manual is only required for the JAR-FCL or Part-FCL Licence, not for the NPPL or LAPL, although the theoretical knowledge examination may include some of this material for any of the licences.

Operational Information

For safe flight operations it is essential that all pilots refer to current operational information. This basically involves using latest issues of aeronautical charts, and amended flight information publications, circulars and NOTAM (Notices to Airmen).

In the UK, the primary source of operational information is the UK Aeronautical Information Publication (AIP), a large, frequently amended manual produced to an international standard by the Civil Aviation Authority. Your flying school and Air Traffic Services (ATS) units should have amended copies of the UK AIP available for reference, although these days they are often available on a CD rom, or directly on the internet from the AIS website: www.ais.org.uk. As the AIP is a formidable and bulky set of documents for a PPL holder (because the majority covers airline-type instrument flight procedures), there is also available a conveniently sized publication known as *Pooley's Flight Guide*, which is revised regularly. You will find references to both *Pooley's Flight Guide* and the UK AIP throughout *The Air Pilot's Manual*. Note that these references are no substitute for referring to current, amended documents. If you are ever in any doubt about operational information, in *Pooley's Flight Guide* or the UK AIP, refer to an amended copy of the AIP and current air legislation documents; and **always** check the latest AIRACs (which detail AIP updates), AIP Supplements and Aeronautical Information Circulars (AICs) and NOTAM prior to flight.

Section **One**

Basic Navigation Theory

Chapter 1

The Pilot/Navigator 3

Chapter 2

Speed 15

Chapter 3

Direction 31

Chapter 4

Wind Side of the Navigation Computer 49

Chapter 5

Calculator Side of the Navigation Computer 81

Chapter 6

Vertical Navigation 103

Chapter 7

Time 131

Chapter 8

The Earth 149

Chapter 9

Aeronautical Charts 163

The Pilot/Navigator

Air Navigation

All air navigation involves basic principles that apply to all aeroplanes, from the simplest trainer to the most sophisticated passenger jets. These basic principles are discussed in this manual.

Since *The Air Pilot's Manual* is a training programme for the Private Pilot's Licence (PPL), we will concentrate on accurate navigation of a light aircraft, flown by a single pilot, in visual conditions.

PPL holders, when flying cross-country, act as pilot, navigator and radio operator. They must:

- **primarily fly the aeroplane** safely and accurately;
- **navigate correctly;**
- **operate the radio** and attend to other duties in the cockpit.

In short, they must 'aviate, navigate and communicate'.

To conduct a cross-country flight efficiently, navigation tasks must be coordinated with (and not interfere with) the smooth flying of the aeroplane. It is most important that the pilot/navigator clearly understands the basic principles underlying navigation so that correct techniques and practices can be applied quickly and accurately without causing distraction or apprehension.

Prepare Soundly

Being properly prepared prior to a cross-country flight is essential if it is to be successful. Always flight plan meticulously. This establishes an accurate base against which you can measure your in-flight navigation performance.

Pre-flight consideration should be given to navigation items such as:

- the **serviceability** of your watch or aircraft clock – time is vital to accurate navigation;
- the **contents** of your 'nav bag' – pencils, navigation computer, protractor and scale (or a plotter), suitable maps and charts, and relevant flight information publications;
- the **preparation** of the appropriate maps and charts;
- the **desired route**;
- the **terrain en route**;
- the **airspace en route** (uncontrolled, controlled, special rules, advisory etc.);
- the **suitability** of the destination and any alternate aerodromes.

Sound preparation is the basis for a confident navigation exercise.

- the forecast weather en route and at the destination and alternate aerodromes (plus any actual reports that are available);
- the calculation of accurate headings and groundspeeds;
- consideration of fuel consumption, and accurate fuel planning.

It sounds like a lot, but each item considered individually is simple to understand. After considering them one by one in separate chapters, we will put them all together and see how they fit into a normal cross-country flight.

In Flight, Fly Accurate Headings

Once the aeroplane is in flight, flying a reasonably accurate heading (which involves reference to both the direction indicator and outside cues) is essential if the aeroplane is to track towards the desired destination. Maintaining cruise airspeed, and comparing your progress and times of arrival at various fixes with those estimated at the flight-planning stage, will normally ensure a pleasant and drama-free journey.

Navigation Tasks are Additional to Flying the aeroplane

Our objective in this volume of *The Air Pilot's Manual* is to show you navigation techniques that will not increase your workload in the cockpit to an unacceptable degree, but will still allow time to fix your position and navigate the aeroplane safely to your desired destination.

We make the assumption that you already know how to fly the aeroplane; the idea here is to add to these flying skills the basic principles of air navigation. Other aspects that have a bearing on the conduct of a cross-country flight are covered in their own sections (for instance, airspace, radio procedures and meteorology in Vol. 2).

The Earth

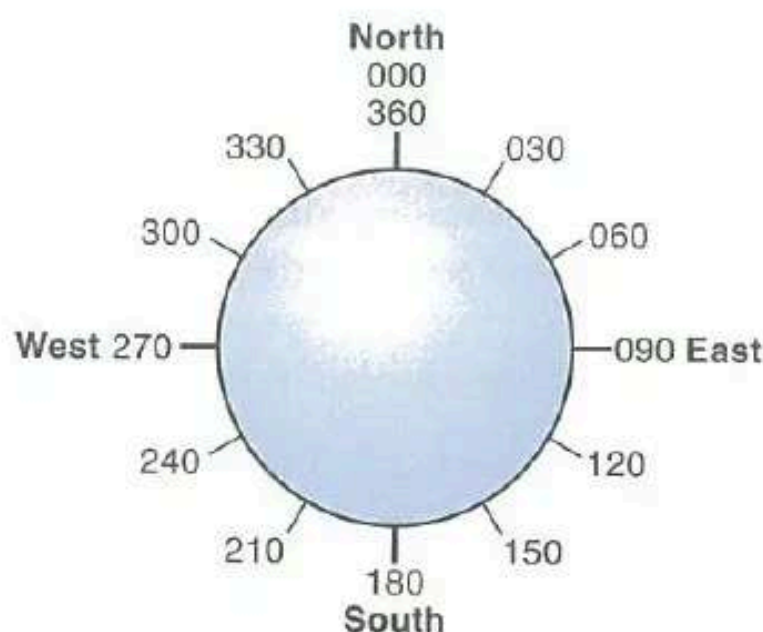
All navigation is done with reference to the surface of the earth – starting from the elementary exercise of ‘navigating’ the aeroplane around the circuit during your initial training (which requires visual reference to ground features such as the runway and points ahead of the aeroplane for tracking) and progressing to the large passenger jets using sophisticated instrument navigation techniques to cover vast distances around the earth.

Direction on Earth

Direction is the angular position of one point to another without reference to the distance between them. It is expressed as the angular difference from a specified reference direction. In air navigation this reference direction is either:

- north (for *true* or *magnetic* bearings); or
- the heading (or the nose) of the aircraft (for *relative bearings*).

The simplest means of describing direction is to consider a circle laid flat and then divided into 360 units, called degrees ($^{\circ}$). These units are numbered clockwise from 000 in the reference direction all the way around the circle to 360.

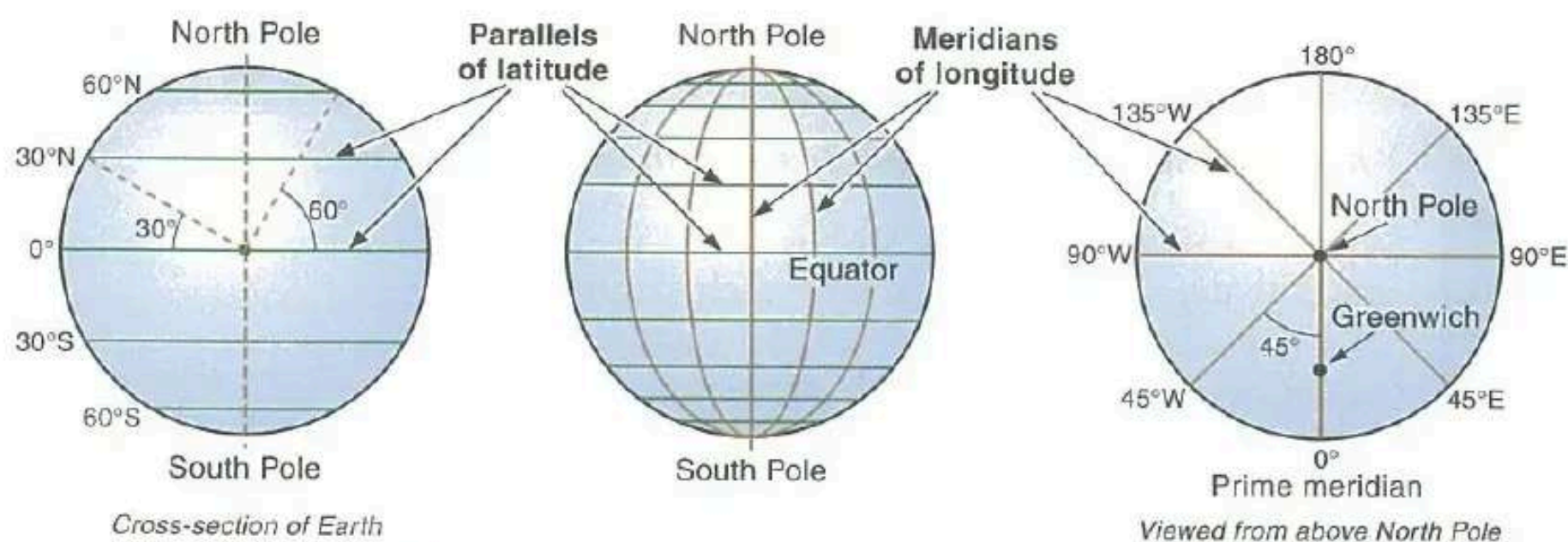


■ Figure 1-1 To measure direction, a circle is divided into 360 degrees ($^{\circ}$)

It is usual to refer to direction as a three-figure group to prevent any misunderstanding in the transmission of messages. For example, north is referred to as 360. East is referred to as 090, south-west as 225.

Position on Earth

The main method of specifying the position of a place on the surface of the earth is the *latitude* and *longitude* system. This involves covering the surface of a reduced earth with an evenly spaced *graticule* of lines – north–south lines joining the North and South Poles, and east–west lines parallel with the equator. The north–south lines are known as **meridians of longitude** and the east–west lines are called **parallels of latitude**.



■ Figure 1-2 Position on earth is usually specified with reference to meridians of longitude and parallels of latitude

The position of any place on the surface of the earth can then be specified with reference to the equator and a *datum* (or prime) meridian of longitude. The universal base longitude used throughout the world – longitude 0° – is the meridian drawn (north–south) through Greenwich, near London, known as the **prime meridian**.

Distance on Earth

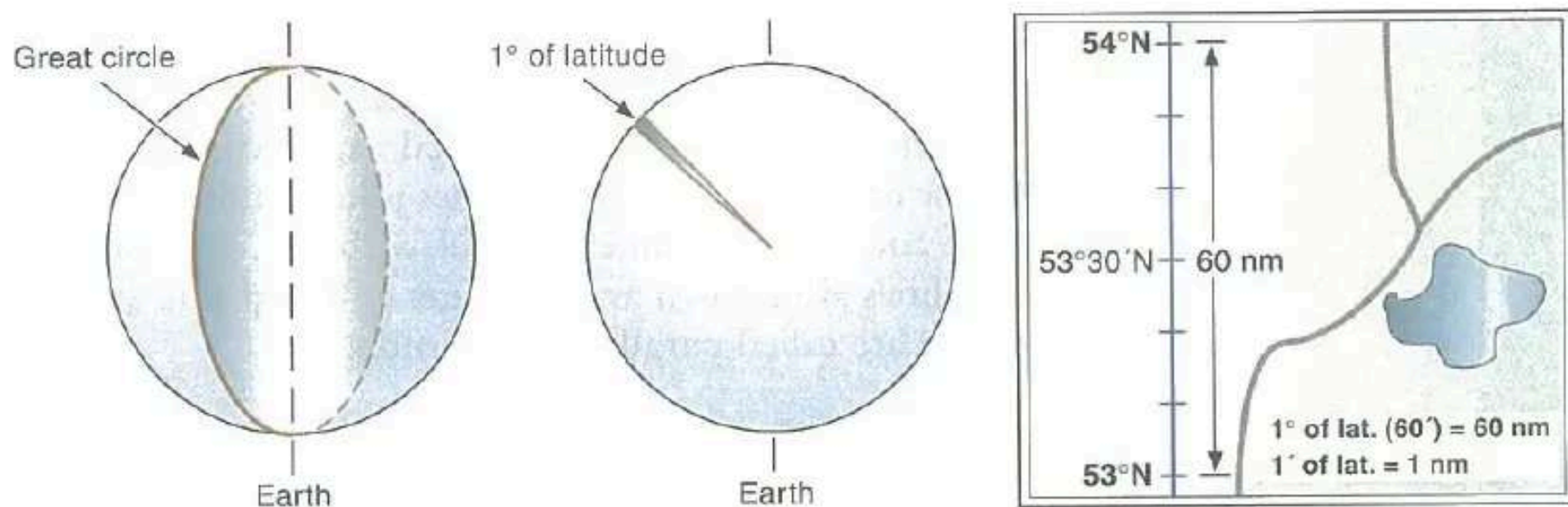
The separation of two points on earth is called **distance** and is expressed as the length of the shortest line joining them.

*The standard unit of distance in navigation is the **nautical mile (nm)**.
1 nm = 1,852 metres (1.852 km).*

The nautical mile is related to the size of the earth in that it is the length of 1 minute of latitude. It is slightly longer than the familiar statute mile; 1 nautical mile (nm) measures 6,076 ft compared to 5,280 ft in the statute mile (sm).

1 nautical mile equals
1 minute of latitude.

One minute of latitude is measured down the side of a chart, i.e. along a meridian of longitude, which is a **great circle**. A great circle is one whose centre lies at the centre of the earth. All meridians of longitude and the equator are great circles. Thus 1 minute of arc of any great circle will be 1 nautical mile. This is explained in more detail in Chapter 8, *The Earth*.



■ Figure 1-3 **A great circle, and 1° of latitude on the earth and on a chart**

There are 360 degrees in a circle and each degree has 60 minutes, i.e. a circle has $(360 \times 60) = 21,600$ minutes – which makes the circumference of the earth approximately 21,600 nautical miles.

If an aeroplane travels 1 nautical mile through an air mass, we refer to this as 1 **air nautical mile (anm)**. As well as the aeroplane moving through the air mass, the air mass will be moving across the ground (in the form of a 'blowing' wind) and will carry the aeroplane along with it. The wind velocity adds an extra effect to

the passage of the aeroplane over the ground. If an aeroplane travels 1 nautical mile over ground or water, we refer to this as 1 **ground nautical mile (gnm)**.

NOTE While navigation distances are measured in nautical miles, other shorter horizontal distances, such as runway length or horizontal distance from cloud, may be referred to in metres.

In air navigation we are concerned not only with *horizontal navigation* but also with *vertical navigation* (see Chapter 6). The traditional and standard unit for vertical distance, or height, is the foot (ft).

Speed

Speed is the rate at which distance is covered; in other words, speed is distance per unit time.

*The standard unit for speed is the **knot (kt)**.
1 knot = 1 nautical mile per hour.*

Direction and Speed Combined

An aeroplane flies in the medium of air. Its motion relative to the air mass is specified by its:

- **direction** (known as **heading**); and
- **speed** through the air mass (**true airspeed**).

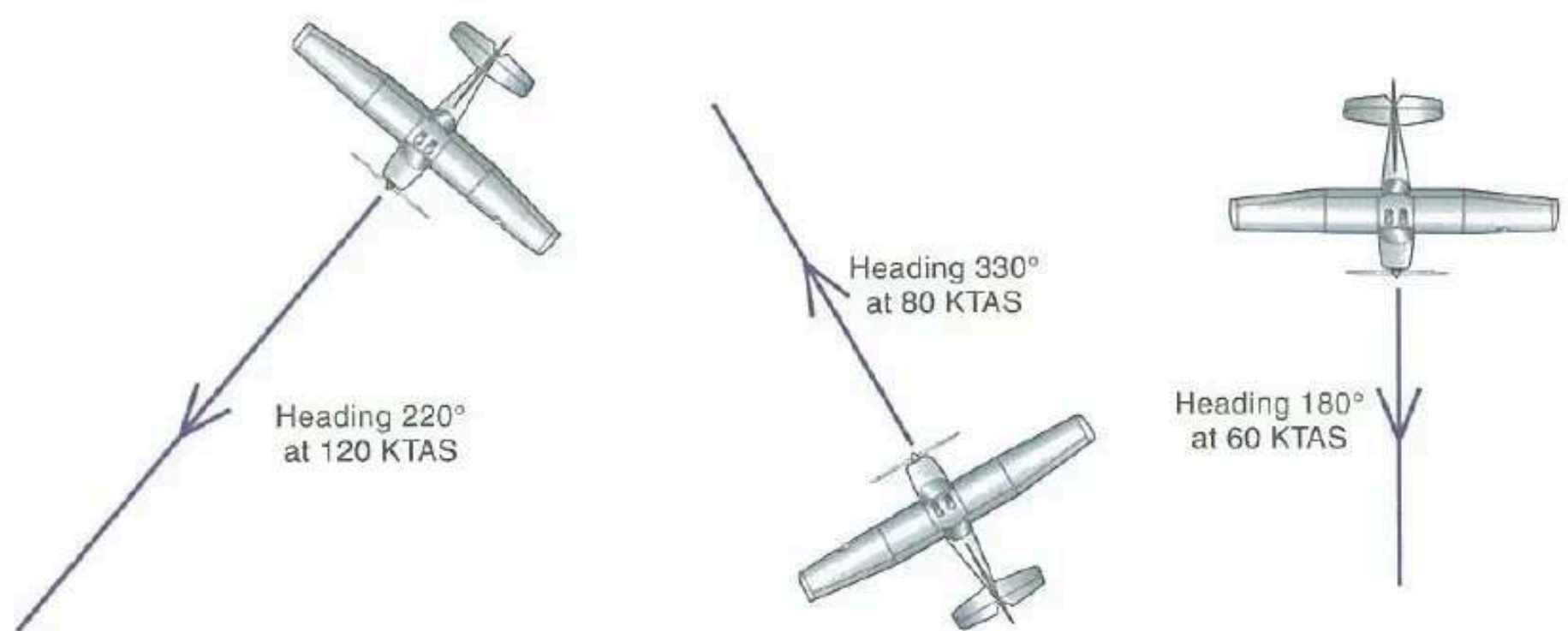
HEADING (HDG). When flown in balance (as it normally is) the aeroplane will travel *through* the air in the direction in which it is heading. If the aeroplane is heading east (090), then its passage relative to the air mass will be easterly (090) also.

TRUE AIRSPEED (TAS). This is the actual speed of the aeroplane *relative* to the air mass. True airspeed is normally abbreviated to TAS, but occasionally to *V* when used in aerodynamic formulae. (See *Principles of Flight* in Vol. 4 of *The Air Pilot's Manual*.)

When considered together, HDG/TAS constitute a **vector** quantity, which requires both *magnitude* (in this case TAS) and *direction* (here HDG) to be completely specified. HDG/TAS is the velocity (direction and speed) of the aeroplane through the air.

*The **HDG/TAS** vector fully describes the motion of the aeroplane relative to the air mass.*

HDG/TAS is symbolised by a single-headed arrow ; the direction of the arrow indicates the direction of movement along the vector line.



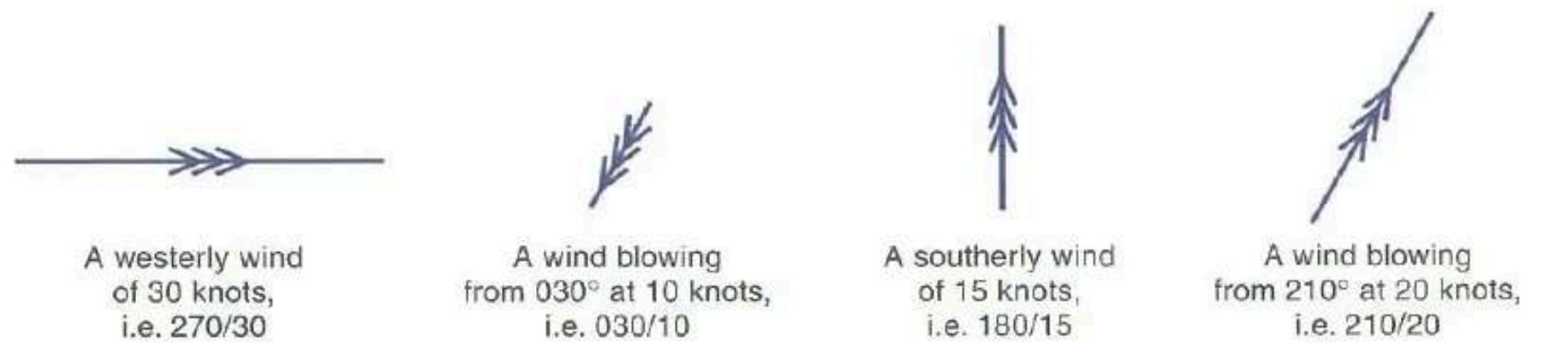
■ Figure 1-4 Examples of the HDG/TAS vector

**An Air Mass can Move Relative to the Ground
(a Wind can ‘Blow’)**

The general movement of air relative to the ground is called **wind velocity** and is abbreviated to **W/V**. Like HDG/TAS, W/V is a vector quantity because both direction and magnitude are specified.

By convention, the wind direction is expressed as the direction *from* which it is blowing. For example, a northerly wind blows from the north towards the south. W/V is symbolised by a triple-headed arrow $\longrightarrow \ggg$.

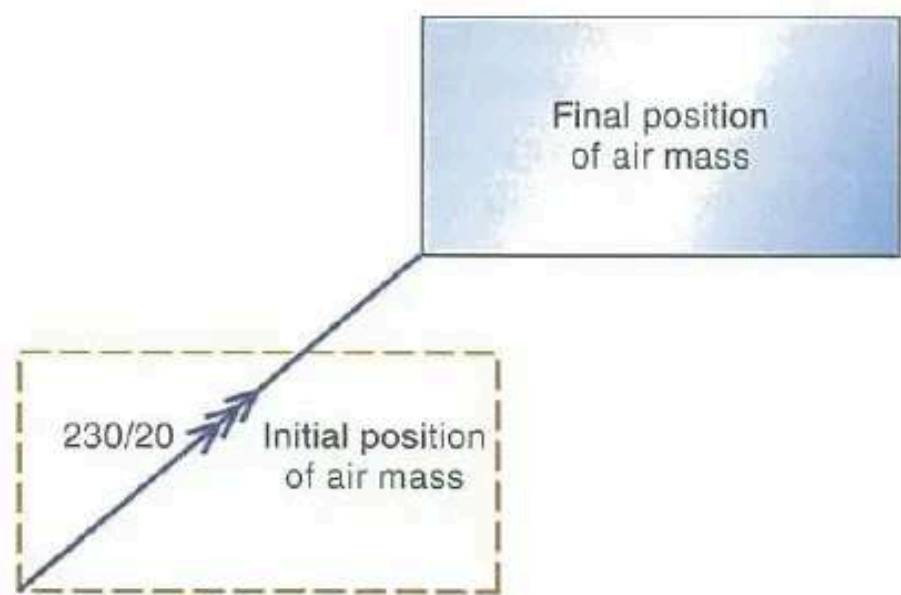
The **W/V** vector fully describes the **horizontal motion of the air mass relative to the earth’s surface**.



■ Figure 1-5 Examples of the wind vector

With a W/V of 230/20, the air mass will be moving relative to the earth’s surface from a direction of 230° at a rate of 20 nautical miles per hour.

In a 6 minute period, for example, the air mass will have moved 2 nm (6 minute = $\frac{1}{10}$ hour; $\frac{1}{10}$ of 20 nm = 2 nm) from a direction of 230°, and therefore towards $(230 - 180) = 050^\circ$.



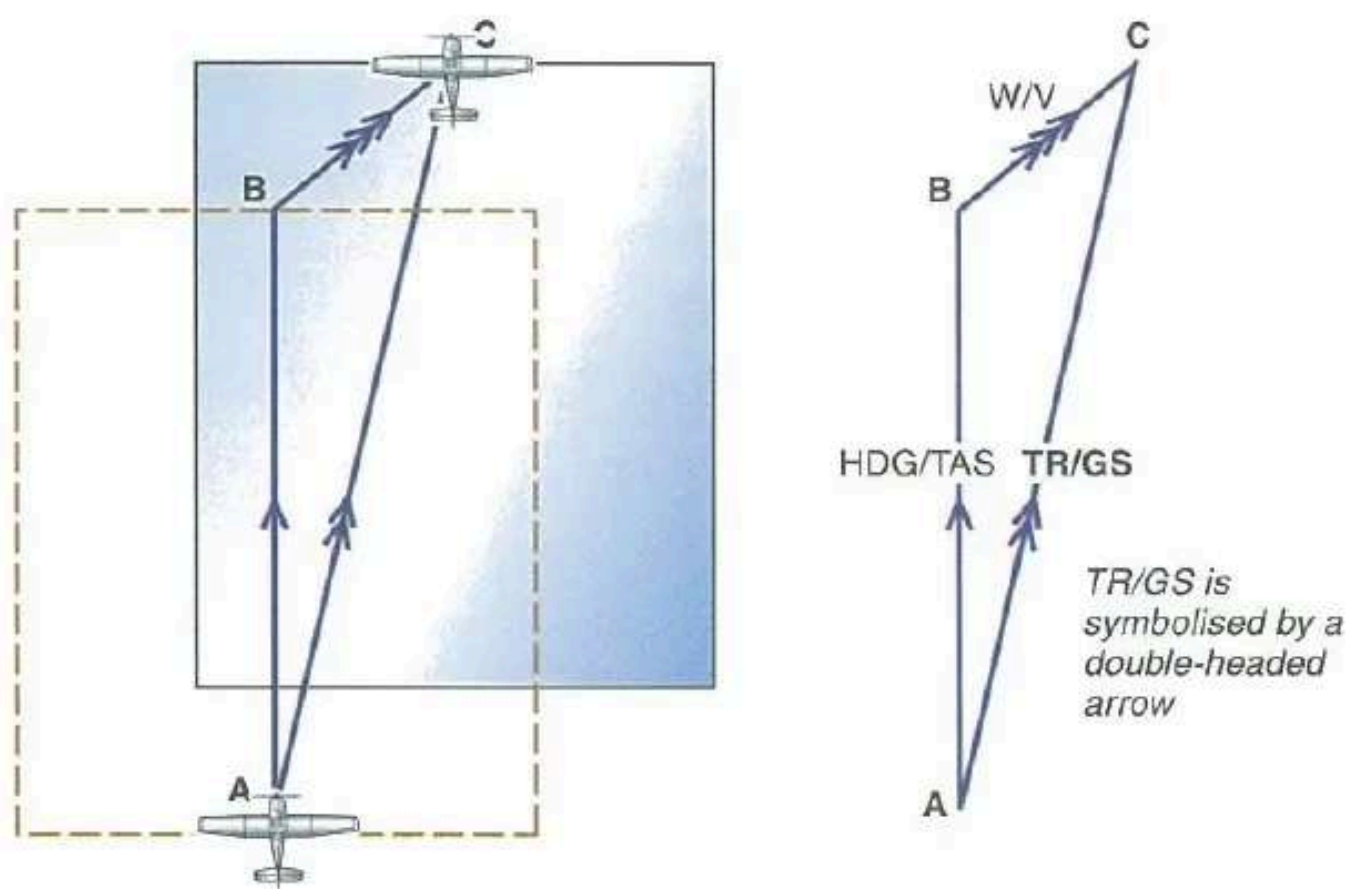
■ Figure 1-6 **A wind of 230/20**

The motion of the aeroplane relative to the surface of the earth is made up of two velocities:

- the **aeroplane** moving relative to the air mass (HDG/TAS); and
- the **air mass** moving relative to the surface of the earth (W/V).

Adding these two together gives the resultant vector of:

- the **aeroplane** moving relative to the surface of the earth. This is the track and groundspeed (TR/GS), which is symbolised by a double-headed arrow $\longrightarrow\longrightarrow$.



■ Figure 1-7 **HDG/TAS + W/V = TR/GS**

An aeroplane flying through an air mass is in a similar situation to you swimming across a fast flowing river. If you dive in at A and head off through the water in the direction of B, the current will carry you downstream towards C. To an observer sitting overhead in the branch of a tree, you will appear to be swimming a little bit ‘sideways’ as you get swept downstream, even though in fact you are swimming straight through the water.

In the same way, it is common to look up and see an aeroplane flying somewhat 'sideways' in strong wind situations. Of course the aeroplane is not actually flying sideways through the air, rather it is flying straight ahead relative to the air mass (HDG/TAS). It is wind velocity (W/V) which, when added to the aeroplane's motion through the air (HDG/TAS), gives it the resultant motion over the ground (track/groundspeed).

To fly from A to C in the above situation, the pilot must fly on a HDG of A-B through the air, i.e. maintain the nose of the aeroplane in a direction parallel to A-B. The wind will have the effect of B-C. The combined effect of these, known as the resultant, will give the aeroplane a track over the ground of A-C.

The TR/GS vector fully describes the motion of the aeroplane relative to the earth's surface.

The Triangle of Velocities

The two velocities:

- HDG/TAS: the aeroplane moving through the air mass; and
- W/V: the air mass moving over the ground;

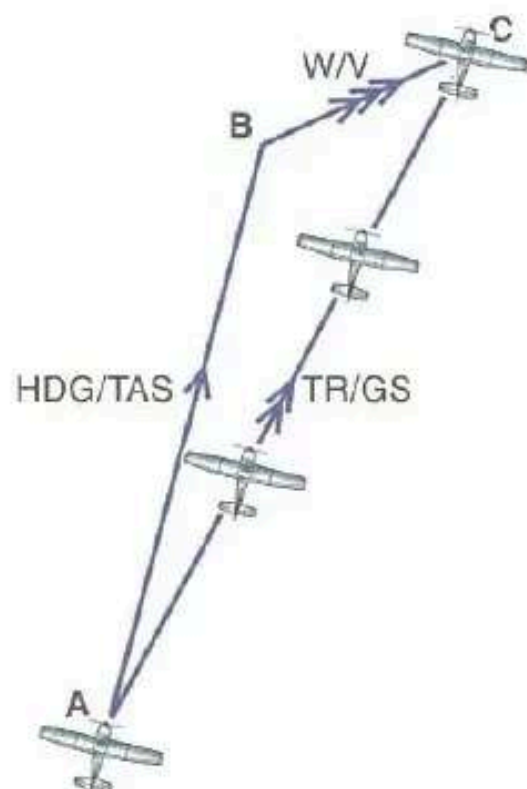
when added together as vectors, give the resultant:

- TR/GS (track/groundspeed) – the aeroplane moving over the ground.

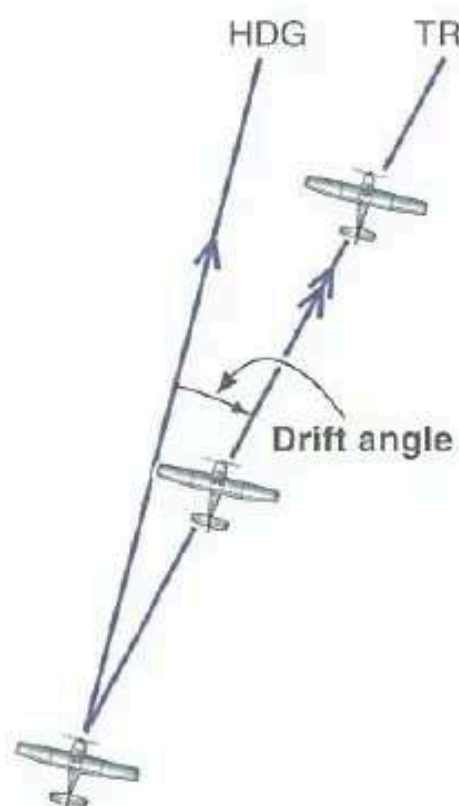
These three vectors form the **triangle of velocities**. It is a pictorial representation of the vector addition:

$$\text{HDG/TAS} + \text{W/V} = \text{TR/GS}$$

i.e. the combined effect of HDG/TAS plus W/V will give the resultant TR/GS (Figure 1-8).



■ Figure 1-8 **The triangle of velocities**

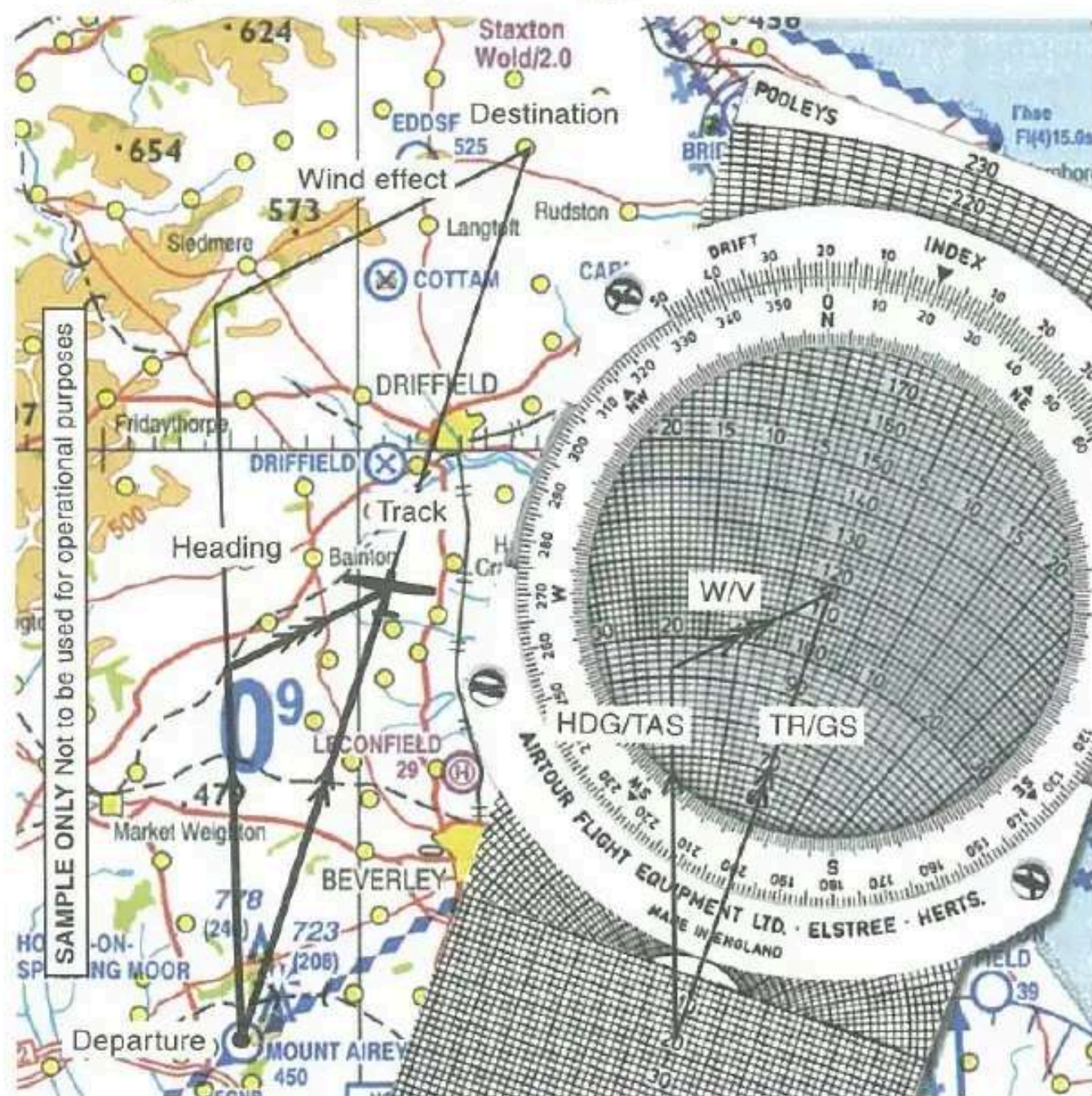


■ Figure 1-9 **Drift is the angle between heading and track**

We add the two vectors for HDG/TAS and W/V 'head to tail' i.e. starting from A, the head of the HDG/TAS vector at B is the starting point for the tail of the W/V vector which then ends up at C.

The resultant effect of the two combined is the TR/GS vector starting at A and finishing at C. This is the path that the aeroplane would fly over the ground. The angle between the HDG and the track (TR) is called the **drift angle** (Figure 1-9).

You may have already seen this triangle of velocities illustrated on a navigation computer, as in Figure 1-10.



■ Figure 1-10 **The triangle of velocities laid on the wind side of a navigation computer**

Do not be put off by the apparently complicated appearance of the navigation computer. It is a marvellous device designed to make navigation tasks easier. Chapter 4 describes using the wind side of the computer in detail, so it will become quite clear.

At the flight planning stage:

- you will know the desired track (track required); and
- will obtain a forecast wind velocity.

Using the known true airspeed, you will be able to calculate:

- the heading required to ‘make good’ the desired track; and
- the expected groundspeed.

Later on during the flight you may find that, even though you have flown the HDG/TAS accurately, your actual track made good (TMG) over the ground differs from your desired track; in other words there is a track error. It is most likely to be caused by the actual wind being different from the forecast wind that you used at the flight planning stage. You will then have to make some adjustments to the HDG to carry out the navigation task of rejoining your desired track and continuing to the destination.

This is what air navigation is all about. The essential principles are simple and have now been covered. All we have to do is expand on them in the following chapters and combine them into practical navigation operations.

Summary of Terminology

HDG/TAS: Heading (HDG) is the actual heading of the aeroplane in degrees steered by the pilot. It may be related to true north, magnetic north or compass north.

True airspeed (TAS) is the actual speed of the aeroplane through the air. It will differ significantly from the airspeed indicated on the airspeed indicator (the indicated airspeed) due to the air being less dense the higher the aeroplane flies. The pilot will need to complete a small calculation to convert indicated airspeed (IAS) to true airspeed (TAS) when flying at altitude.

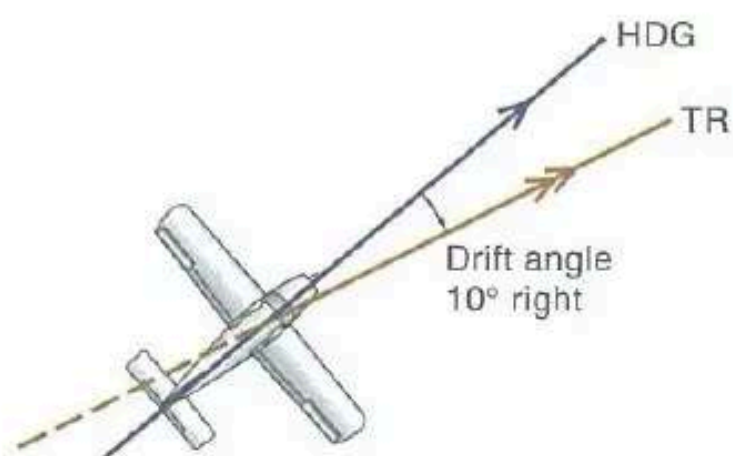
The normal unit for airspeed is the knot. IAS is useful for aerodynamics, but TAS is necessary for navigation. The normal unit of distance for navigation is the nautical mile (nm) and if it is distance relative to the air, we call it an air nautical mile (anm).

TR/GS: Track (TR) is the path of the aeroplane over the surface of the earth, and is usually expressed in degrees true or magnetic.

Groundspeed (GS) is the actual speed of the aeroplane over the ground and is measured in knots. A GS of 120 kt means that 120 ground nautical miles would be covered in 1 hour at that GS.

DRIFT is the difference between the HDG steered by the pilot and the track of the aeroplane over the ground. The wind blows the aeroplane from its HDG/TAS through the air onto its TR/GS over the earth’s surface.

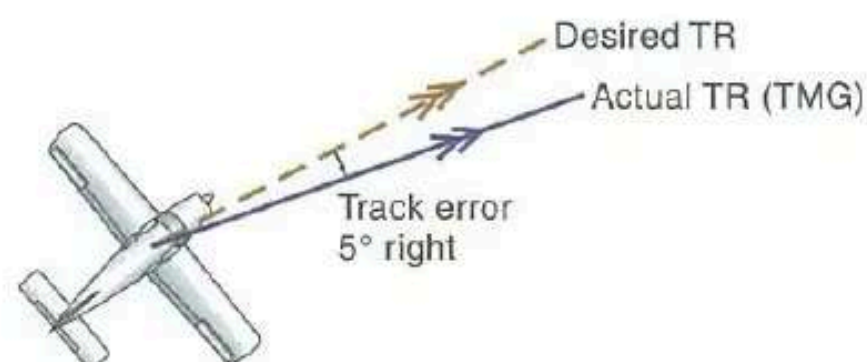
Drift is measured from the HDG (the nose of the aeroplane) to the TR, and is specified in degrees left (port) of HDG or right (starboard) of HDG.



■ Figure 1-11 **Drift is the angle between heading and track**

W/V: Wind direction is expressed in degrees *true* or *magnetic* and is the direction *from* which the wind is blowing. Wind speed is measured in knots (kt). 1 kt = 1 nm per hour.

TRACK ERROR: The actual track made good (TMG) over the ground will often differ from the *desired track*. The angular difference between the desired track and the TMG is called **track error** and is specified in degrees left (port) or right (starboard) of the desired track.



■ Figure 1-12 **Track error is the angle between desired track and the track made good (TMG)**

NOTE Track error is totally different from *drift*.

LATITUDE: The distance of a place north or south from the equator, measured in degrees.

LONGITUDE: The distance of a place from the prime meridian (0°) through Greenwich, also measured in degrees.

NAUTICAL MILE: The length of 1 minute of latitude measure along a meridian, i.e. down the side of a chart.

KNOT: Unit of speed equal to 1 nautical mile per hour.

GREAT CIRCLE: Intersection of the earth's surface and a plane passing through the earth's centre.

Now complete **Exercises 1 – The Pilot/Navigator**.

Exercises and Answers are at the back of the book.

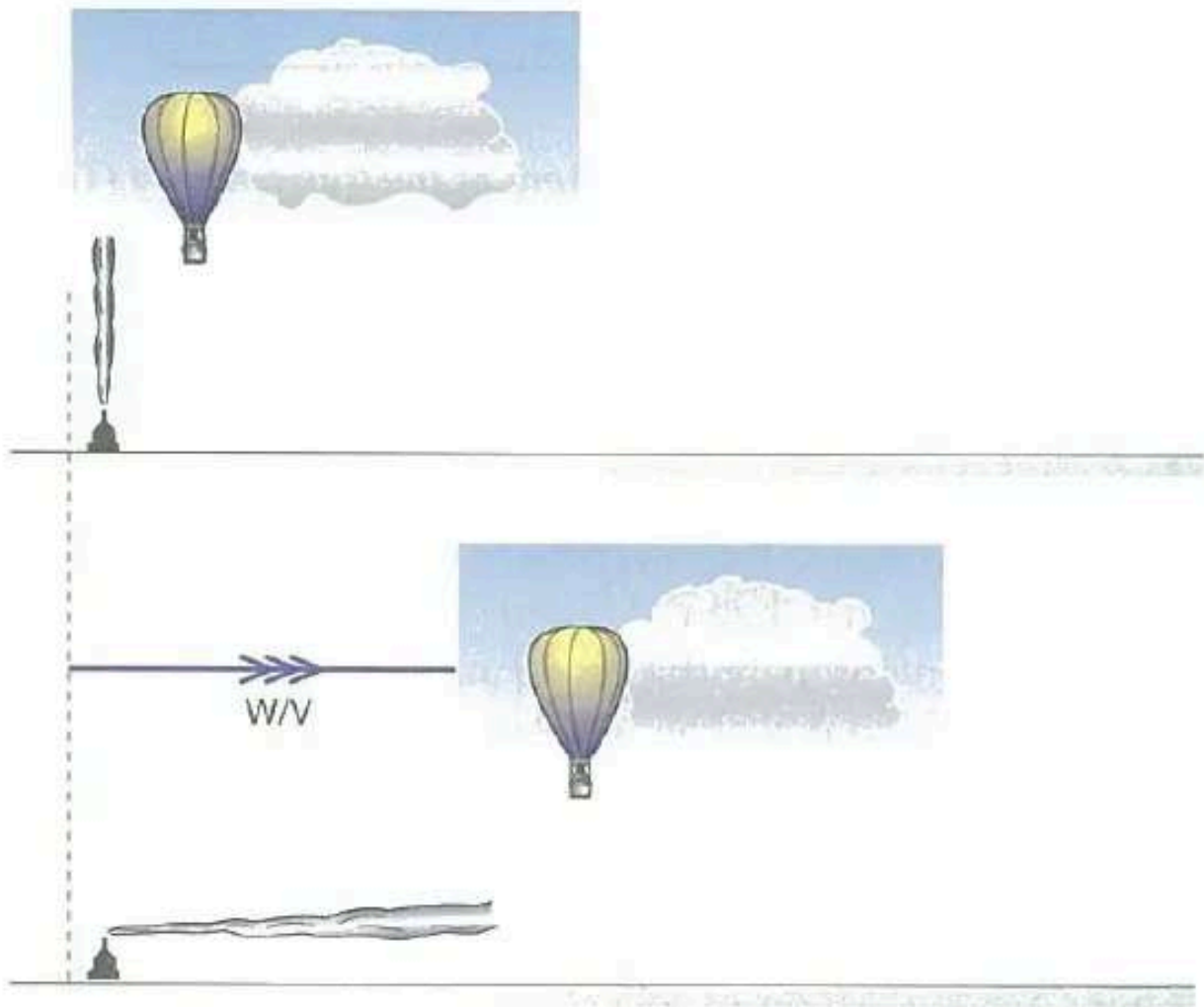


Speed

Airspeed

A sound understanding of the factors involved in *airspeed* is important if you are to become a competent pilot or navigator. The **true airspeed (TAS)** of an aircraft is its rate of progress or speed through the air mass in which it is flying. Whether the air mass is moving over the ground or is stationary is irrelevant to the true airspeed. TAS is simply the speed of the aircraft through the air.

In contrast, a hot-air balloon or a cloud has no horizontal driving force of its own and so just hangs in the air. This means the TAS of a balloon or a cloud is zero because it is not moving *relative* to the air mass.



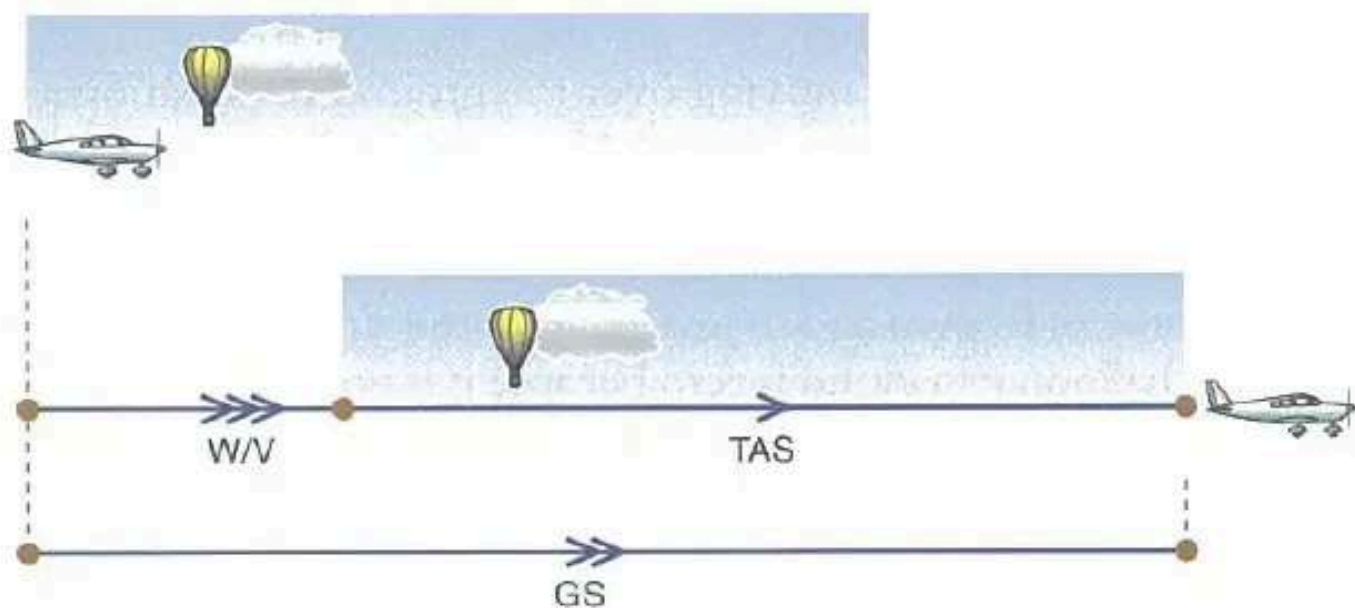
■ Figure 2-1 **An air mass can be stationary or move as wind**

If the air mass is moving relative to the ground (i.e. the wind velocity is other than zero), then the balloon or cloud will be carried by the air mass across the ground. Being static in the air mass, the balloon or the cloud could theoretically be used as a point against which to measure the true airspeed (TAS) of an aircraft. In other words, an aircraft will fly past a balloon, or a cloud, at its true airspeed.

The actual speed of an aircraft relative to the ground is called the **groundspeed (GS)**. The resultant groundspeed is a combination of:

- the **true airspeed (TAS)** – the movement of the aircraft relative to the air mass); and
- the **wind velocity (W/V)** – the movement of the air mass relative to the ground).

This is familiar from the ‘triangle of velocities’ in Chapter 1.



■ Figure 2-2 The groundspeed is the resultant of the true airspeed (TAS) and the wind velocity (W/V)

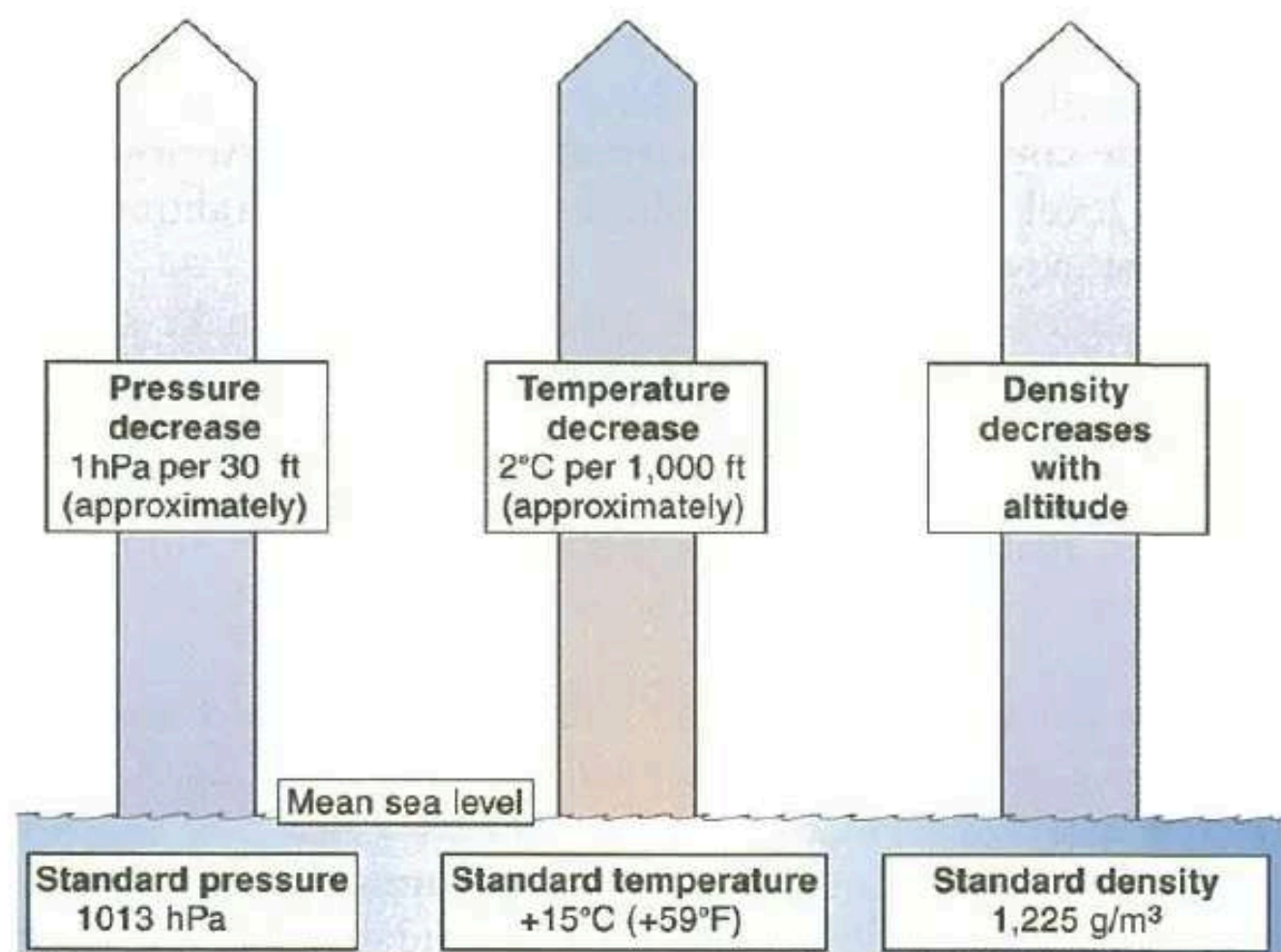
At this stage we are only interested in airspeed – the speed of the aircraft through the air. (Groundspeed comes later in the book.)

International Standard Atmosphere (ISA)

A *standard atmosphere* has been defined as a ‘measuring stick’ against which we can compare the actual atmosphere that exists at a given place on a given day. The standard atmosphere has:

- A standard mean sea level (MSL) pressure of 1013.25 hectopascals (hPa), which decreases by about 1 hPa for each 30 ft of altitude gained. For practical purposes, we use 1013 hPa.
- A standard MSL temperature of + 15°C, which decreases by about 2°C for each 1,000 ft of altitude gained.
- The ISA MSL air density is 1,225 gm/cubic metre, and this also decreases as altitude is gained.

NOTE The **hectopascal (hPa)**, a standard unit of pressure for aviation, equivalent to millibars, has been adopted in many countries. Because 1 mb = 1 hPa, only the name change is significant. The UK recently adopted the hPa but many places still use millibars (mb) and the USA still uses inches and you will see hPa in the Republic of Ireland and in Europe.



■ Figure 2-3 The International Standard Atmosphere (ISA)

Speed Measurement

To measure the speed of an aircraft is a little more complicated than you might expect. The basic instrument used is the **airspeed indicator (ASI)** which is a pressure-operated instrument. The airspeed displayed is given the logical name **indicated airspeed (IAS)**.

Due to the nature of the atmosphere – in which air pressure and air density decrease with altitude – and the design of the airspeed indicator, the indicated airspeed (IAS) is usually *less than* the true airspeed (TAS).



■ Figure 2-4 Airspeed indicator with a TAS correction scale

The indicated airspeed shown on the airspeed indicator in the cockpit and the true airspeed of the aeroplane through the air will only be the same value when International Standard Atmosphere mean sea level (ISA MSL) conditions exist. Such conditions are usually not experienced.

In conditions other than ISA MSL, pilots must make simple calculations (either mentally or by navigation computer) to convert the IAS they read on the airspeed indicator to the TAS needed for navigation.

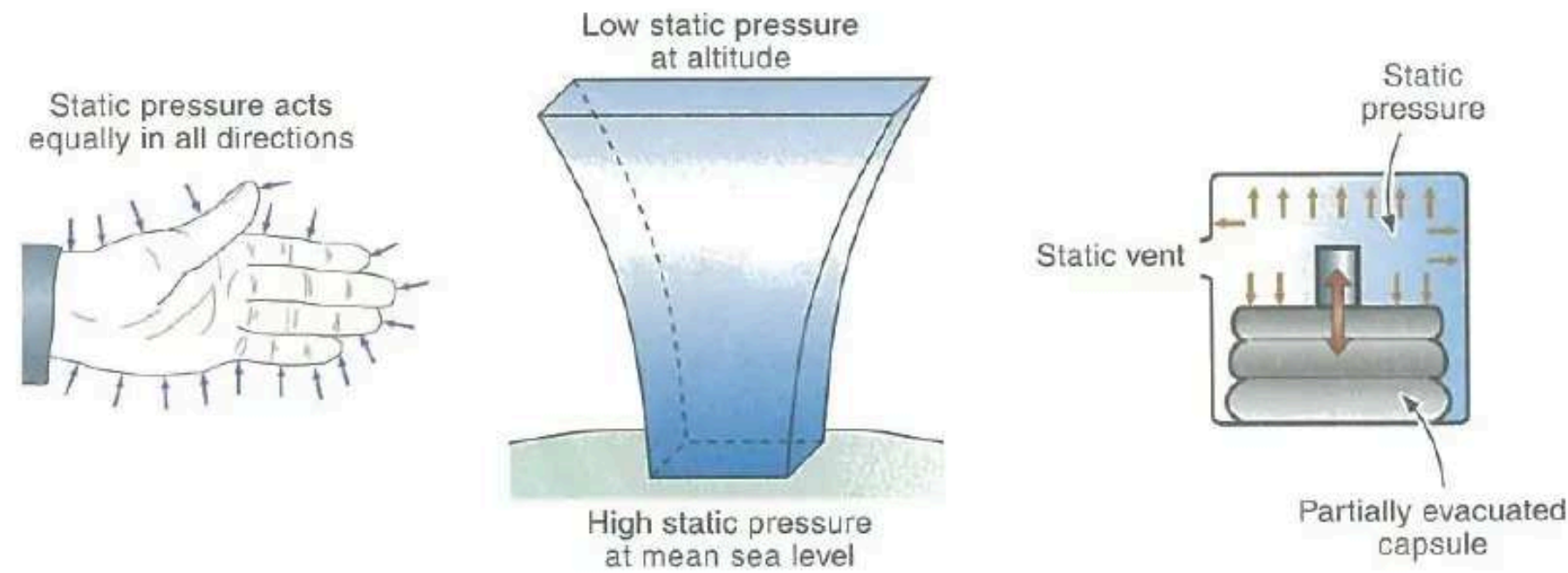
The fact that the word *airspeed* has a number of meanings in aviation may be confusing at first but you must understand the differences.

- Performance of the aeroplane is related to indicated airspeed (IAS) (i.e. whether the aircraft will stall or not, its rate of climb performance, lift/drag ratio etc.), and is a function of IAS. Indicated airspeed is related to dynamic pressure.
- Navigation and flight planning depend on true airspeed (TAS), wind velocity (W/V) and groundspeed (GS). True airspeed is the actual speed of the aeroplane through the air.

To understand the difference between the two basic airspeeds: indicated airspeed (IAS), and true airspeed (TAS), we need to consider briefly certain properties of the atmosphere and the principles of fluid flow.

Static Pressure

Static pressure at any point in the atmosphere is exerted equally in all directions. It is a result of the weight of all the molecules composing the air above that point. At this very moment, static pressure of the atmosphere is being exerted at all points on the skin of your hand.



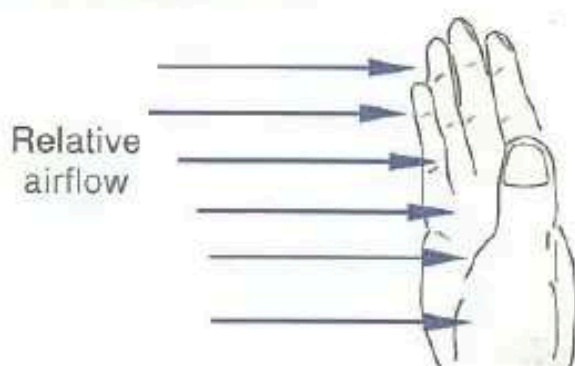
■ Figure 2-5 **Static pressure**

As its name implies, static pressure does not involve any motion of the body relative to the air.

Dynamic Pressure

If you hold your hand up in a strong wind or out of the window of a moving car, then an extra wind pressure, or ‘moving pressure’, is felt due to the air striking your hand.

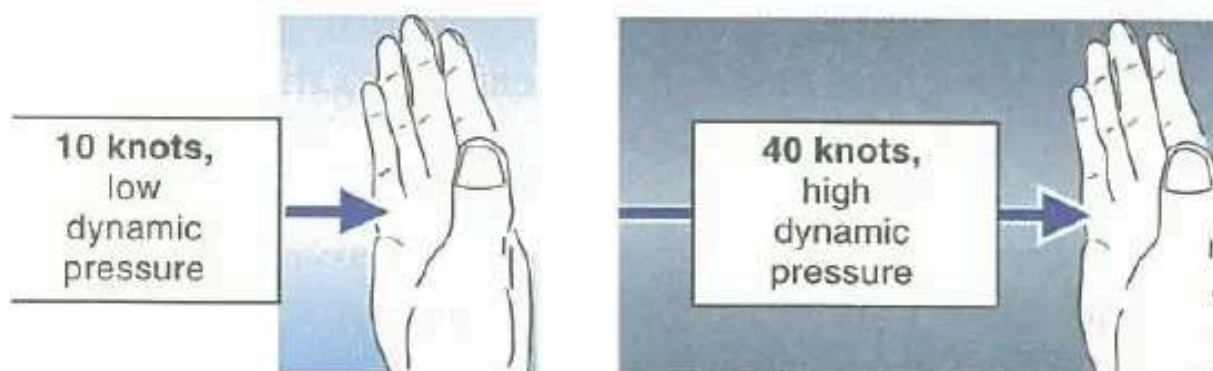
This extra pressure, over and above the static pressure which is always present, is called **dynamic pressure**, or pressure due to relative movement. It is felt by a body that is moving relative to the air, i.e. it could be moving through the air, or the air could be flowing past it.



■ Figure 2-6 **Dynamic pressure**

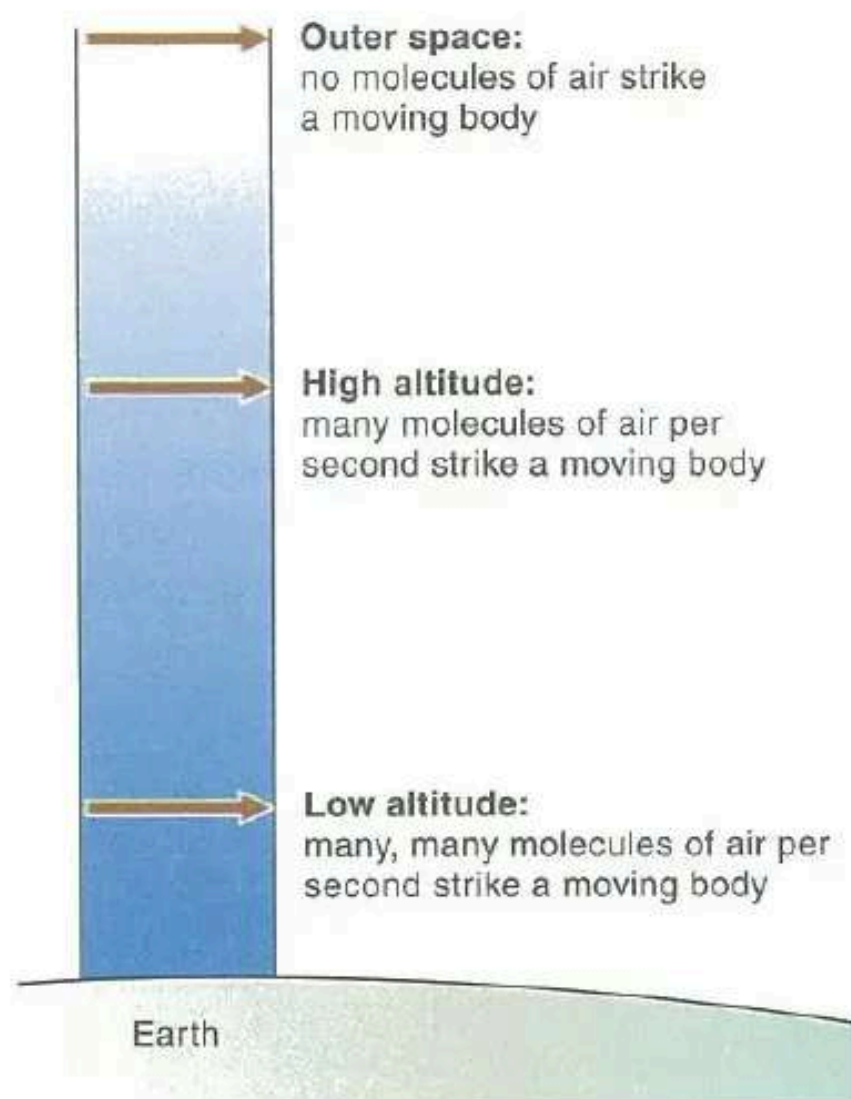
Just how strong dynamic pressure is depends on a number of things, the two main ones being:

1. **The speed of the body relative to the air.** The faster the car drives or the faster the wind blows, then the stronger the extra dynamic pressure that you feel on your hand. This is because of the greater number of air molecules that strike it per second.



■ Figure 2-7 **Dynamic pressure increases with airspeed**

2. **The density of the air.** In outer space, no matter how fast you travelled, you would not feel any dynamic pressure because there are practically no molecules to strike you. In contrast, at sea level, where the atmosphere is densest, your hand would be struck by many molecules per second – certainly many more than in the upper regions of the atmosphere. Even though you might be travelling at the same speed, you will feel a much lower dynamic pressure in the higher levels of the atmosphere, where the air is less dense, than in the lower levels.



■ Figure 2-8 **Air density decreases with altitude**

So, for an aircraft moving at a constant true airspeed, less dynamic pressure is experienced the higher the altitude. The actual measure of dynamic pressure is written:

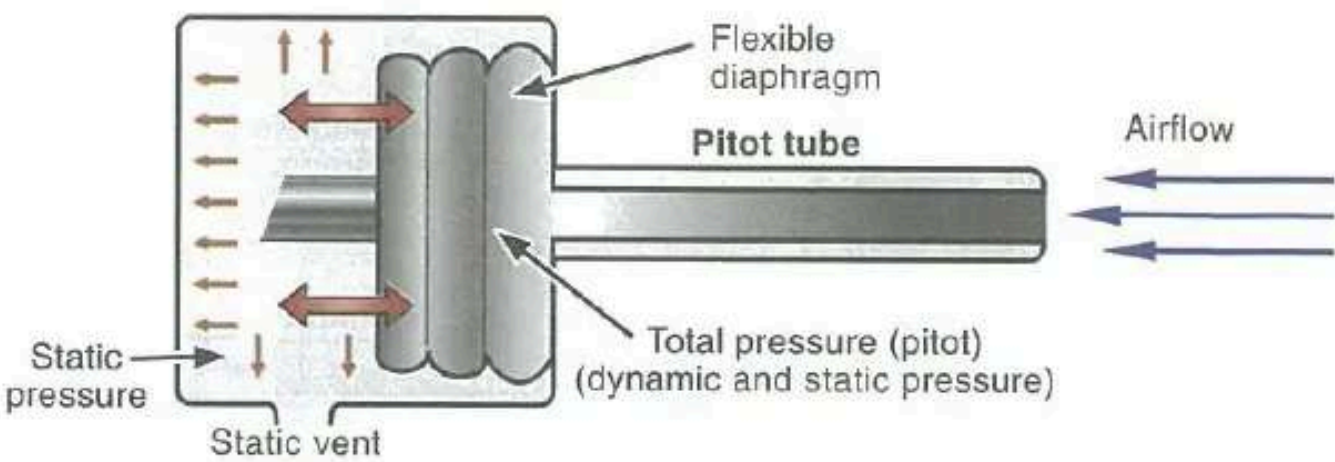
$$\text{Dynamic pressure} = \frac{1}{2} \times \rho \times V\text{-squared}$$

- ρ represents air density, which decreases with altitude.
- V represents the speed of the body relative to the air, i.e. the true airspeed. (It does not matter whether the body is moving through the air, or the air blowing past the body, or a combination of both – as long as they are moving relative to one another there will be an airspeed and a dynamic pressure.)

Total Pressure

In the atmosphere some static pressure is always exerted, but only if there is motion of the body relative to the air will any dynamic pressure (due to relative motion) be felt by the surface exposed to the airflow. Thus:

Total pressure consists of static pressure plus dynamic pressure.



■ Figure 2-9 **Total pressure is measured by a pitot tube**

Much of this theory about pressure was developed by the Swiss scientist Daniel Bernoulli, and is expressed in Bernoulli’s equation, which, in simplified form, is:

Static pressure measured by static line (barometer or altimeter)	+	Dynamic pressure $\frac{1}{2} \rho \times V\text{-squared}$	=	Total pressure measured by pitot tube
---	---	---	---	--

An expression for dynamic pressure can be obtained by subtracting the term static pressure from both sides of this equation:

Dynamic pressure = total pressure – static pressure

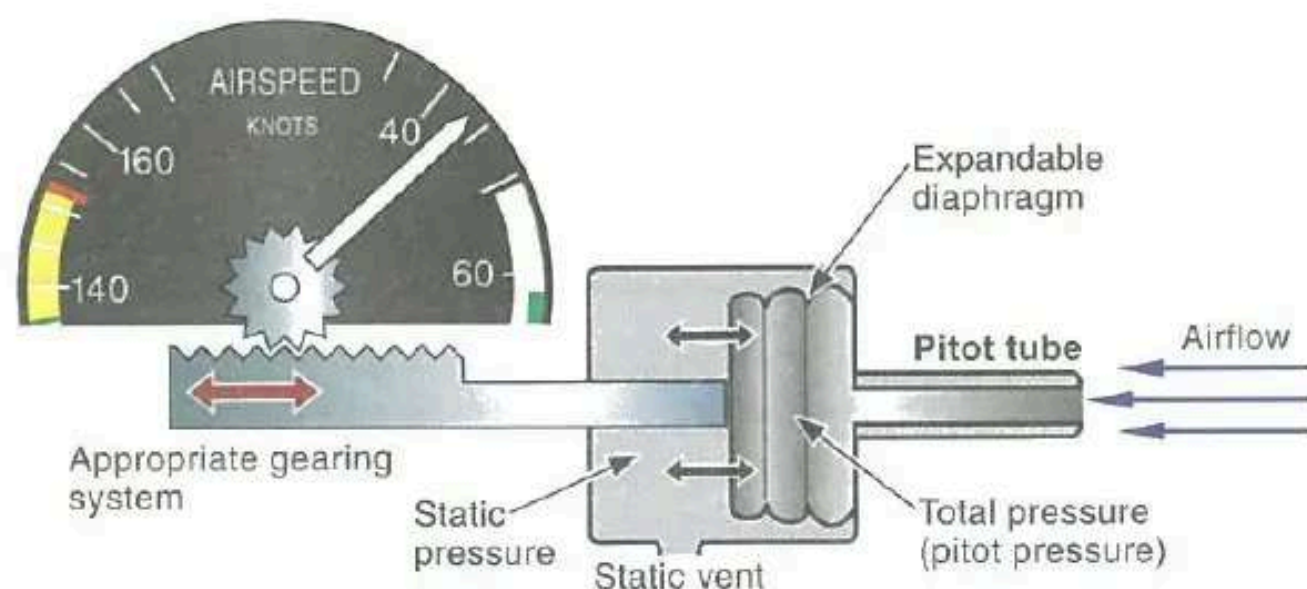
Indicated Airspeed (IAS)

A measure of dynamic pressure can be found by starting with the total (pitot) pressure, and subtracting the static pressure from it. This is done using a diaphragm with total pressure from the pitot tube fed onto one side, and static pressure from the static line fed onto the other side.

The diaphragm in the airspeed indicator (ASI) system positions itself according to the difference between the total pressure and the static pressure, i.e. according to the dynamic pressure. A pointer connected to the diaphragm through a gearing mechanism then moves around the ASI scale as the diaphragm responds to these pressure variations.

If we assume that the density of air (*rho*) remains constant at its mean sea level value (which it does not), the scale around which the pointer moves can be graduated in units of speed. This results in an airspeed indicator that displays the airspeed accurately only under ISA MSL conditions, i.e. when the air density is 1,225 grams per cubic metre (the same as at + 15°C, pressure altitude zero).

If the air density (*rho*) is precisely 1,225 gm/cubic metre, then the airspeed indicator will show an indicated airspeed that is the same as the true airspeed of the aeroplane through the air.



■ **Figure 2-10** *The flexible diaphragm in the airspeed indicator drives the pointer to display indicated airspeed (IAS)*

NOTE Airspeed indicators are usually calibrated in knots but you may see indicators graduated in statute miles per hour, the familiar mph.

Indicated airspeed (IAS) is what we read on the airspeed indicator (ASI).

Rectified Airspeed (RAS)

A particular pitot-static system and its cockpit airspeed indicator (ASI) will experience some small errors. The main two are:

1. **Instrument error** – resulting from poor design and construction of the ASI itself, or from friction within it.
2. **Position error** – resulting from sensing errors inherent in the position on the aircraft of the static vent and the pitot tube. Their position with respect to the airflow is critical and may lead to somewhat incorrect readings when the airflow pattern is disturbed at certain airspeeds, angles of attack, or wing flap settings.

The pilot can correct the reading of indicated airspeed shown on the ASI by using a calibration table (found in the Pilot's Operating Handbook for the aeroplane) to obtain a value known as **rectified airspeed (RAS)** or **calibrated airspeed (CAS)**.

NOTE Rectified airspeed is the term commonly used in the United Kingdom, whereas calibrated airspeed is used in the United States of America, many European countries, Australia and New Zealand. Navigation computers may be labelled with either RAS or CAS, or both.

The calculated RAS figure is what the ASI would read if the particular airspeed indicator system was perfect. RAS is therefore more accurate than IAS and, if you have taken the trouble to calculate RAS, it should be used in preference to IAS in navigation calculations.

The instrument and position errors of an airspeed indicator system are usually no more than a few knots and, for our purposes at PPL level, we can generally assume that indicated airspeed (IAS) and rectified airspeed (RAS) are equal. To remind you we will occasionally write IAS (RAS).

Now complete **Exercises 2 – Speed-1**.



Relating True Airspeed to Indicated Airspeed

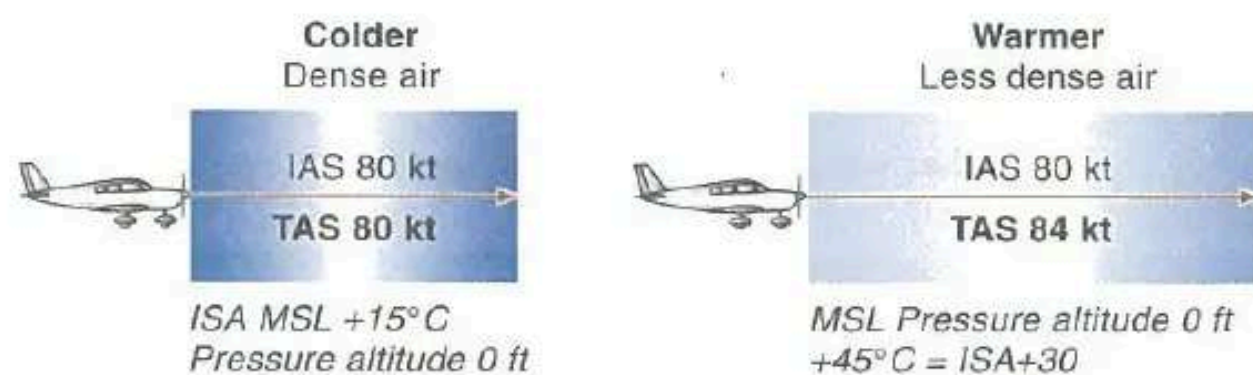
The aeroplane will rarely be flying in an air mass that has the same density as that under ISA MSL conditions (1,225 gm/cubic metre), the basis of the calibration of the airspeed indicator. Generally an aeroplane flying at altitude will be experiencing an air density significantly less than this, because air density (ρ) decreases with altitude. This will also be the case when there is an increase in temperature.

The indicated airspeed (even if it has been corrected for instrument and position errors to give rectified airspeed) will need to be further corrected for density error if the pilot is to know the exact speed at which the aeroplane is moving through the air – the true airspeed.

Whereas the position and instrument error (if any) will be different for each ASI system, the density error applies equally to all systems because it is a function of the atmospheric conditions at that time and place.

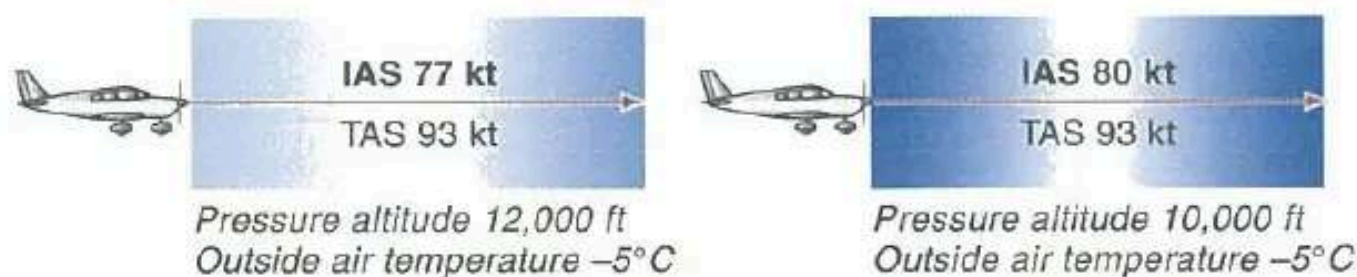
Air density varies for two main reasons:

1. **Temperature.** Cold air is dense, warm air is less dense, so on a warm day an aircraft must travel faster through the air for the same number of molecules per second to strike it, and for the same IAS to be indicated. TAS varying with temperature (for a constant IAS) is one reason why, on a warm day, an aeroplane requires longer take-off and landing distances. The TAS is higher to give you the same IAS, and the IAS is what you 'fly by'.



■ Figure 2-11 **Constant IAS (RAS): TAS varies with air temperature**

2. **Pressure.** The greater the pressure altitude (i.e. the lower the air pressure), the fewer the molecules per unit volume. For two aircraft with the same true airspeed (TAS), the higher aircraft will have a lower indicated airspeed (IAS) because it will strike fewer molecules of air per second than the lower aircraft.



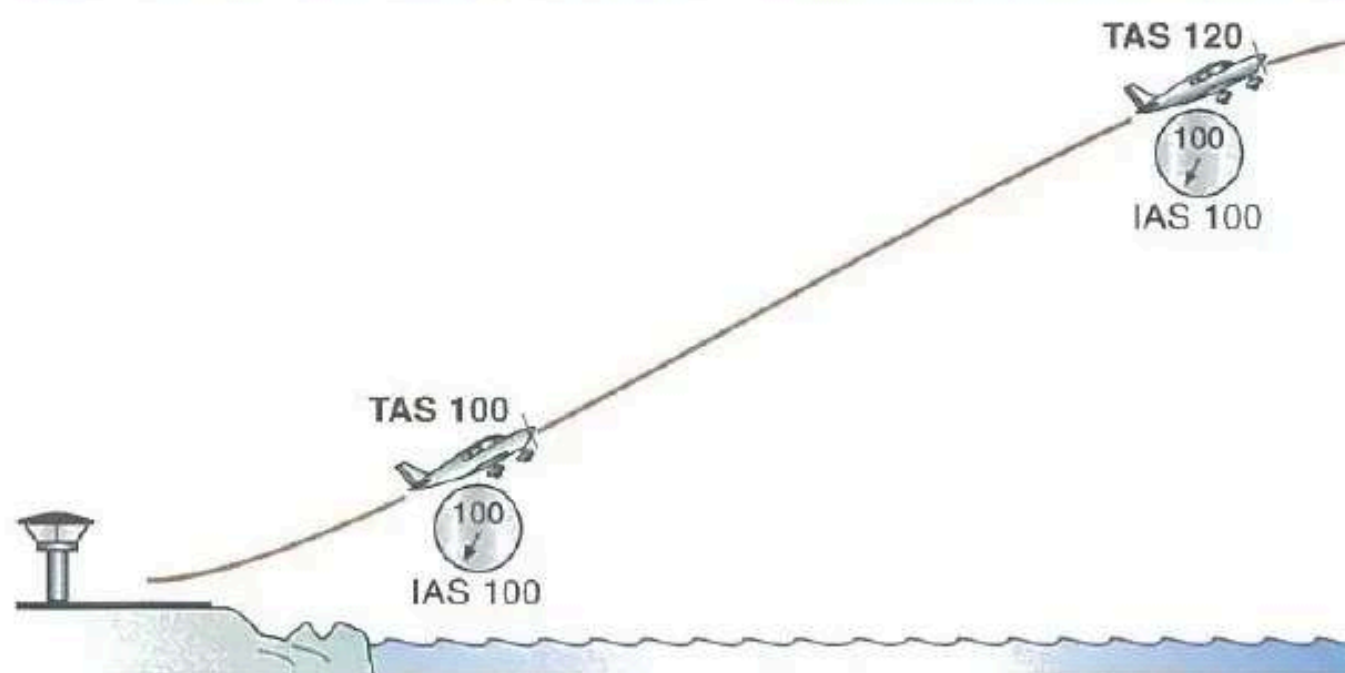
■ **Figure 2-12 Same TAS: the aircraft in less dense air has a lower IAS (RAS)**

Remember that IAS (RAS) is only equal to TAS under ISA MSL (International Standard Atmosphere mean sea level) conditions. At higher altitudes the IAS (or RAS) will be less than the TAS because the aircraft will be flying through the thinner air with an airspeed well in excess of that indicated on the ASI.

What Happens When We Climb at a Constant IAS?

As an aeroplane gains altitude, the air density (ρ) decreases. If we adopt the usual climb technique of maintaining a constant IAS (a constant dynamic pressure ' $\frac{1}{2} \times \rho \times V^2$ '), the decrease in ρ is made up by an increase in V (the true airspeed).

The higher we climb, when flying at a constant IAS, the greater the TAS.



■ **Figure 2-13 The higher we climb, the greater the TAS for a constant IAS (RAS)**

Using the Navigation Computer to Find TAS from IAS

Finding TAS from IAS is simple with a navigation computer. The principles illustrated here apply to most types available.

On the calculator side of most navigation computers is an **airspeed correction window**, which allows us to:

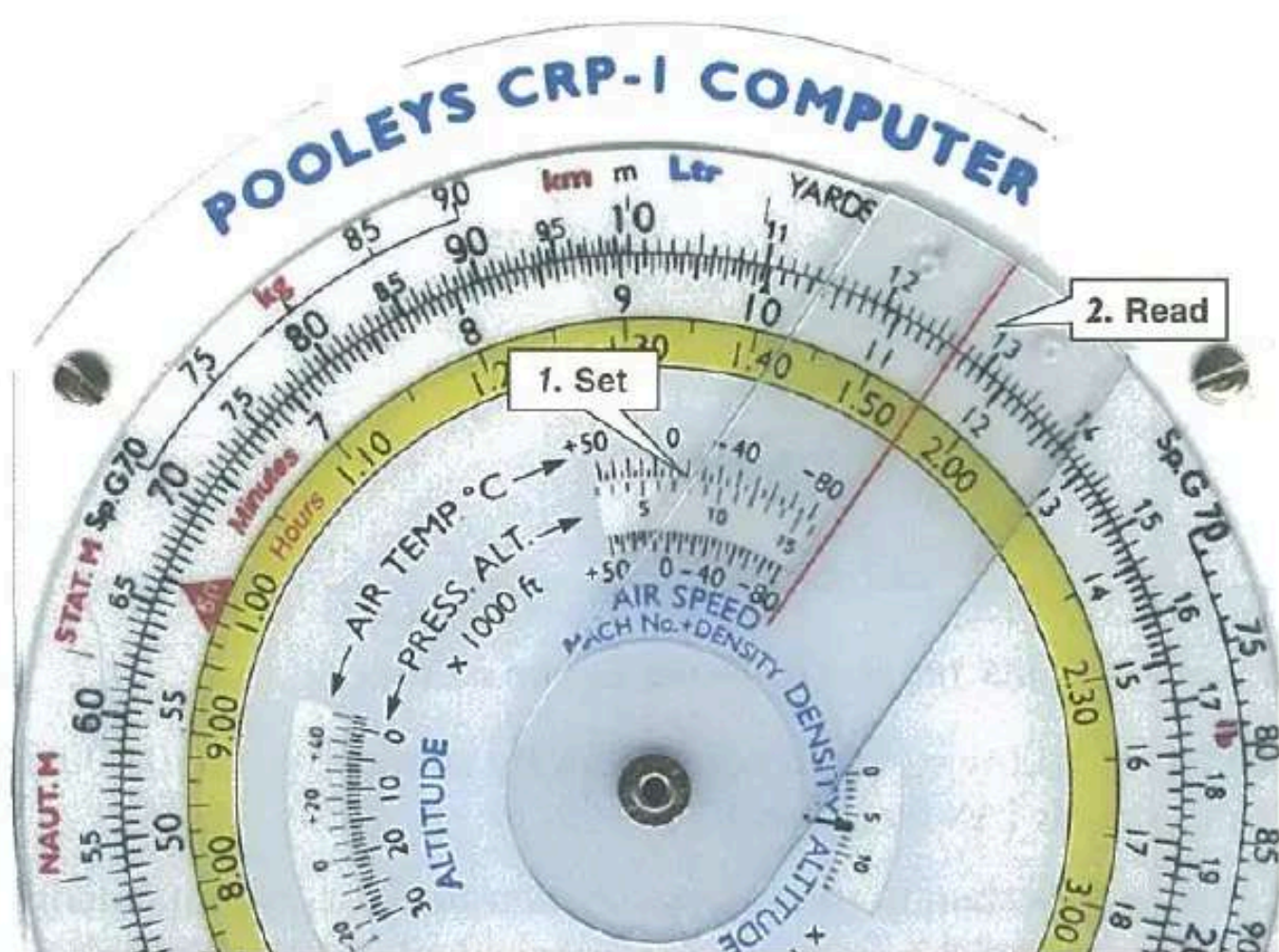
- match up the *pressure altitude* and the *air temperature* (the main factors determining density); and then
- from IAS (or RAS/CAS) on the inner scale, read off TAS on the outer scale.

NOTE On many navigation computers the inner scale is labelled RAS or CAS (or IAS) and the outer scale TAS. Check your own computer.

Ensure that you use the Celsius temperature scale, as all temperatures in UK meteorology forecasts (and those for most other countries) are given in degrees Celsius (formerly centigrade).

EXAMPLE 1

1. Temperature is -10°C at pressure altitude 8,000 ft.
2. RAS (CAS) 115 kt gives TAS 127 kt.



■ Figure 2-14 **Example 1, finding TAS from IAS (RAS) and air temperature on the navigation computer**

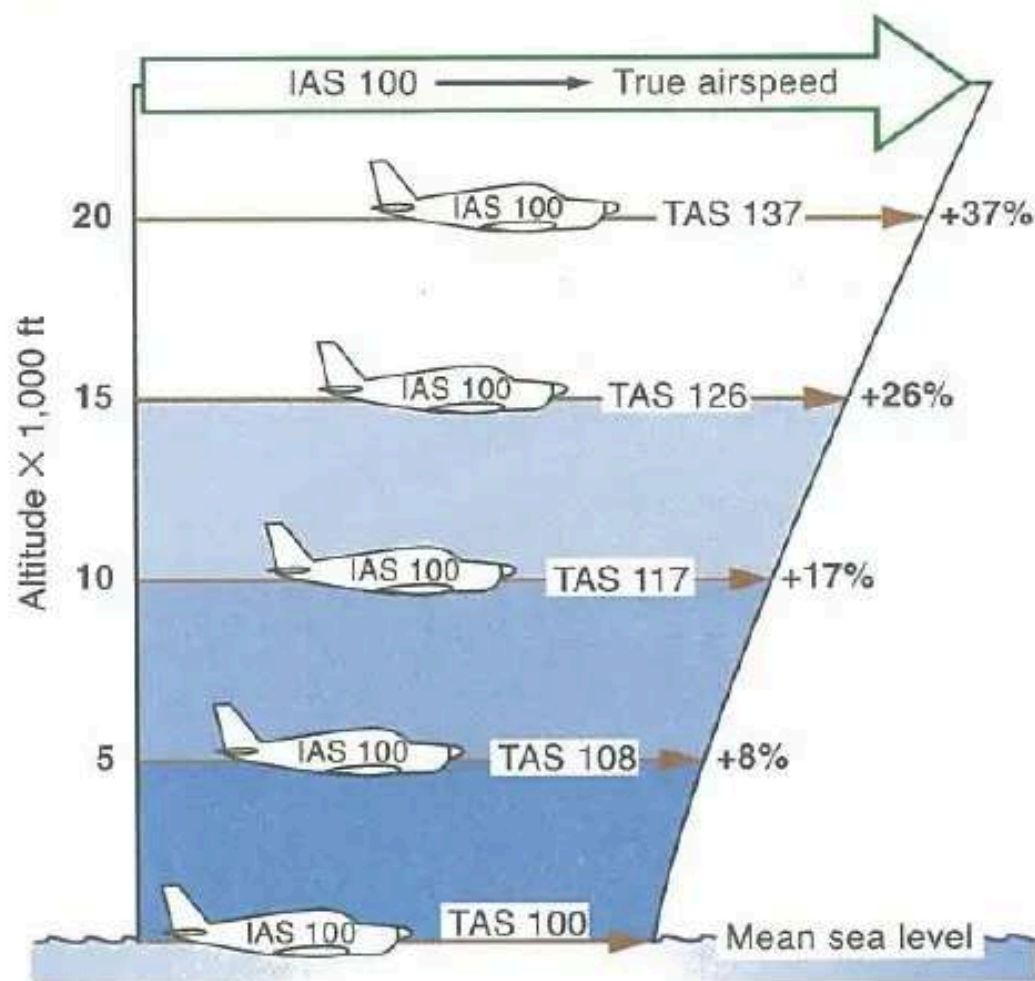
As a further example, line up the ISA MSL conditions of $+15^{\circ}\text{C}$ and pressure altitude 0. The computer will then show that under these conditions IAS (inner scale) and TAS (outer scale) are the same.

Variation of TAS with Altitude

Assume that the recommended climb speed for your aeroplane is 100 kt IAS. Using your navigation computer, see if you can come up with similar answers for the TAS as we have in Figure 2-15, for

a climb at IAS 100 kt from MSL to 20,000 ft. (Assume standard atmosphere conditions, where temperature decreases by 2°C for each 1,000 ft climbed.)

Pressure altitude	Temp.	IAS/RAS	TAS
20,000 ft	−25°C	100 kt	137 kt
15,000 ft	−15°C	100 kt	126 kt
10,000 ft	−5°C	100 kt	117 kt
5,000 ft	+5°C	100 kt	108 kt
ISA MSL	+15°C	100 kt	100 kt



■ Figure 2-15 IAS 100 kt: TAS increases with altitude

NOTE At 5,000 ft, TAS exceeds IAS by about 8%. At 10,000 ft, TAS exceeds IAS by about 17%.

These are handy figures to remember for rough mental calculations and also for when experienced pilots are talking about the speeds at which their aeroplanes ‘true-out’. If you are cruising at 5,000 ft with IAS 180 kt showing on the airspeed indicator, then your TAS will be approximately 8% greater (8% of 180 = 14), i.e. 194 kt TAS.

Now complete **Exercises 2 – Speed-2**.



For a Constant TAS, What IAS is Required?

It is interesting to compare what indicated airspeed will be shown in the cockpit if a constant true airspeed is required at various

levels. See if you can obtain the same answers as we do for IAS with a constant true airspeed of 200 kt at various pressure altitudes. Once again, assume a standard atmosphere to be present.

Pressure Altitude	Temp.	TAS	IAS/RAS
20,000 ft	−25°C	200 kt	146 kt
15,000 ft	−15°C	200 kt	159 kt
10,000 ft	−5°C	200 kt	172 kt
5,000 ft	+5°C	200 kt	185 kt
ISA MSL	+15°C	200 kt	200 kt

To fly the same TAS, IAS will decrease with higher altitudes as the air density decreases.

So, for the same TAS, the greater the pressure altitude, the lower the IAS.

The higher an aircraft flies, the more the TAS exceeds the IAS.

Variation of TAS with Outside Air Temperature (OAT)

Temperature at the one level in the atmosphere will vary from place to place and from time to time. Since temperature affects air density, it will also affect the relationship between IAS and TAS.

1. The mean sea level situation if temperature varies:

	Temp.	IAS/RAS	TAS	
Pressure altitude 0 ft	ISA+20 = +35°C	100 kt	104 kt	(less dense air)
	ISA+10 = +25°C	100 kt	102 kt	
	ISA = +15°C	100 kt	100 kt	
	ISA−10 = +5°C	100 kt	98 kt	
	ISA−20 = −5°C	100 kt	96 kt	(more dense air)

The less dense the air, the greater the TAS, compared to IAS (RAS).

2. The situation at pressure altitude 10,000 ft, if temperature varies:

	Temp.	IAS/RAS	TAS	
Pressure altitude 10,000 ft	ISA+20 = +15°C	100 kt	121 kt	(less dense air)
	ISA+10 = +5°C	100 kt	119 kt	
	ISA = −5°C	100 kt	117 kt	
	ISA−10 = −15°C	100 kt	115 kt	
	ISA−20 = −25°C	100 kt	113 kt	(more dense air)

True airspeed is important for navigation and flight planning because TAS is the actual speed of the aeroplane through the air mass.

More IAS to TAS Computer Calculations

EXAMPLE 2 At FL70, OAT -5°C , IAS (RAS) 105 kt. Find the TAS.

Working:

Set the pressure altitude 7,000 ft against -5°C OAT in the true airspeed (TAS) window.

Then, against RAS 105 kt on the inner scale, read off TAS 115 kt on the outermost scale.

TAS is 115 kt.

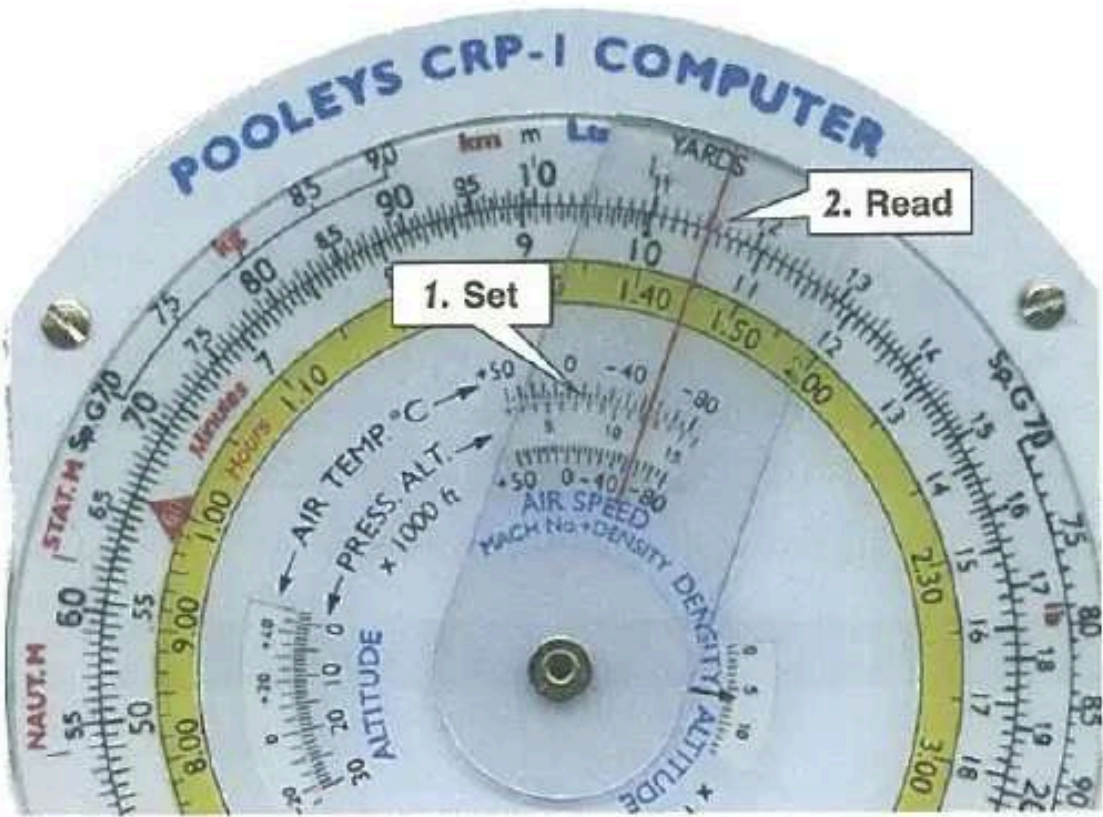


Figure 2-16 Example 2

In some situations the required pieces of information are not always given directly, but have to be derived first from other information which is provided. We will take an example from a high flight level to illustrate the widening gap between IAS (RAS) and TAS as an aeroplane climbs.

EXAMPLE 3 A turboprop plans on flying at flight level 280 (FL280), where the temperature is forecast to be $\text{ISA}+10^{\circ}\text{C}$. If its rectified airspeed (RAS) will be 150 kt, what TAS can be expected?

Working:

FL280 is pressure altitude 28,000 ft.

$$\begin{aligned} \text{At FL280, ISA} &= +15 - (2 \times 28) \\ &= +15 - 56 \\ &= -41^{\circ}\text{C} \\ \text{so ISA}+10 &= -41 + 10 \\ &= -31^{\circ}\text{C} \end{aligned}$$

In the computer *airspeed* window, set *pressure altitude* 28,000 against OAT -31°C .

Against RAS 150 kt, read off TAS 240 kt on the outer scale.

Finding the Required IAS to Achieve a Particular TAS

The usual in-flight problem is to determine the TAS from the indicated airspeed read off the ASI. Sometimes, however, you need to be able to work these problems in reverse, say to achieve a certain desired TAS or GS for flight planning purposes, when you will start with these and work back to find the IAS (RAS/CAS) necessary to achieve this.

EXAMPLE 4 Cruising at FL100 and temperature ISA -10 , what is the required RAS (CAS) to give you a true airspeed of 200 kt?

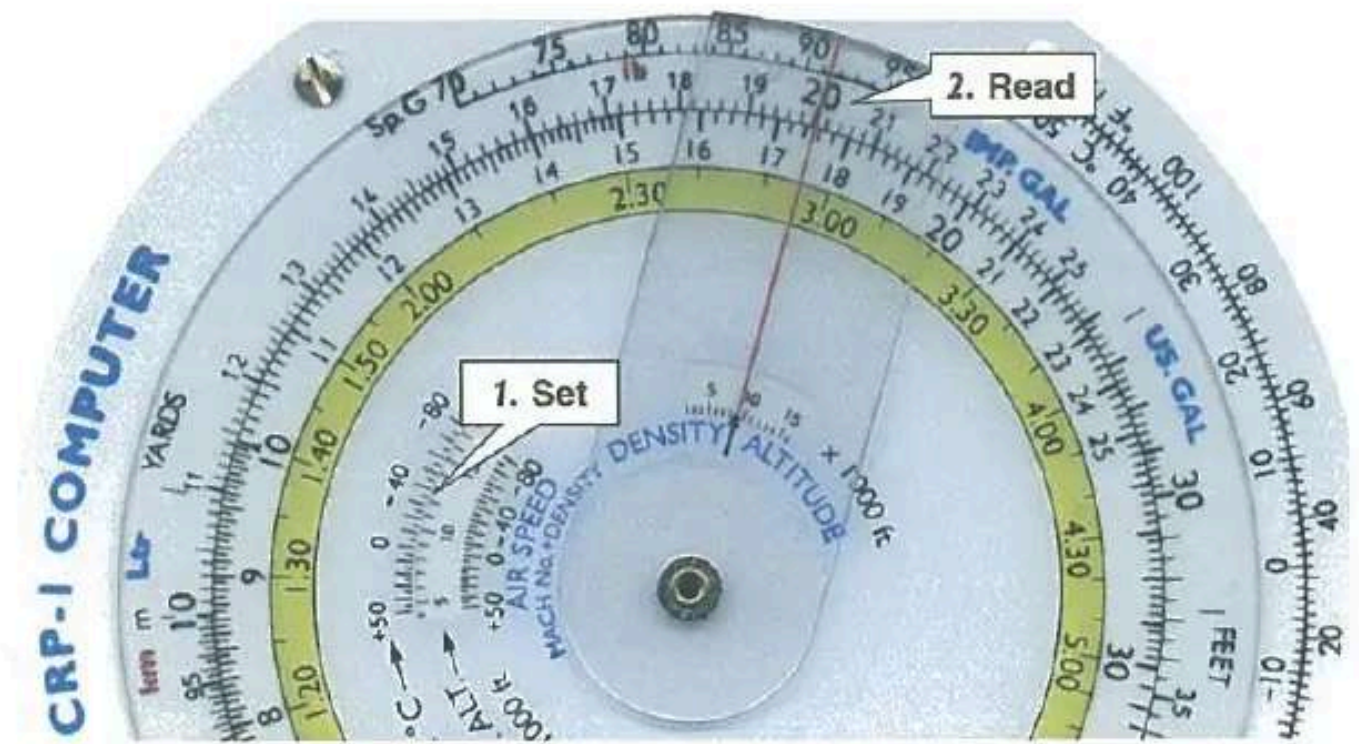
Working:

Pressure altitude is 10,000 ft,

where $\text{ISA} = +15 - (2 \times 10)$
 $= +15 - 20$
 $= -5^{\circ}\text{C}$
so $\text{ISA}-10 = -5 - 10$
 $= -15^{\circ}\text{C}$

In the computer *airspeed* window, set *pressure altitude* 10,000 against OAT -15°C .

Against TAS 200 on the outer scale, read off RAS 175 kt on the inner scale.



■ Figure 2-17 Example 4

Airspeed Terminology

IAS: Indicated airspeed.

RAS OR CAS: Rectified airspeed or calibrated airspeed. Generally approximately equal to indicated airspeed.

TAS: True Airspeed. At higher altitudes, TAS is usually greater than IAS/RAS/CAS).

Now complete **Exercises 2 – Speed-3**.



Direction

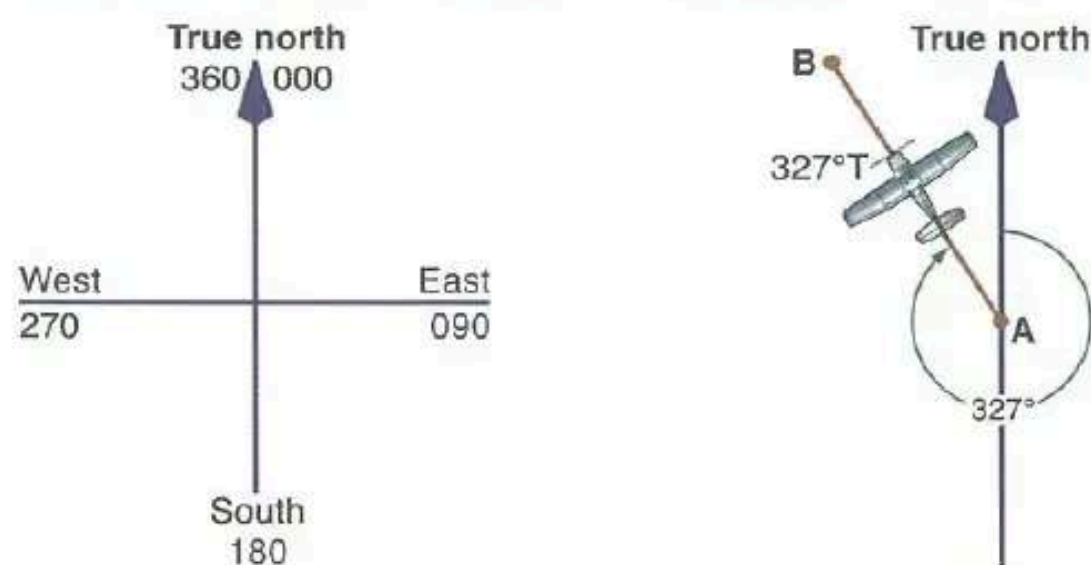
Direction is obviously of prime importance to accurate navigation. As aircraft navigate with reference to the earth's surface, we will begin with a brief look at the earth itself.

There is a geographical axis passing through two physical points on the surface of the earth about which the planet rotates. These points are the *geographic* North and South Poles. Any 'straight' line drawn around the earth's surface joining these two points is aligned in a true north–south direction.

By convention, the basic reference direction is **north**, and other directions are measured clockwise from this reference in degrees ($^{\circ}$). Since there are 360° in a circle, **east** is described as 090° , **south** as 180° , **west** as 270° , and **north** as 000° or 360° . Any direction (be it the desired track of an aeroplane, or the direction from which the wind is blowing) can be defined in this way.

True Direction

If direction is described with reference to **true north** (the direction to the geographic North Pole), it is called the **true direction**, symbolised by 'T'. East is therefore written as 090°T or 090T . The track between town A and town B illustrated below is 327°T .



■ Figure 3-1 **True direction**

A more approximate means of describing direction is using the **cardinal points**, which are the four chief directions of north, south, east and west – further divided by the **quadrantal points** north-east, south-east, south-west and north-west. If necessary, these can be divided even further to give, for instance, NNW (nor-nor-west). Obviously, the 360° method is superior for aeronautical navigation.

True direction, however, is a problem for pilots, because most aeroplanes do not have an instrument that can determine the

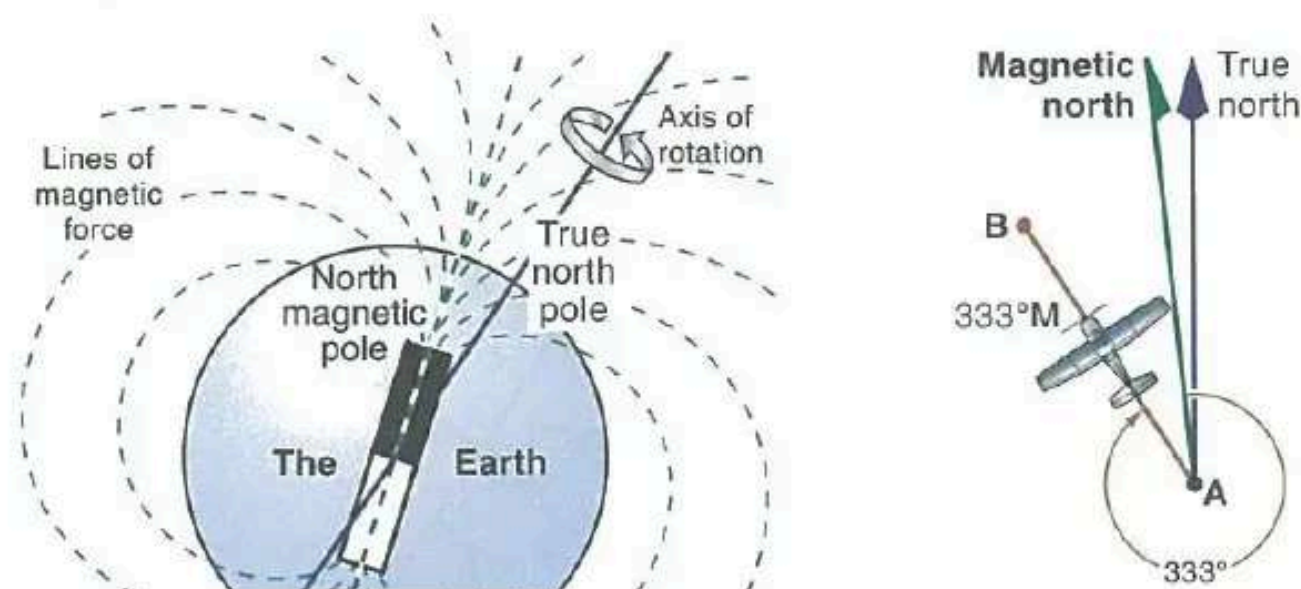
direction of true north. The magnetic compass, the prime source of directional information in the cockpit, aligns itself with **magnetic north**, rather than with true north.

NOTE As you will see, this statement does not hold true if there are extraneous magnetic fields, say due to radios or nearby magnetic materials, that are strong enough to affect the magnet within the compass. At this stage, we will assume that the magnet is influenced only by the earth's magnetic field and none other.

Magnetic Direction

Near to the true (geographic) North Pole is an area from which the earth's magnetic field emanates, known as the **north magnetic pole** to avoid confusion with the geographic pole. Similarly, there is a **south magnetic pole** located near the true South Pole.

A small magnet that is suspended and free to move will seek to align itself with these roughly north–south lines of magnetic force. This is the basis of the **magnetic compass**. If a compass card is attached to a magnetic 'needle', then the magnetic heading of an aeroplane can be read-off against a **lubber line**, or index, on the compass face.



■ Figure 3-2 **Magnetic direction**

A direction defined by reference to the north-seeking end of a magnetic compass is known as a **magnetic direction**. In Figure 3-2, the direction between the same two towns, A and B, is now described as 333°M .

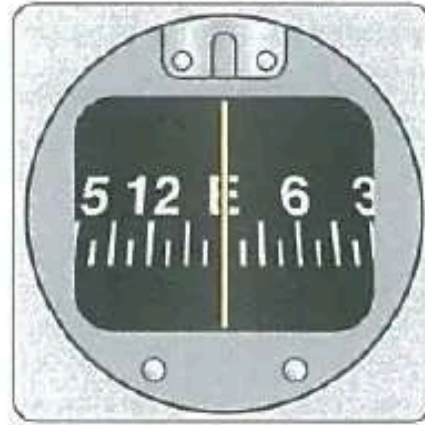
The actual direction between the two towns of course has not changed, only our description of it has, because of the two different reference directions, TN and MN. In the above case, 327°T and 333°M are the same physical direction described differently.

Direction Related to Magnetic North

Why introduce the complication of degrees related to magnetic north? Because the simple magnetic compass is the most reliable source of directional information. Instruments that display direction relative to true north are both complicated and

expensive, and subject to certain operational limitations not associated with the conventional magnetic compass. Even in the most sophisticated aircraft flying today, such as the Boeing 767, Airbus A320 and BAe 146, a simple magnetic compass is installed.

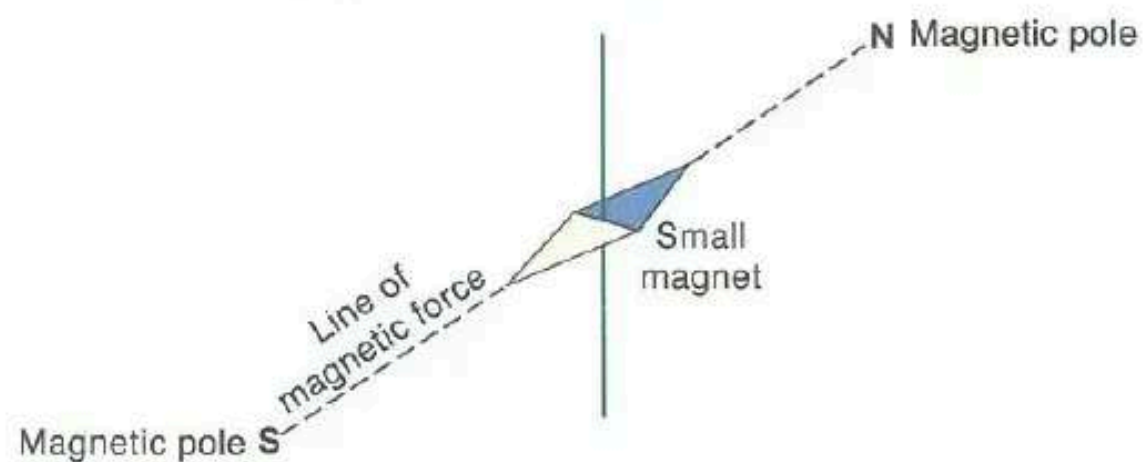
In most light aircraft, the magnetic compass is the primary source of directional information, to which other heading or direction indicators (often gyroscopic) are aligned.



■ Figure 3-3 **The magnetic compass**

To obtain accurate directional information from the magnetic compass, you must understand how it operates, and also its inaccuracies while the aeroplane is turning or changing speed. This is covered fully in the *Flight Instruments* section of Vol. 4 of this series. A summary follows here.

A bar magnet that is freely suspended horizontally will swing so that its axis points roughly north–south. The end of the magnet that points towards the earth’s north magnetic pole is called the **north-seeking pole** of the magnet.



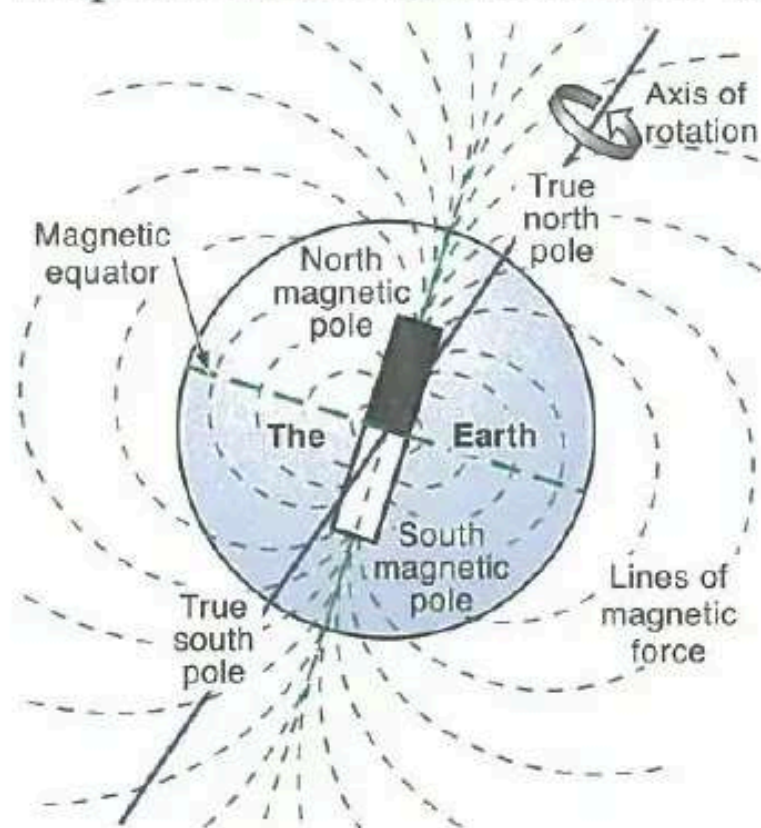
■ Figure 3-4 **Simple bar magnet**

The Earth’s Magnetic Field (Terrestrial Magnetism)

The earth acts like a very large and weak magnet. Its surface is covered by a weak magnetic field – lines of magnetic force that begin deep within the earth near Hudson Bay in Canada and flow towards a point deep within the earth near South Victoria Land in Antarctica. Because of their proximity to the *geographic* North and South Poles, the magnetic poles are referred to as the **north magnetic pole** and the **south magnetic pole**.

Variation

The latitude-longitude grid shown on charts is based on the geographic poles at either extremity of the earth's axis of rotation, so the meridians of longitude run true north and true south, and the parallels of latitude run true east and true west.



■ Figure 3-5 **The earth has a magnetic field**

Our small compass magnet, however, does not point exactly at true north and true south. A magnetic compass, if it is working perfectly and is influenced only by the earth's magnetic field, will point at the north *magnetic* pole, near Hudson Bay in Canada. At most points on the earth this is a different direction from true north. The angular difference between true north and magnetic north at any particular point on the earth is called the **magnetic variation** at that point.

If the magnet points slightly east of true north, the variation is *east*. If the magnet points to the west of true north, the variation is *west*. West variation is experienced over the entire UK.

Variation at any point on the earth is measured from true north to magnetic north. For example, a magnetic compass in London will point 2° west of true north, i.e. the magnetic variation is 2°W , since magnetic north lies 2° west of true north. In Liverpool the magnetic variation is $3\frac{1}{2}^\circ\text{W}$. In the area of the two towns A and B illustrated in Figure 3-2 it is 6°W ; 327°T and 333°M are one and the same direction when the magnetic variation is 6°W .

Variation is the angular difference from true north to magnetic north.

NOTE Because the earth's magnetic poles are not stationary, variation changes over time. In the British Isles variation reduces by 7–8 minutes annually (about 1° every 8 years).

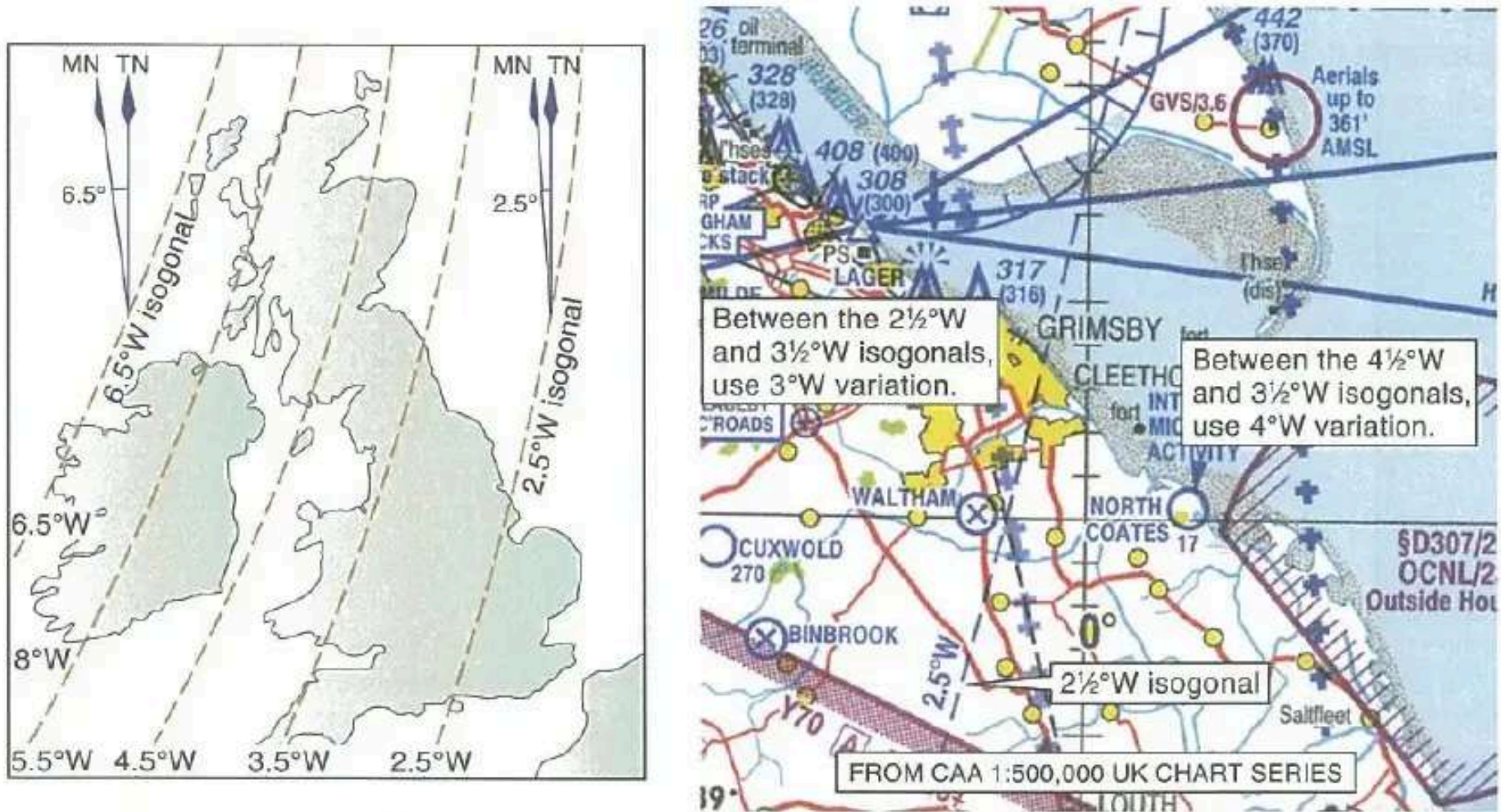
Isogonals

As well as the lines forming the latitude-longitude grids, maps have other lines joining places that have the same magnetic variation. These lines are known as **isogonals** or **isogonic lines**. On the UK 1:500,000 aeronautical chart, the isogonals are shown as dashed lines coloured blue.

The 2° west isogonal joins all the places having a variation of 2° west (e.g. Grimsby, Coningsby, Cranfield, Ryde). If you are anywhere on this line, then the message that your compass is giving you about magnetic north can be related to true north; your compass will point at magnetic north, which will be 2° west of true north.

Isogonals are lines on a chart joining places of equal magnetic variation.

To assist you in choosing the magnetic variation in your area, CAA 1:500,000 UK aeronautical charts show half-degree isogonals. Between these lines you use the appropriate whole number of variation, as shown in Figure 3-6.

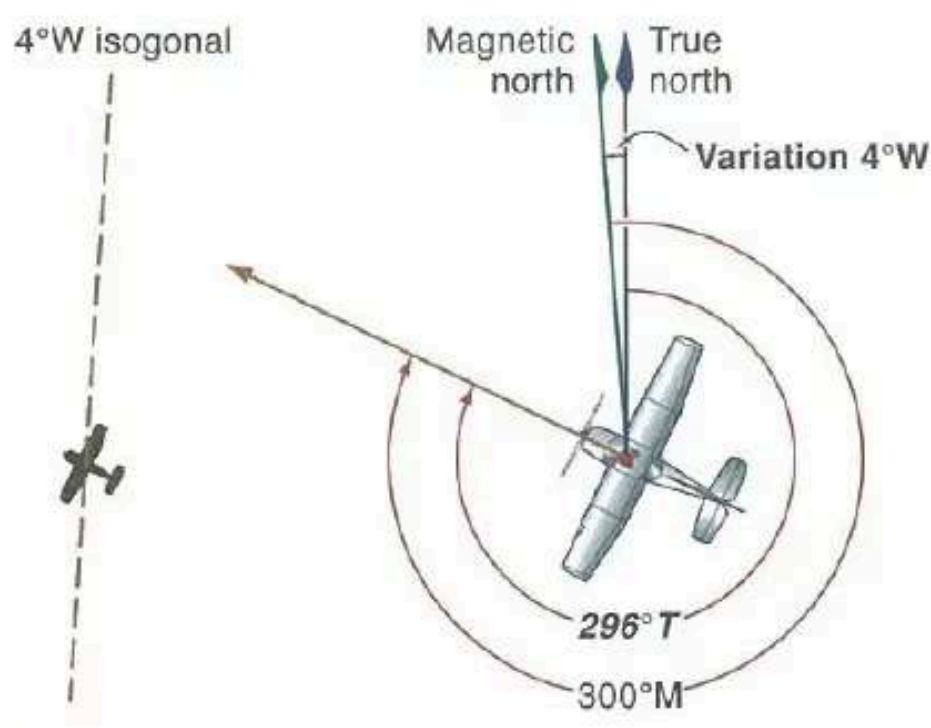


■ Figure 3-6 **Variation is the angle between true and magnetic**

If magnetic north is to the west of true north (west variation), then °M will exceed °T. Conversely, if magnetic north is to the east of true north (east variation), then °M will be less than °T. An easy way to remember the relationship between true and magnetic is:

Variation west, magnetic best. Variation east, magnetic least.

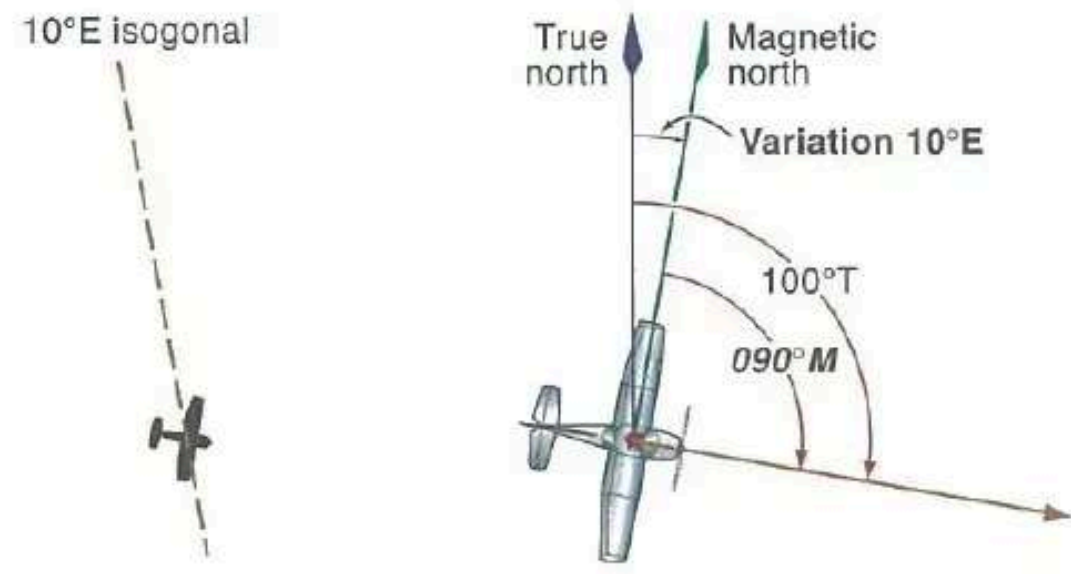
EXAMPLE 1 While flying on the Continent, you are steering your aeroplane on a heading of 300°M with reference to the magnetic compass. From an aeronautical chart you determine that magnetic variation in the vicinity is 4°W . What is the aeroplane's heading in $^{\circ}\text{true}$?



■ Figure 3-7 Variation west, magnetic best; answer 296°T

EXAMPLE 2 Convert 100°T to a magnetic direction in an area where variation is 10°E .

$$\begin{array}{r} 100^{\circ}\text{T} \\ -10^{\circ}\text{E (Variation east, magnetic least)} \\ \hline 090^{\circ}\text{M (Answer)} \end{array}$$



■ Figure 3-8 Variation east, magnetic least; answer 090°M

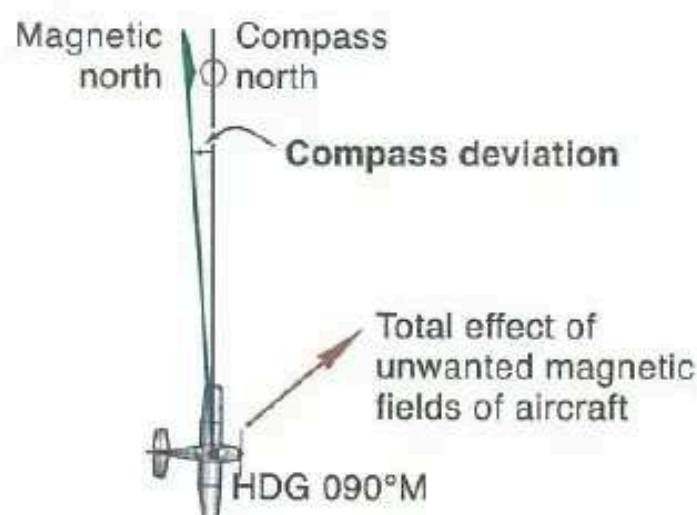
The Agonic Line

The isogonal that joins places that have zero variation (i.e. magnetic north and true north coincide) is called the **agonic line**. It passes through Europe.

Compass Deviation

Unfortunately, the magnet in the magnetic compass is affected not only by the magnetic field of the earth, but by any magnetic field that exists in its vicinity, such as the magnetic fields surrounding

the metal structure of the aeroplane, rotating parts in the engine, the radios, etc. The effect of these additional magnetic fields in a particular aeroplane is to deviate or deflect the compass from indicating magnetic north precisely. This imprecision is known as **compass deviation**.



■ **Figure 3-9 Compass deviation**

Deviation varies according to the heading that the aeroplane is on, since these unwanted extra magnetic fields are related to the aeroplane itself. If their resultant is diagonal to the longitudinal axis of the aeroplane (Figure 3-10) then, when the aeroplane is steering 045° , or its reciprocal 225° , it will be aligned with the earth's magnetic field and will not cause the compass needle to deviate. In other words, on these headings, compass deviation is zero.



■ **Figure 3-10 Compass deviation nil in this aeroplane on these headings**

If, on the other hand, the aeroplane is heading east, the alignment of the unwanted magnetic fields will deviate the compass as shown by the deviation card in Figure 3-13. The compass needle will then point towards a *compass north* that is slightly to the east of magnetic north in this case (by 1°). Even though the magnetic heading might be 090°M , the compass will indicate 089° .

An easy way to remember the relationship between magnetic and compass directions is:

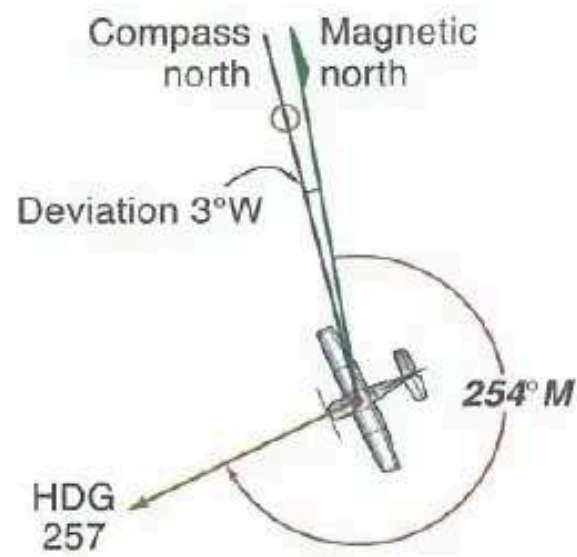
Deviation east, compass least. Deviation west, compass best.

EXAMPLE 3 An aeroplane is flying with a heading of 257° indicated on the magnetic compass in the cockpit. If, on that heading, deviation is 3°W, what is the aeroplane’s magnetic heading?

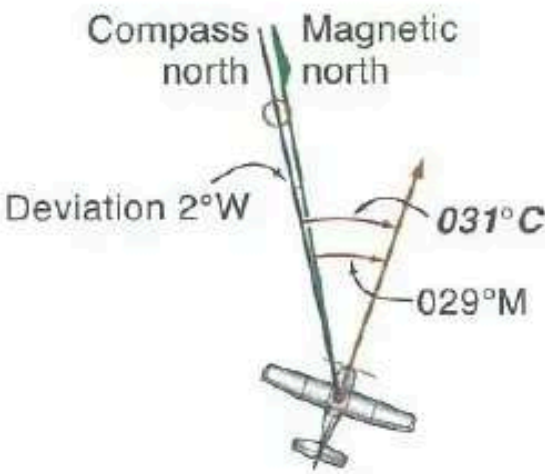
257°C (compass)

-3 (Deviation west, compass best)

254°M (Answer)



■ Figure 3-11 Example 3



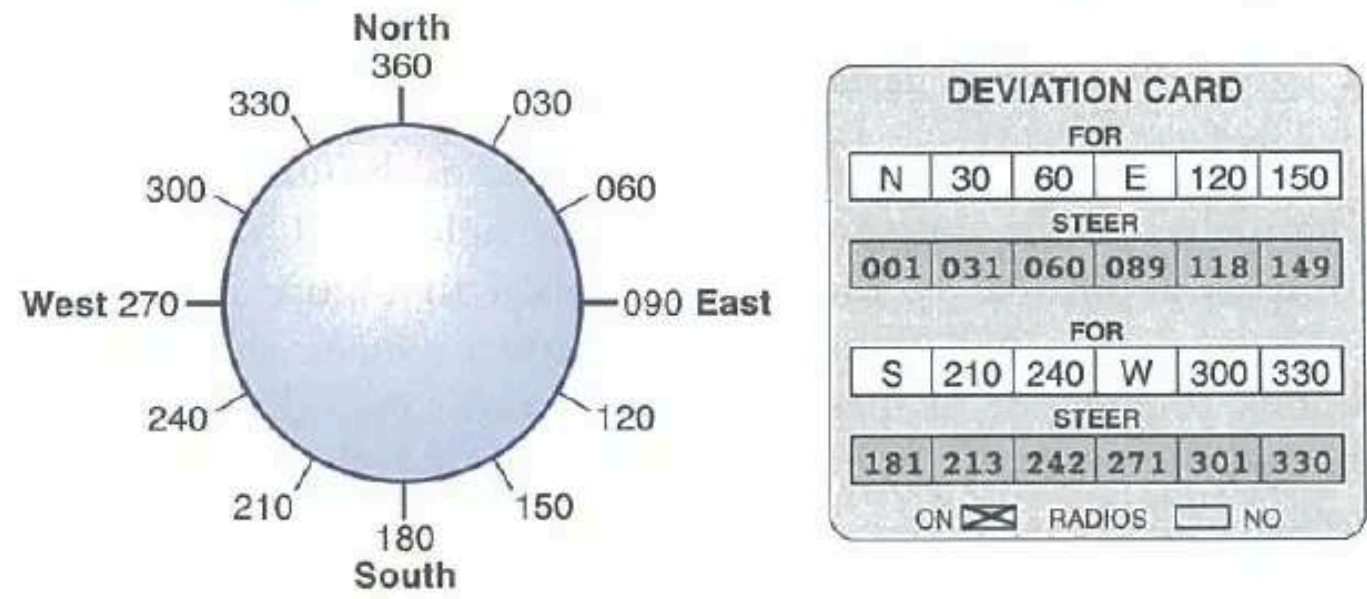
■ Figure 3-12 Example 4

EXAMPLE 4 What compass direction must be steered to achieve a magnetic heading of 029°M, if the compass deviation is 2° west?

Deviation west, compass best: $029^{\circ}\text{M} + 2 = 031^{\circ}\text{C}$ (Answer)

The Compass Deviation Card

Rather than continually having to carry out deviation corrections to the compass headings, a simpler approach is for each aircraft to have a small placard known as the deviation card displayed near the compass. This card shows the pilot what corrections need to be made to the actual magnetic compass reading (described as °C, for compass) in order to obtain the desired magnetic direction in °M. This correction usually involves no more than a few degrees (and in fact, the correction may be so small that the pilot does not apply it).



■ Figure 3-13 A compass rose and compass deviation card. To achieve a magnetic heading of 270° using this compass, steer 271°C.

Deviation is the angular distance from magnetic north to compass north for that particular compass, with the aircraft on that particular heading.

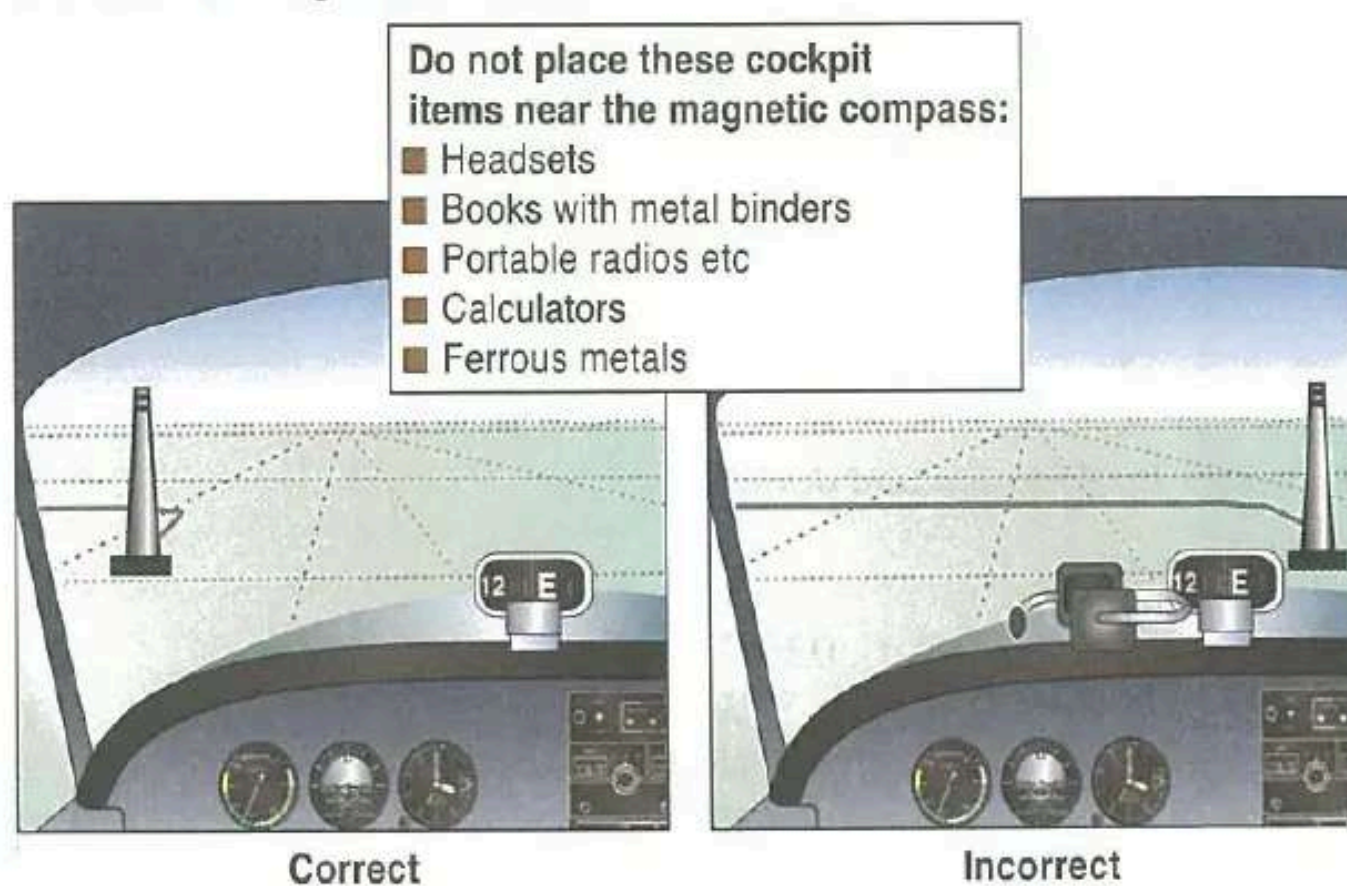
Precautions When Carrying Magnetic or Metal Goods

The compass deviation card is filled out by an engineer who has checked the compass in that particular aeroplane with it pointing in different directions. It may be done with electrical services off, or with them on (which is the normal in-flight situation). Electrical services, such as radios, often generate their own magnetic fields and may affect the compass indication.

The compass deviation correction card allows only for the magnetic influences in the aeroplane that were present when the engineer calibrated the compass in a procedure known as *swinging the compass*.

Any other magnetic influences introduced into the aeroplane at a time following the swinging of the compass will not be allowed for, even though they can significantly affect the compass. Therefore, as pilot, ensure that no metal or magnetic materials, such as metal pens, clipboards, books with metal binders, key rings, headsets, electronic calculators, transistor radios, or other devices that generate magnetic fields are placed anywhere near the compass.

Such magnetic or metal materials placed near the compass may introduce large and unpredictable errors. Many pilots have been lost or 'temporarily uncertain of their position' as a result of random deviations in the compass readings caused by these extraneous magnetic fields.

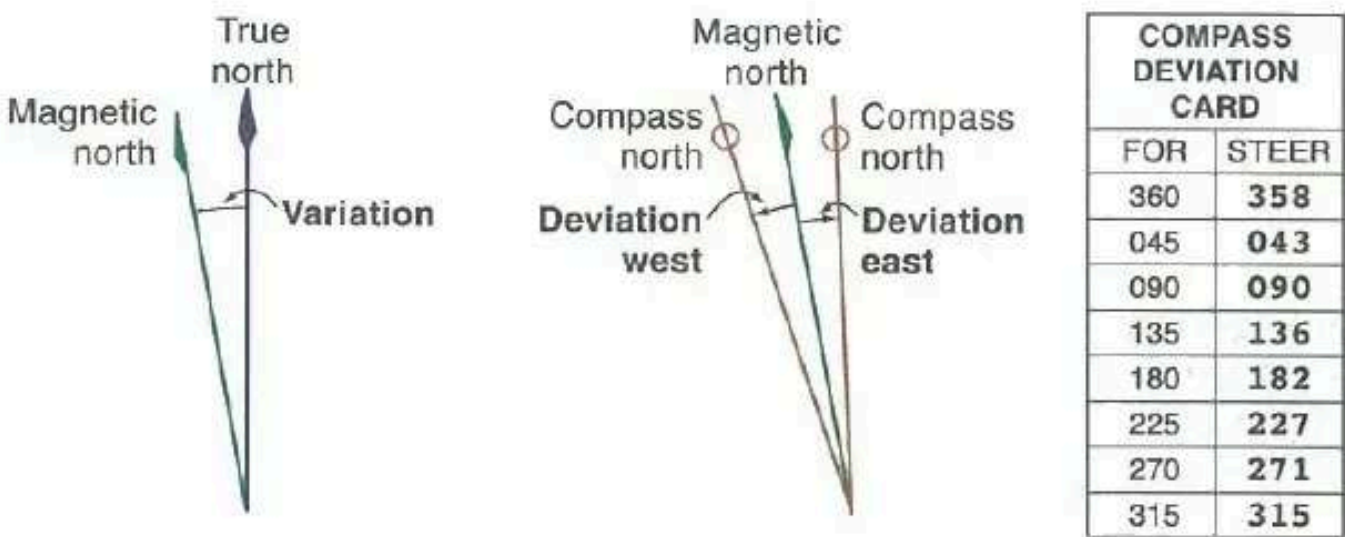


■ Figure 3-14 **Keep foreign objects away from the magnetic compass**

Relating True, Magnetic and Compass Headings

- 1. HDG(C) is the actual heading that you observe on the magnetic compass in the cockpit. It relates to compass north for that particular compass in that particular aeroplane on that particular heading.
- 2. Either refer to the deviation card, or apply 'Deviation east, compass least; deviation west, compass best', to convert HDG(C) to HDG(M). The aeroplane has not changed its direction in space, but its heading is now related to *magnetic north* rather than *compass north* (since the peculiarities of that particular compass have been accounted for and corrected). Usually this correction for deviation is insignificant and is often disregarded.
- 3. Apply **variation** (found on a chart) to HDG(M) to convert it to HDT(T), which is the heading of the aeroplane related to the geographic poles, known as the North and South Poles or the *true poles*. The correction for variation can be large and should always be applied (and in the right sense – applying 6°W variation as if it were 6°E variation will give you a 12° error). Remember: 'Variation east, magnetic least; variation west, magnetic best.'

The above process takes you from *degrees compass* to *degrees magnetic* to *degrees true*. It is just as easy to carry out the reverse process to go from °T to °M to °C.



■ Figure 3-15 Variation and deviation

EXAMPLE 5

Aircraft heading	020°T	
Variation	7°W	
Aircraft heading	027°M	
Deviation	2°E	
Aircraft heading	025°C	(Answer)

EXAMPLE 6

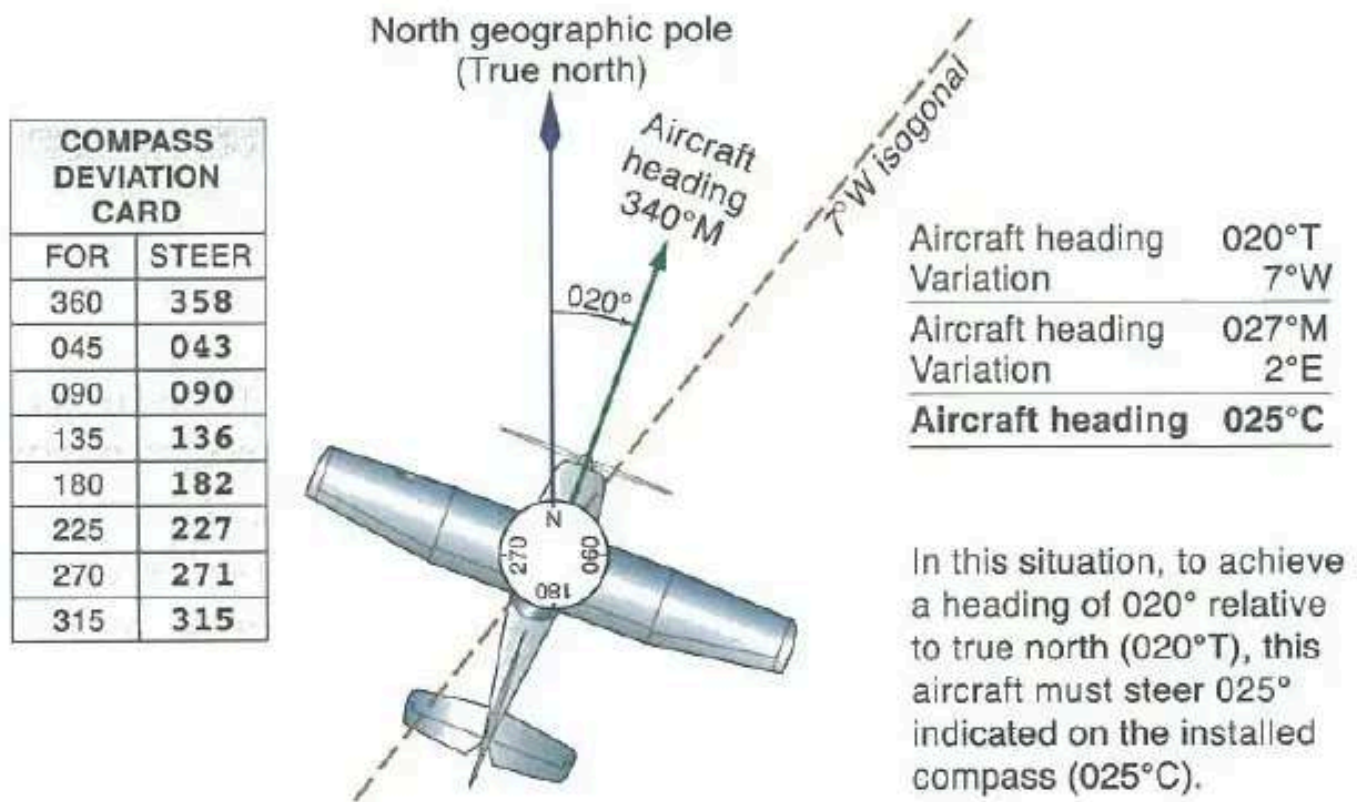
Aircraft heading025°C

Deviation2°E

Aircraft heading027°M

Variation7°W

Aircraft heading020°T (Answer)

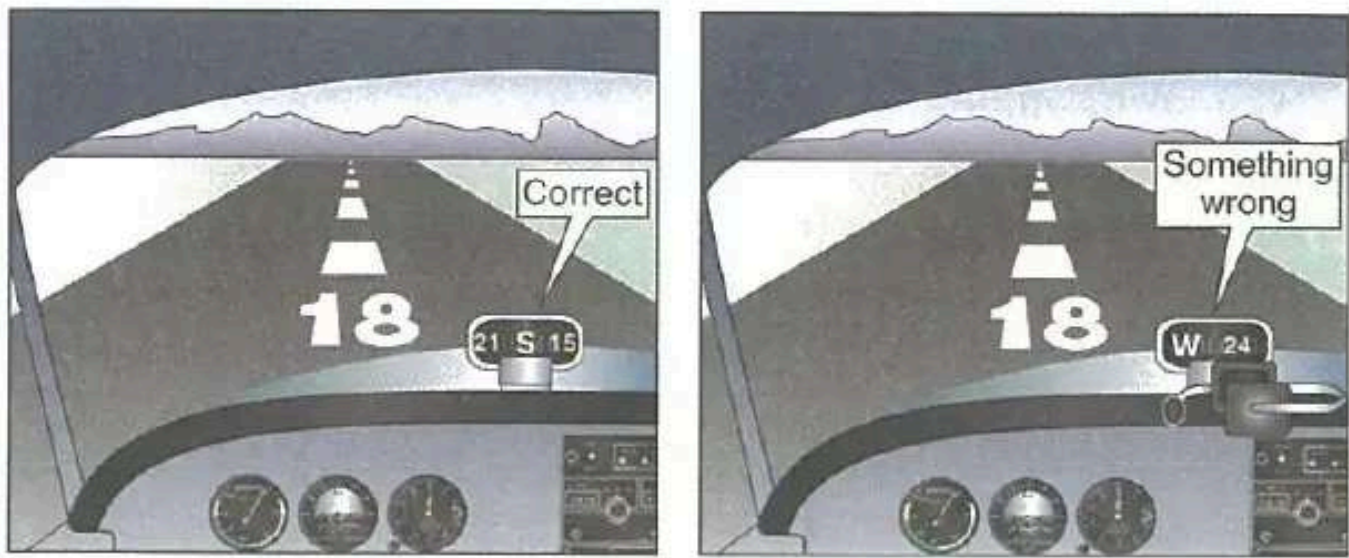


■ Figure 3-16 The relationship of compass heading, magnetic heading and true heading with variation and deviation

The maximum accuracy we consider practical in navigation is 1°, hence there is no need for us to consider the further subdivision of 1° into 60 minutes and each of these minutes into 60 seconds. Only apply a variation or deviation correction accurate to the nearest degree.

Pilot Serviceability Checks on the Magnetic Compass

- Pre-flight, check that the compass is securely installed and can be read easily. The liquid in which the magnet is suspended should be free of bubbles and should not be discoloured. The glass should not be broken, cracked or discoloured, and it should be secure.
- Check the position of the compass deviation card.
- Check that the compass indication is at least approximately correct. Runways are named according to their magnetic direction (e.g. a runway pointing 243°M is called Runway 24), so when pointing in the same direction as this runway, your compass should indicate this, at least approximately.
- When taxiing out prior to take-off, turn the aircraft left and right and check that the response of the magnetic compass is correct. Remember that the magnet should remain in the same north–south direction, and the aeroplane turn around it.

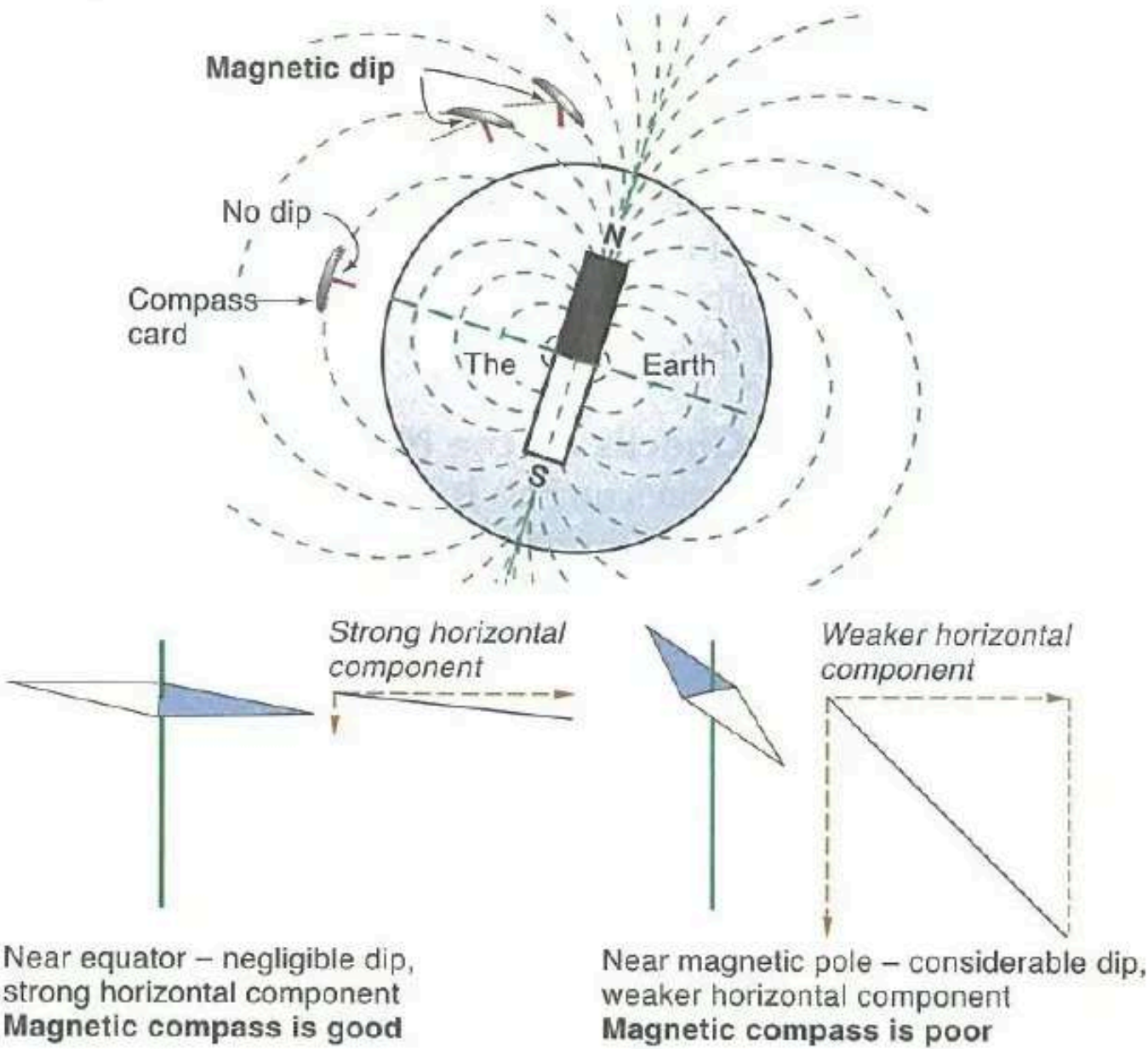


■ Figure 3-17 Always cross-check compass direction

Magnetic Dip and Compass Errors

The earth’s magnetic field is weak, and varies in strength and direction over the entire surface of the earth. The strength of the magnetic field can be resolved into two components:

- a horizontal component parallel to the surface of the earth, which is used to align the compass needle with magnetic north; and
- a vertical component, which causes the magnetic needle to dip down.



■ Figure 3-18 The horizontal component of the earth’s magnetic field is strong near the equator and weak near the poles

A magnetic compass indicates direction more accurately at middle and low latitudes than near the poles.

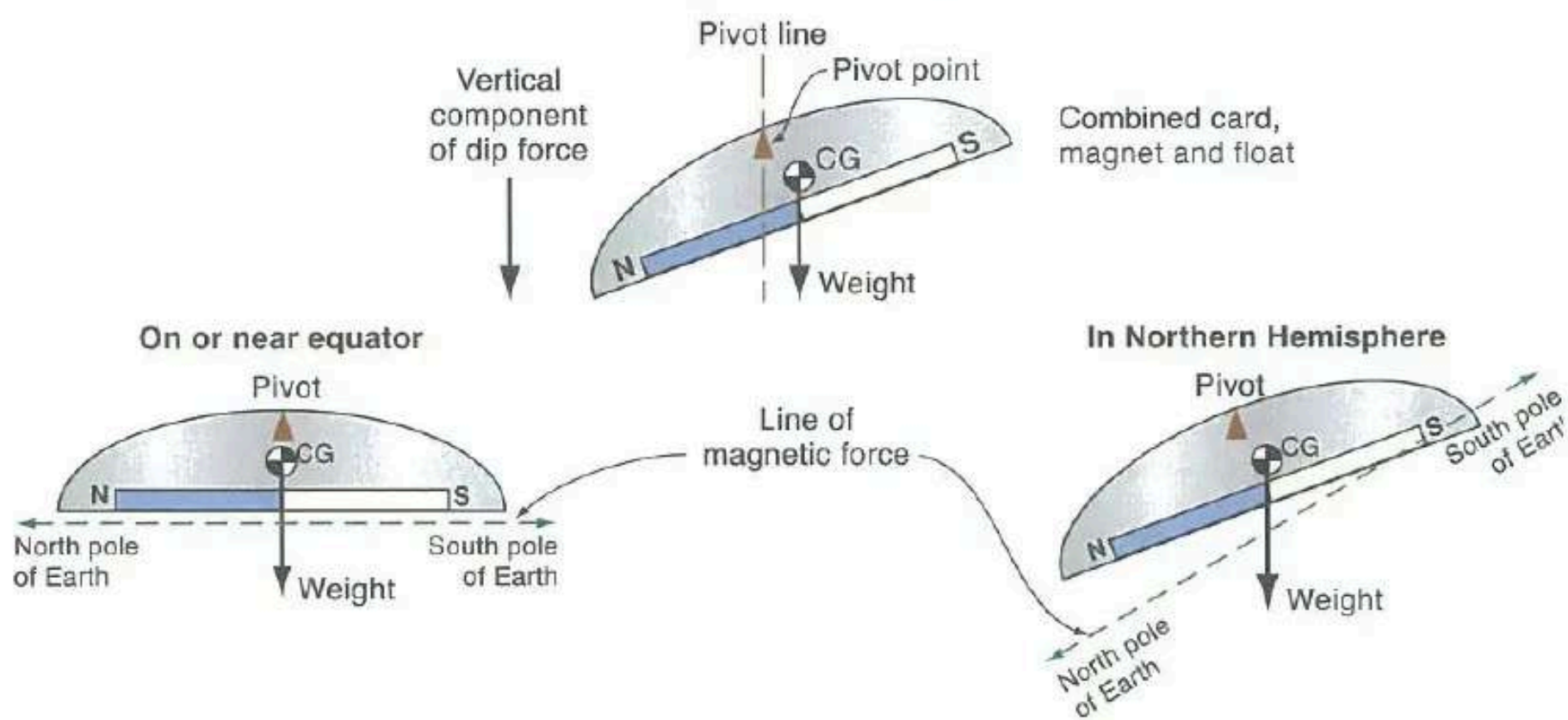
At the so-called ‘magnetic equator’ (roughly mid-way between the magnetic poles), the lines of magnetic force are parallel with the earth’s surface (i.e. they are horizontal). Consequently, the horizontal component of the earth’s magnetic field is at its strongest here and so the magnetic compass is stable and accurate in these areas.

At the higher latitudes near the magnetic poles, where the line of magnetic force run in through the earth’s surface, the vertical component of the earth’s magnetic field causing dip is stronger and the horizontal component parallel to the surface of the earth is weaker. This makes the compass less effective as an indicator of horizontal direction in the polar regions compared with its performance at the lower latitudes.

At latitudes higher than 60 degrees north or south (i.e. close to the poles than 60°N or S), the magnetic compass is not very reliable at all.

As a means of avoiding the compass needle *dipping* down in line with the magnetic force, it is suspended in a manner that displaces its centre of gravity (CG) from the pivot point at which it is suspended (and indirectly attached to the aeroplane structure).

The greater the dip, the more the needle dips down toward the nearer magnetic pole, and the more its CG is displaced. This causes the weight force to *balance* the dip force and to keep the needle approximately horizontal.



■ Figure 3-19 To minimise the effect of magnetic dip, the needle’s CG is displaced

Turning and Acceleration Errors

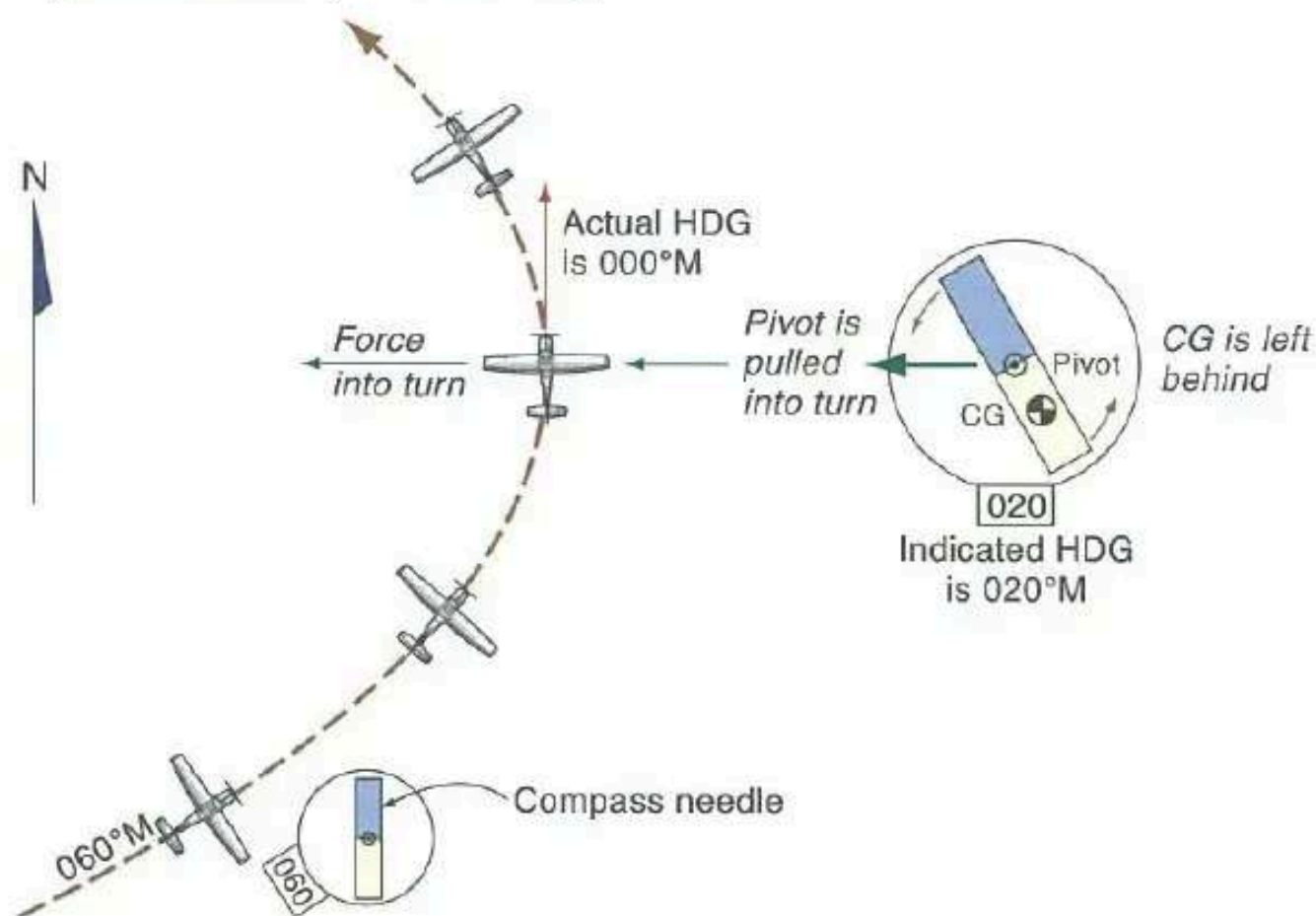
Any acceleration of the aeroplane will be transmitted to the compass needle via its pivot. The needle’s CG will tend to continue at its previous velocity and so will be left behind in a

acceleration, and will move ahead in a deceleration. In a turn, the aeroplane (and the pivot) is accelerating towards the centre of the turn, with the CG trying to 'fly off at a tangent'.

Indication errors in the magnetic compass and what causes them is covered in detail in Vol. 4 of *The Air Pilot's Manual*. It will suffice here to summarise the effect of these errors on the magnetic compass indications (which you should know for the examination in this subject):

TURNING ERRORS. are greatest when turning through headings of magnetic north or south (and zero when turning through headings of east or west).

When heading towards the nearer magnetic pole, the magnetic compass is 'sluggish' and will under-indicate the amount of turn (for both left and right turns). You should stop the turn before the magnetic compass indicates your desired heading. Once settled into steady straight and level flight, the compass will settle down and (hopefully) indicate your desired heading. If not, make minor adjustments to your heading.



■ Figure 3-20 **The compass is sluggish and lags behind when turning through north (in the northern hemisphere)**

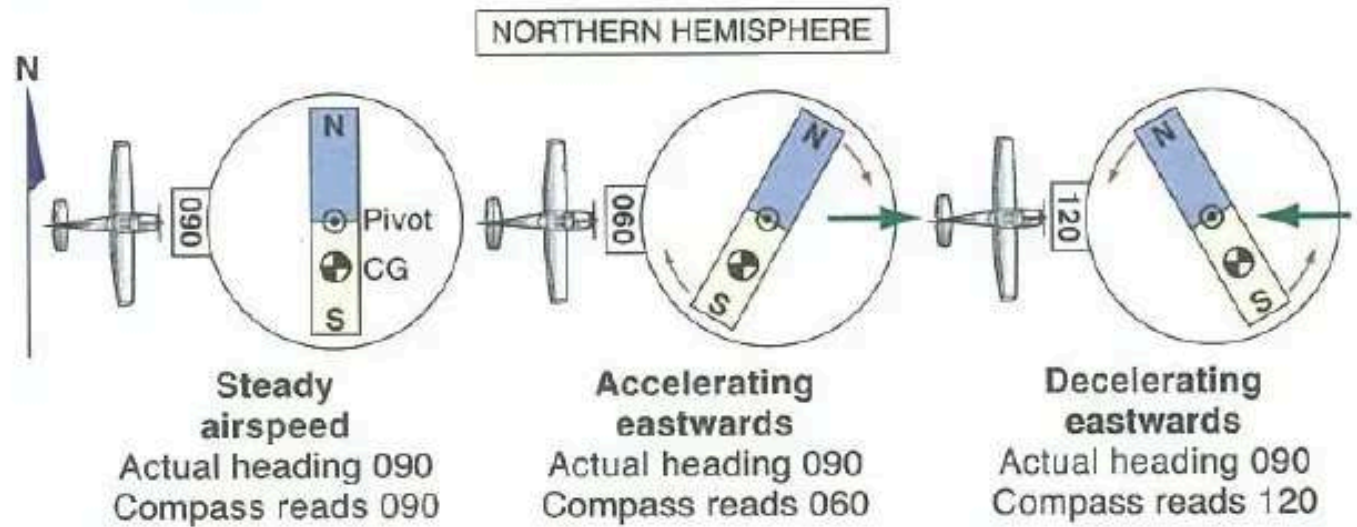
In the case illustrated in Figure 3-20, an aeroplane flying initially on a heading of 060°M is turning left through north. The CG tends to fly off at a tangent and so the compass card rotates anticlockwise, thereby under-indicating the amount of turn. For example, when the aeroplane is turning through 000°M (magnetic north), the compass is only indicating 020°M.

When heading towards the more distant magnetic pole, the magnetic compass is 'lively' and will over-indicate the amount of

turn. You should continue the turn through your desired heading as indicated on the magnetic compass during the turn. Once settled into steady straight and level flight, the compass will settle down and (hopefully) indicate your desired heading. If not, make minor adjustments to your heading.

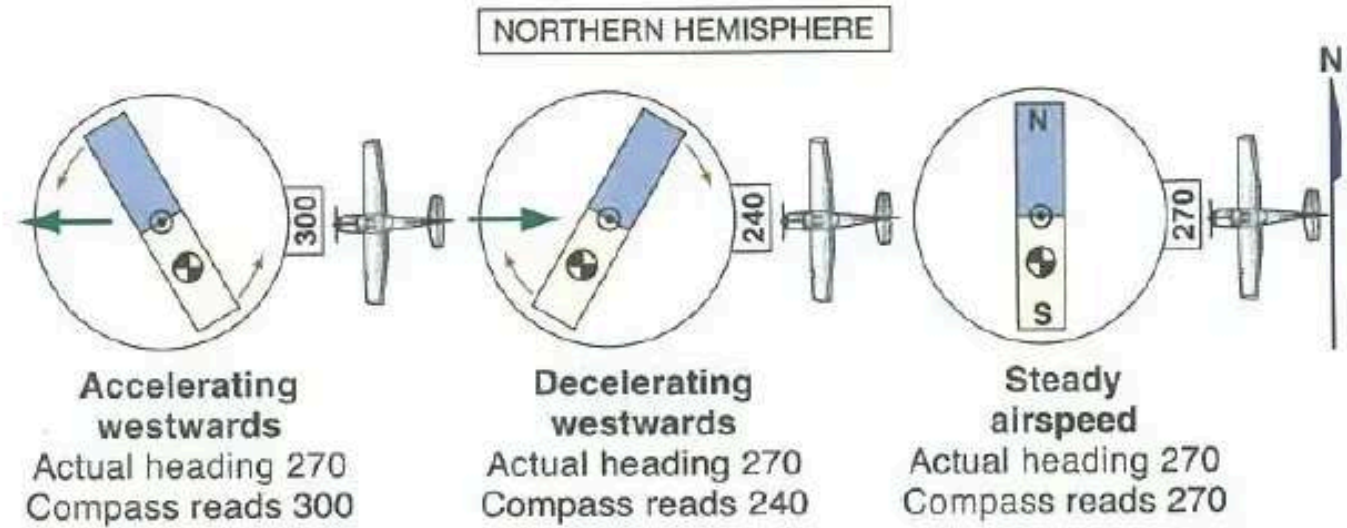
ACCELERATION ERRORS are maximum on east and west magnetic headings (and zero on north and south headings).

Acceleration produces a false indication of turning towards the nearer magnetic pole (i.e. towards north in the northern hemisphere). Increasing speed (accelerating) towards the east will cause the compass needle and its attached card to rotate clockwise causing a false indication of say 080° (instead of 090°). Another way of looking at this is that, on accelerating in an easterly direction, the centre of gravity is left behind, causing the compass card to rotate to the right. This gives a *false* indication of a turn towards north.



■ *Figure 3-21 Acceleration east produces a false indication of a turn towards the north*

Considering an acceleration on a westerly heading, again the centre of gravity is left behind, the compass card in this case turns anticlockwise and indicating (incorrectly) a turn toward north.



■ *Figure 3-22 Acceleration west produces a false indication of a turn towards north*

Conversely, deceleration on an easterly or westerly heading produces a false indication of turning towards the further magnetic pole (i.e. towards south in the northern hemisphere).

If the aeroplane is heading north or south, the pivot supporting the compass needle and the needle's centre of gravity are in line, and so the needle will not be displaced by accelerations or decelerations in the north–south directions.

On intermediate headings, the acceleration error will increase with the proximity of the aeroplane's heading to east or west.

EXAMPLE 7 You are flying in the UK. When turning from 150°M through south onto a heading of 220° using the magnetic compass, because it will over-indicate the amount of turn, you should continue the turn beyond a compass indication of 220 (say by 10°) and then level the wings and allow the compass to settle down.

EXAMPLE 8 You are flying in Scotland and accelerating from 80 kt to 150 kt on an easterly heading. Even though, by reference to a point on the horizon, you are still heading east, the needle of the magnetic compass will swing in a clockwise direction and indicate an apparent turn to the north. Once you attain a steady speed and allow the compass to settle down, it should indicate the correct heading again.

The Direction or Heading Indicator

Most light aircraft instrument panels include a gyroscopic direction indicator (DI). This instrument is also known as the heading indicator (HI) or directional gyro (DG).

Being a gyro-based instrument, its indication is steady compared to that of the magnetic compass but, due to the fact that the earth is rotating (at 15° per hour) and the direction indicator's axis is fixed in space by the gyroscope, the direction indicator has to be re-aligned with a known reference direction at regular intervals.

The magnetic compass is used as the reference for the direction indicator and so, when aligned with the compass, the DI will indicate the heading of the aircraft in degrees magnetic.

NOTE Do not align the direction indicator (DI) with the magnetic compass if you are changing speed or direction, as the magnetic compass will be experiencing acceleration or turning errors, i.e. keep the wings level and maintain a constant speed when aligning the DI with the compass.

One of the advantages of a direction indicator is that it is not subject to turning or acceleration errors. Its accuracy depends on it being correctly aligned with magnetic north, so this must be done when the magnetic compass is indicating correctly.

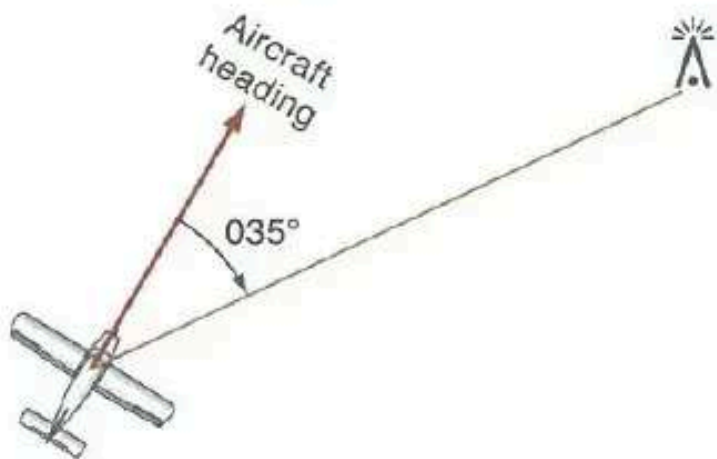


■ Figure 3-23
Direction indicator (DI)

Relative Bearings

It is usual to define the direction of an object from an aeroplane in terms of its **relative bearing**, or its direction relative to the nose (or heading) of the aeroplane.

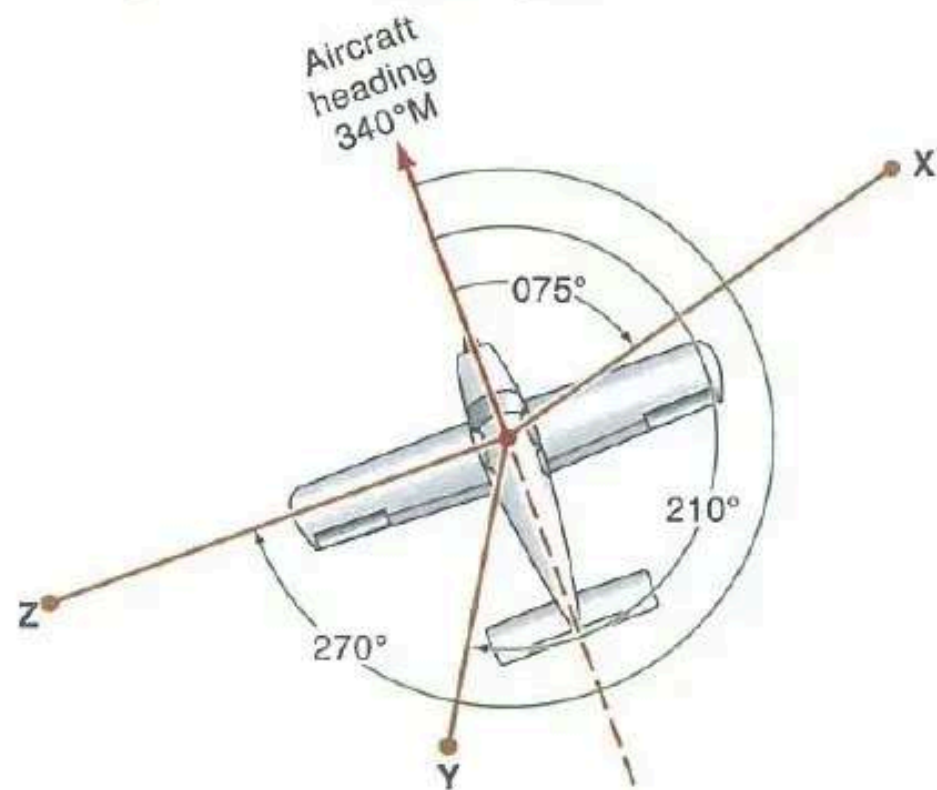
A relative bearing of an object from an aeroplane is its angular distance from the aircraft’s heading measured clockwise from the nose of the aeroplane from 000°REL through to 360°REL.



■ Figure 3-24 The radio mast bears 035° relative from the aeroplane

Converting Relative Bearings to Magnetic Bearings

To convert a relative bearing to a magnetic bearing (and vice versa), simply add the relative bearing of the object to the magnetic heading of the aircraft to obtain the magnetic bearing of the object from the aeroplane.



	AIRCRAFT HEADING		RELATIVE BEARING		MAGNETIC BEARING
Point X	340°M	+	075°	=	055°M
Point Y	340°M	+	210°	=	190°M
Point Z	340°M	+	270°	=	250°M

■ Figure 3-25 Relative bearings and magnetic bearings

Aircraft magnetic heading	±	Relative bearing of object from aircraft	=	Magnetic bearing of object from aircraft
--	----------	---	----------	---

If the answer works out to be in excess of 360°, then this means you have gone more than once around the complete circle. To achieve a usable answer simply deduct 360°, e.g. 372° is the same direction as 012°.

These are two different ways of describing the position of an object seen from the aeroplane:

- the **relative bearing** is related to the nose (HDG) of the aeroplane; and
- the **magnetic bearing** is related to magnetic north.

You will come across relative bearings later in your flying training when you study radio navigation aids in preparation for obtaining an Instrument Meteorological Conditions (IMC) rating. In radio navigation:

- a **fixed-card radio compass** uses relative bearings;
- a **radio magnetic indicator** (which has a radio compass needle superimposed on a compass card that indicates directions relative to magnetic north) uses magnetic bearings.

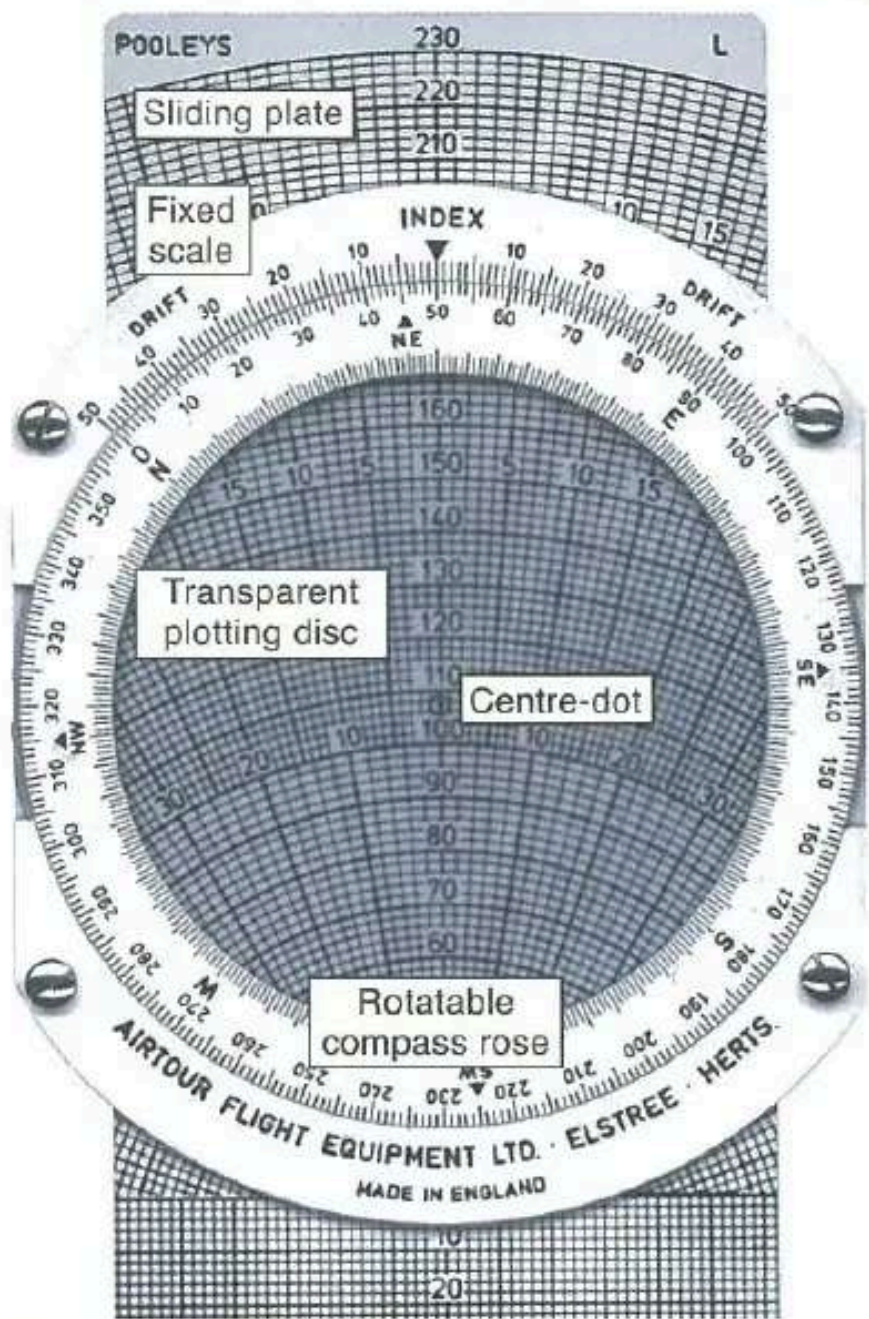
Now complete **Exercises 3 – Direction**.



Wind Side of the Navigation Computer

The slide navigation computer is a wonderful invention that allows you to handle navigation problems involving the triangle of velocities quickly and accurately. The three vectors in the triangle can be marked on the plotting disc so that they appear in the same relationship, one to the other, as in flight, making it easier to visualise the situation and check that the vectors have been applied correctly.

- Components of the wind side of a navigation computer are:
- A circular, rotatable compass rose (or azimuth) set in a fixed frame which is marked with an index at the top.
 - A transparent plastic plotting disc attached to the rotatable compass rose, marked with a centre-dot.
 - A slide plate printed with concentric *speed arcs* and radial *drift lines*. This plate slides through the frame and compass rose assembly, hence the name 'slide' navigation computer.



■ Figure 4-1 Wind side of a slide navigation computer

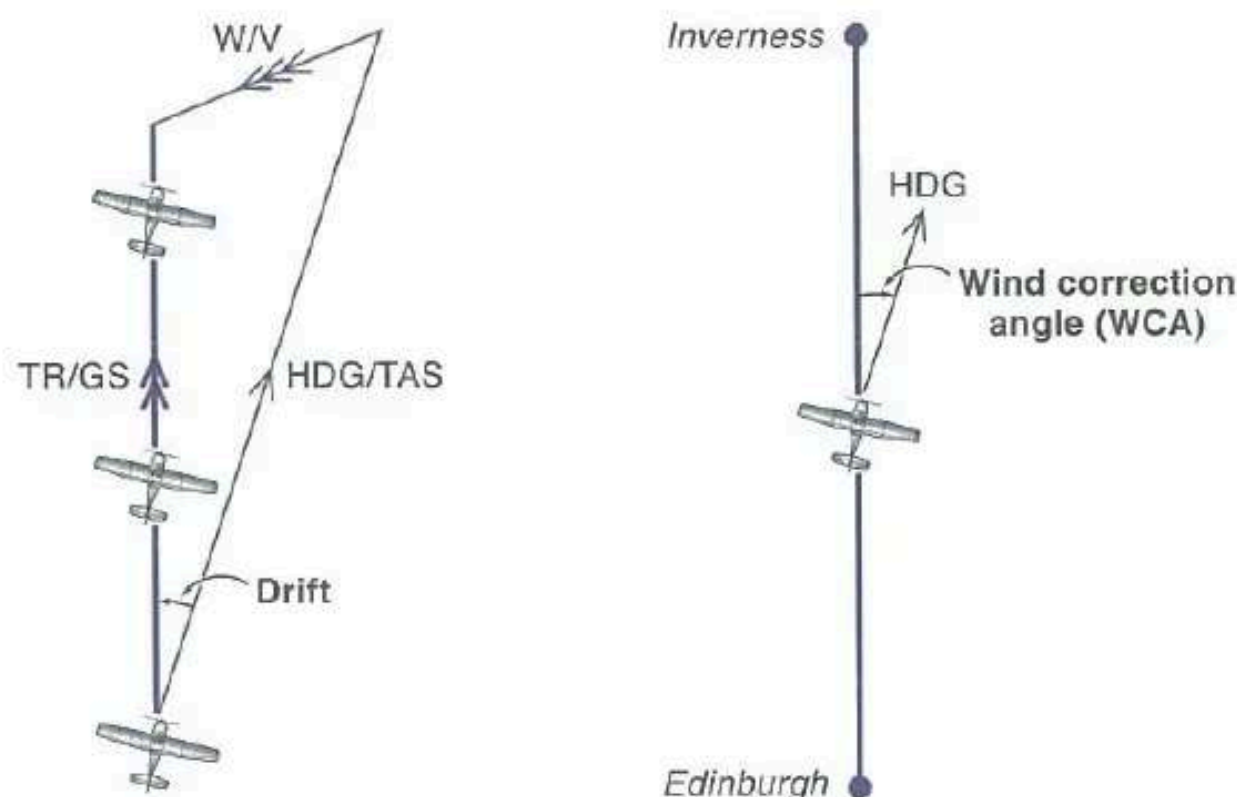
The Triangle of Velocities

Before you can use the navigation computer effectively, you must understand the triangle of velocities which, of course, has three sides. The first two are:

1. The motion of the aeroplane through the air – heading/true airspeed.
2. The motion of the air over the ground – wind velocity (direction and speed).

When these two are added together, the third side of the triangle, which is their resultant effect, is:

3. The motion of the aeroplane over the ground – track/groundspeed.



■ Figure 4-2 **The triangle of velocities and wind correction angle**

Note that the wind always blows the aeroplane from its heading to its track. In other words, the wind vector must start at the HDG/TAS vector and end at the TR/GS vector. If you have this firmly fixed in your mind, then computer solutions will follow easily and correctly.

A wind from the right will carry an aeroplane onto a track that is to the left of its heading. This is known as left drift or port drift. The drift angle is measured from HDG to TR. In Figure 4-2, 10° left drift is shown.

To achieve a particular track, say between two towns, the aeroplane must be steered into-wind on a heading that allows for the drift. In the above illustration, to achieve the track shown, the aeroplane is being steered on a steady heading 10° to the right of track. This can be described as a wind correction angle of 10° right. The wind correction angle is measured from TR to HDG and is, of course, equal and opposite to the drift angle.

Each of the three vectors in the triangle of velocities has two aspects: magnitude (size) and direction. This means that in the triangle of velocities there are six components:

VECTOR	MAGNITUDE	DIRECTION
HDG/TAS	TAS	HDG
W/V	Wind speed	Wind direction (from)
TR/GS	GS	TR

Typical navigation problems involve knowing four of these six elements and finding the other two. It is as simple as that!

Marking the Vector on the Plotting Disc

If you can sketch a triangle of velocities with the three vectors placed properly, then the wind side of the computer will create no difficulties at all.

While you are still learning how to use the computer, it is a good idea to mark each of the vectors with their arrowheads on the plotting disc to ensure that the triangle of velocities is portrayed correctly. Marks on the plotting disc can be removed quite easily at the end of each problem with a moist finger.

Once you become familiar with the use of the computer, however, the actual drawing in of each vector becomes unnecessary, and just one single mark (known as the wind-mark) to illustrate the extent of the wind velocity is all that is needed.

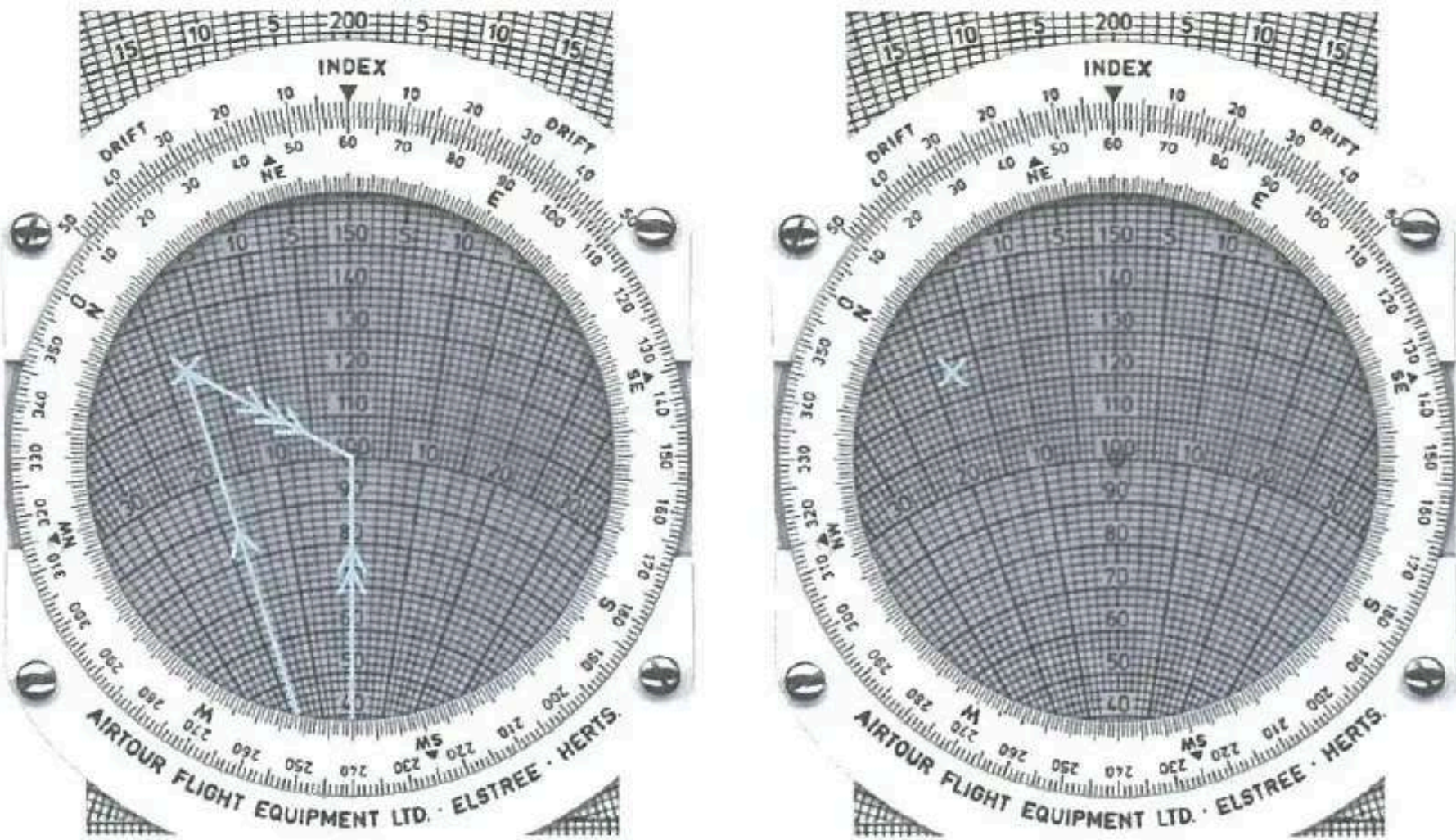


Figure 4-3 Initially show the vectors (left); with practice you need only show the wind-mark

Work Totally in Degrees True (or Totally in Degrees M)

It is most important that, when working out the vector triangle, the directions are all related to the one datum. In the UK, it is traditional for PPL pilots to work in degrees true. You may work in degrees magnetic; the eventual results are identical but, whatever you do with the wind side of your computer, ensure that you stay either totally in degrees true or totally in degrees magnetic. Do not mix °T and °M in the one triangle of velocities!

Our initial examples are worked in °T.

Choice of Method

The navigation computer is such that each problem can be solved in a number of ways, each method producing the correct results. You will find a variety of methods in common use – choose the method that suits you best.

We recommend that you first learn your instructor’s preferred method and disregard the others. Later in your training you may like to consider the other methods but, in the early stages, use one method only.

Most use of the navigation computer is made at the flight planning stage on the ground prior to flight, when the pilot already knows:

- the track in °T, measured on an aeronautical chart;
- the wind direction and velocity in °T/kt, found on the weather forecast; and
- the true airspeed that the aeroplane will achieve.

The wind side of the computer can then be used to calculate:

- the heading to steer; and
- the groundspeed that will be achieved.

The flight planning problem can be summarised as:

KNOWN:	FIND:
TR, W/V and TAS	HDG and GS

Our preferred method for wind side calculations finds HDG and GS very efficiently. We consider it first as Method A, beginning on page 53.

Method B is another commonly used approach to solving the triangle of velocities, and is known as the ‘wind-mark-down’ method. It is considered second, beginning on page 65.

Now proceed to the method recommended by your instructor.

Wind Side Method A

Determining HDG and GS When Flight Planning

EXAMPLE 1 Find HDG and GS

KNOWN:		FIND:
Required track (TR) measured off chart	150°	HDG and GS
True airspeed (TAS) known, or calculated from expected RAS/CAS	100 kt	
Wind velocity (W/V) stated on forecast	360°T/30	
Magnetic variation shown on chart	5°W	

An efficient way to record navigation data is to use a flight log, like that in Figure 4-4. The known information can be entered first as shown, with the results of your calculations following at a later stage.

From/To	Safety ALT	ALT Temp	RAS	TAS	W/V	TR °T	Drift	HDG °T	Var	HDG °M	GS	Dist	Time	ETA	HDG °C
				100	360/30	150			5°W						

Figure 4-4 Typical flight log

STEP 1. PLACE THE W/V ON THE PLOTTING DISC.

Wind direction and wind speed are known, i.e. both aspects of the W/V vector are known.

Rotate the compass rose until the wind direction 360 (north) is under the index. Mark the start of the W/V vector 30 kt above the centre-dot.

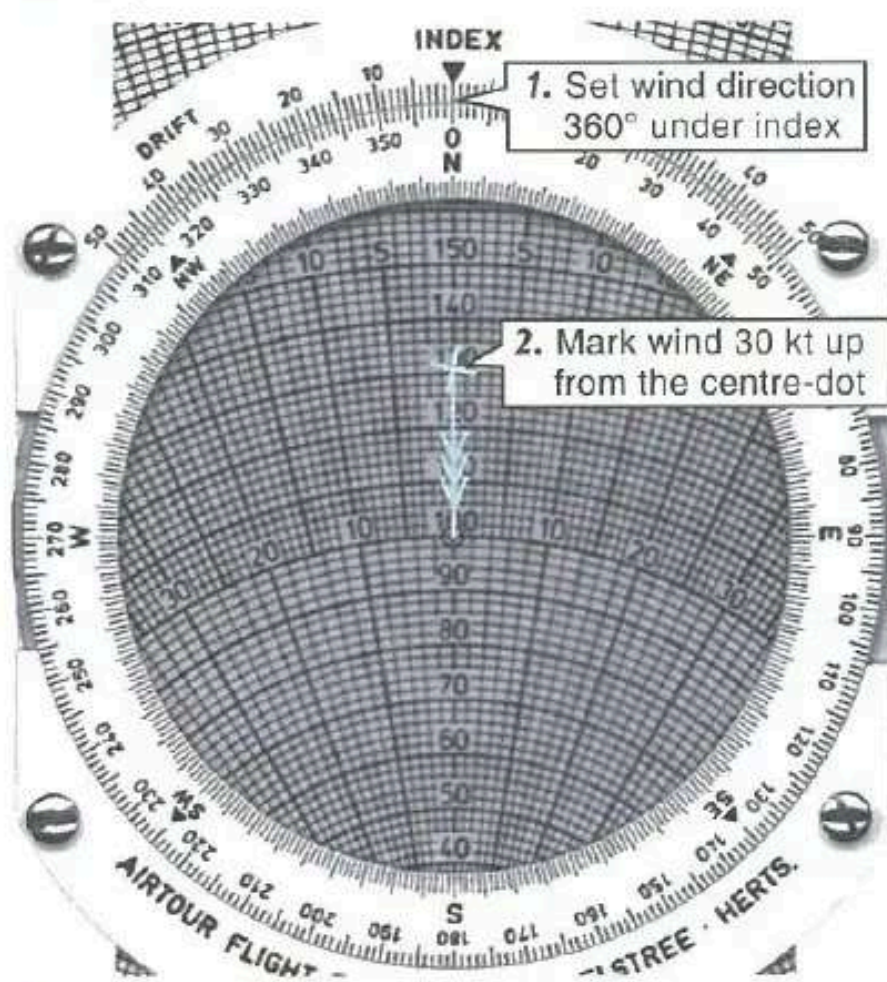
Since the track is known, show the W/V blowing towards the centre-dot, i.e. the wind vector will come from direction 360° and end at the centre-dot. Marking the starting point of the wind 30 kt above the centre-dot is most easily achieved by first setting one of the labelled wind arcs under the centre-dot, and then placing the wind-mark 30 kt above it.

At this early stage in your training, mark in the full W/V, showing the three arrowheads of the W/V vector pointing down towards the centre-dot. This will give you a clear picture as the whole triangle of velocities is developed.

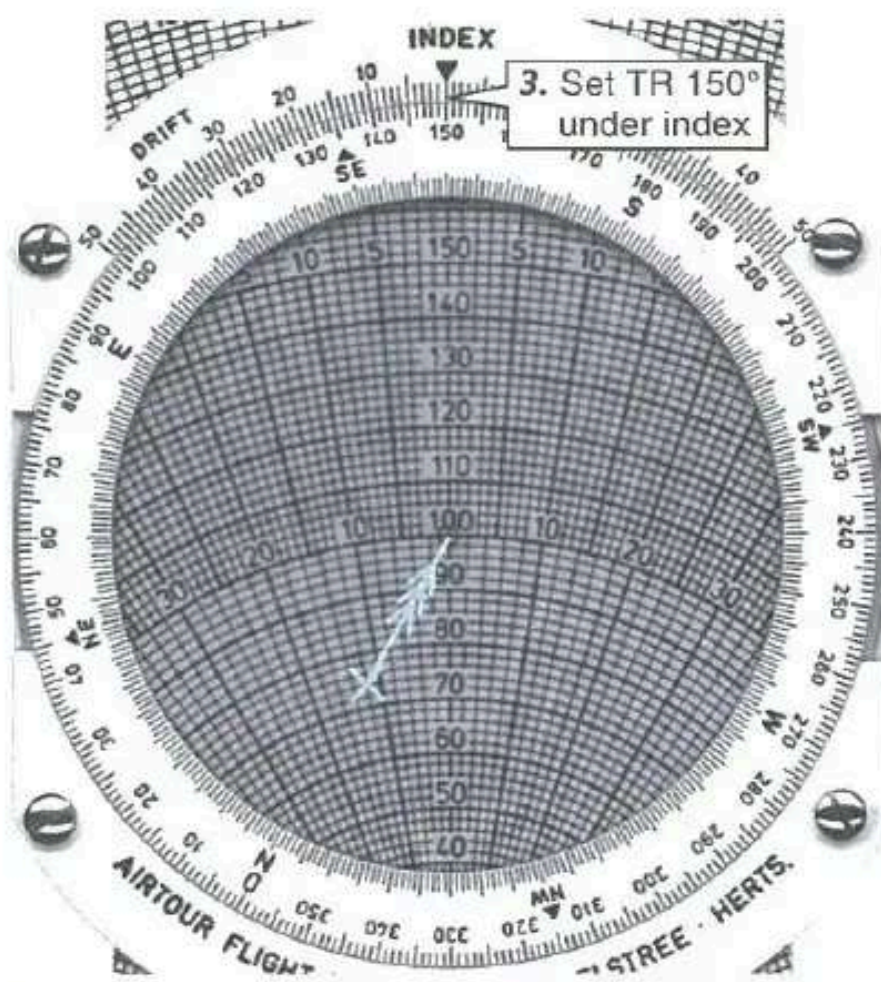
STEP 2. PLACE THE TR/GS VECTOR ON THE PLOTTING DISC.

Track is known (having been measured on the chart); ground-speed is not known. In this case, only one aspect of the TR/GS vector is known – its direction, but not its magnitude.

Rotate the compass rose until the required TR of 150°T is under the index. (We cannot position the appropriate ground-speed arc under the centre-dot because we do not yet know its value.)



■ Figure 4-5 Wind velocity 360/30 set on the computer at the flight planning stage: Example 1, Step 1



■ Figure 4-6 Example 1, Step 2 completed

STEP 3. PLACE THE HDG/TAS VECTOR ON THE PLOTTING DISC.

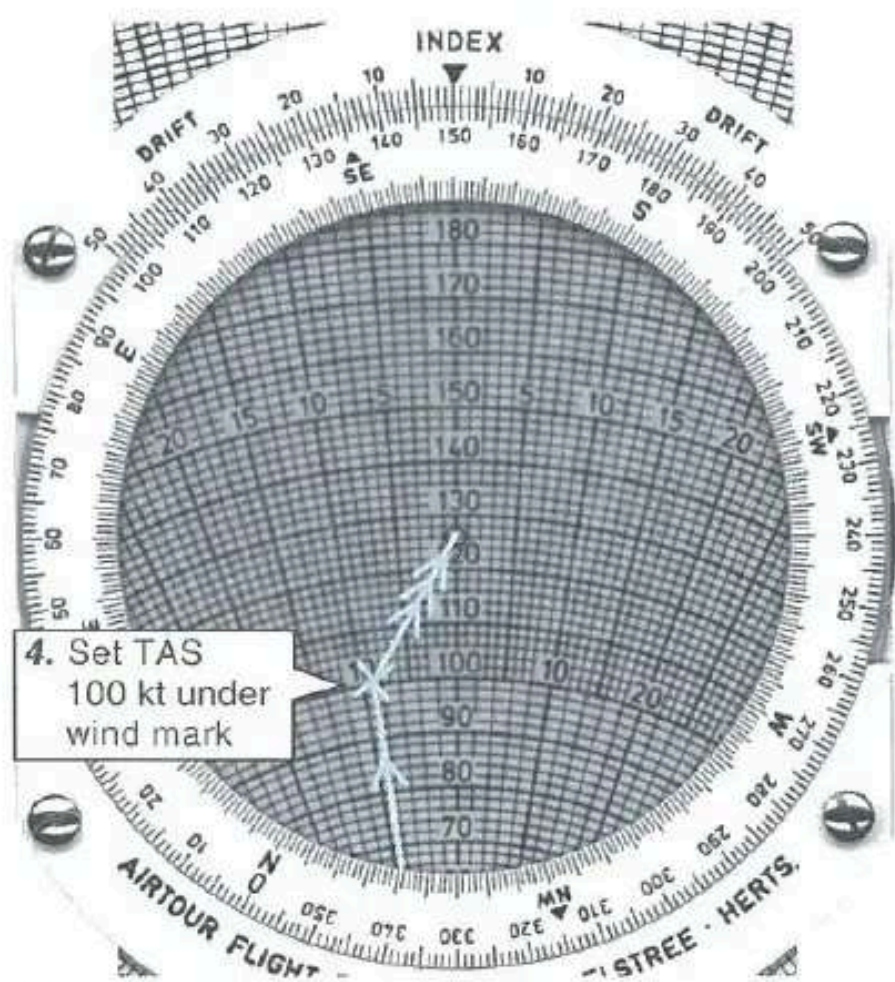
True airspeed is known, but the heading is not. In this case, only one aspect of the HDG/TAS vector is known – its magnitude.

Move the slide and place the TAS 100 kt speed arc under the wind-mark (which is the starting point of the W/V vector). This is where a clear understanding of the triangle of velocities is most important! The end of the HDG/TAS vector is where the W/V begins.

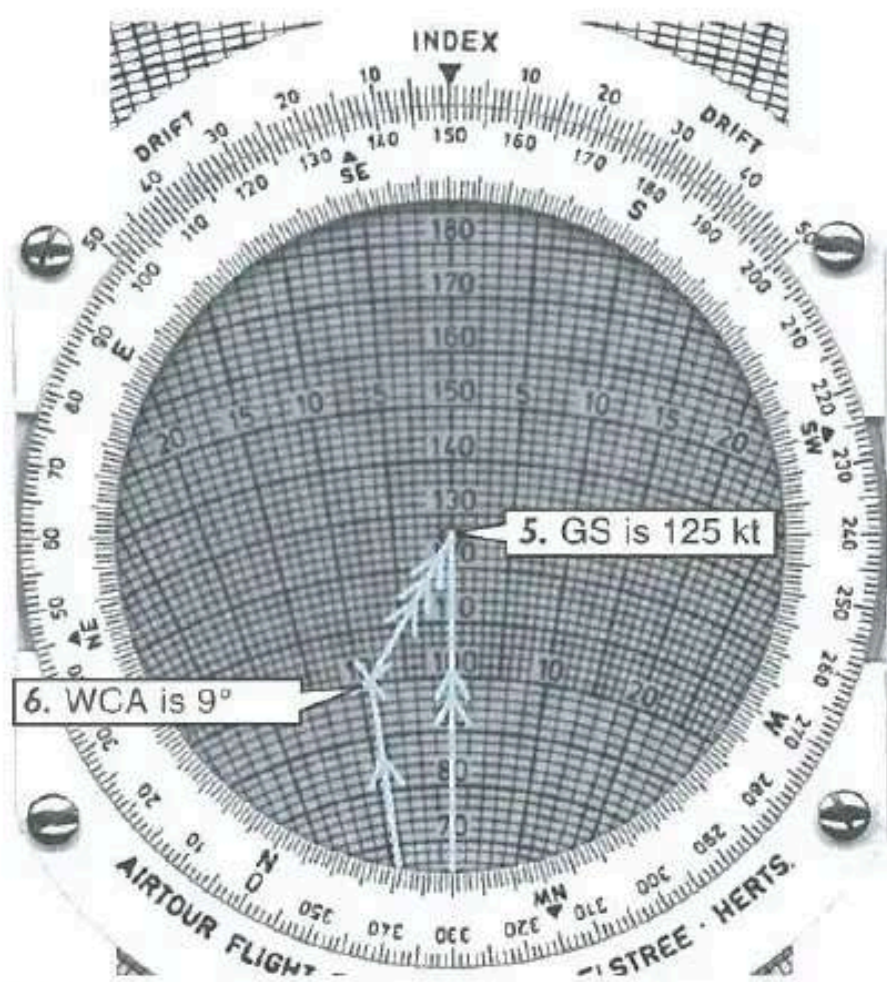
STEP 4. READ OFF THE ANSWERS FOR HDG AND GS.

The GS of 125 kt appears under the centre-dot. From the drift lines, the wind correction angle is 9° to the left of the track. This means to achieve TR 150°T , the aeroplane must be pointed 9° into-wind and flown on a HDG of 141°T to allow for the 9° of right (starboard) drift.

The arithmetic (for determining HDG 141 from TR 150 and a wind correction angle of 9° into-wind) can be checked on the scale near the index mark.



■ Figure 4-7 Example 1, Step 3



■ Figure 4-8 Example 1, Step 4

ANSWER HDG 141°T, GS 125 kt

NOTE If magnetic heading is required, it is simply a matter of applying variation. If variation is 5°W, then 141°T is 146°M (variation west, magnetic best).

From/To	Safety ALT	ALT Temp	RAS	TAS	W/V	TR °T	Drift	HDG °T	Var	HDG °M	GS	Dist	Time	ETA	HDG °C
				100	360/30	150	9°R	141	5°W	146	125				

■ Figure 4-9 Example 1

EXAMPLE 2 Find HDG and GS

Given a true airspeed of 174 kt, a forecast W/V of 240°T/40 kt, and a desired track of 290°T, calculate the true heading required and the groundspeed that will be achieved. What is the magnetic heading if variation in the vicinity of the flight is 6°W?

KNOWN:		FIND:
TAS.	174 kt	HDG and GS
Required TR	290°T	
W/V	240°T/40	

From/To	Safety ALT	ALT Temp	RAS	TAS	W/V	TR °T	Drift	HDG °T	Var	HDG °M	GS	Dist	Time	ETA	HDG °C
				174	240/40	290			6°W						

■ Figure 4-10 Example 2

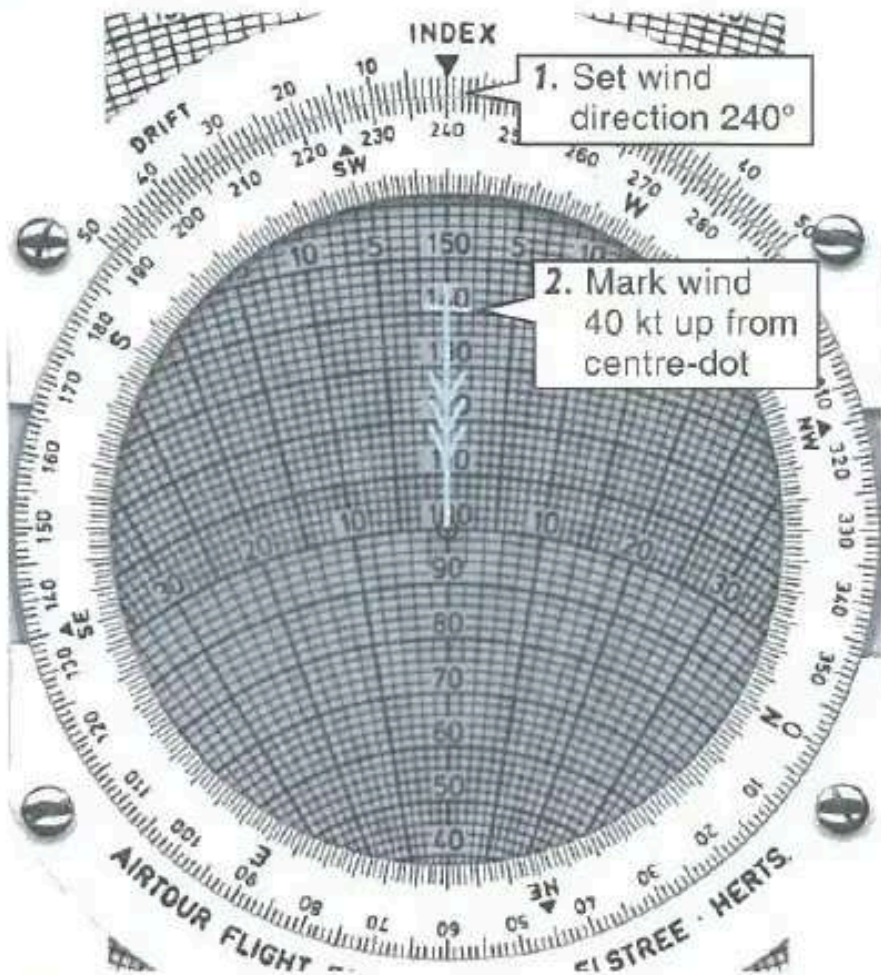
STEP 1. PLACE THE W/V ON THE PLOTTING DISC.

Both wind speed and wind direction are known. Since the required TR is known, we show the W/V blowing down from the index towards the centre-dot.

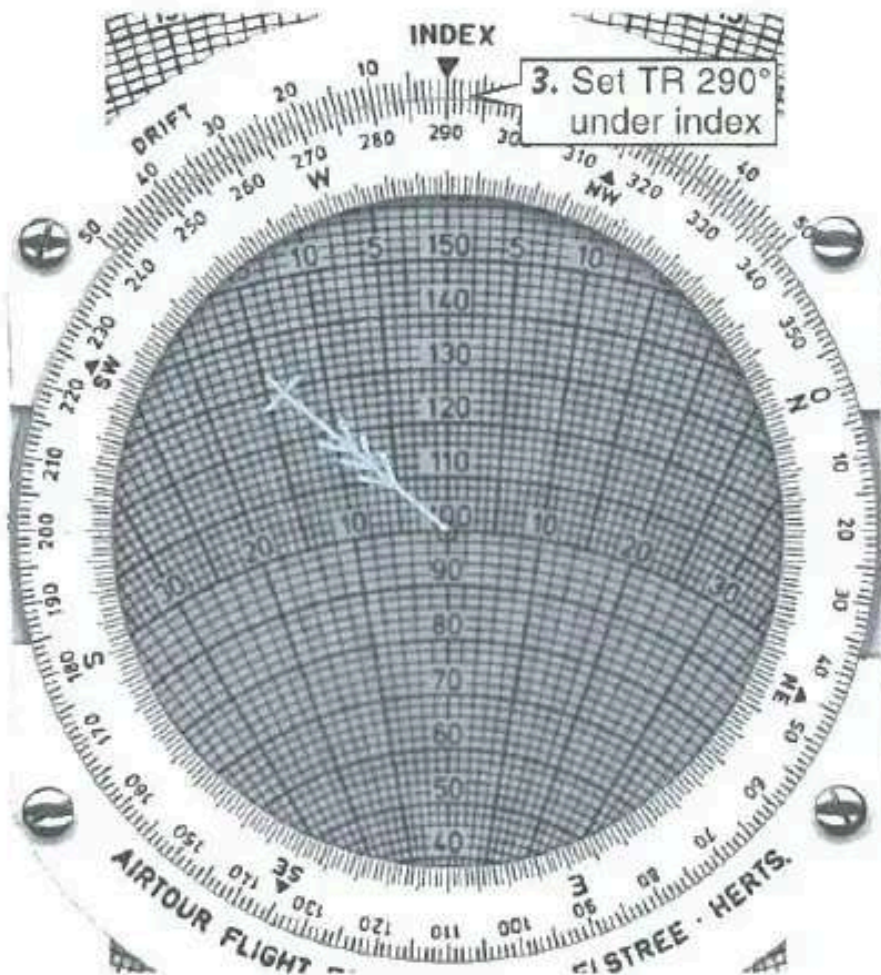
Set wind direction 240 under the index. Mark in the wind strength 40 kt above the centre-dot.

STEP 2. PLACE THE TR/GS VECTOR ON THE PLOTTING DISC.

TR is known. GS is not known. Rotate the compass rose until the required TR 290°T is under the index. (We cannot position the appropriate speed arc under the centre-dot because we do not yet know the GS.)



■ Figure 4-11 Example 2, Step 1



■ Figure 4-12 Example 2, Step 2

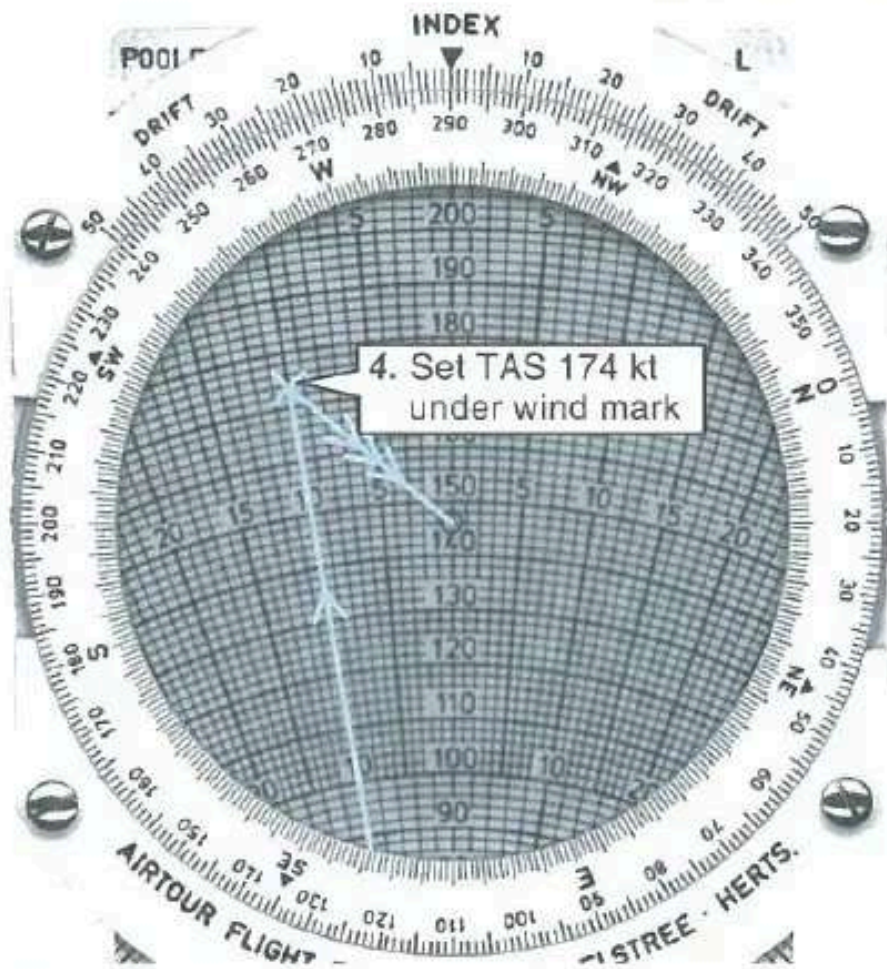
STEP 3. PLACE THE HDG/TAS VECTOR ON THE PLOTTING DISC.

TAS is known. HDG is not known. Move the slide to place the TAS 174 kt under the wind-mark, which is starting point of the W/V vector (since the W/V vector begins where the TAS/HDG vector ends).

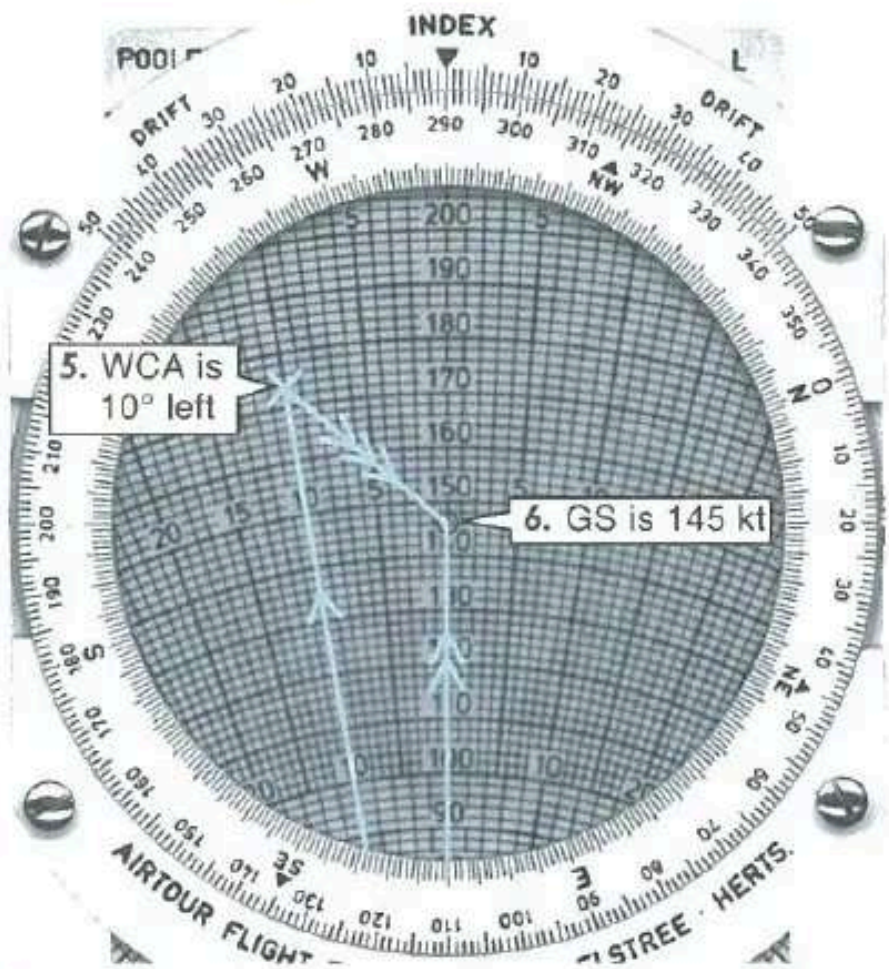
STEP 4. READ OFF THE ANSWERS FOR HDG AND GS.

The wind correction angle of 10° left can now be read off the drift lines, i.e. the aeroplane must be steered 10° into wind (i.e. left of the desired track to allow for the 10° right, or starboard, drift). In this case, to achieve the desired track of 290°T, the aeroplane must be steered on a heading of 280°T.

The groundspeed 145 kt lies under the centre-dot.



■ Figure 4-13 Example 2, Step 3



■ Figure 4-14 Example 2 almost completed: HDG 280°T, GS 145 kt

Magnetic variation in the area is 6°W, so HDG 280°T is 286°M.

ANSWER HDG 280°T, variation 6°W, HDG 286M; GS 145 kt

From/To	Safety ALT	ALT Temp	RAS	TAS	WIV	TR °T	Drift	HDG °T	Var	HDG °M	GS	Dist	Time	ETA	HDG °C
				174	240/40	290	10°R	280	6°W	286	145				

■ Figure 4-15 Example 2

EXAMPLE 3 Finding HDG/TAS to Achieve a Particular TR/GS

Occasionally you may want to achieve a particular groundspeed, for instance to arrive overhead the destination at a particular time. This example is a variation on the previous one.

We want to know what true airspeed is required to achieve a groundspeed of 120 kt in Example 2.

KNOWN:		FIND:
TR	290°T	TAS and HDG
W/V	240°/40 kt	
GS	120 kt	

From/To	Safety ALT	ALT Temp	RAS	TAS	W/V	TR °T	Drift	HDG °T	Var	HDG °M	GS	Dist	Time	ETA	HDG °C
					240/40	290			6°W		120				

Figure 4-16 Example 3

STEP 1. PLACE W/V ON THE PLOTTING DISC.

Set wind direction 240° under the index. Place wind-mark 40 kt above the centre-dot, wind blowing down.

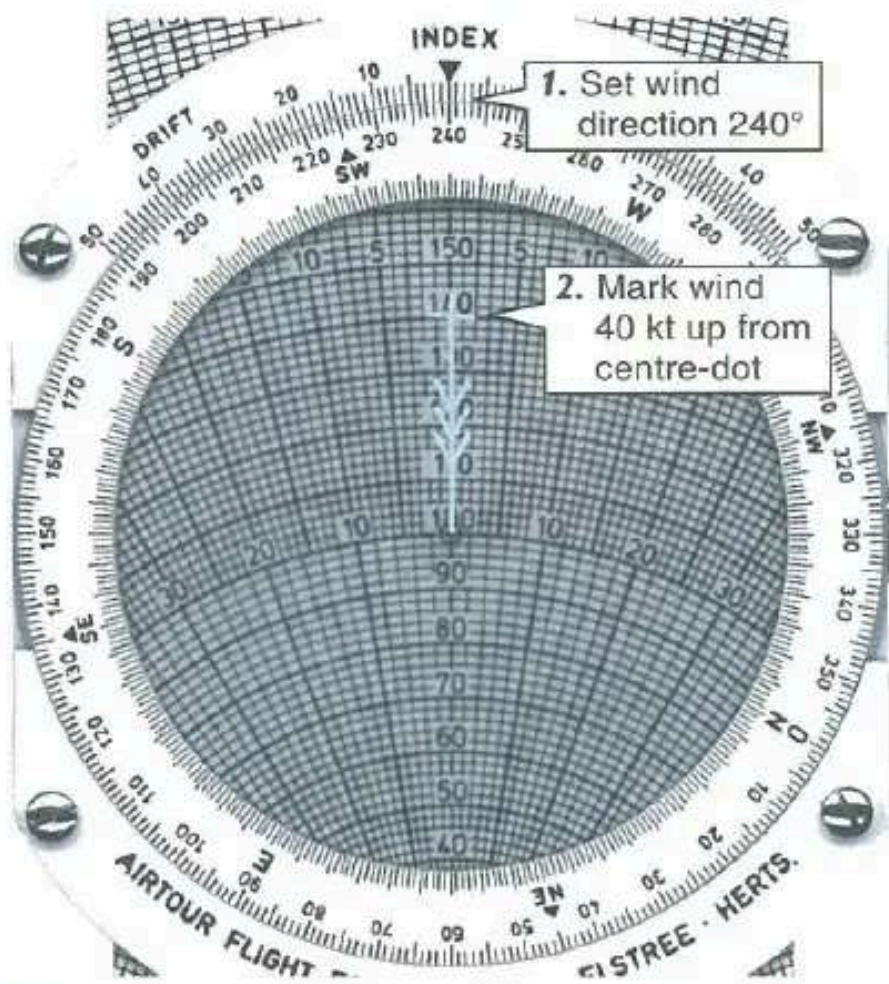


Figure 4-17 Example 3, Step 1

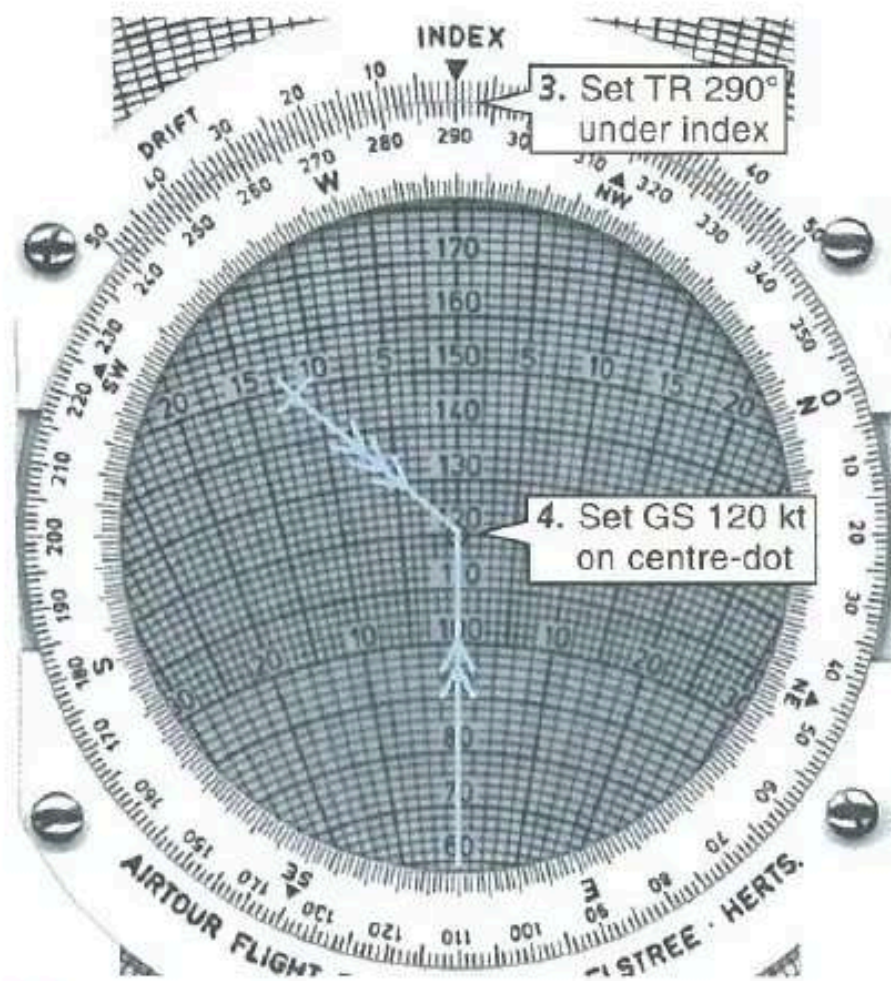
STEP 2. PLACE THE TR/GS VECTOR ON THE PLOTTING DISC.

In this case both TR 290°T and GS 120 kt are known. Rotate the compass rose and set TR 290°T under the index. Set GS 120 kt under the centre-dot.

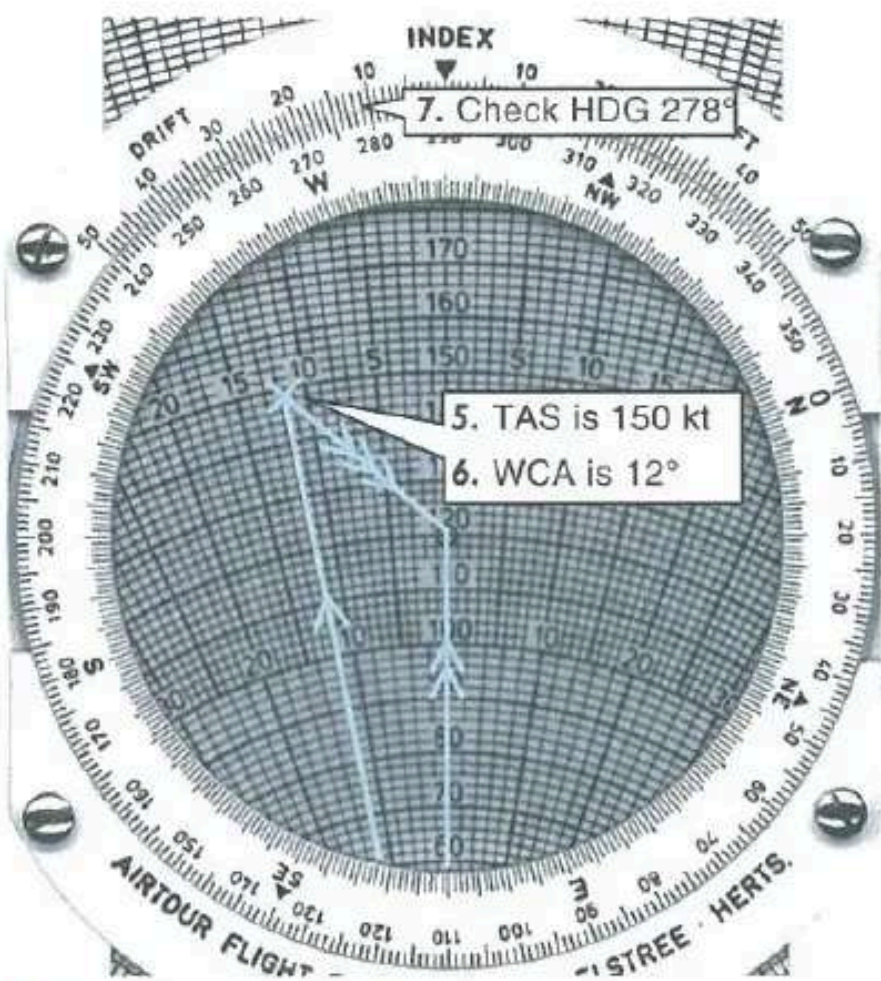
STEP 3. READ OFF TAS, AND DETERMINE HDG FROM DRIFT.

Read off TAS 150 kt under the wind-mark. Read off drift of 12°, therefore HDG is 12° left of TR. HDG is 278°T.

NOTE The wind is from the left, so the wind correction angle is 12° left of the desired track to allow for the expected 12° right (starboard) drift. You can check your arithmetic against the small scale near the index mark.



■ Figure 4-18 Example 3, Step 2



■ Figure 4-19 Example 3, completed

ANSWER TAS 150 kt, HDG 278°T

From/To	Safety ALT	ALT Temp	RAS	TAS	WIV	TR °T	Drift	HDG °T	Var	HDG °M	GS	Dist	Time	ETA	HDG °C
				150	240/40	290	12°R	278	6°W	284	120				

■ Figure 4-18 Example 3

Now complete **Exercises 4 – Wind Side-1.**



Calculating the Wind Velocity In Flight

After obtaining two position fixes en route, it is possible to determine the actual W/V, which can then be used in further calculations rather than the less accurate forecast wind.

The two position fixes enable you to determine the track made good (TMG) and the groundspeed achieved during the time you have maintained a reasonably steady heading and true airspeed.

When in-flight, the pilot knows:

- the heading maintained from the compass or direction indicator;
- true airspeed;
- the track made good (TMG) by observation of the ground and map;
- the groundspeed by comparing distance covered with the time it takes.

These are four of the six items in the triangle of velocities and, using them, you can find the other two – wind direction and wind speed (i.e. wind velocity). This problem can be summarised as:

KNOWN:	FIND:
HDG/TAS and TR/GS	W/V

Remember that in the UK it is usual for PPL pilots to work in degrees true.

EXAMPLE 4 Finding the W/V In Flight

KNOWN:		FIND:
HDG	143°M	W/V
Variation	5°W, (HDG 138°T)	
TAS	120 kt	
TR	146°T	
GS	144 kt	

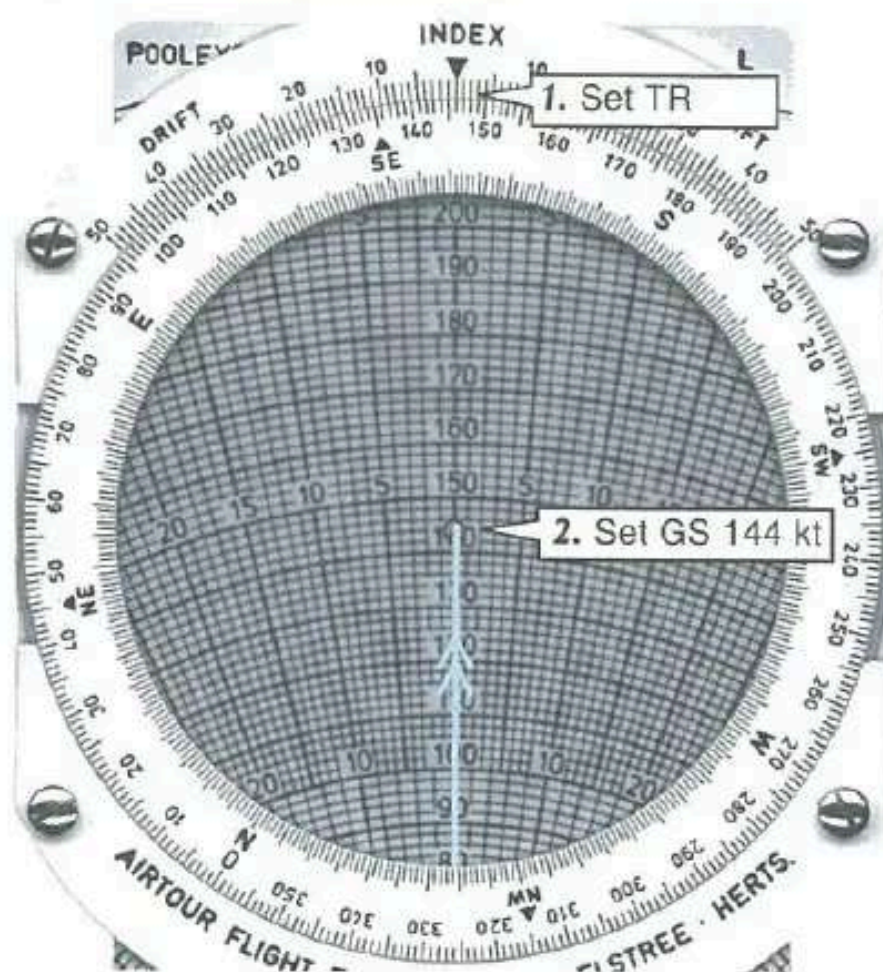
NOTE Another way of asking the same question is to give HDG 138°T and right drift 8° (rather than TR 146°T).

From/To	Safety ALT	ALT Temp	RAS	TAS	W/V	TR °T	Drift	HDG °T	Var	HDG °M	GS	Dist	Time	ETA	HDG °C
				120		146		138	5°W	143	144				

■ Figure 4-19 Example 4

STEP 1. PLACE THE TR/GS VECTOR UNDER THE CENTRE-DOT.

Rotate the compass rose and set track 146°T under the index. Set GS 144 kt under the centre-dot.



■ Figure 4-20 Example 4, Step 1

STEP 2. PLACE THE HDG/TAS VECTOR ON THE PLOTTING DISC.

From HDG 138°T and TR 146°T , the drift is 8° to the right; the HDG is 8° to the left of TR (i.e. the wind correction angle is 8° left).

Mark in the HDG direction as the 8° drift line to the left of track. Mark the TAS where the 120 kt speed arc intersects the drift line; this indicates the HDG/TAS vector.

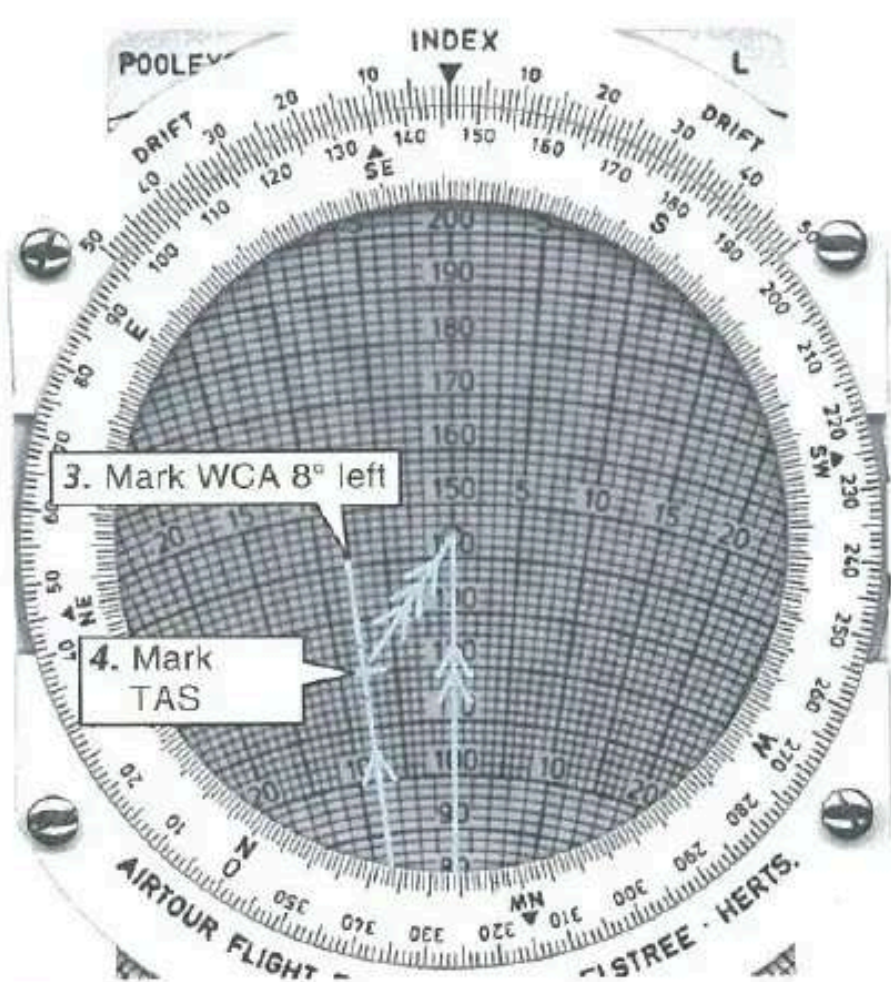
The W/V blows the aircraft from HDG to TR.

STEP 3. DETERMINE THE W/V.

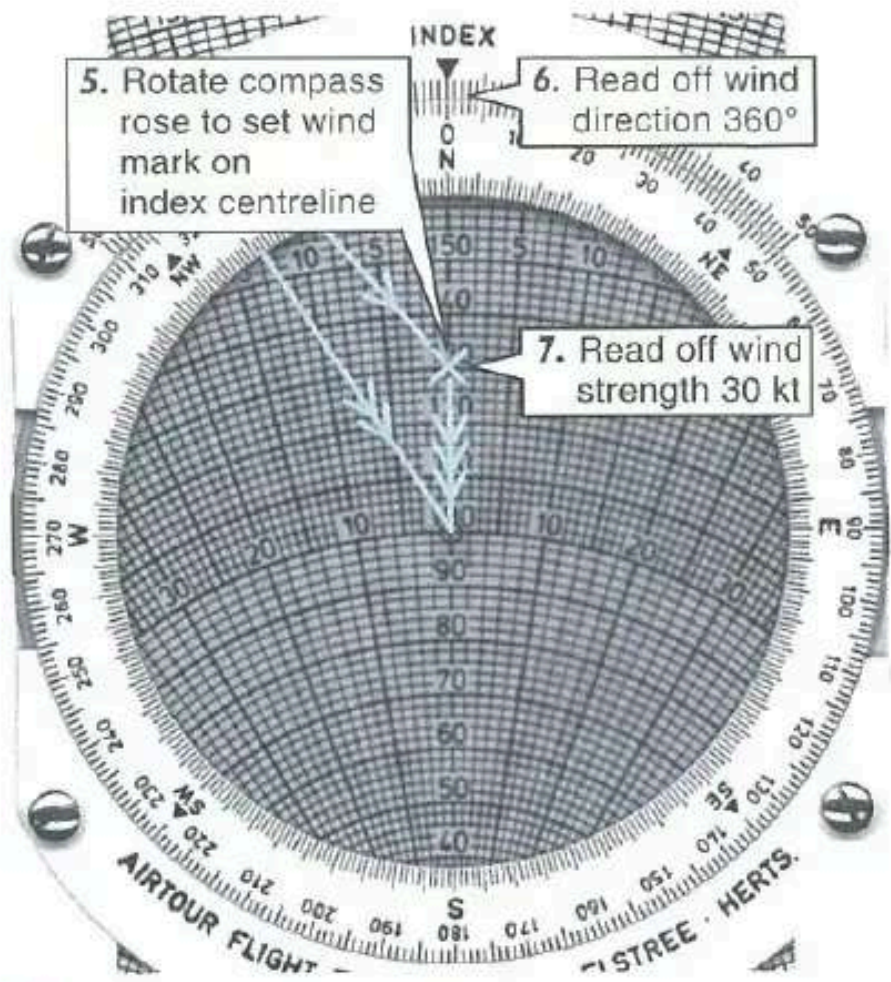
Rotate the compass rose until the wind-mark is on the index line with the arrows pointing down towards the centre-dot. The direction the wind is blowing from, 360°T , is now indicated under the index.

Read off the wind strength of 30 kt. (Setting a definite speed arc under the centre-dot, 100 kt in this case, makes the wind strength easier to read.)

ANSWER W/V $360^{\circ}\text{T}/30$ kt



■ Figure 4-21 Example 4, Step 2



■ Figure 4-22 Example 4, Step 3

From/To	Safety ALT	ALT Temp	RAS	TAS	W/V	TR °T	Drift	HDG °T	Var	HDG °M	GS	Dist	Time	ETA	HDG °C
				120	360/30	146	8°R	138	5°W	143	144				

■ Figure 4-23 Example 4

Now complete Exercises 4 – Wind Side-2.



Finding TMG and GS In Flight

The usual means of determining track made good and groundspeed is by visually fixing your position over ground features. On a flight over water or featureless terrain or in poor visibility, however, where you are unable to fix your position, it is possible to estimate a ‘dead reckoning’ track and groundspeed using:

- heading from the compass (corrected to HDG in degrees true, using variation if required);
- TAS (calculated from the rectified/calibrated airspeed if necessary);
- W/V (speed and direction, known from the forecast or previous calculations).

EXAMPLE 5 Find TR and GS

KNOWN:		FIND:
HDG	057°M	TR/GS
TAS	120 kt	
W/V	115°T/40	
VAR	7°W	

From/To	Safety ALT	ALT Temp	RAS	TAS	W/V	TR °T	Drift	HDG °T	Var	HDG °M	GS	Dist	Time	ETA	HDG °C
				120	115/40			050	7°W	057					

Figure 4-24 Example 5

Since the HDG is known, but the TR is not, the HDG/TAS vector will be set under the centre-dot and the W/V will emanate from there. In this case, with the wind direction set under the index, the wind will start at the centre-dot and blow down the slide, hence the wind-mark will be shown down from the centre-dot.

STEP 1. PLACE THE W/V VECTOR ON THE PLOTTING DISC.

Rotate the compass rose until the wind direction 115°T is under the index. Mark the end of the W/V vector 40 kt below the centre-dot.

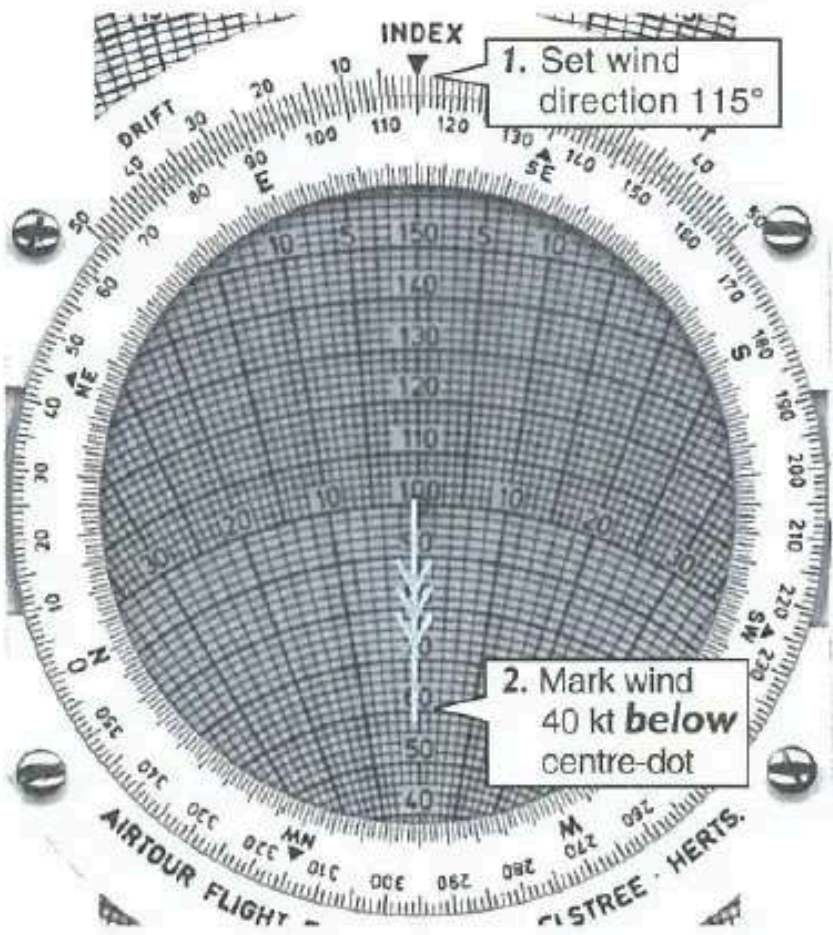


Figure 4-25 Example 5, Step 1

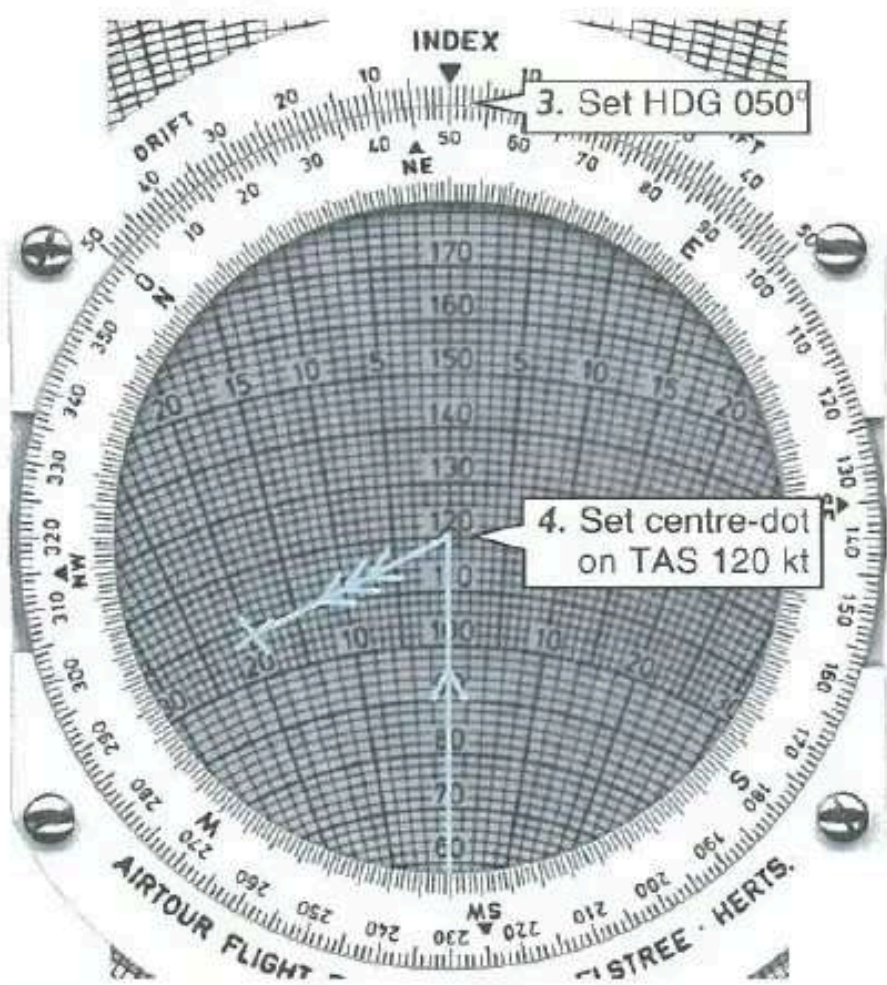
STEP 2. PLACE THE HDG/TAS VECTOR UNDER THE CENTRE-DOT.

Set HDG 050°T under the index. Move the slide so that TAS 120 kt is under the centre-dot.

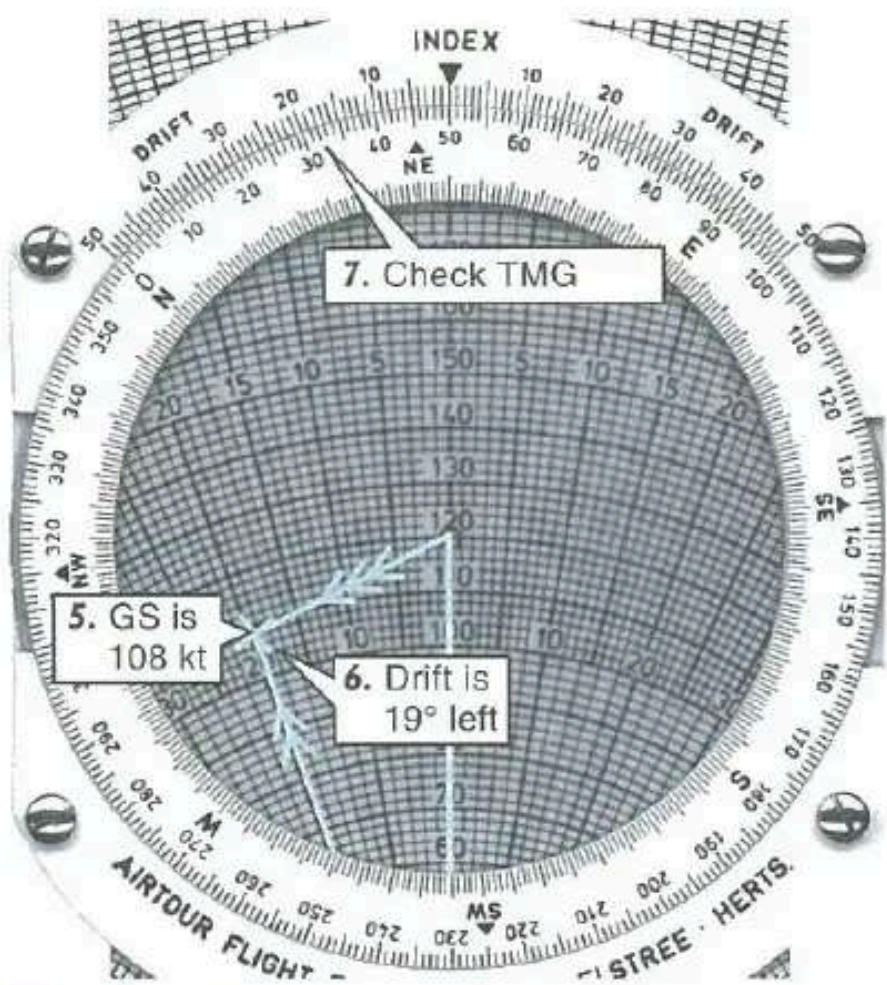
STEP 3. READ OFF THE GS AND DRIFT.

The HDG/TAS vector is now set under the centre-dot and the W/V vector acts out from the centre-dot. The wind-mark on the outer end of the W/V vector lies over the TR/GS vector.

The GS of 108 kt appears under the wind dot. The drift lines indicate 19° of drift left of HDG 050°, giving a TMG of 031°T. (You can check your arithmetic against the small scale near the index mark.)



■ Figure 4-26 Example 5, Step 2



■ Figure 4-27 Example 5, completed

From/To	Safety ALT	ALT Temp	RAS	TAS	W/V	TR °T	Drift	HDG °T	Var	HDG °M	GS	Dist	Time	ETA	HDG °C
				120	115/40	031	19L	050	7°W	057	108				

■ Figure 4-28 Example 5

ANSWER Drift 19° left (port), TR 031°T; GS 108 kt.

Now complete **Exercises 4 – Wind Side-3.**



If you are satisfied with the above method of using the wind side of your computer, then it is not necessary to work through the next part (Method B), as the calculations are a repeat of what you have just completed.

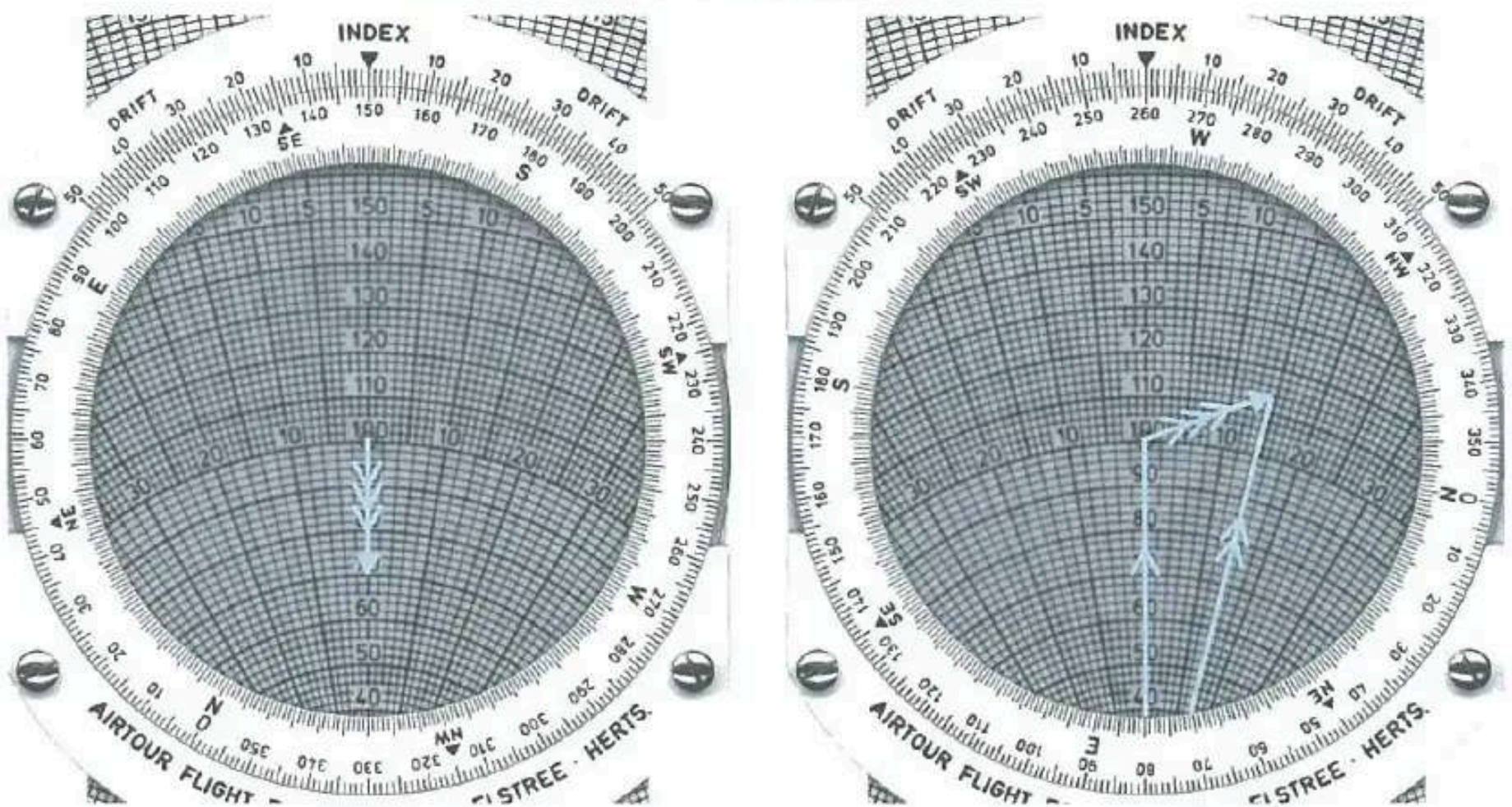
You should, however, complete the final part of this chapter, beginning on page 76, which shows how to work out wind components. This is important information for take-off and landing operations on runways where the wind is not blowing directly along the runway (often the case).

Wind Side Method B

Use Method B if your flying instructor recommends it. It is called the ‘wind-mark-down’ method because:

- the wind direction (from which it blows) is placed under the index;
- the centre-dot is used as the starting point of the W/V vector; and
- the wind-mark, in this case representing the end of the W/V vector, is drawn beneath the centre-dot (‘down’ from the centre-dot).

Since the W/V blows the aeroplane from its heading to its track, and the heading vector ends where the W/V vector starts, the heading/TAS vector should be placed up the centre of the slide so that it ends at the centre-dot.



■ Figure 4-29 The ‘wind-mark-down’ method

The simplest type of problem to solve using the wind-mark-down method is the in-flight situation of knowing the HDG, TAS and W/V, and then having to find the TR and GS. This situation could arise on an over-water flight, or a flight over barren terrain or in poor visibility, where it is difficult to obtain two position fixes to determine TR and GS. HDG is found from the compass, TAS is calculated from the indicated airspeed, and forecast W/V (in °T) is used.

Remember that all calculations on the computer must be totally in °T, or totally in °M. Do not mix them! In the UK, it is usual for PPL pilots to work in °T, so this will mean converting the compass reading into °T, using magnetic variation (and deviation if required).

Finding Track and Groundspeed In Flight

EXAMPLE 6 Find TR and GS

KNOWN:		FIND:
HDG	050°T	TR and GS
TAS	120 kt	
W/V	140°T/30 kt	

In practice, you would probably enter this data on a flight log to keep it neat and orderly.

From/To	Safety ALT	ALT Temp	RAS	TAS	W/V	TR °T	Drift	HDG °T	Var	HDG °M	GS	Dist	Time	ETA	HDG °C
				120	140/30			050							

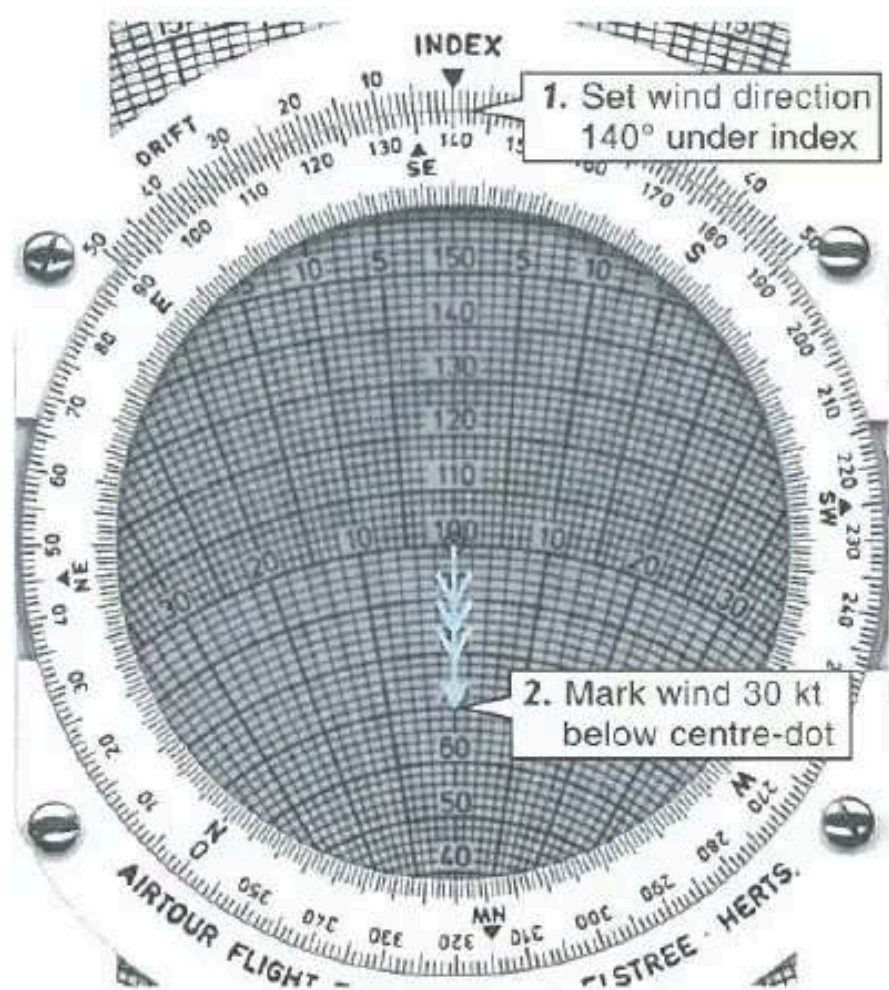
Figure 4-30 Example 6

STEP 1. PLACE THE W/V ON THE PLOTTING DISC.

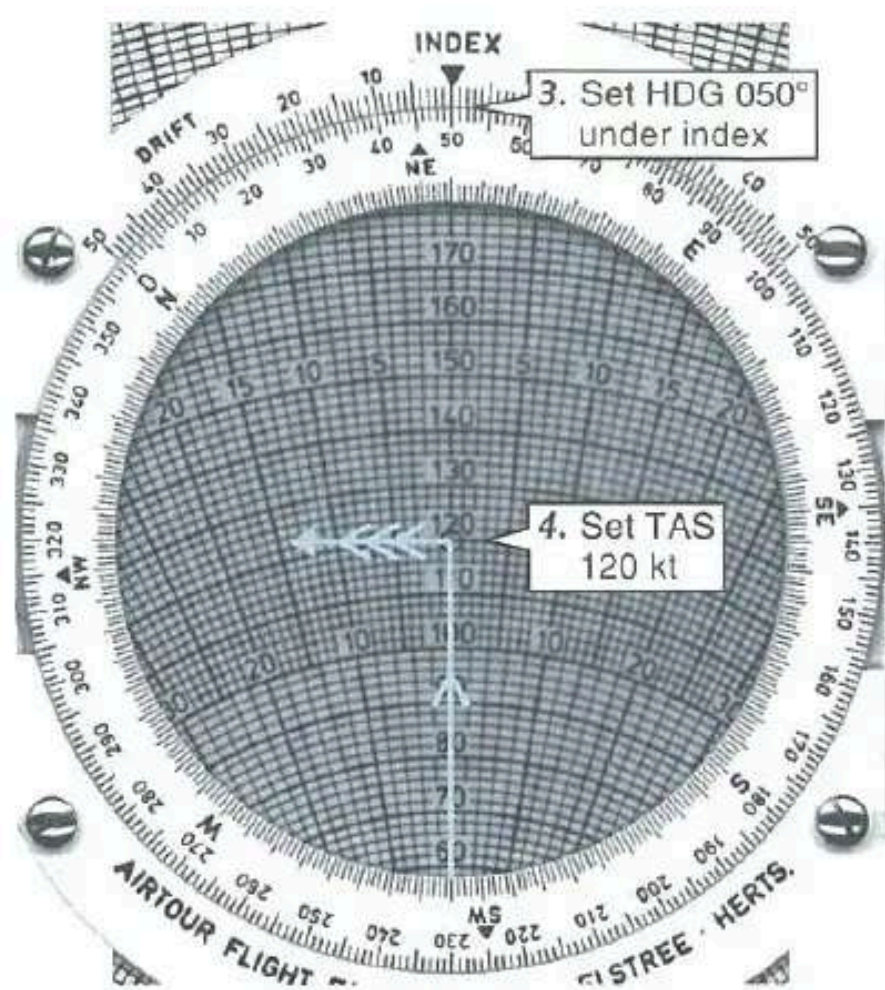
Rotate the compass rose until the wind direction 140°T is under the index. Set the centre-dot (in transparent window) on an easily noted value (100 in the illustration) and mark the end of the W/V vector 30 knots down from the centre-dot.

STEP 2. PLACE THE HDG/TAS VECTOR UNDER THE CENTRE-DOT.

Rotate the compass rose until the HDG 050°T is under the index. Move the slide so that the TAS 120 kt is under the centre-dot.



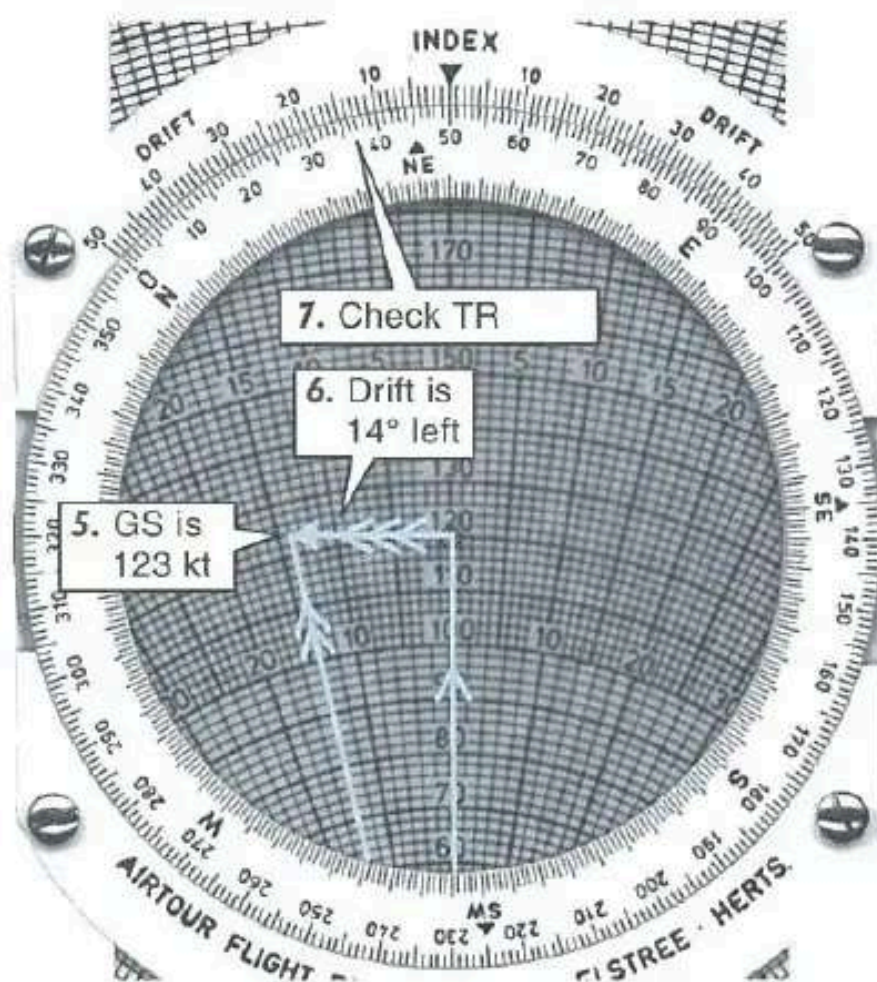
■ Figure 4-31 Example 6, Step 1



■ Figure 4-32 Example 6, Step 2

STEP 3. READ OFF GS AND DRIFT, THEN CALCULATE TRACK.

The wind-mark now lies over the groundspeed 123 kt. The wind-mark lies over the 14° drift line, which means the wind blowing from the right will cause a HDG of 050°T to result in a track of $(050 - 14) = 036^{\circ}\text{T}$. (You can check your arithmetic against the small scale near the index mark.)



■ Figure 4-33 Example 6, Step 3

ANSWER GS 123 kt, TMG 036°T added to the flight log.

From/To	Safety ALT	ALT Temp	RAS	TAS	WIV	TR °T	Drift	HDG °T	Var	HDG °M	GS	Dist	Time	ETA	HDG °C
				120	140/30	036		050			123				

■ Figure 4-34 Example 6

Now complete **Exercises 4 – Wind Side-3**.
(Wind Side-2 and Wind Side-1 follow.)



Finding the Wind Velocity In Flight

EXAMPLE 7 Find WIV

This is a typical in-flight situation, where HDG and TAS are known, and two position fixes allow you to determine the track made good and the groundspeed.

KNOWN:		FIND:
HDG	160°T	WIV (direction and strength)
TAS	145 kt	
TMG	168°T	
GS	157 kt	

From/To	Safety ALT	ALT Temp	RAS	TAS	WIV	TR °T	Drift	HDG °T	Var	HDG °M	GS	Dist	Time	ETA	HDG °C
				145		168		160			157				

■ Figure 4-35 Example 7

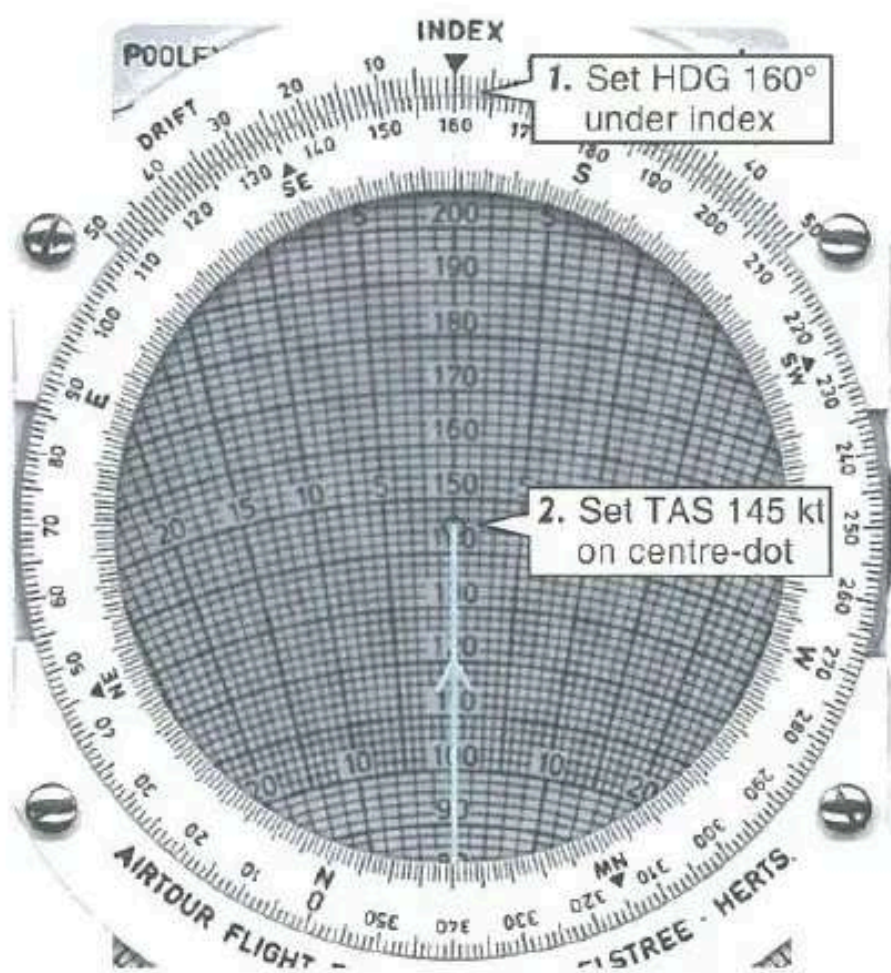
NOTE The same question could have been asked differently, with HDG 160°T and drift 8° right given, rather than TR 168°T.

STEP 1. PLACE THE HDG/TAS VECTOR UNDER THE CENTRE-DOT.

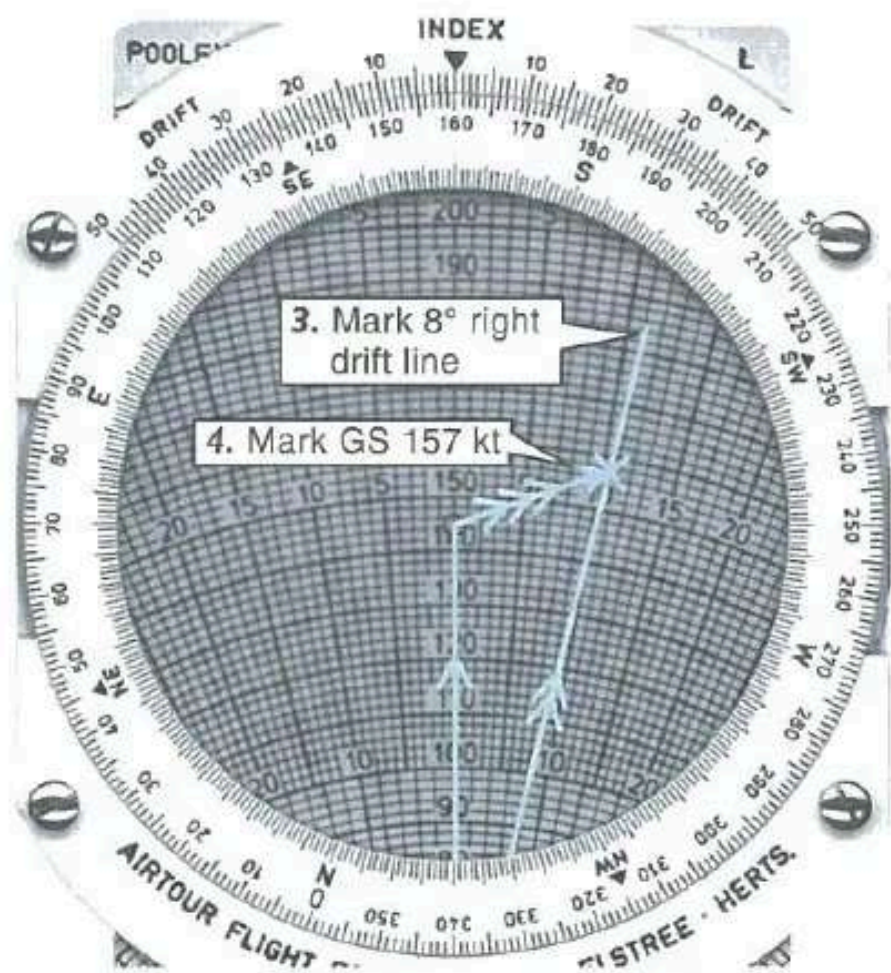
Rotate the compass rose until the HDG 160°T appears under the index. Move the slide until TAS 145 kt is under the centre-dot.

STEP 2. PLACE THE TMG/GS ON THE PLOTTING DISC.

Mark in the 8° right drift line, since the drift is 8° to the right, calculated from the HDG 160°T and the TMG 168°T. Mark the point along this line where the GS is 157 kt. This point is the end of the TR/GS vector. (The wind has blown the aeroplane from the centre-dot to this point marked on the plotting disc.)



■ Figure 4-36 Example 7, Step 1

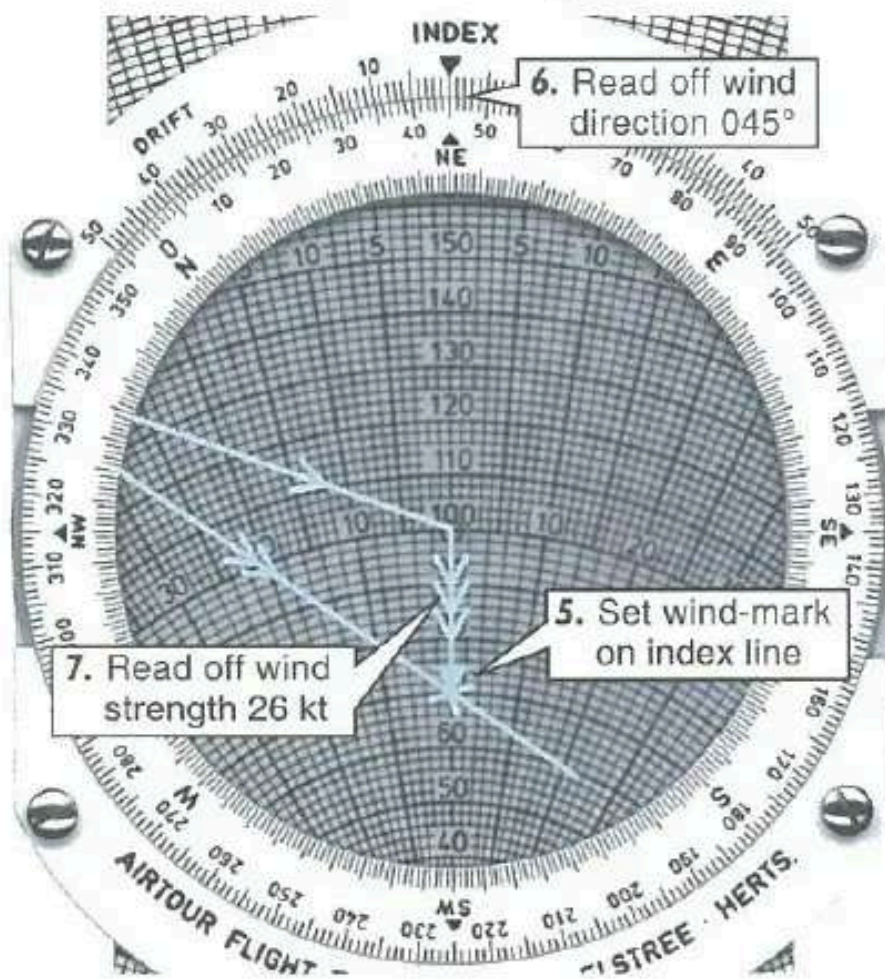


■ Figure 4-37 Example 7, Step 2

STEP 3. READ OFF THE W/V.

Rotate the compass rose until the wind dot appears directly down from the centre-dot. The direction from which the wind is blowing appears under the index, i.e. 045°T.

The wind strength of 26 kt can be read from the distance beneath the centre-dot. (You can adjust the slide to make reading this easier.)



■ Figure 4-38 Example 7, Step 3

ANSWER W/V is 045°T/26 kt

From/To	Safety ALT	ALT Temp	RAS	TAS	W/V	TR °T	Drift	HDG °T	Var	HDG °M	GS	Dist	Time	ETA	HDG °C
				145	045/26	168		160			157				

Figure 4-39 Example 7

Now complete Exercises 4 – Wind Side-2.



The Flight-Planning Situation

EXAMPLE 8 Find HDG and GS, Knowing TR, TAS and W/V

This is the typical situation prior to flight. Since it is at the flight-planning stage when the navigation computer is used most, you must become adept at this sort of problem.

From the aeronautical chart, you can measure the desired track (and distance). The wind is known from the forecast. The aeroplane’s performance in terms of true airspeed is known, or can be found from the Flight Manual or Pilot’s Operating Handbook.

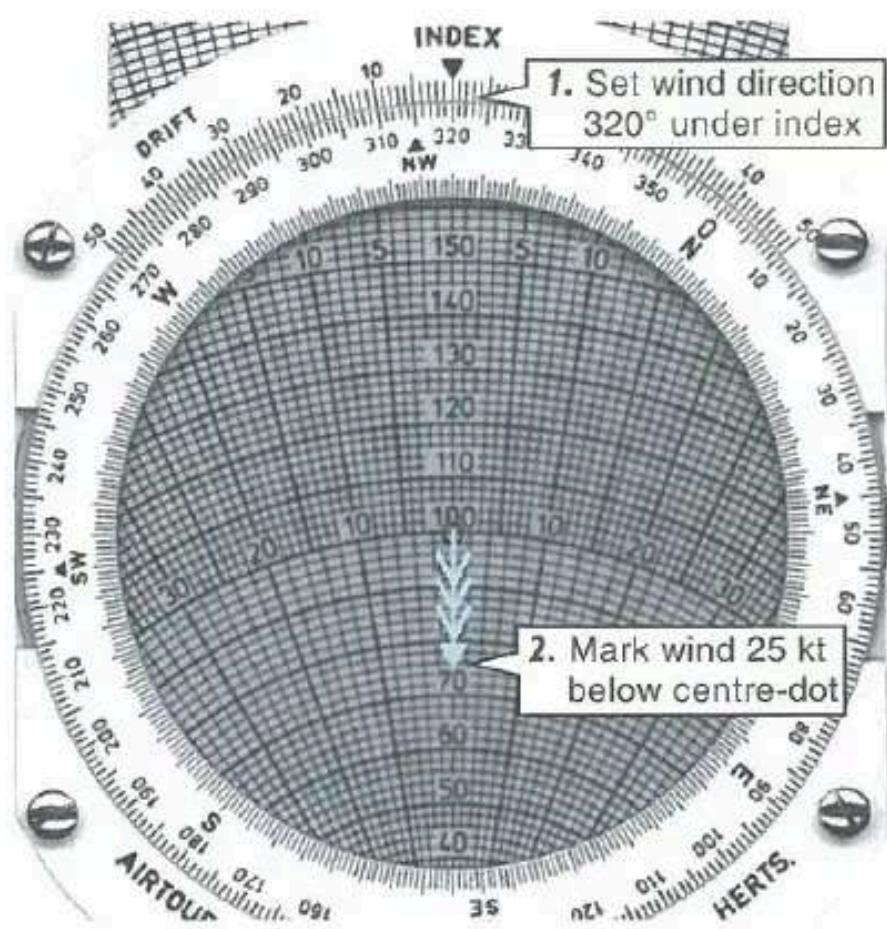
KNOWN:			FIND:
Desired track	295°T		HDG and GS
TAS	97 kt		
Forecast W/V	320°T/25		

From/To	Safety ALT	ALT Temp	RAS	TAS	W/V	TR °T	Drift	HDG °T	Var	HDG °M	GS	Dist	Time	ETA	HDG °C
				97	320/25	295									

Figure 4-40 Example 8

STEP 1. PLACE THE W/V ON THE PLOTTING DISC.

Rotate the compass rose until the direction from which the wind is blowing is under the index, i.e. 320°T. Put a wind-mark 25 kt down from the centre-dot.



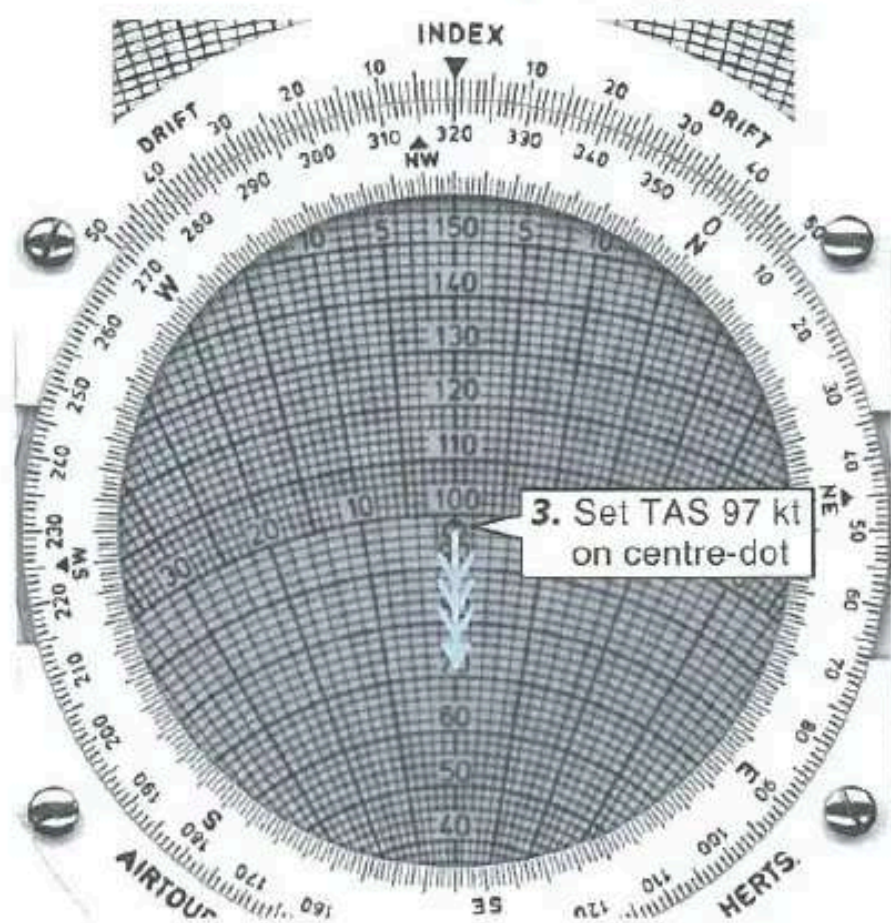
■ Figure 4-41 Example 8, Step 1

STEP 2. PLACE THE HDG/TAS UNDER THE CENTRE-DOT.

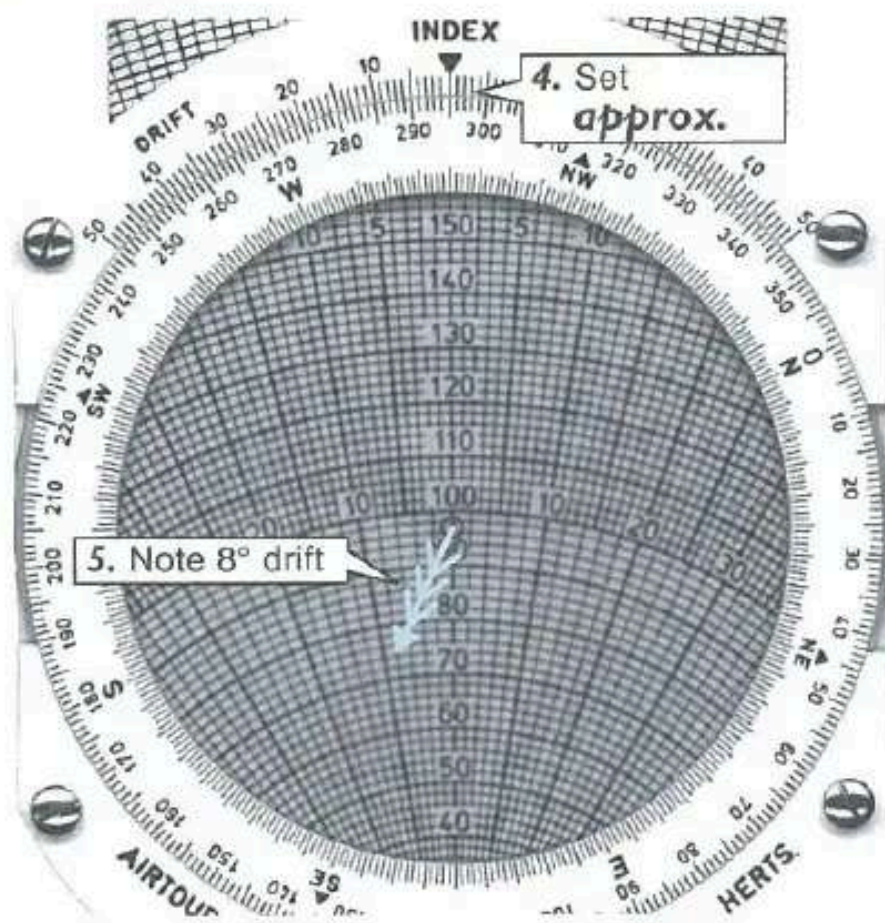
Since the W/V starts at the centre-dot and blows away from it, place the end of the HDG/TAS vector under the centre-dot.

Move the slide until TAS 97 kt appears under the centre-dot. Set approximate HDG 295°T under the index.

When we come to set the HDG under the index, we are faced with a problem – we do not know the HDG! To get started, use the desired track as an approximate HDG; i.e. assume that the HDG is approximately 295°T and rotate the compass rose until 295°T appears under the index.



■ Figure 4-42 Example 8, Step 2a



■ Figure 4-43 Example 8, Step 2b

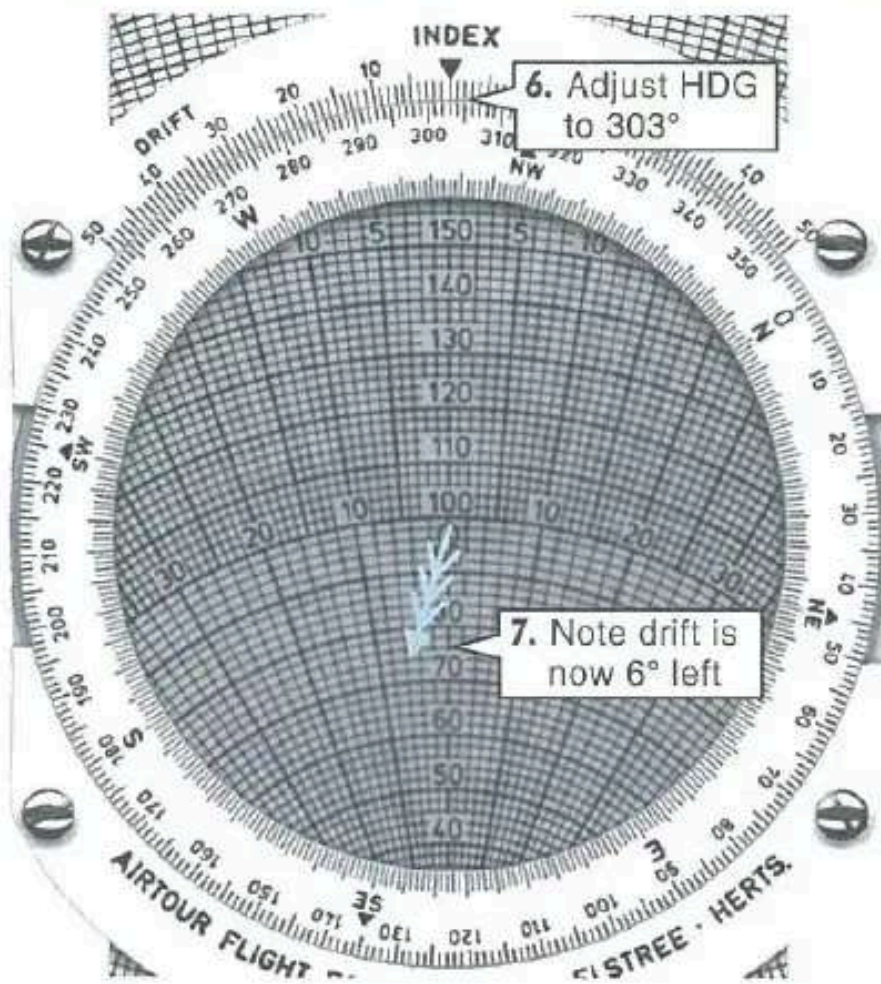
STEP 3. ADJUST THE APPROXIMATE HDG TO ALLOW FOR DRIFT.

The W/V dot lies over the 8° left drift line, so, on a HDG of 295°T, the aeroplane should achieve a track 8° to the left of this.

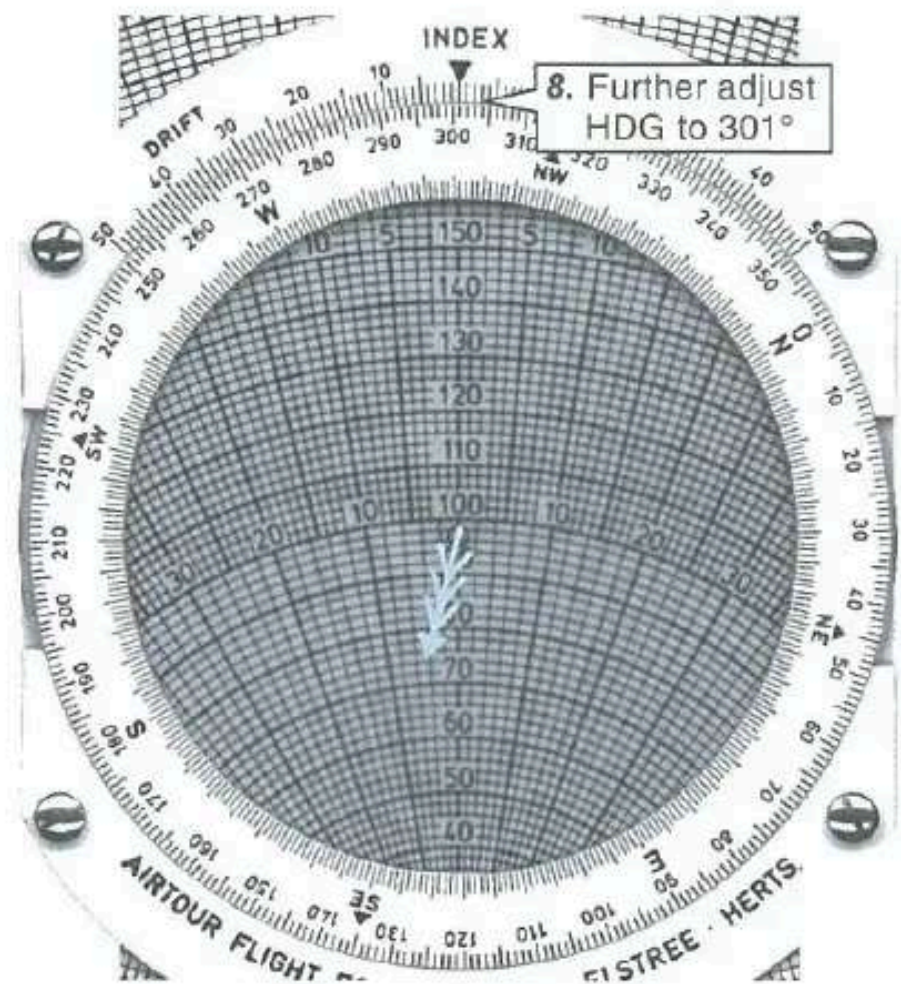
Since the desired track is 295°, the aeroplane should be heading approximately 8° to the right of this to allow for the left drift, i.e. on a HDG of 303°T, with a wind correction angle of 8° right.

Rotate the compass rose until 303°T appears under the index, and check the drift (and calculate the track). The drift is now indicated to be only 6° (not the original 8°), so adjust the HDG to allow for only 6°, i.e. an adjusted HDG is 301°T.

Set the adjusted HDG 301°T under the index. By checking the position of the wind-mark, we see that this minor ‘jiggle’ of the compass rose has not appreciably altered the 6° drift. (In other words, the adjusted HDG and the expected 6° left (port) drift will allow us to achieve the desired track of 295°T.)



■ Figure 4-44 Example 8, Step 3a



■ Figure 4-45 Example 8, Step 3b

STEP 4. READ OFF THE HDG AND GS.

The HDG 301°T appears under the index. The GS 73 kt appears under the wind-mark.

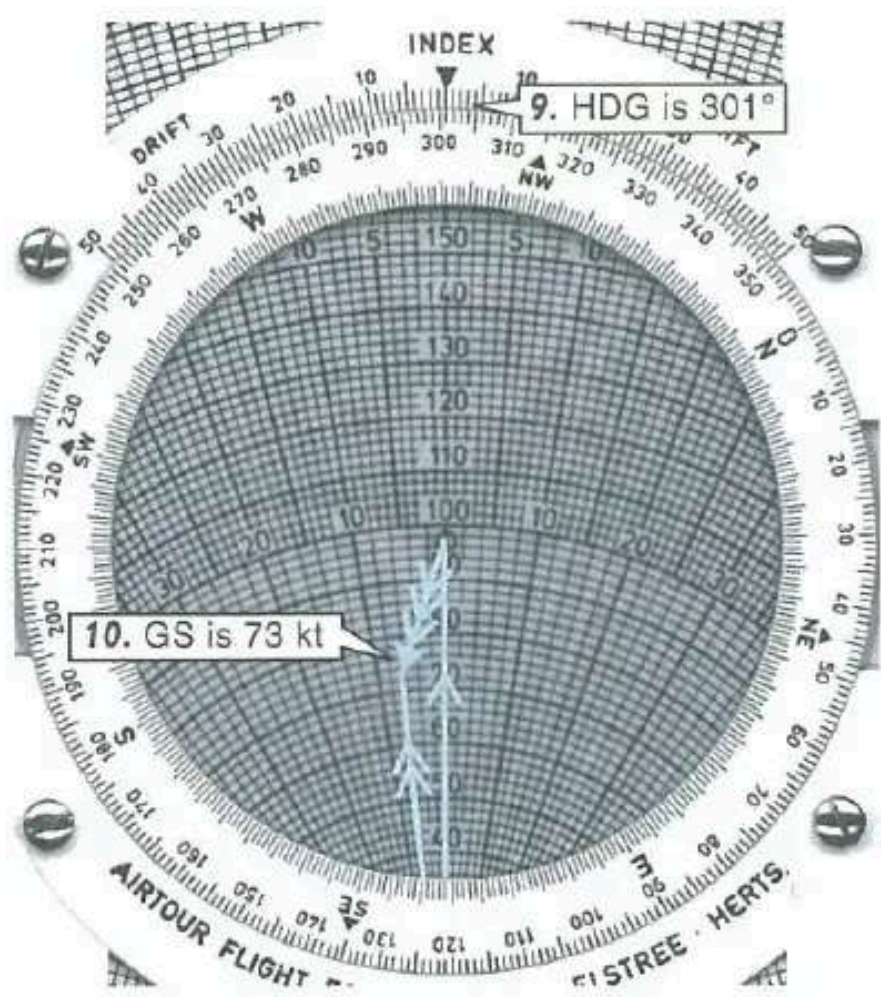


Figure 4-46 Example 8, Step 4

ANSWER HDG 301°T, GS 73 kt

From/To	Safety ALT	ALT Temp	RAS	TAS	W/V	TR °T	Drift	HDG °T	Var	HDG °M	GS	Dist	Time	ETA	HDG °C
				97	320/25	295		301			73				

Figure 4-47 Example 8

NOTE Because there have to be some adjustments to the initial HDG by rotating the compass rose a few degrees this way and that, the above method is sometimes also known as the ‘jiggle’ method, as well as the ‘wind-mark-down’ method. It still gives accurate answers.

To solve a slightly unusual flight-planning problem such as “What TAS is required to achieve a groundspeed of 80 kt?” the same procedure as above can be used, except that the wind-mark is placed over the appropriate GS arc, and then the TAS and HDG can be read off under the centre-dot.

Now complete Exercises 4 – Wind Side-1.



EXAMPLE 9 Find HDG and GS, Knowing TR, TAS and WIV

This variation on the ‘wind-mark-down’ method avoids having to jiggle the compass rose when calculating HDG and GS at the flight-planning stage. It repeats the previous example. Check if your instructor recommends this method.

KNOWN:		FIND:
TR	295°T	HDG and GS
TAS	97 kt	
WIV	320°T/25 kt	

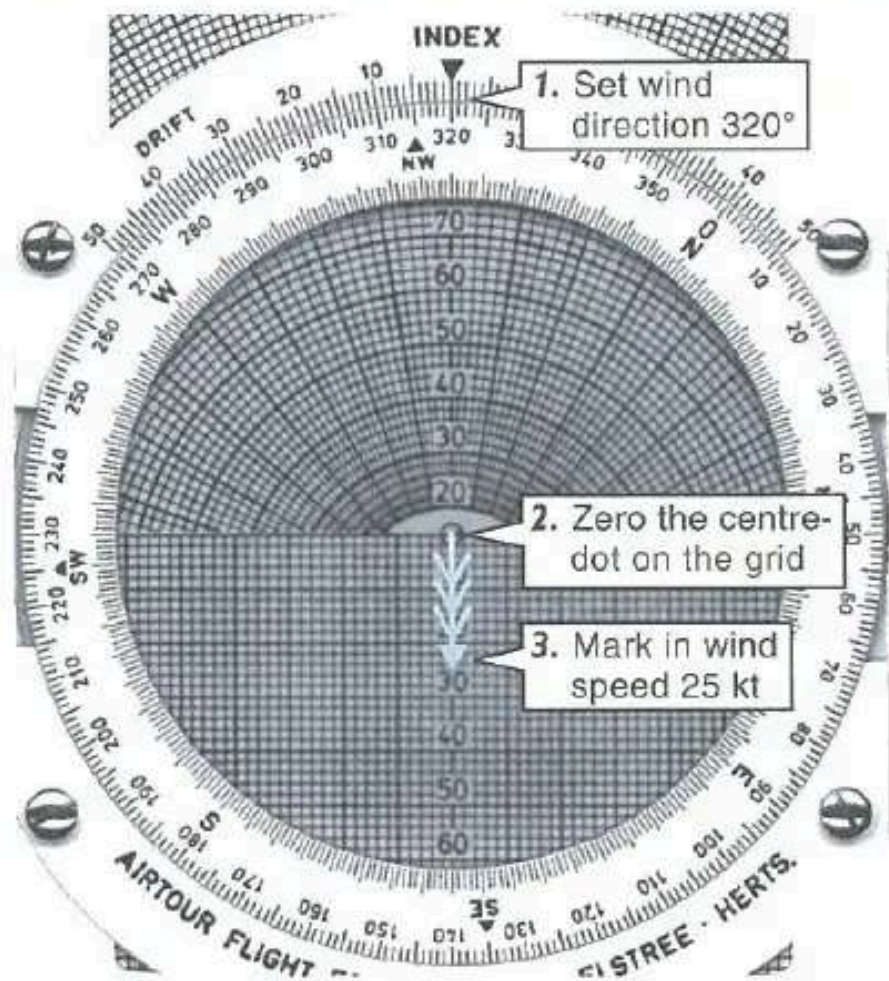
From/To	Safety ALT	ALT Temp	RAS	TAS	WIV	TR °T	Drift	HDG °T	Var	HDG °M	GS	Dist	Time	ETA	HDG °C
				97	320/25	295									

■ Figure 4-48 Example 9

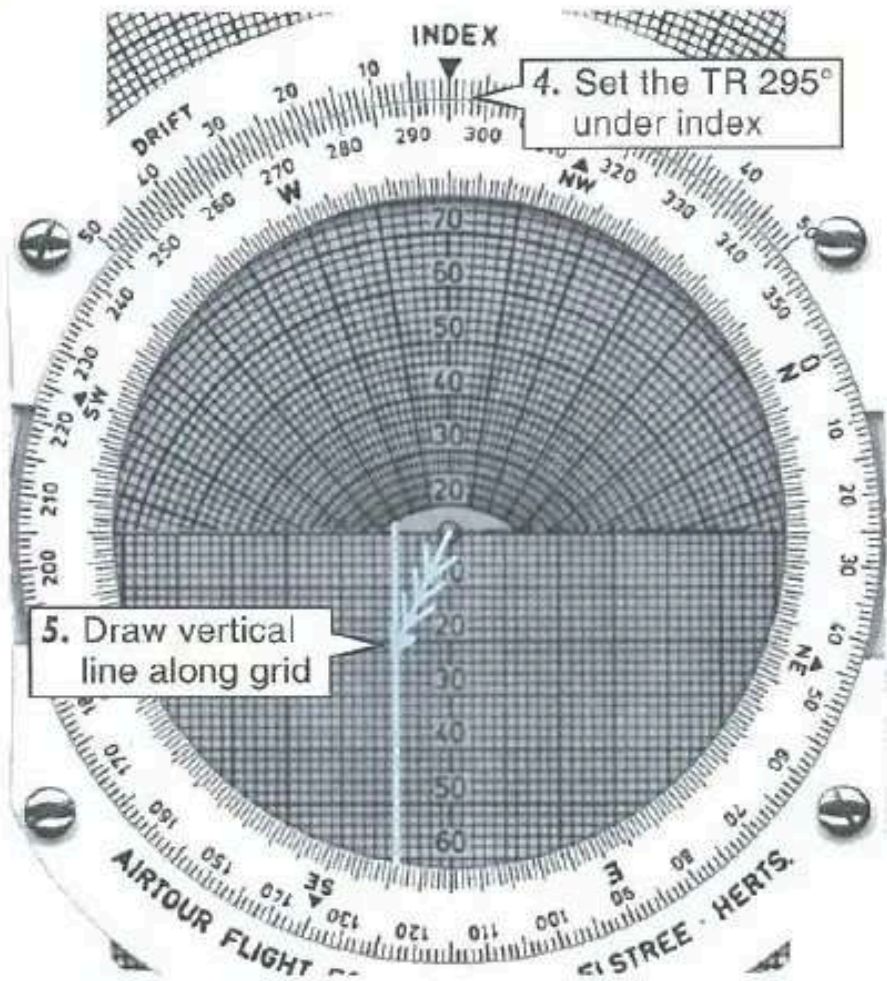
STEP 1. SET W/V ON THE PLOTTING DISC USING THE SQUARE GRID.

Rotate the compass rose until the wind direction 320°T appears under the index.

Set the zero point of the square grid under the centre-dot. Put a wind-mark 25 kt down from the centre-dot.



■ Figure 4-49 Example 9, Step 1



■ Figure 4-50 Example 9, Step 2

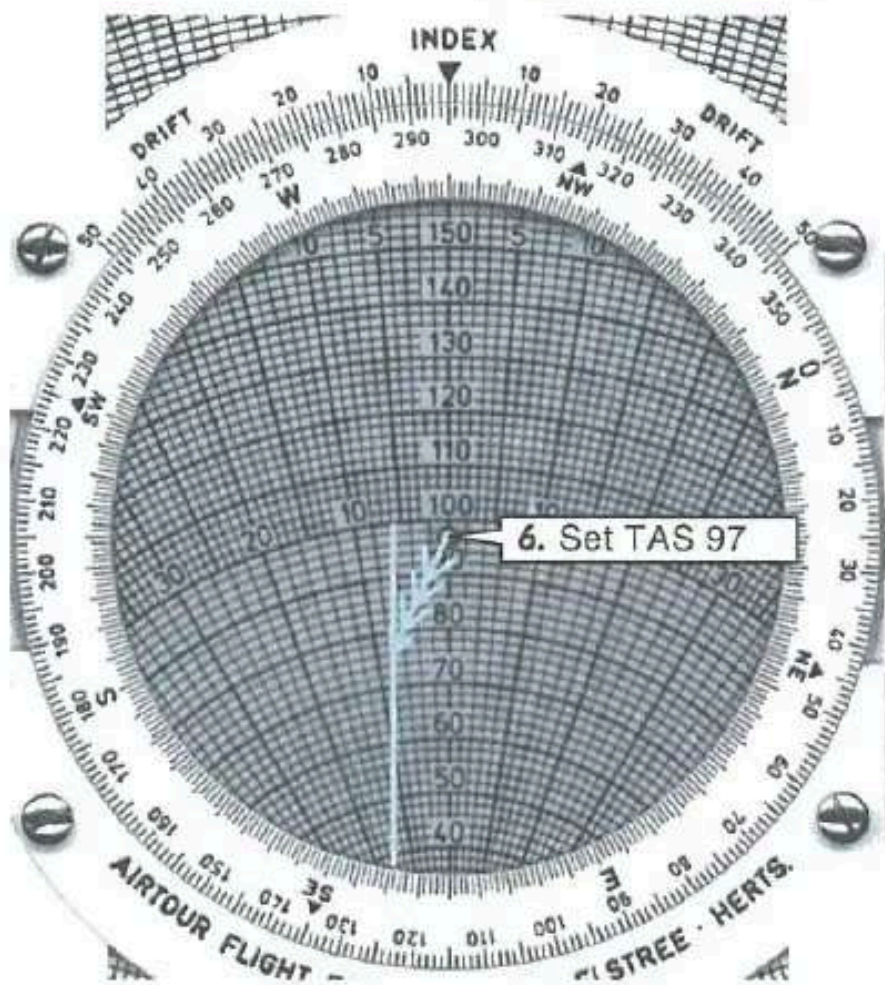
STEP 2. SET TRACK, AND MARK IN CROSSWIND LINE.

Rotate the compass rose until the TR 295°T appears under the index. Run a vertical line down through the wind-mark, which is a 10 kt crosswind component. It is this crosswind component from the right that causes the left drift.

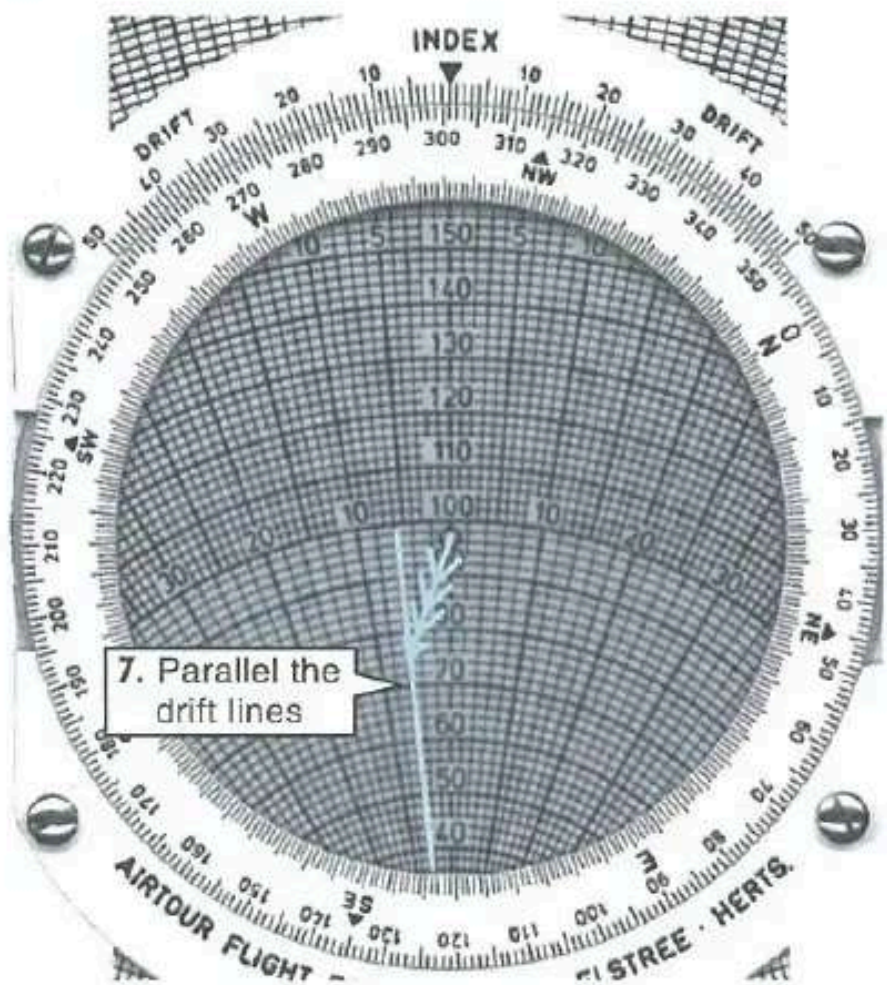
STEP 3. SET HDG/TAS ON THE PLOTTING DISC.

Move the slide and place TAS 97 kt under the centre-dot. Note that the crosswind effect line does not parallel the drift lines at this stage.

Rotate the compass rose until the crosswind effect line is parallel with one of the drift lines.



■ Figure 4-51 Example 9, Step 3a

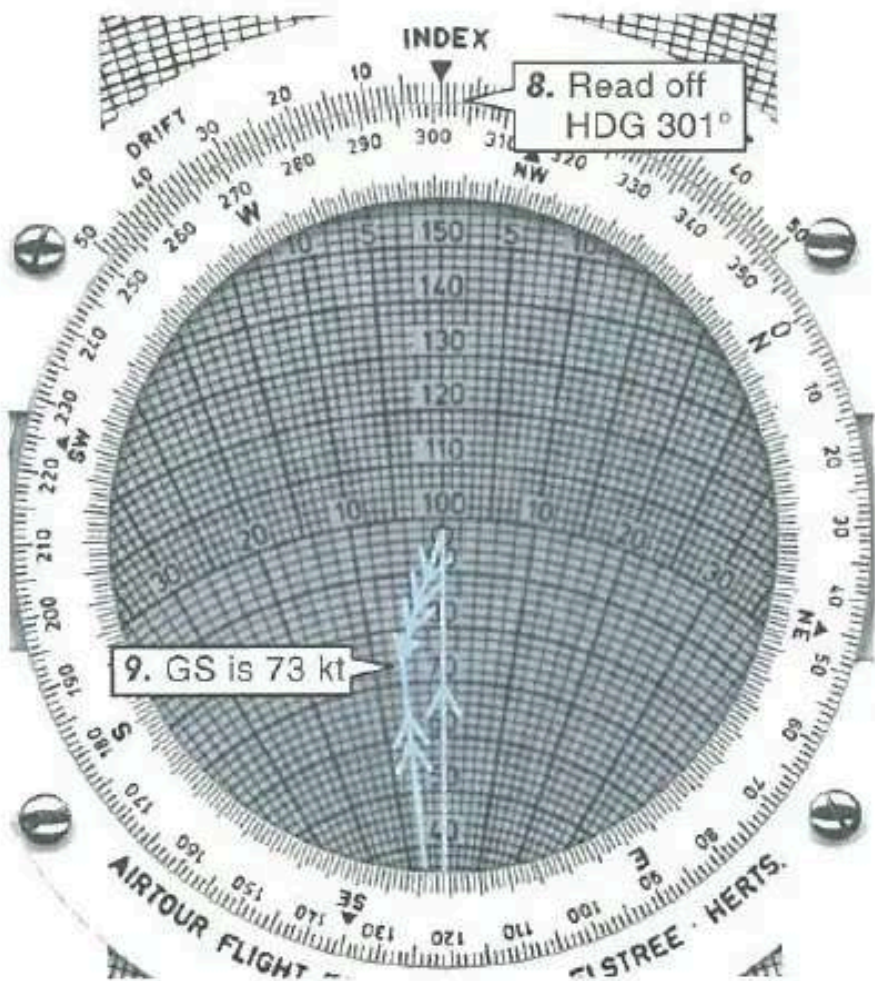


■ Figure 4-52 Example 9, Step 3b

STEP 4. READ OFF HDG AND GS.

HDG/TAS lies under the centre-dot. The TR/GS vector ends where the W/V vector ends.

Read off HDG 301°T under the index. Read off GS 73 kt under the wind-mark.



■ Figure 4-53 Example 9, Step 4

ANSWER HDG 301°T, GS 73 kt

From/To	Safety ALT	ALT Temp	RAS	TAS	WIV	TR °T	Drift	HDG °T	Var	HDG °M	GS	Dist	Time	ETA	HDG °C
				97	320/25	295		301			73				

■ Figure 4-54 Example 9

Now complete **Exercises 4 – Wind Side-1** using this method.



Wind Components

Often a wind needs to be broken down into its two components:

- the headwind or tailwind component; and
- the crosswind component.

This is especially the case when taking off and landing, because:

1. For performance reasons, you often need to know the headwind or tailwind component to determine the take-off or landing distance required.
2. For safe aeroplane handling, you always need to know (at least approximately) the crosswind component on the runway that you intend using. The crosswind component is very important and every aeroplane has a maximum crosswind limit

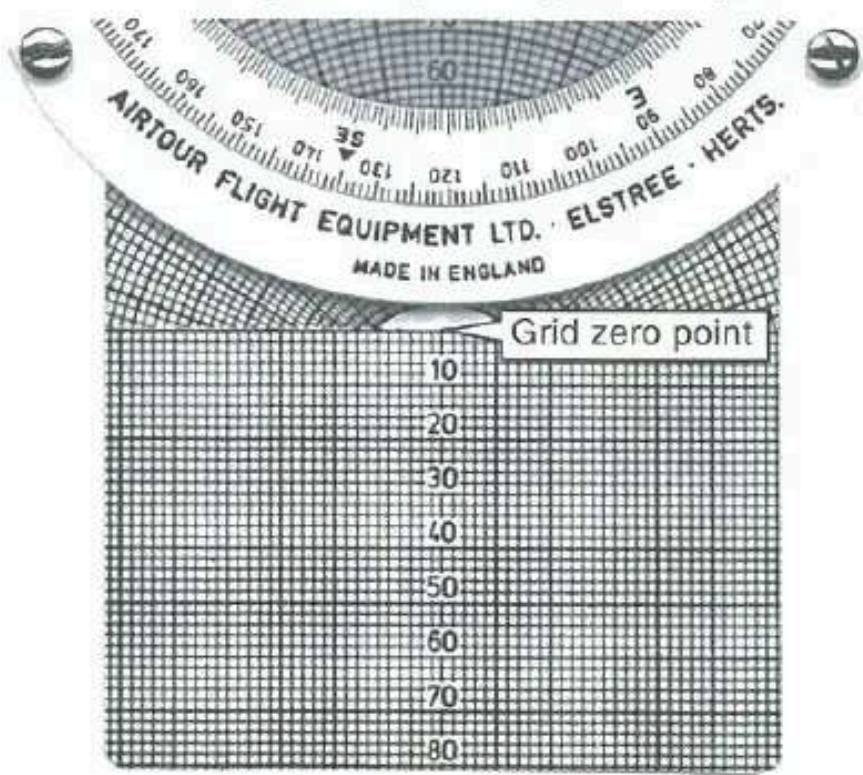
specified in its Flight Manual. This crosswind limit should never be exceeded.

Winds found on forecasts, which are most likely to be used for flight-planning purposes, are given in degrees true, whereas the wind in take-off and landing reports broadcast by Air Traffic Control are given in degrees magnetic, so that they can be easily related to runway direction, which is always in °M. This applies to the direction of the wind given to you by the Tower, or as broadcast on the Automatic Terminal Information Service (ATIS).

A runway whose centreline lies in the direction 074°M will be designated Runway 07. A runway whose centreline lies in the direction 357°M will be designated Runway 36. A wind of 350°M/25 kt would favour RWY 36, which is almost directly into-wind. RWY 07 would experience a strong crosswind from the left; the pilot should determine just how strong before using this particular runway.

Since both wind direction and runway direction are measured from the same datum (magnetic north), there is no need to convert into degrees true for your computer manipulations. When using your computer, work either totally in °T or totally in °M.

In Vol. 4 of *The Air Pilot's Manual*, when discussing take-off and landing performance, we illustrate a simple way of determining wind components mentally. Here we show you how to do it using the square grid on your navigation computer.



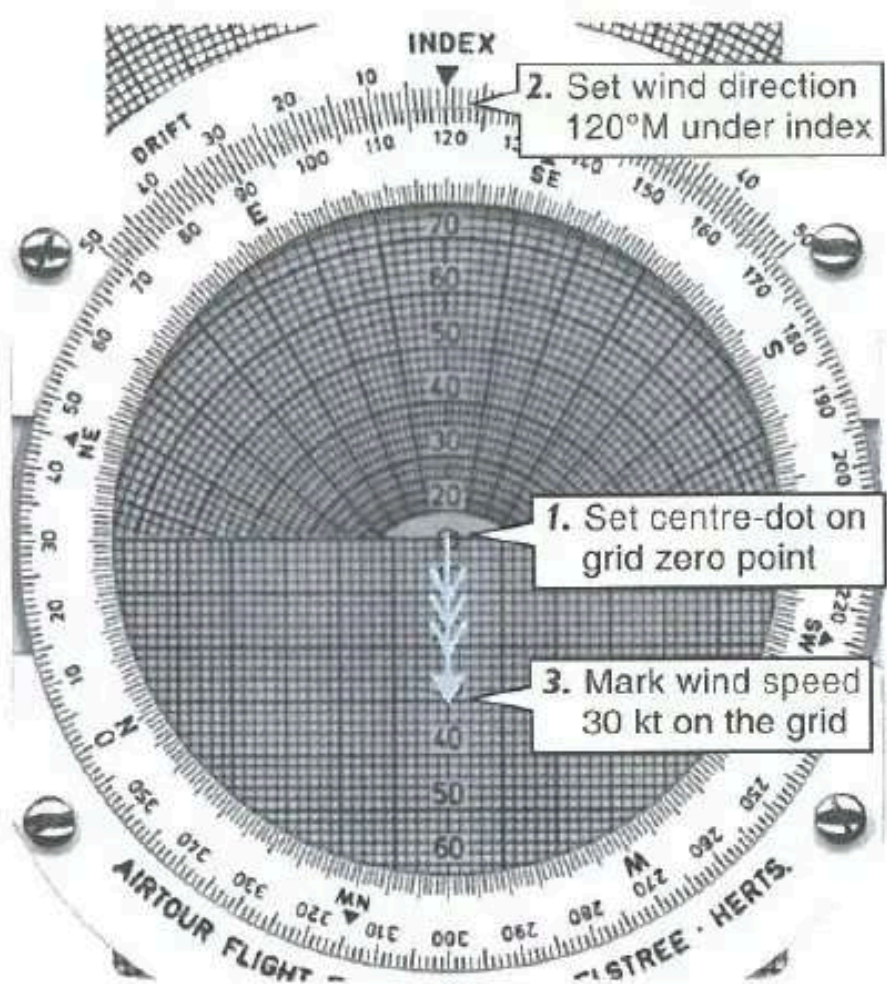
■ Figure 4-55 The square grid on the computer slide

Finding Wind Components

What crosswind and headwind components exist on Runway 18 if the wind broadcast by the Tower is 120°M/30? (Runway 18 means that the runway direction is approx 180°M.)

STEP 1. SET UP THE W/V ON THE SQUARE GRID.

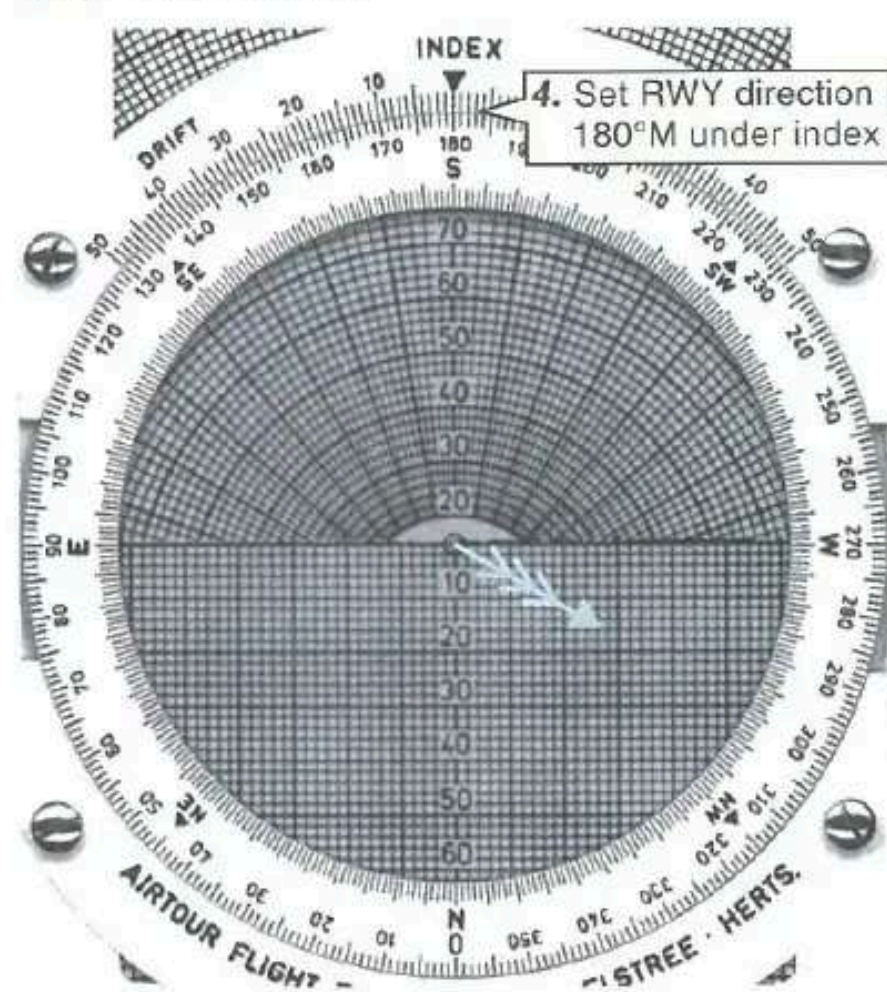
Set the zero point of the square grid under the centre-dot. Rotate the compass rose and set wind direction 120° under the index. Put a wind-mark 30 kt down from the centre-dot.



■ Figure 4-56 Wind components, Step 1

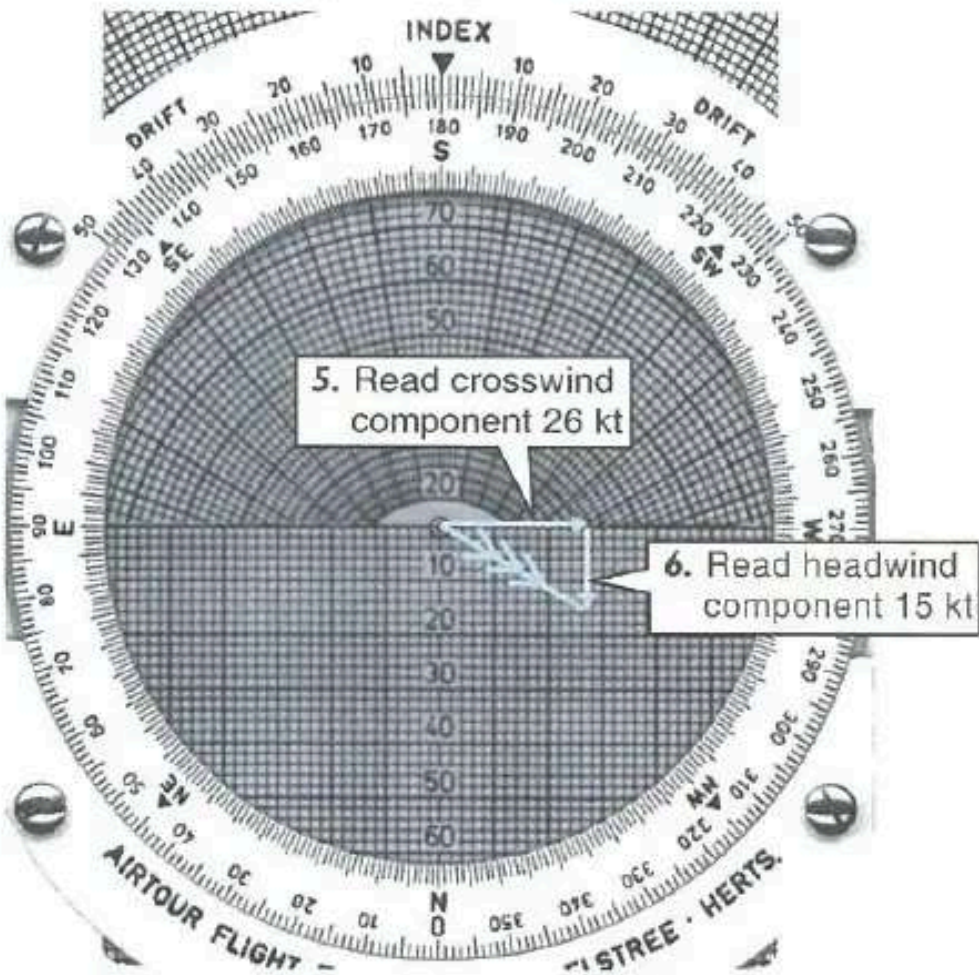
STEP 2. SET RUNWAY DIRECTION.

Rotate the compass rose until the runway direction 180°M is under the index.



■ Figure 4-57 Wind components, Step 2

STEP 3. READ OFF HEADWIND AND CROSSWIND COMPONENTS.
Drop a vertical line to the wind-mark. The length of this line is the headwind component, 15 kt.
The horizontal distance of this line from the centre-dot is the crosswind component, 26 kt.



■ Figure 4-58 **Wind components, Step 3**

ANSWER Headwind (H/W) 15 kt, crosswind (X/W) 26 kt

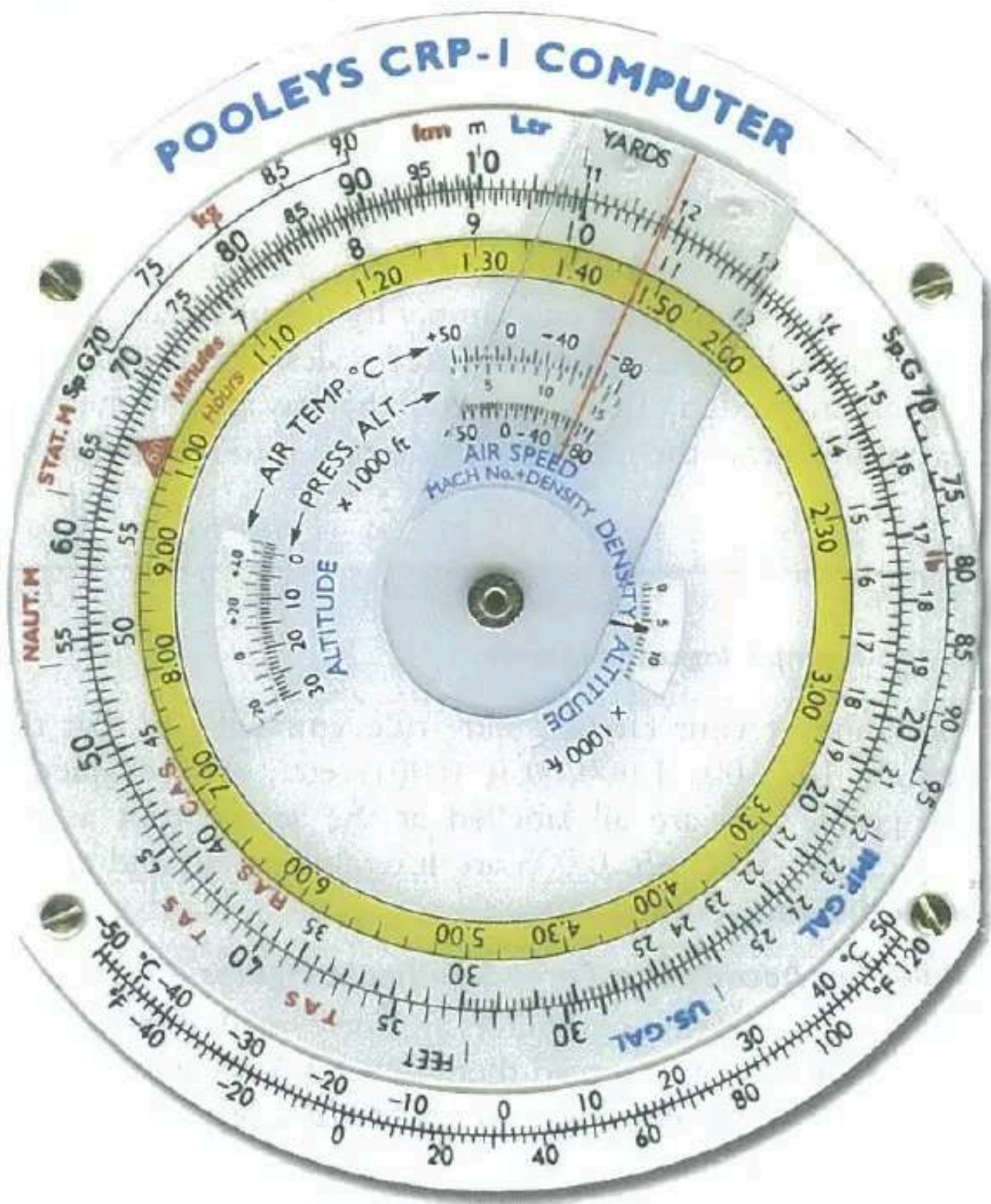
NOTE If, when the runway direction is set under the index, the end of the wind vector is above the horizontal line through the centre-dot, then there is a tailwind component and it is advisable to consider changing runways. The precise strength of the downwind component can be found by placing the reciprocal to the runway direction under the index (360 in the above case), which allows you to find what is a headwind component for a take-off in that direction.

Now complete **Exercises 4 – Wind Side-4.**



Calculator Side of the Navigation Computer

The calculator side of the navigation computer looks complicated at first, but once you become familiar with it you will find its use straightforward. A little work now learning how to use it will be repaid many times over, as it simplifies your calculations and saves lots of time. The calculator side is useful for solving quickly and accurately the numerous small calculations involved in air navigation.



■ Figure 5-1 **Calculator side of a navigation computer**

Various computer types differ in minor ways on the calculator side but the way they are used is the same. As you go through this chapter we suggest that you have your computer close by and follow our examples with it.

Use of the computer for solving the altitude and airspeed problems on the small scales in the mid-area of the calculator side is covered in Chapter 2, Speed and Chapter 7, Time. In this chapter we cover what can be done with the two outside scales, which form a circular slide rule.

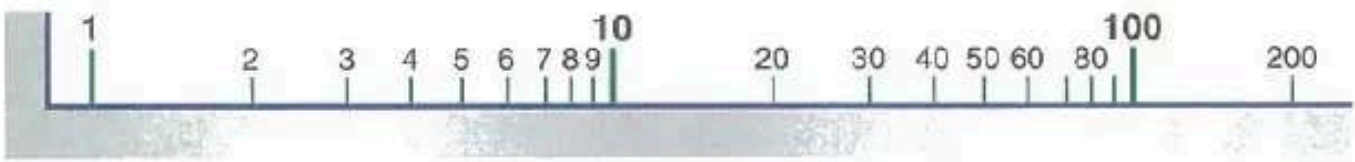
The Circular Slide Rule

Simple multiplication and division can be done mentally, or, if more complicated, with electronic calculators, or with *logarithms*. A logarithm is one of a series of numbers, set out in tables, that make it possible to work out problems by adding and subtracting numbers, instead of multiplying and dividing. A slide rule is just a pictorial means of using logarithms.

To avoid the need for pilots to carry long, straight slide rules, a circular slide rule has been devised. It is fast, never has flat batteries, and is accurate enough for our purposes. The circular slide rule has two scales:

- an inner rotary scale; and
- an outer fixed scale.

Both scales are marked with logarithmic graduations, making it possible to multiply and divide simply by the physical addition or subtraction of lengths of the graduated scales. The numbers on the scales are marked in order from 10 all the way around the scale to 100, but the larger the number the closer it is to its neighbour.

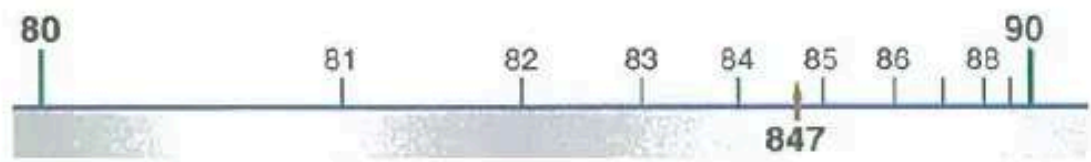


■ Figure 5-2 A simple logarithmic scale

If you look at your circular slide rule you will see that the numbers 1, 10, 100, 1,000, 0.1, 0.001, etc., are completely interchangeable and are all labelled at the same point as 10. Similarly, 5, 50, 500, 0.5, 0.005 are interchangeable and are all labelled at the same point as 50.

Slide Rules are Accurate to Three Significant Figures

Slide rules and navigation computers are fairly small in size, so the accuracy to which you can read them depends on your eyesight. Most slide rule scales allow you to read to an accuracy of three digits.



■ Figure 5-3 This scale reads 847

Unfortunately the scale does not tell us where the decimal point belongs in relation to the three significant figures. For example, 847 could be written any of the following ways:

847.0
84.7
8.47
0.847
8,470.0

To know where to place the decimal point you must estimate the approximate answer to the question by quick mental arithmetic. The mental calculations give an approximate answer (allowing us to place the decimal point correctly); the slide rule manipulations give us that answer accurate to three significant figures.

Calculations

When using the slide rule part of the navigation computer, always follow this procedure:

1. Rough mental calculation.
2. Slide rule manipulation to obtain three significant figures.
3. Place the decimal point (from rough mental calculation).

As well as telling you where to place the decimal point, the rough mental calculation is a check that your computer manipulations were at least approximately correct and your answer not grossly in error. First doing the mental check results in fewer errors, and is good practice for the many times in flight and on the ground when mental calculations are necessary.

Multiplication

EXAMPLE 1 Multiply 3.25×4.29

Step 1. Rough mental calculation

$$3 \times 5 = 15$$

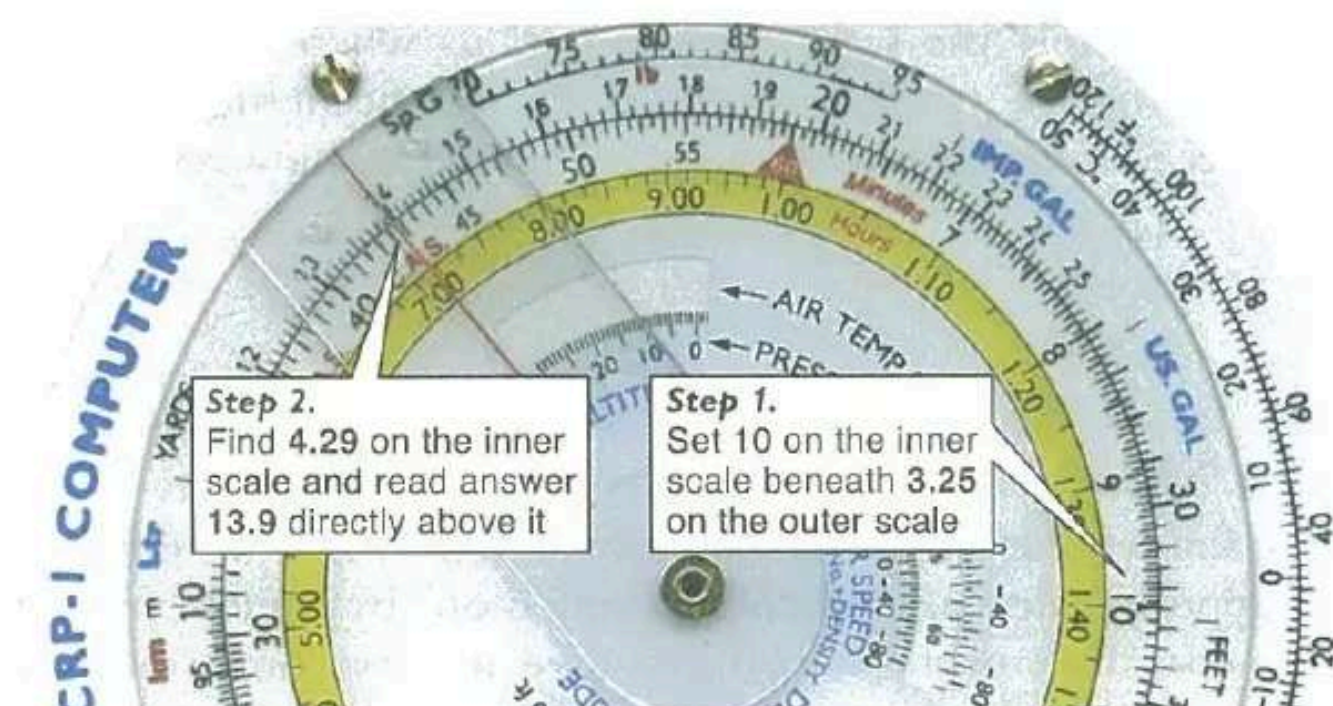
Step 2. Computer manipulation

Treat 10 on the outer scale as the starting point and find 3.25. Set 10 on the inner scale below 3.25 on the outer (or 32.5 or 325 or 3,250). Find 4.29 on the inner scale and mark directly above it, i.e. 139.

Step 3. Place the decimal point

Determine position of decimal point by referring back to the rough calculation $3 \times 5 = 15$, so the answer is 13.9. It is obviously not 1.39 or 139.0.

ANSWER $3.25 \times 4.29 = 13.9$



■ Figure 5-4 The multiplication $3.25 \times 4.29 = ?$ on the circular slide rule

Division

EXAMPLE 2 The division 36 divided by 12 ($36 \div 12$) is an easy one to begin with. We simply need to subtract the logarithm of 12 from the logarithm of 36.



■ Figure 5-5 The division $36 \div 12$ on the circular slide rule

Step 1. Rough calculation

$$36 \div 12 = 3$$

Step 2. Computer manipulation

Treat 10 on the outer scale as the starting point; the logarithm of 36 is the distance from 10 to 36 on the outer scale. Beneath 36 place 12 (the logarithm of 12 is the distance from 10 to 12 on the inner scale).

Then locate 10 on the inner scale and above it find the answer, 30. (Do not confuse 10 with 1:00 which symbolises *1 hour = 60 minutes* on some navigation computers.) The log of 12 has thus been subtracted from the log of 36.

Step 3. Place the decimal point

To determine the position of the decimal point is easy in this case, because we know that $36 \div 12 = 3$, and so the answer is not 30 or 300 but 3.0.

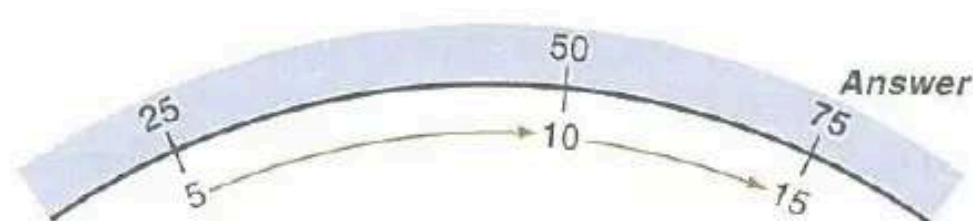
ANSWER $36 \div 12 = 3.0$

Combined Multiplication and Division

Occasionally calculations involve both multiplication and division, and the circular slide rule is ideal for this. The best approach is to

begin with a division and follow with the multiplication. This process can be repeated as many times as necessary.

EXAMPLE 3 Calculate 25 times 15, divided by 5. (Another simple one that can be done mentally, but a good example to begin with.)



■ Figure 5-6 The calculation $\frac{25 \times 15}{5}$ on the circular slide rule

Step 1. Rough check

$25 \div 5 = 5$ which, multiplied by 15, gives $5 \times 15 = 75$.

Step 2. Computer manipulation

Do the division $25 \div 5$ by placing 5 on the inner scale under 25 on the outer scale. (The answer to this appears on the outer scale above the 10 on the inner scale, and the setting does not have to be altered to do the multiplication of this answer by 15.)

Do the multiplication by finding 15 on the inner scale and marking the answer above it on the outer scale, i.e. 75. (You will determine the position of the decimal point by doing a quick mental check $25 \div 5 = 5$, which is multiplied by 15 to give 75.0.)

Step 3. Place the decimal point

Reference to our rough check confirms that the decimal point is placed to give 75.0.

ANSWER 75.0

EXAMPLE 4 Calculate $\frac{25.7 \times 3.96}{5.12}$

Step 1. Rough calculation

$$\frac{25 \times 4}{5} = 20$$

Step 2. Computer manipulation

Set 25.7 on the outer scale against 5.12 on the inner scale. Against 3.96 on the inner scale read off on the outer scale the result 19.9.



■ Figure 5-7 Calculation of $\frac{25.7 \times 3.96}{5.12}$

Step 3. Place the decimal point

From our mental calculation, the decimal point should be placed thus: 19.9.

ANSWER 19.9

Ratios

$D = \frac{25 \times 4}{5}$ may be expressed slightly differently as a ratio:

$$\frac{D}{4} = \frac{25}{5} \quad \text{and find } D.$$

A rough mental check indicates that D is approximately 20. The set-up on the circular slide rule is shown in Figure 5-8. The answer is of course 20.0. This could be the answer to a number of navigation questions:

EXAMPLE 5 If an aeroplane travels 25 nautical miles (nm) in 5 minutes, how far would it travel in a further 4 minutes?

ANSWER 20 nm



■ Figure 5-8 Finding ratios on the circular slide rule

EXAMPLE 6 If an aeroplane burns 25 litres of fuel in 5 minutes, how many litres would the aeroplane have burned at the same rate in 4 minutes?

ANSWER 20 litres

EXAMPLE 7 If an aeroplane travels 25 nm in 5 minutes, how far would it travel in 40 minutes?

ANSWER 200 nm

EXAMPLE 8 If you descend 2,500 feet in 5 minutes, how far would you expect to descend in the next 4 minutes? What is your rate of descent?

ANSWER 2,000 ft, and 500 ft/min (1 minute is 10 on the inner scale)

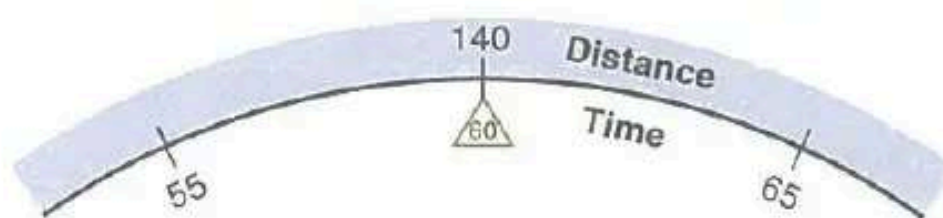
These are all common navigation calculations and you must be able to perform them with speed and accuracy.

Solving Speed, Distance, Time and Ratio Problems

Speed is the ratio of distance/time. As there are 60 minutes in 1 hour, a speed of 140 knots is the same as travelling a distance of 140 nautical miles over the ground in 60 minutes. We can set this up on the circular slide rule by placing the 60 minutes (sometimes written as 1:00 hour) on the inner TIME scale against the 140 on the outer DISTANCE scale.

On most computers the inner scale is marked TIME and the outer scale DISTANCE (usually somewhere near the 60 mark). This

is important to keep in mind – that TIME is always on the inner scale and DISTANCE/SPEED on the outer.



■ Figure 5-9 A typical distance-time problem

The circular slide rule is now set up to answer many questions, such as:

EXAMPLE 9 At a groundspeed (GS) of 140 kt, how far will you travel in 30 minutes?

Step 1. Rough Check

30 minutes is half an hour which, at 140 kt, is 70 nm.

Step 2.

Find 30 minutes on the inner TIME scale and read off the answer 70 nm on the outer DISTANCE scale.

ANSWER 70 nm

EXAMPLE 10 At a groundspeed (GS) of 140 kt, how far will you travel in 15 minutes?

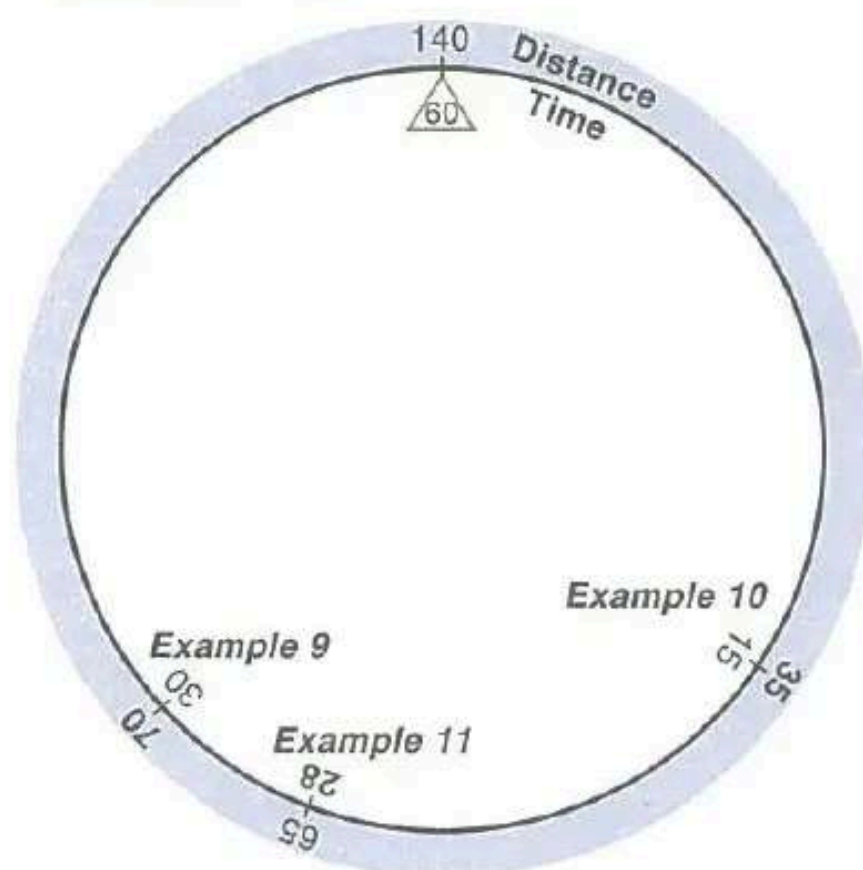
Step 1. Rough Check

15 minutes is $\frac{1}{4}$ hour which, at 140 kt, gives $140 \div 4 = 35$ nm.

Step 2.

Find 15 minutes on the inner TIME scale and then read off the answer 35 nm on the outer DISTANCE scale.

ANSWER 35 nm



■ Figure 5-10 Speed, time and distance problems on the circular slide rule

EXAMPLE 11 At a GS of 140 kt, how long will it take you to travel 65 nm?

Step 1. Rough Check

65 nm is slightly less than $\frac{1}{2}$ of 140 nm which, at 140 kt, will take slightly less than 30 minutes to cover.

Step 2.

Find 65 nm on the outer DISTANCE scale and read off the answer 28 minutes (approximately) on the inner TIME scale.

ANSWER 28 minutes (approximately)

Now complete **Exercises 5 – Calculator-1.**



Further Problems

The computer can be used to help us solve many different types of problem involving multiplication, division or ratios. Another sort of calculation we can make is:

EXAMPLE 12 If we cover 16 nm over the ground in 10 minutes, what is our groundspeed?

Step 1. Rough check

10 minutes is $\frac{1}{6}$ hour. We cover 16 nm in 10 minutes, so we will cover six times this distance in an hour, i.e. $6 \times 16 = 96$ nm. The speed is therefore 96 kt.

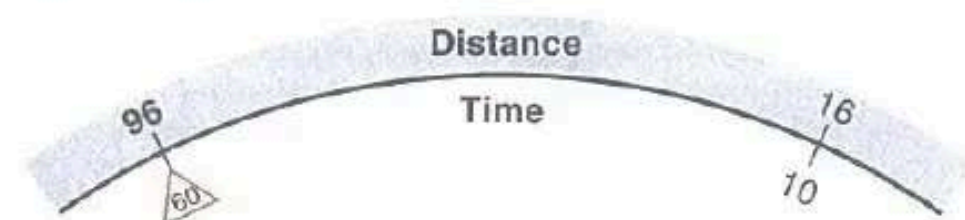
Step 2.

Set up 16 nm on the outer DISTANCE scale against 10 minutes on the inner TIME scale.

Step 3.

Against 60 minutes (1:00 hour) on the inner TIME scale read off on the outer DISTANCE scale the distance you would travel in that time, which of course is 96 nm; that is, the GS is 96 kt.

ANSWER 96 kt



■ Figure 5-11 **Finding groundspeed**

The computer is now set up to answer other questions relevant to this situation, such as:

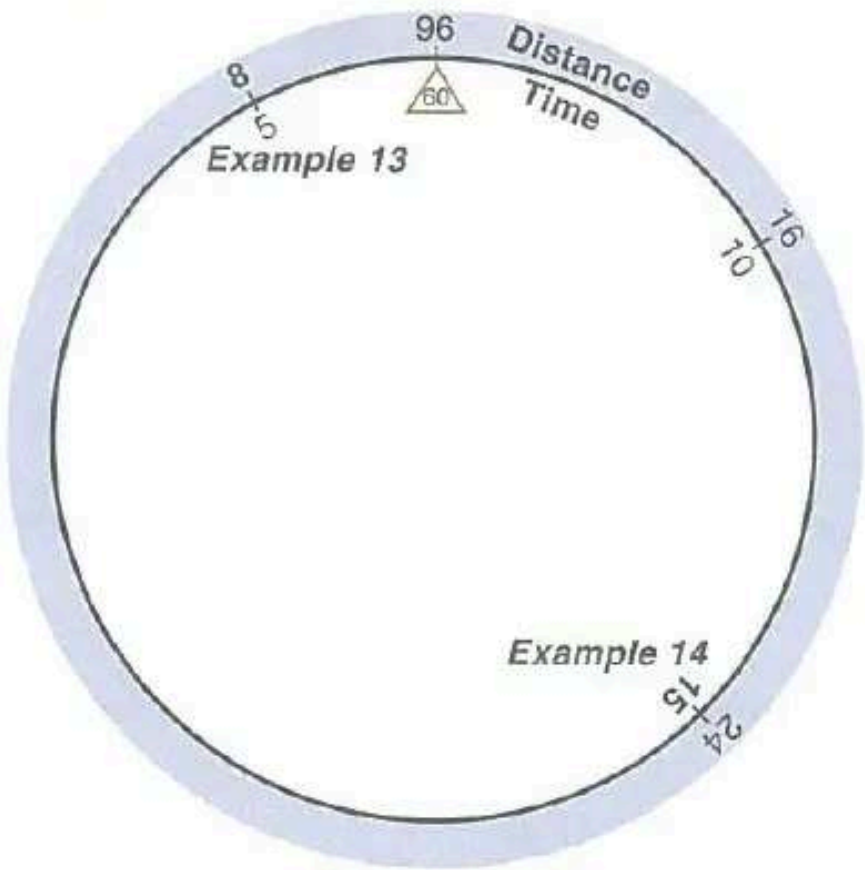
EXAMPLE 13 How far will the aeroplane then travel in a further 5 minutes?

ANSWER 8 nm

EXAMPLE 14 How long will it take to travel 24 nm?

ANSWER 15 minutes

In each case, a rough mental check will confirm that the answers are 8 nm (and not 80 or 0.8) and 15 minutes (and not 150 or 1.5).



■ Figure 5-12 Time-speed-distance set-up

Now complete **Exercises 5 – Calculator-2**.



Fuel Consumption Problems

EXAMPLE 15 If an aeroplane is burning fuel at the rate of 30 litres per hour, what fuel burn-off can you expect in 8 minutes?

Step 1. Rough check

30 litres per hour is $\frac{1}{2}$ litre per minute which, for 8 minutes, will give a burn-off of 4 litres.

Step 2.

Set 60 minutes (1:00 hour) on the inner TIME scale against 30 (litres) on the outer scale.

Step 3.

Against 8 minutes on the inner TIME scale read off 4 (litres) on the outer scale. (Our rough mental calculation indicates that the answer is 4 and not 40 or 400 or 0.4).

ANSWER 4 litres



■ Figure 5-13 Example 15, a typical fuel consumption problem

EXAMPLE 16 If we have burned 4 litres in 10 minutes, how much will we burn in the next 25 minutes?



■ Figure 5-14 **Example 16**

Step 1. Rough check

$25 = 2.5 \times 10$, so expect to use $2.5 \times 4 = 10$ litres.

Step 2.

Set up 10 minutes on the inner TIME scale against 4 litres on the outer scale.

Step 3.

Against 25 minutes on the inner TIME scale read off 10 litres on the outer scale.

ANSWER 10 litres (not 1 or 100 or 1,000)

The circular slide rule is now set up to answer other problems relevant to this situation, such as:

1. What is the rate of fuel consumption in litres/hour?

Answer: 24 litres/hour

2. How long would it take to burn 28 litres?

Answer: 70 minutes

EXAMPLE 17 If your aeroplane has 26 US gallons of usable fuel in the tanks, and the average consumption rate is 5.5 USG/hour, what is the safe flight endurance if you wish to land with 1 hour's reserve unused?

Step 1. Rough check

Subtracting the reserve (5.5 USG in this case) leaves 20.5 USG as flight fuel. A rough mental check (20 USG at 6 USG/hour) = $3\frac{1}{3}$ hours.

Step 2.

By computer, with 60 minutes on the inner scale set against 5.5 USG on the outer, read against 20.5 USG on the outer scale the answer of 224 minutes on the inner.

ANSWER 3 hours 44 minutes (224 minutes)

Remember the Rough Mental Check

The rough mental check will avoid gross errors being made. For example, an answer of 22.4 minutes to the above question may not seem in error to a pilot who does not back up his computer manipulations with mental checks, yet it is in error by a factor of 10. As well as allowing you to place the decimal point correctly, a mental check also allows an approximate check of the numbers involved.

Do not neglect the mental check in order to save time or energy. What is the point of calculating the fuel to be 139 and ordering 1.39 litres or 13.9 litres rather than 139 litres?

It seems ridiculous, but rough estimates to back up computer calculations could have prevented a number of accidents, including some involving large jets. Learn the value of mental checks early in your flying career; professional pilots make mental checks all the time.

Now complete **Exercises 5 – Calculator-3.**



‘Off-Track’ Problems

EXAMPLE 18 If an aeroplane is 5 nm off-track after travelling 20 nm, how far off-track will it be after travelling 45 nm?

Step 1. Rough check

This is a simple ratio problem of $\frac{5}{20} = \frac{?}{45}$

$\frac{5}{20} = \frac{10}{40}$ Answer will be slightly greater than 10.

Step 2.

- 1. Set up 5 against 20.
- 2. Find the answer 11.2 against 45.



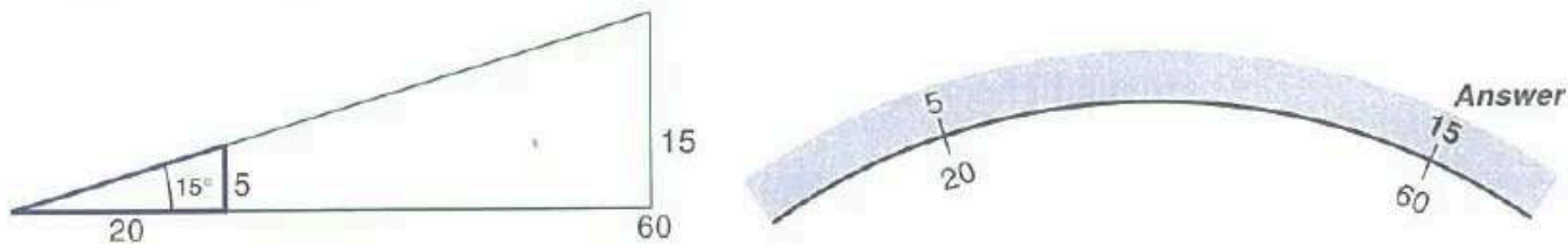
■ Figure 5-15

ANSWER 11.2 nm off-track

The computer is now set up to find answers to problems such as:

1. If you are 5 nm off-track after travelling 20 nm, how far off-track will you be after travelling a total of 60 nm?

Answer: 15 nm

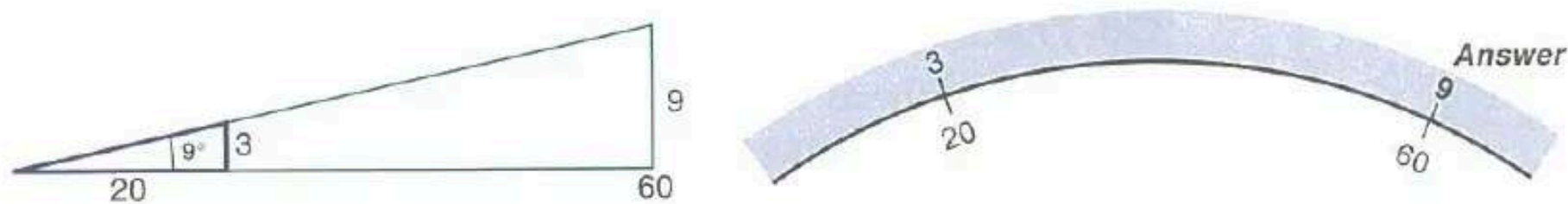


■ Figure 5-16 **An off-track problem**

These numbers are particularly significant as they relate to the *1-in-60 rule*. If you measure it out you will find that:

- 1 nm off-track in 60 nm is a track error of 1° .
- 15 nm off-track in 60 nm is a track error of 15° .
- 5 nm off-track in 20 nm is a track error of 15° , because 5 nm off-track in 20 nm is the same as 15 nm off-track in 60 nm.

EXAMPLE 19 If an aeroplane is 3 nm off-track after travelling 20 nm, what is its track error?



■ Figure 5-17 **Example 19**

ANSWER 3 nm off-track in 20 nm is 9° track error using the 1-in-60 rule.

NOTE The 1-in-60 method of calculating angles is sufficiently accurate up to about 20° . It is explained fully in Section 3, *En Route Navigation Techniques*.

Now complete **Exercises 5 – Calculator-4**.

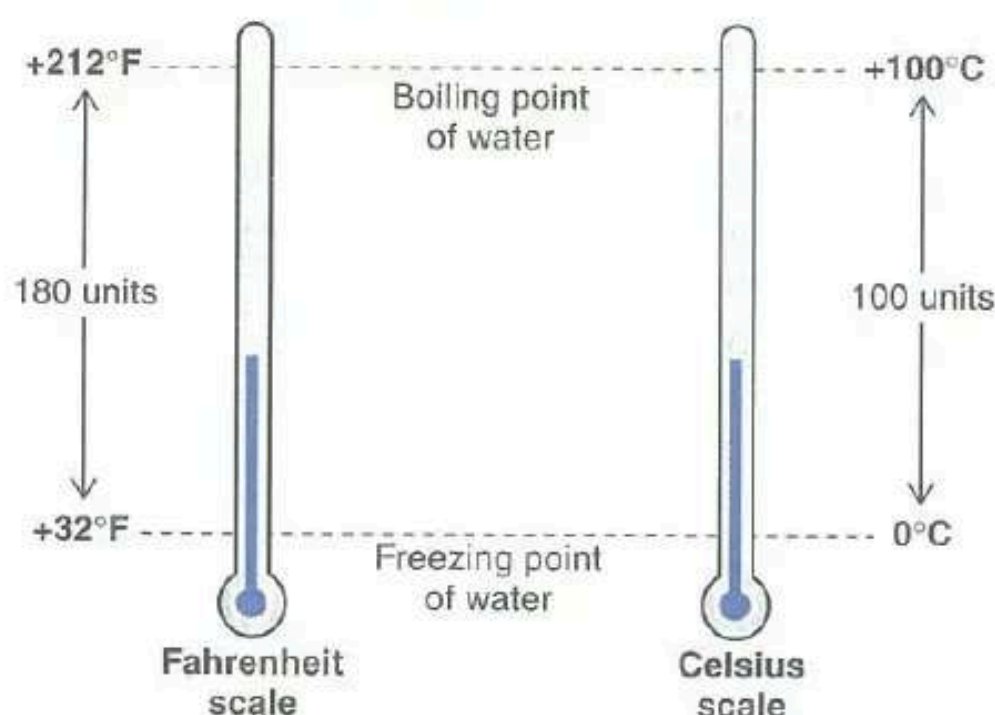


Conversions

You will often have to convert from one unit to another. Some of the common conversions are labelled on the scales of the circular slide rule, but they vary from computer to computer so check your own computer handbook.

Temperatures

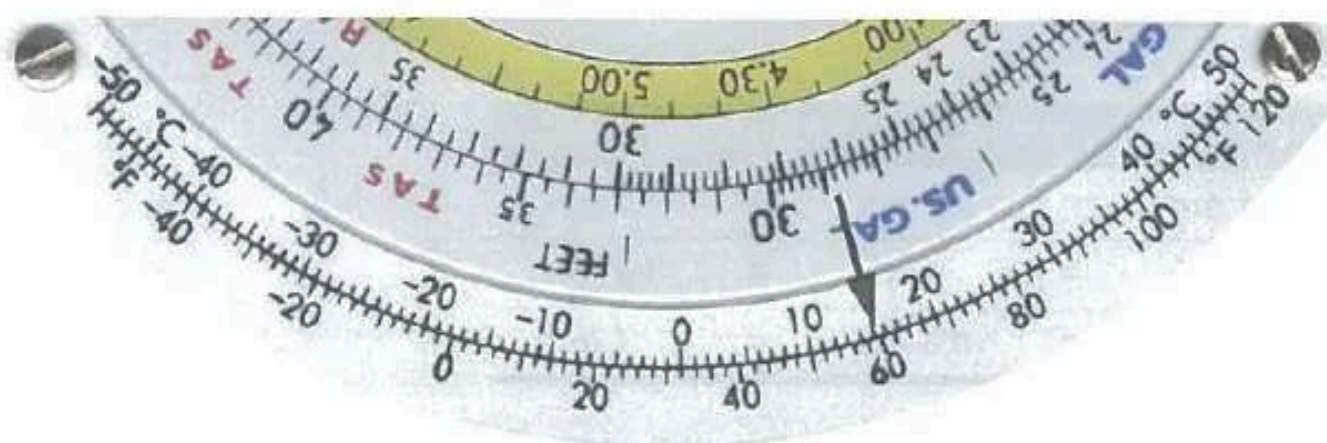
The standard unit for temperature is the degree Celsius ($^{\circ}\text{C}$), except in the USA and a few other places where they still use degrees Fahrenheit ($^{\circ}\text{F}$). Both systems are based on the freezing and boiling points of water.



■ Figure 5-18 *Temperature scales are based on the freezing and boiling points of water*

Some computers have a separate F/C scale that allow simple conversions from one to the other. The reason for this separate scale is that we cannot convert from Fahrenheit to Celsius, or vice versa, by a simple arithmetical conversion, because the zero on one scale is not zero on the other. $0^{\circ}\text{C} = +32^{\circ}\text{F}$.

EXAMPLE 20 Convert $+15^{\circ}\text{C}$ to $^{\circ}\text{F}$.



■ Figure 5-19 $+15^{\circ}\text{C} = +60^{\circ}\text{F}$

NOTE There are formulae that convert $^{\circ}\text{C}$ to $^{\circ}\text{F}$ and vice versa, but the computer conversion is much easier.

$$^{\circ}\text{F} = \left(\frac{9}{5} \times ^{\circ}\text{C} \right) + 32 \quad ^{\circ}\text{C} = \frac{5}{9} \times (^{\circ}\text{F} - 32)$$

Now complete **Exercises 5 – Calculator-5**.



Distances

Nautical Miles, Statute Miles and Kilometres

It is an unfortunate fact of life that we have to deal with the same physical distance being measured in different units. For most navigation purposes we use the **nautical mile**. Nautical miles are important because of their relationship to the *angular* measurement of latitude on the earth.

In the UK long distances are measured in **statute miles**, which are shorter than nautical miles. A statute mile is an arbitrary distance laid down by Royal Statute during the reign of Elizabeth I.

In the metric system, which is based on the number 1,000, the unit of distance is the **metre**. Kilometres (1,000 metres) are used for long distances; centimetres ($\frac{1}{100}$ metre) and millimetres ($\frac{1}{10,000}$ metre) for short distances.

A metre is $\frac{1}{10,000,000}$ (one ten millionth) of the distance from the equator to a pole, which makes a kilometre $\frac{1}{10,000}$ of the distance from the equator to a pole, so the average distance from the equator to a pole is 10,000 km.

The kilometre is much shorter than the nautical mile. It is the standard unit for distance in most of Europe and many other parts of the world. The UK is in the process of changing over from imperial and statute standards to metric standards, so, eventually, the kilometre may officially replace the statute mile.

Pilots need to understand the relationship between these units of distance and be able to convert from one to the other. The relationship between navigation, statute and metric units is:

1 nautical mile = 1.15 statute miles = 1.852 kilometres (1,852 metres).

FOR QUICK MENTAL CHECKS, REMEMBER:				
1 nm	=	1.2 sm	=	2 km
0.9 nm	=	1 sm	=	1.5 km
0.5 nm	=	0.6 sm	=	1 km

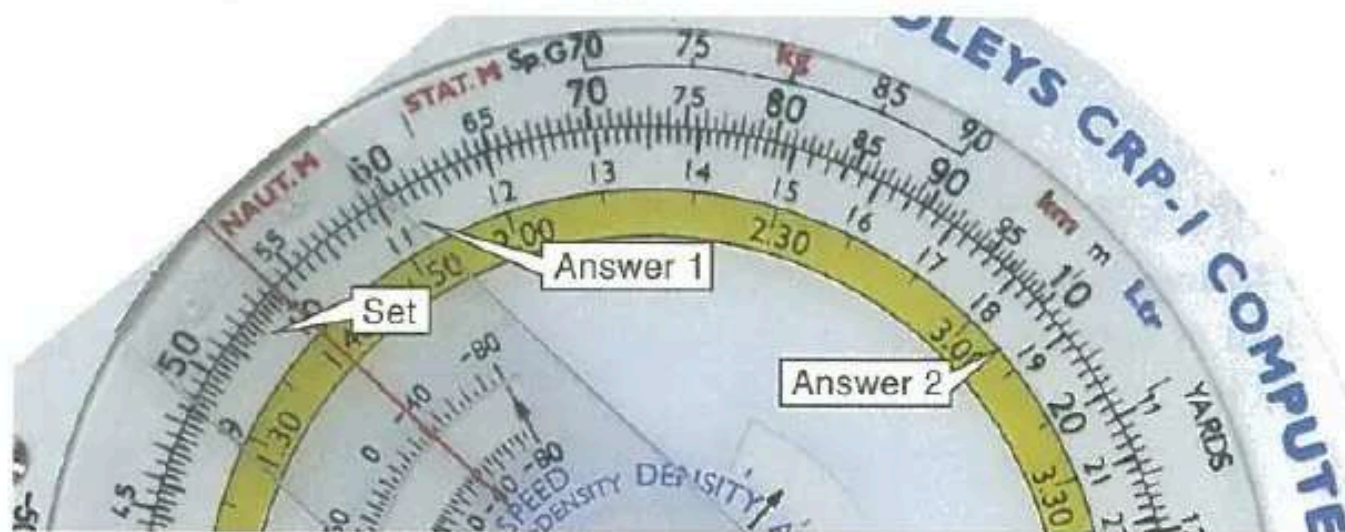
The navigation computer can provide accurate and straight-forward unit conversions. It has indices for nautical miles, statute miles and kilometres marked on the outer scale.

How to Convert from One Unit to Another

1. As with all computer calculations, do a rough mental check.
2. Set the known quantity on the inner scale of the computer against its index on the outer scale.
3. Against the index of the required unit on the outer scale, read off the answer on the inner scale.

EXAMPLE 21 Convert 10 nm to sm and km.**Step 1. Rough check:**

1 nm = approx. 1.2 sm = approx. 2 km
 10 nm = approx. 12 sm = approx. 20 km



■ Figure 5-20 **Example 21**

ANSWER 10 nm = 11.5 sm = 18.5 km

This method may also be used to convert speeds:

EXAMPLE 22 Convert 231 kph to mph and kt.**Step 1. Rough check**

approx. 0.5 nm = approx. 0.6 sm = 1 km
 231 kph = approx. 0.6×231 mph = approx. 138 mph
 231 kph = approx. 0.5×231 kt = approx. 115 kt

ANSWER 231 kph = 144 mph = 125 kt

Now complete **Exercises 5 – Calculator-6.**

**Converting Feet to Metres and Vice Versa**

The standard unit for altitude is the **foot** in most countries, but many Eastern countries use metres. Pilots flying in both these areas need to be able to convert quickly from one system to the other (although the aeroplanes are usually fitted with separate altimeters – one showing feet and the other showing metres).

UK pilots visiting European countries will find topographical charts with the elevation of high ground marked in metres. Some of these charts cover parts of the UK, but the elevations are still shown in metres. A hill with a spot height marked as 1,000 on a metric chart will be 3,280 feet high. Beware!

If you use a chart with heights *above mean sea level* marked in metres, but you are flying an altitude in feet, then you must be able to convert confidently between these two units. For approximate checks, use 1 metre = 3.3 feet.

Also, many older pilots (as well as American pilots) may be used to thinking of runway lengths in feet rather than metres. If you are

contemplating flying in the USA or even if you read American aviation books, conversion of feet to metres and vice versa is necessary. It is also in the syllabus for the navigation examination.

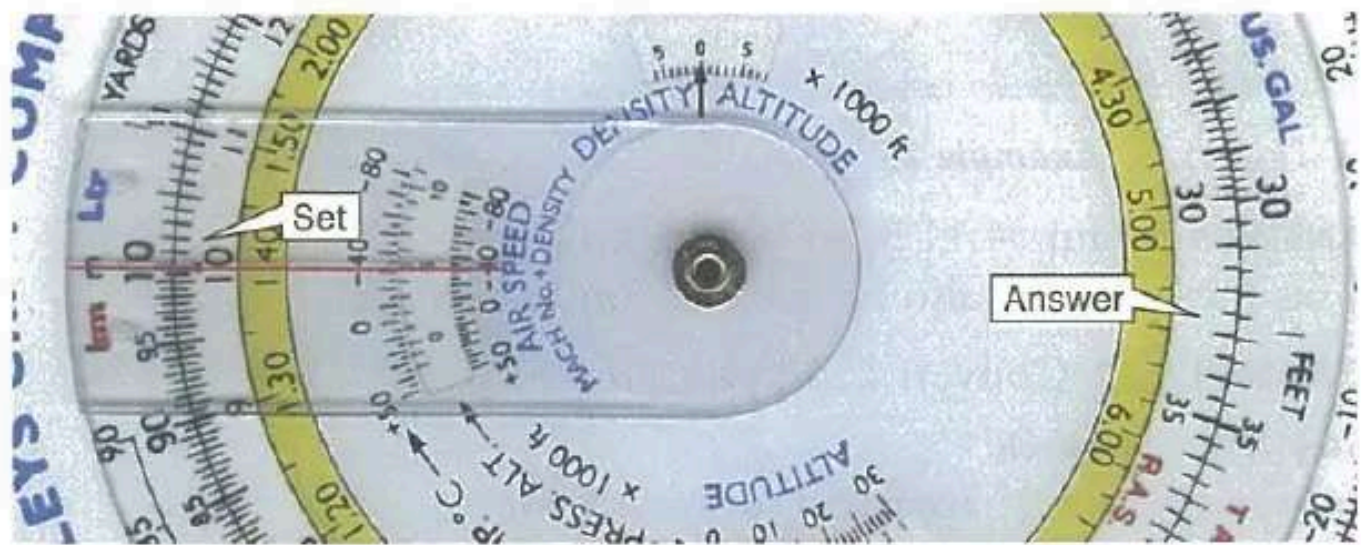
$1 \text{ metre} = 3.28 \text{ feet approximately.}$

The conversion is very simple using the red indices for feet and metres on the outer scale of the computer.

EXAMPLE 23 Convert 1,000 metres to feet on the computer.

Step 1. Rough check

$1 \text{ m} = \text{approx. } 3.3 \text{ ft}$
 $1,000 \text{ m} = \text{approx. } 3,300 \text{ ft}$



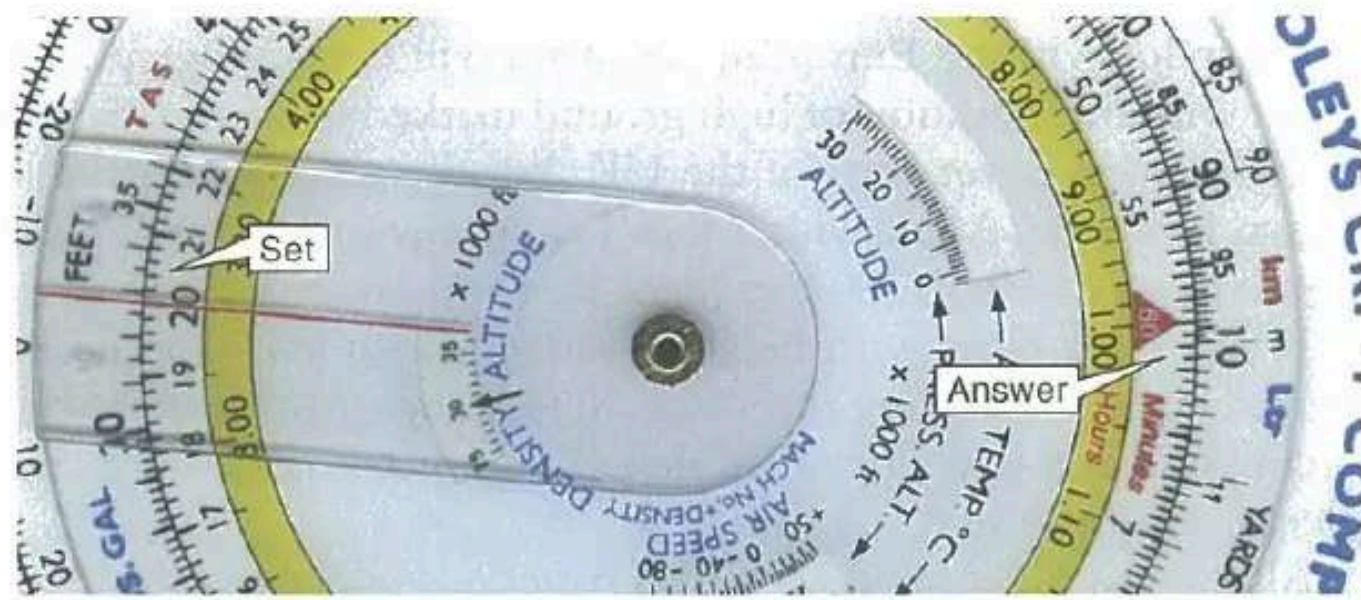
■ Figure 5-21 **Example 23**

ANSWER $1,000 \text{ m} = 3,280 \text{ ft}$

EXAMPLE 24 Convert 2,000 feet to metres.

Step 1. Rough check

$1 \text{ ft} = \text{approx. } \frac{1}{3.3} \text{ m}$
 $2,000 \text{ ft} = \text{approx. } 2,000 \times \frac{1}{3.3} \text{ m}$
 $= \frac{2,000}{3.3} \text{ m} = \text{approx. } 600 \text{ m}$



■ Figure 5-22 **Example 24**

ANSWER 2,000 ft = 610 metres

SUMMARY OF LENGTH UNITS	
1 nm = 1.15 sm	1 sm = 5,280 ft
1 nm = 6,076 ft	1 km = 3,281 ft
1 nm = 1,852 metres	1 metre = 3.28 ft

Now complete **Exercises 5 – Calculator-7.**



Volume and Weight

In aviation we are faced with three different units for volume: the imperial gallon, the US gallon and the litre. In most general aviation aircraft the fuel gauges are marked in US gallons, yet you order fuel from the fuel agent in litres. There is great potential for confusion here, so you must become proficient in converting fuel quantities from one unit to another.

The weight of the fuel on board, as well as its volume, concerns the pilot for two main reasons:

- weight and balance (loading of the aircraft); and
- energy content of the fuel (which depends on weight, rather than volume).

So you must be able to convert fuel from a volume to a weight with accuracy.

100 octane (or higher) Avgas weighs only about 0.71 or 0.72 times the weight of an equal volume of water. We can describe this by saying that the **specific gravity (SG)** of Avgas = 0.71 or 0.72. The precise value depends to some extent on temperature and you should consult your instructor as to which value to use in your usual flying conditions.

Relative density is the modern term and it clearly describes its purpose, which is to compare the density of a particular fluid to the density of water. Most navigation computers, however, are marked with *Sp. G*, so we will continue to use the term *specific gravity*. It means the same thing.

The conversion from one unit of volume to another (imperial gallons, US gallons or litres), or to a weight (kg or lb) is easy on the *CRP-1* computer.

Weight Conversions

FOR QUICK MENTAL CHECKS, REMEMBER:		
1 kilogram (kg)	=	2.2 pounds (lb)

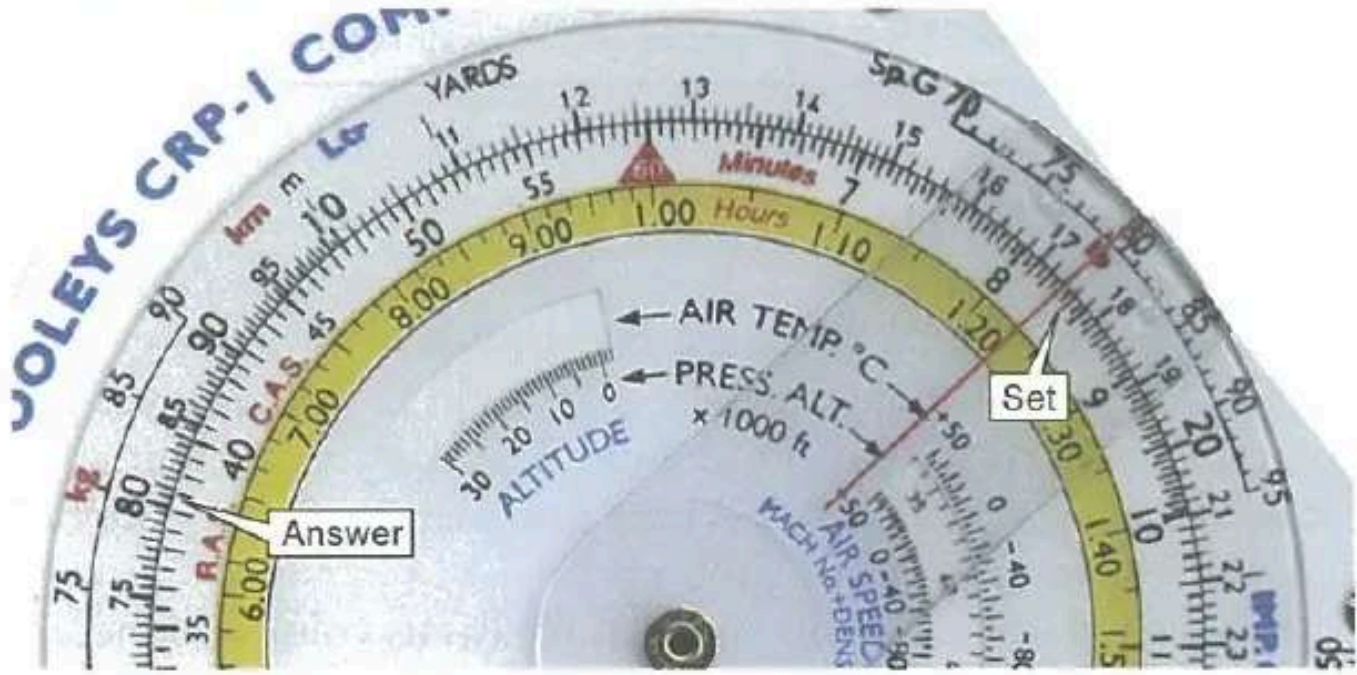
EXAMPLE 25 Convert 83 pounds (lb) to kilograms (kg).

Step 1. Rough calculation

1 kg = approx. 2.2 lb
83 lb = approx. $83 \div 2.2$ kg = approx. 40 kg

Step 2.

Set 83 on the inner scale against *lb* on the outer scale. Against *kg* on the outer scale read off 38 on the inner scale.



■ Figure 5-23 **Example 25 answer: 36.** Use the rotating arm to help align the numbers with the index.

ANSWER 36 kg

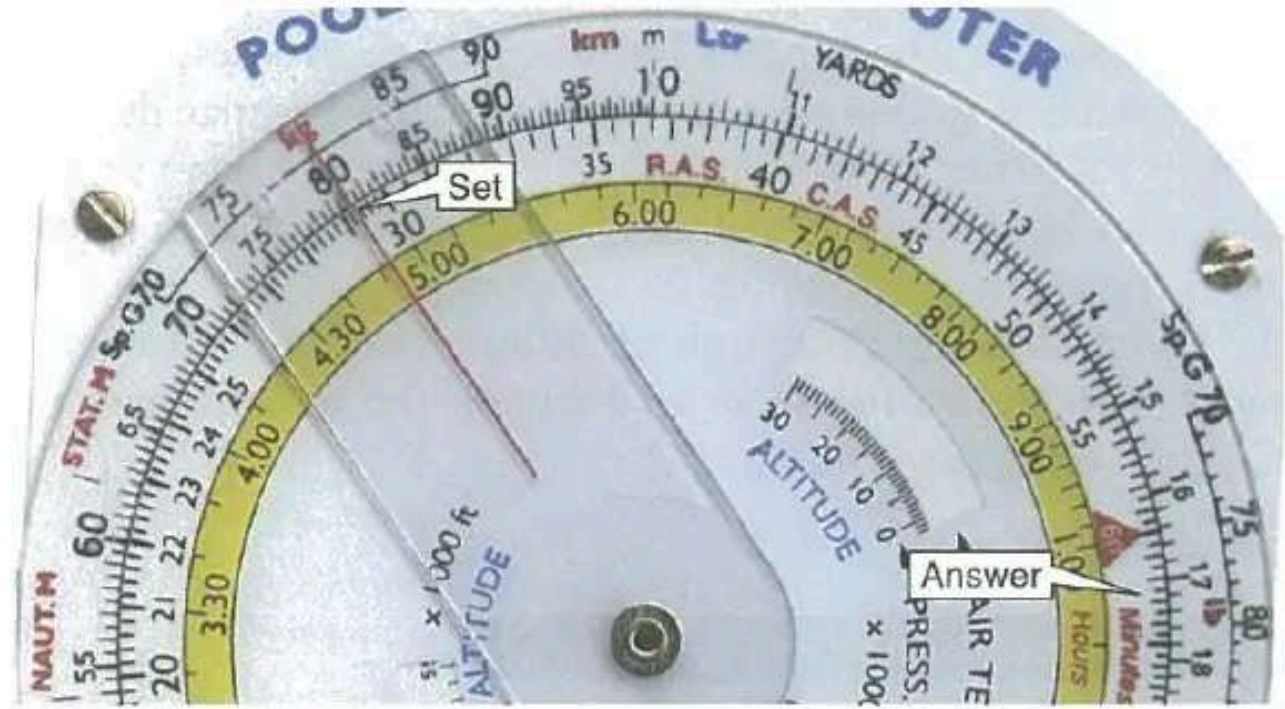
EXAMPLE 26 Convert 293 kg to lb.

Step 1. Rough check

1 kg = approx. 2.2 lb
293 kg = approx. 2.2×300 lb = 660 lb

Step 2.

Set 293 on the inner scale against *kg* on the outer scale. Against *lb* on the outer scale, read off 648.



■ Figure 5-24 **Example 26 answer: 648**

ANSWER 648 lb

Now complete **Exercises 5 – Calculator-8.**



Volume Conversions

FOR QUICK MENTAL CHECKS, REMEMBER:				
1 imperial gallon	=	1.2 US gallons	=	4.5 litres
0.8 imperial gallon	=	1 US gallon	=	4 litres
0.2 imperial gallon	=	0.25 US gallon	=	1 litre

For volume conversions, use the indices marked on the outer scale for *US gal*, *Imp gal* and *ltr*.

- 1. Set the known quantity on the inner scale against its index on the outer scale.
- 2. Read off the answer on the inner scale against the desired index on the outer scale.

EXAMPLE 27 Convert 24 US gallons to imperial gallons and litres.

Step 1. Rough check

- 1 US gal = 0.8 imp gal
- 24 US gal = 0.8 × 24 = approx. 20 imperial gal
- 1 US gal = 4 litres
- 24 US gal = 4 × 24 = approx. 96 litres

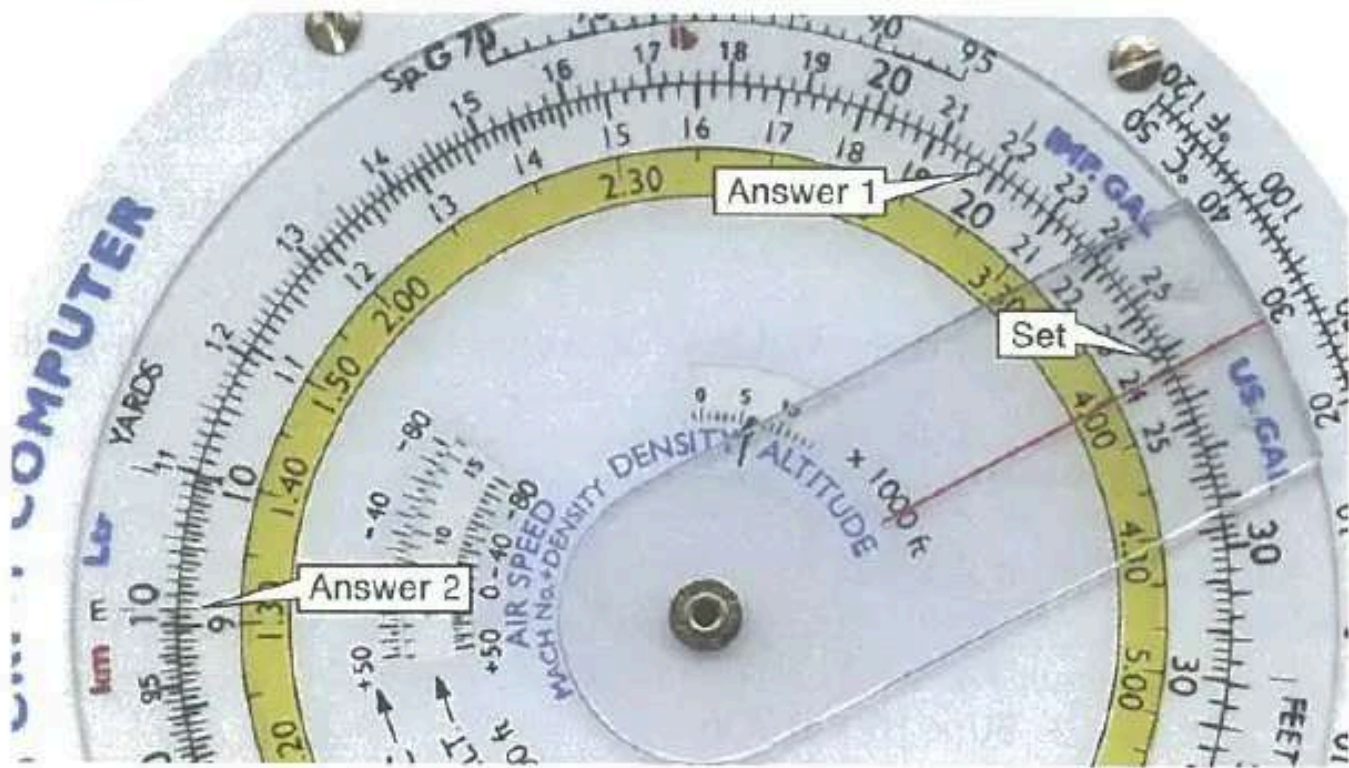


Figure 5-25 **Example 27, Step 2**

ANSWER 24 USG = 20 IG = 91 litres

Volume to Weight Conversions

We need to remember:

Specific gravity (SG) of Avgas = 0.71 or 0.72.

It is useful to know the relationship of volume to weight for water, since water is the standard for relative density (specific gravity). At normal temperatures and pressure:

- 1 litre of water weighs 1 kg;
- 1 imperial gallon of water weighs 10 lb;
- 1 US gallon of water weighs 8.33 lb.

For a rough check on the volume/weight relationship for Avgas, use:

AVGAS: VOLUME/WEIGHT (APPROX.)	
Quantity	Weight
1 litre	1 × SG kg
1 imperial gallon	10 × SG lb
US gallons	first convert to imperial gallons (1 IG = 1.2 USG), then use 10 × SG to find weight in lb

Our navigation computer has two scales (one graduated in *kg* and one in *lb*) to cater for fluids of different specific gravity (SG). This is useful as the SG for Avgas (used in piston engines) is usually about 0.71 or 0.72, and the SG for Avtur (used in turbine engines) is usually about 0.79.

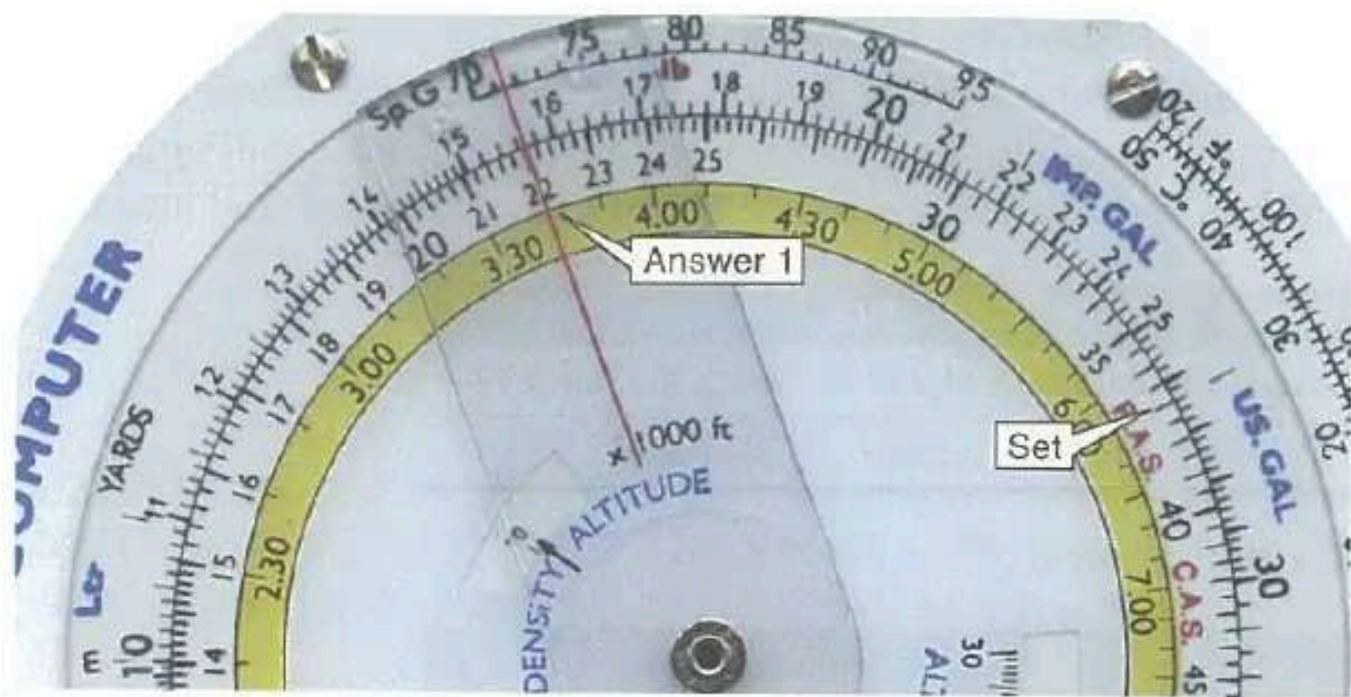
TO CONVERT A VOLUME TO WEIGHT:

1. Set the volume on the inner scale against its index on the outer scale.
2. Against the given SG (on the *kg* or *lb* scale) read off the weight on the inner scale.

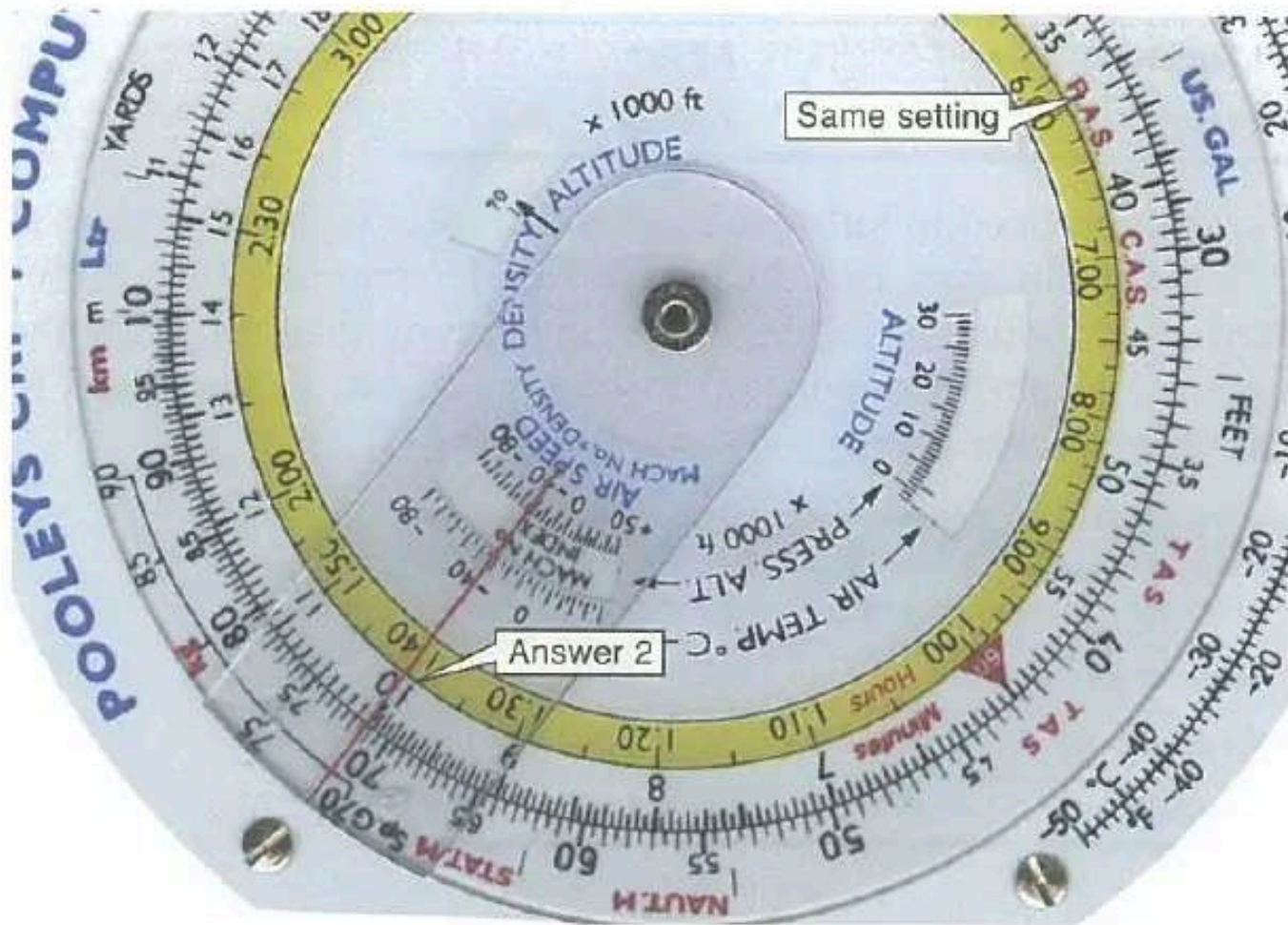
EXAMPLE 28 What does 37 USG of Avgas (SG 0.71) weigh in lb and kg?

Step 1. Rough check

$$\begin{aligned} 37 \text{ US gal} &= 0.8 \times 37 \\ &= \text{approx. 30 IG} \\ &= 30 \times 10 \times 0.7 \text{ lb} \\ &= 210 \text{ lb} \\ &= \frac{210}{2.2} \\ &= \text{approx. 100 kg} \end{aligned}$$



■ Figure 5-26 Example 28, Step 2



■ Figure 5-27 Example 28, Step 3

ANSWER 37 USG (SG 0.71) = 220 lb or 99.5 kg

Now complete **Exercises 5 – Calculator-9.**



Standard Specific Gravity Values

The density of a fluid will vary with temperature; however, for our purposes in light aircraft where only relatively small quantities will be used, we can assume standard SG values for fuel and oil.

Remember:
Specific gravity of Avgas is 0.71 or 0.72.

AVGAS SPECIFIC GRAVITIES	
Avgas 100 octane & higher	SG = 0.71 or 0.72
Avgas less than 100 octane	SG = 0.72

In round figures:

QUANTITY OF AVGAS	WEIGHT
1 litre	0.7 kg
1 imperial gallon	7.2 lb
1 US gallon	2.69 kg

You do not need to know the following specific gravity values at PPL level but they may be of interest. Jet engines burn Avtur (kerosene), which has a higher specific gravity than Avgas. Some aircraft load sheets require oil to be considered, so you need to know its specific gravity.

OTHER SPECIFIC GRAVITIES	
Avtur (for turbine engines)	SG = 0.79
Mineral oil	SG = 0.90
Synthetic oil	SG = 0.96
De-ice fluid	SG = 1.1

Vertical Navigation

Navigating a car or a ship is a two-dimensional activity – horizontal navigation – concerned with speed and direction in the horizontal plane. Navigating an aeroplane requires three-dimensional awareness, and we must also consider height. Vertical navigation is important to pilots for three basic reasons:

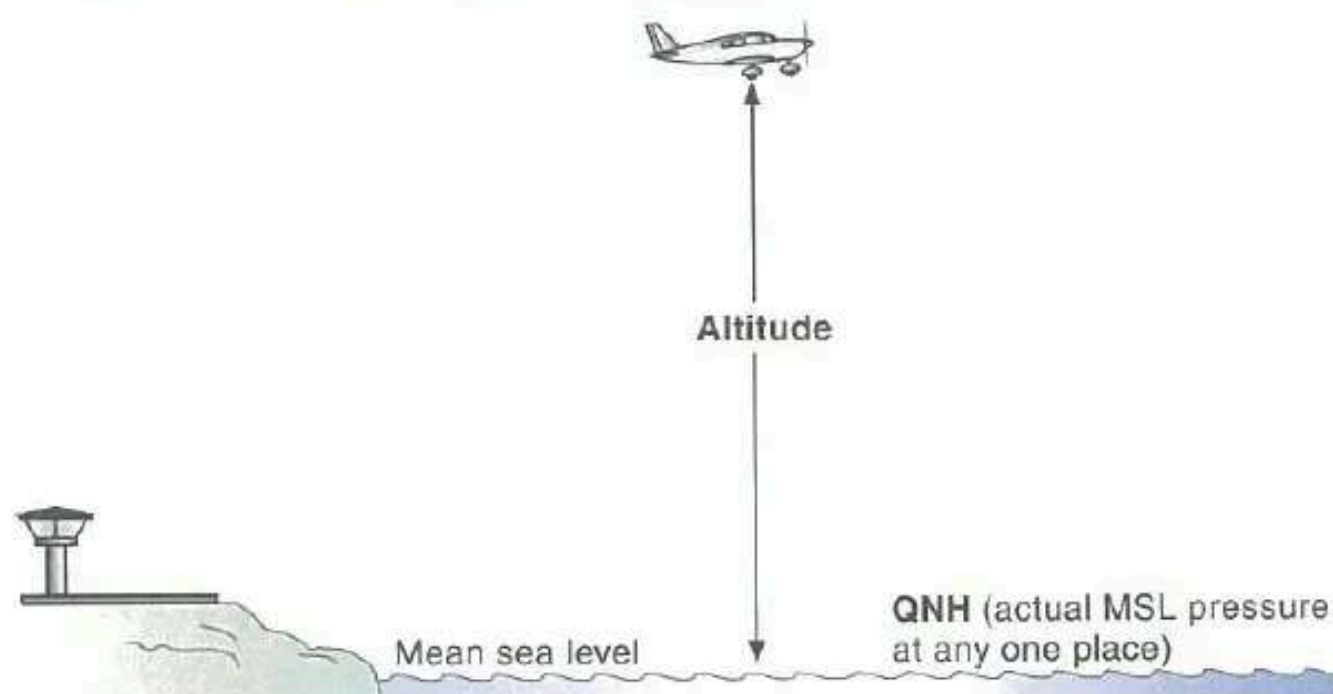
1. For **terrain clearance**, to ensure that the aircraft will not collide with terrain or fixed obstacles on the ground.
2. For **traffic separation**, to allow pilots to cruise at an altitude different from that of nearby aircraft, to ensure safe vertical separation.
3. To **calculate the performance capabilities** of the aircraft and its engine, so as to operate safely and efficiently.

Vertical navigation is the guidance of flight in the vertical plane, and includes the science of measuring vertical distances in the atmosphere, known as *altimetry*.

Measuring vertical distance in the atmosphere is not as simple as it sounds. There are errors in the measuring instrument (the altimeter) and compromises in the principle on which it is built, with the result that the altimeter presents the pilot with approximate vertical information of the aircraft's position.

Altitude

Altitude is the vertical distance of a level, point, or object, measured from *mean sea level* (MSL). This definition appears in the UK Aeronautical Information Publication (AIP). The abbreviation for altitude is *alt* or ALT.



■ Figure 6-1 **Altitude is the vertical distance above mean sea level (amsl)**

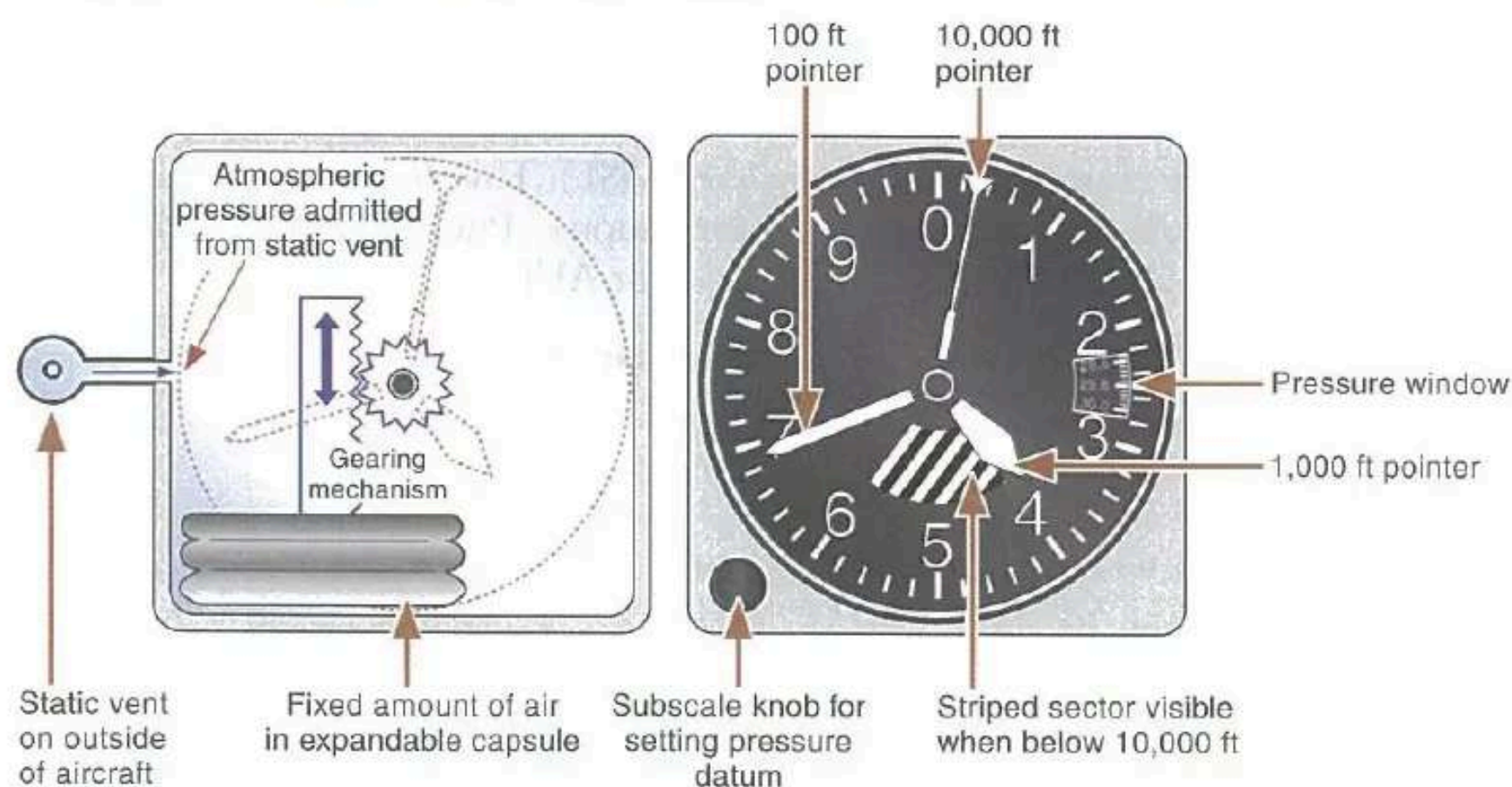
In the UK and the Western world the standard unit of height for aviation purposes is the *foot*. In many Eastern countries it is the *metre*.

The Altimeter

The basic instrument used to measure altitude is the pressure altimeter. This is simply a **barometer** – a device that measures atmospheric pressure. Barometers work on the principle that, in the atmosphere, air pressure decreases as height increases. This means that the higher you are in the earth's atmosphere, the lower the pressure – hence the need for most people to wear oxygen masks when they climb Mount Everest or when they fly above 10,000 ft in unpressurised aeroplanes.

There are various types of pressure altimeter. The most compact and robust type suitable for installation in an aircraft is the **aneroid barometer**, which is similar to those seen in many homes. As the aeroplane goes higher, the atmospheric pressure of the air in which it is flying decreases, and the aneroid, which is an expandable and compressible metal capsule containing a fixed amount of air, is able to expand.

Through a system of linkages, a pointer is driven around a scale. This scale does not read directly in units of pressure, such as *hectopascals*, but rather in *feet*. Calculations by the designer have been made to relate the pressure to the altitude, and the scale reads in units of altitude, which is what pilots need. The altitude indication is known as **indicated altitude**.



■ Figure 6-2 **The altimeter is a pressure-sensitive instrument**

As the aeroplane climbs, the aneroid expands, driving the pointer to indicate a higher altitude. As the aeroplane descends, the static air pressure in the surrounding atmosphere increases, forcing the aneroid to contract and drive the pointer to indicate a lower altitude.

A number of errors are evident in altimeters, and these may be broken into two main types:

1. Errors in the particular altimeter.
2. Errors in the principle on which altimetry is based, i.e. by how much the pressure in the atmosphere decreases with height.

Errors in the Altimetry Principle

The rate at which air pressure decreases with height varies from time to time and from place to place. The simple barometric altimeter cannot cope with this because it has been calibrated according to a **standard atmosphere** in which a particular height **above mean sea level (amsl)** always corresponds with a particular pressure. Any variation of the actual atmosphere from the standard atmosphere will cause the altimeter to indicate an altitude different from the actual height amsl of the aeroplane.

The various gases surrounding the earth and forming its atmosphere are bound to it by gravity. A standard atmosphere is something that does not exist permanently, since this mixture of gases (or *air* as we call it) has constantly changing values of:

- pressure;
- temperature;
- density; and
- water content (humidity).

Thus it is necessary to have some sort of measuring stick or standard, be it only a theoretical one, against which to compare the actual atmosphere in our vicinity. That measuring stick is the International Standard Atmosphere (ISA).

Lapse Rates

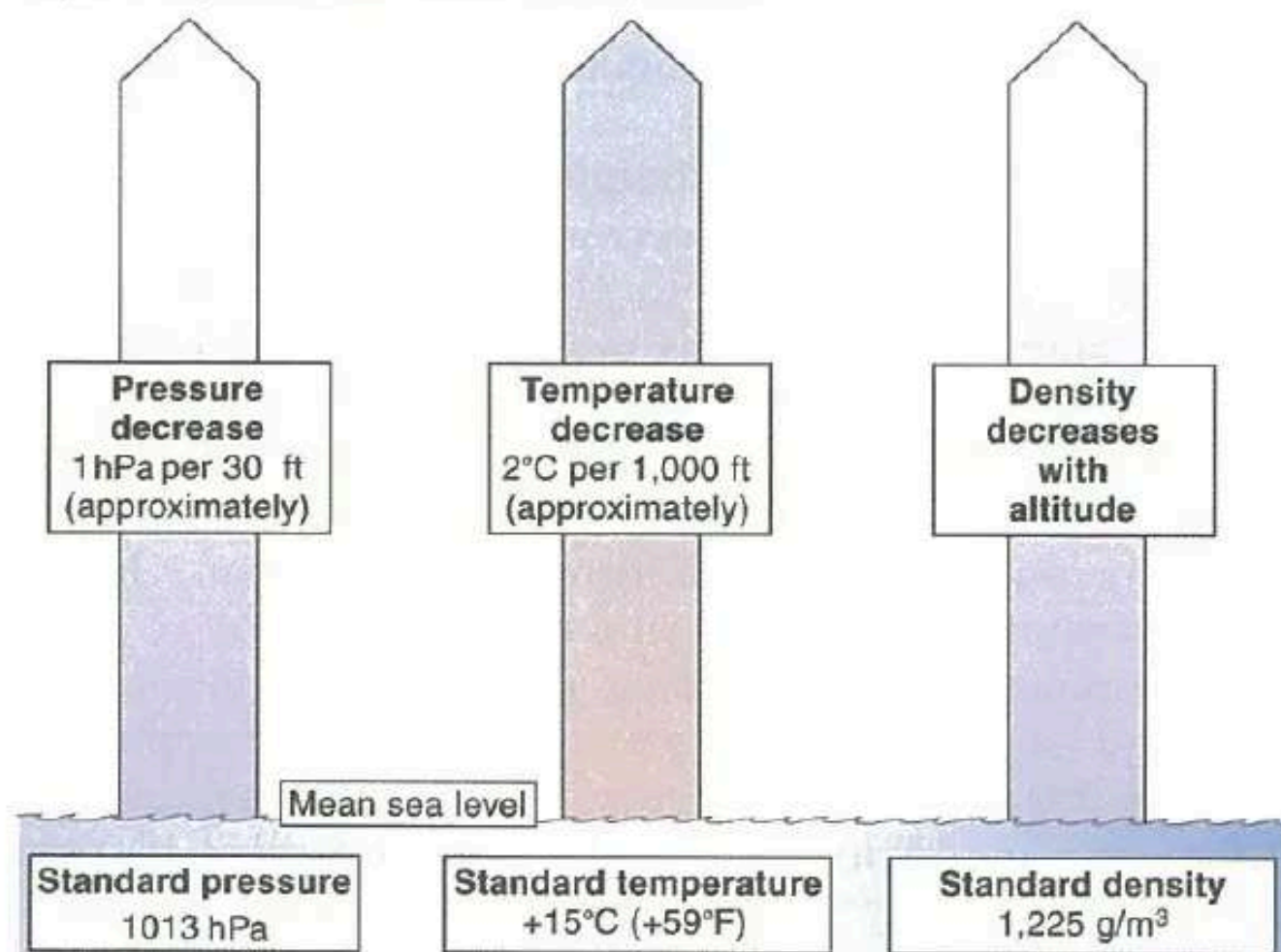
In the International Standard Atmosphere, pressure, temperature and density are defined as decreasing at specified rates with gain in height.

TEMPERATURE. The temperature lapse rate (rate of fall of temperature with increase in height) in the ISA is 1.98°C per 1,000 ft up to 36,090 ft, above which the temperature remains constant at minus 56.6°C (at least in the levels up to which commercial aeroplanes fly). For our purposes, in the ISA:

- temperature falls by 2°C per 1,000 ft up to 36,000 ft; and
- above 36,000 ft, remains constant at -57°C.

PRESSURE. The rate at which pressure in the ISA decreases with height varies, but is approximately 1 hPa per 30 ft up to about 5,000 ft. (This rate drops to about 1 hPa per 70 ft at high levels of the atmosphere, but this need not concern us here.) For our purposes, in the lower levels of the ISA, pressure decreases by 1 hPa for each 30 ft increase in altitude.

DENSITY. Air density decreases with height, so that at 20,000 ft the air density is about one-half of its MSL value, one-quarter at 40,000 ft, and one-tenth at 60,000 ft.



■ Figure 6-3 *The International Standard Atmosphere (ISA)*

The main use of the International Standard Atmosphere is to calibrate altimeters. This is to provide a mathematical relationship between the air pressure measured by the aneroid barometer and the altitude calibrations placed on the scale around which the pointer is driven, so that all aircraft convert pressure to altitude by a standard method.

Pressure Altitude

Pressure altitude (called *pressure height* in some countries), is the height in the ISA above the 1013 hPa pressure datum at which the pressure equals that of the level under consideration. For example, the ISA pressure at a point 600 ft higher than the 1013 pressure level is approximately 993 hPa. If your aeroplane is flying in air whose pressure is 993 hPa, then its pressure altitude is 600 ft.

NOTE 1013.25 hPa is the precise value of standard or ISA MSL pressure. For our purposes, 1013 is sufficiently accurate. There is a Q-code for pressure altitude, QNE, but it is rarely used.

Pressure altitudes are often described in an abbreviated form as **flight levels**, where the final two zeros are omitted. For example, a pressure altitude of 4,500 ft (i.e. 4,500 ft higher than the 1013 pressure level) is also known as flight level 45 (written FL45). Flight levels are used for cruising at higher levels and are usually separated by at least 500 ft, e.g. FL55, FL60, FL65.

A flight level is a pressure altitude.

EXAMPLE 1 35,000 ft in the International Standard Atmosphere above the standard pressure level of 1013 may be referred to as:

- pressure altitude (or pressure height) of 35,000 ft; or as
- flight level 350, FL350 (where the last two zeros are dropped).

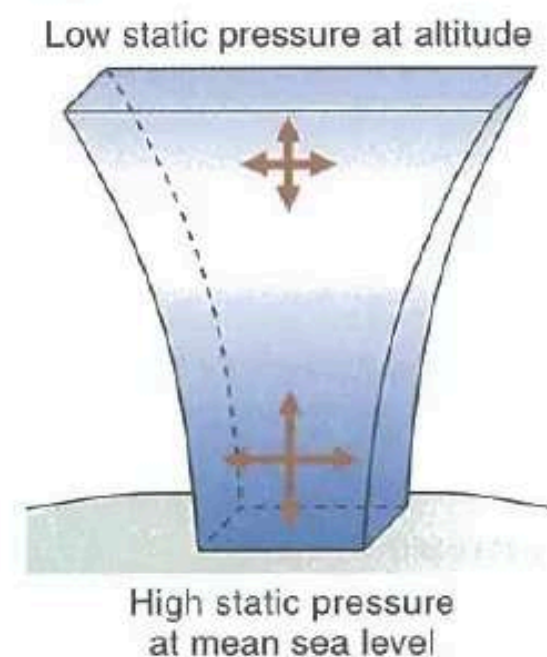
EXAMPLE 2 3,500 ft in the International Standard Atmosphere above the standard pressure level of 1013 may be referred to as:

- pressure altitude (or pressure height) of 3,500 ft, or as
- flight level 35, FL35, (where the last two zeros are dropped).

Air Pressure

The earth's atmosphere consists of countless numbers of molecules all moving at high speed and colliding with any object, be it another molecule or the earth's surface or a person that blocks their path. The force that these molecules exert as they collide gives rise to a *pressure*, or a 'force per unit area'. As these molecules are moving in all directions, the pressure at any point in the atmosphere will be exerted in all directions.

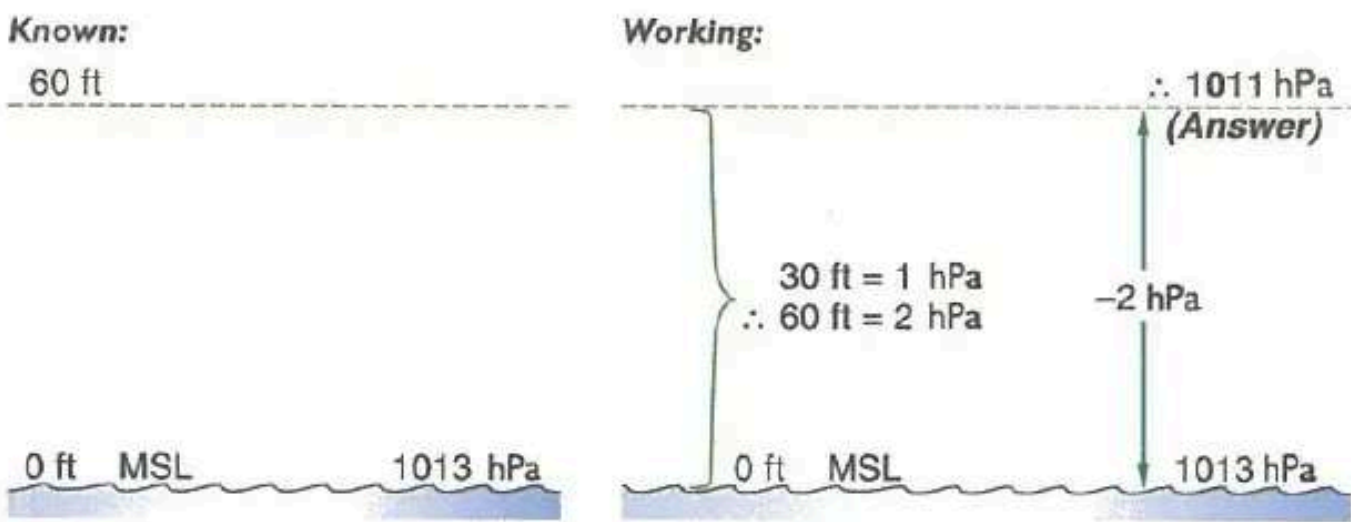
Because of gravity, the ambient pressure at any point in the atmosphere will depend on the weight of air that is above and pressing down. On a 'standard day', the column of air pressing down on the earth's surface exerts a pressure of 1013 hectopascals at sea level.



■ Figure 6-4 **Pressure decreases with increase of altitude**

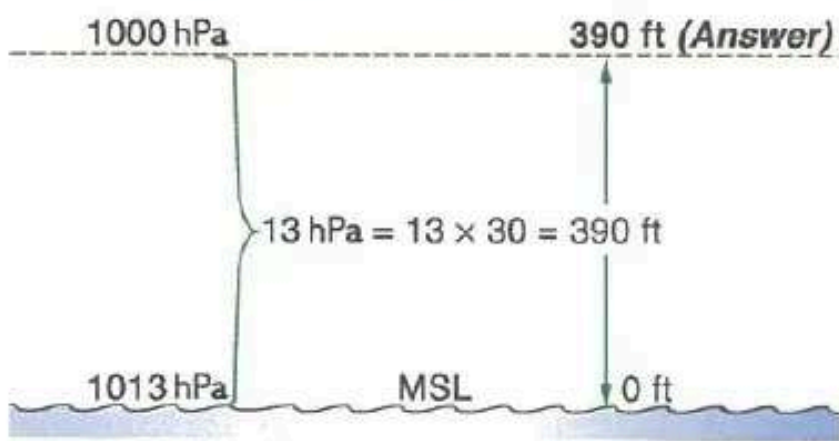
At higher levels in the column of air, the weight pressing down is less and so the pressure will be less. In the lower levels of the atmosphere (up to about 5,000 ft amsl), the pressure drops by about 1 mb for each 30 ft climbed.

EXAMPLE 3 If the pressure at MSL is 1013 (to the nearest hPa), what is the pressure at 60 ft amsl?



■ Figure 6-5 **Example 3 answer: 1011 mb.** Diagrams make altimetry problems simple.

EXAMPLE 4 If the MSL pressure is 1013, at what altitude (height amsl) would you expect the pressure to be 1000 hPa?



■ Figure 6-6 **Example 4 answer: 390 ft**

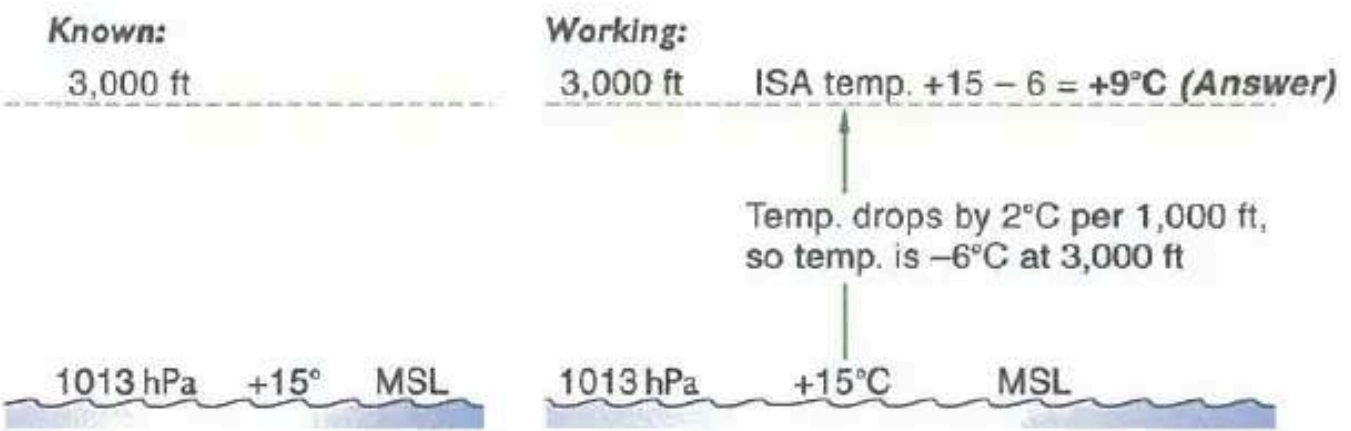
Now complete **Exercises 6 – Vertical Nav-1.**



Air Temperature

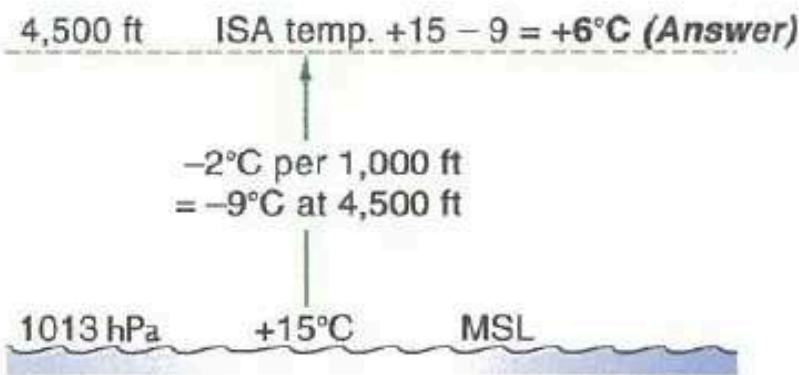
The temperature in any one place varies from hour to hour, from day to day, and from season to season. On an ‘average day’ in a temperate zone, the temperature at MSL (mean sea level) may be +15°C. By convention, +15°C is the standard temperature at sea level in the International Standard Atmosphere. As we climb, the temperature decreases by about 2°C for every 1,000 ft increase in height.

EXAMPLE 5 What temperature exists at 3,000 ft above the 1013 pressure surface (or pressure level) in the International Standard Atmosphere?



■ Figure 6-7 Example 5 answer: +9°C

EXAMPLE 6 What temperature exists at 4,500 ft in the ISA?

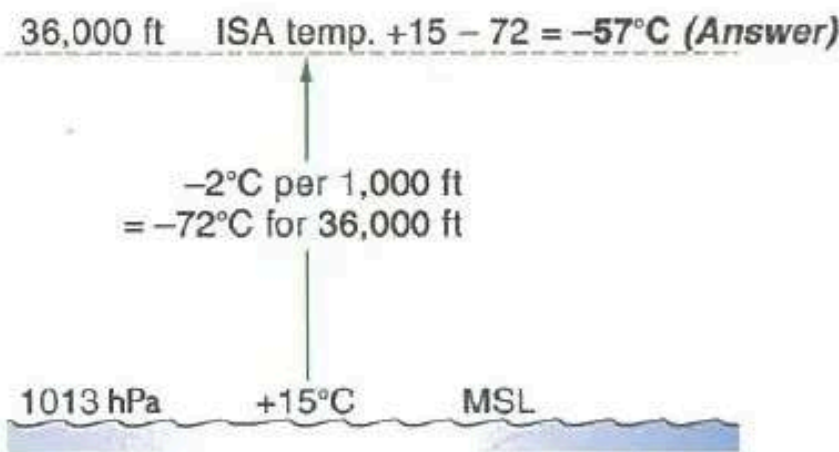


■ Figure 6-8 Example 6 answer: +6°C

EXAMPLE 7 What temperature exists at 45,000 ft in the ISA?

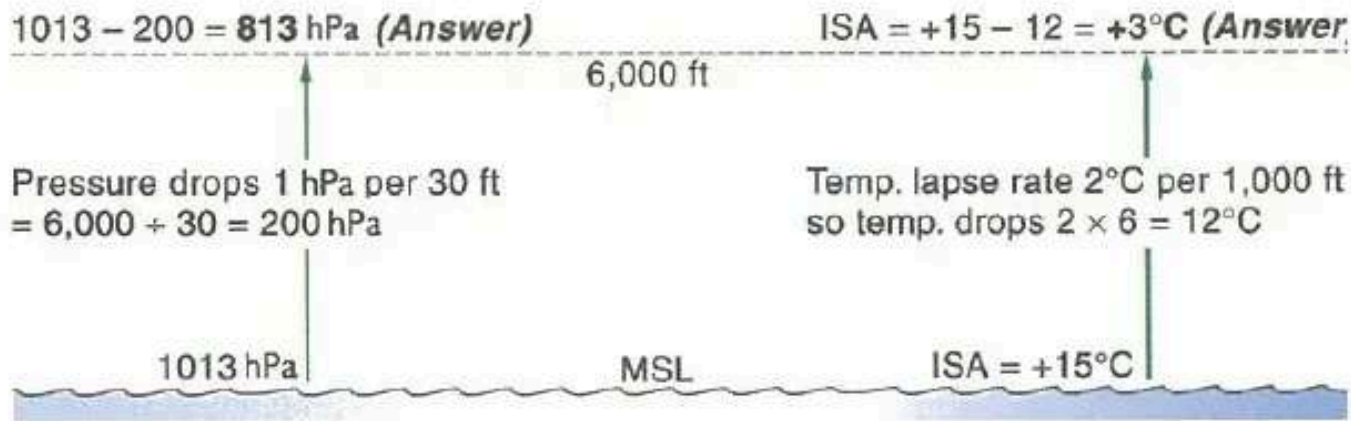
ANSWER -57°C because in the ISA, temperature is constant at -57°C above 36,000 ft.

EXAMPLE 8 What temperature exists at 36,000 ft in the ISA?



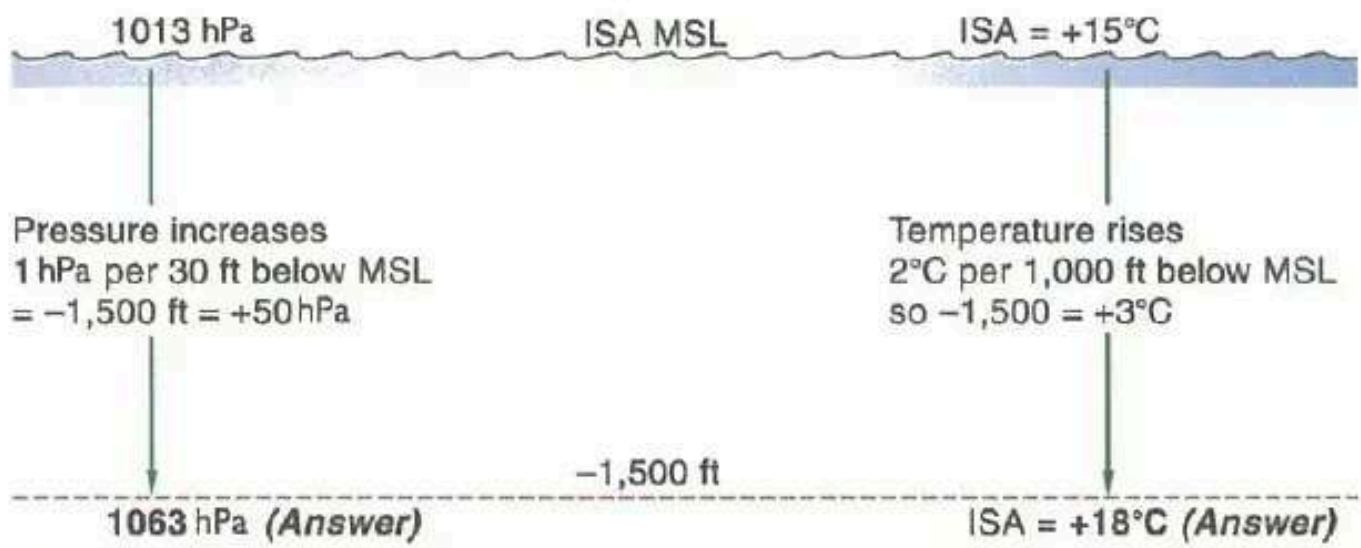
■ Figure 6-9 Example 8 answer: -57°C

EXAMPLE 9 Calculate the ISA values of pressure and temperature for a pressure altitude of 6,000 ft.



■ Figure 6-10 Example 9 answers: 813 mb and +3°C

EXAMPLE 10 Calculate the ISA values of pressure and temperature for a pressure altitude of $-1,500$ ft (i.e. $1,500$ ft below the 1013 pressure level).

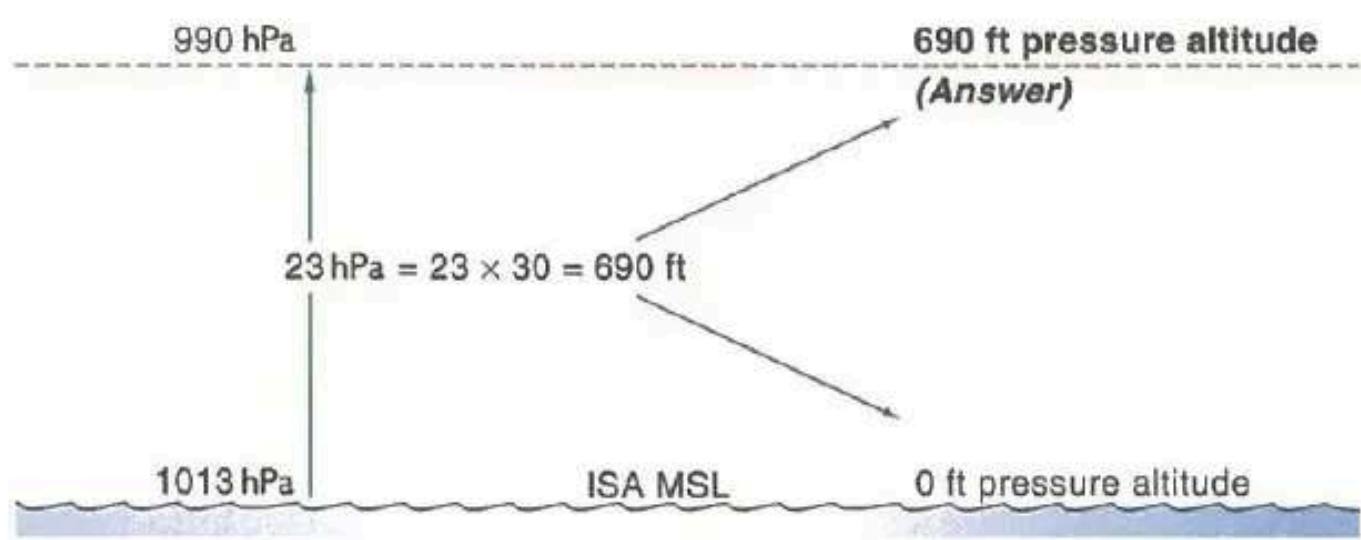


■ Figure 6-11 Example 10 answers: 1063 hPa and $+18^{\circ}\text{C}$

NOTE This sort of calculation for below ISA MSL is sometimes required because:

- 1. There are places on the earth’s surface that are below mean sea level (e.g. Rotterdam, the Dead Sea, Lake Eyre in Australia).
- 2. The actual atmosphere is always different from our so-called standard atmosphere and negative pressure altitudes are not uncommon.

EXAMPLE 11 What is the pressure altitude of the 990 hPa pressure surface?



■ Figure 6-12 Example 11 answer: $690\text{ ft pressure altitude}$

Even though some of these exercises are simple, do not avoid them. A sound understanding of altimetry will stand you in good stead throughout your flying career!

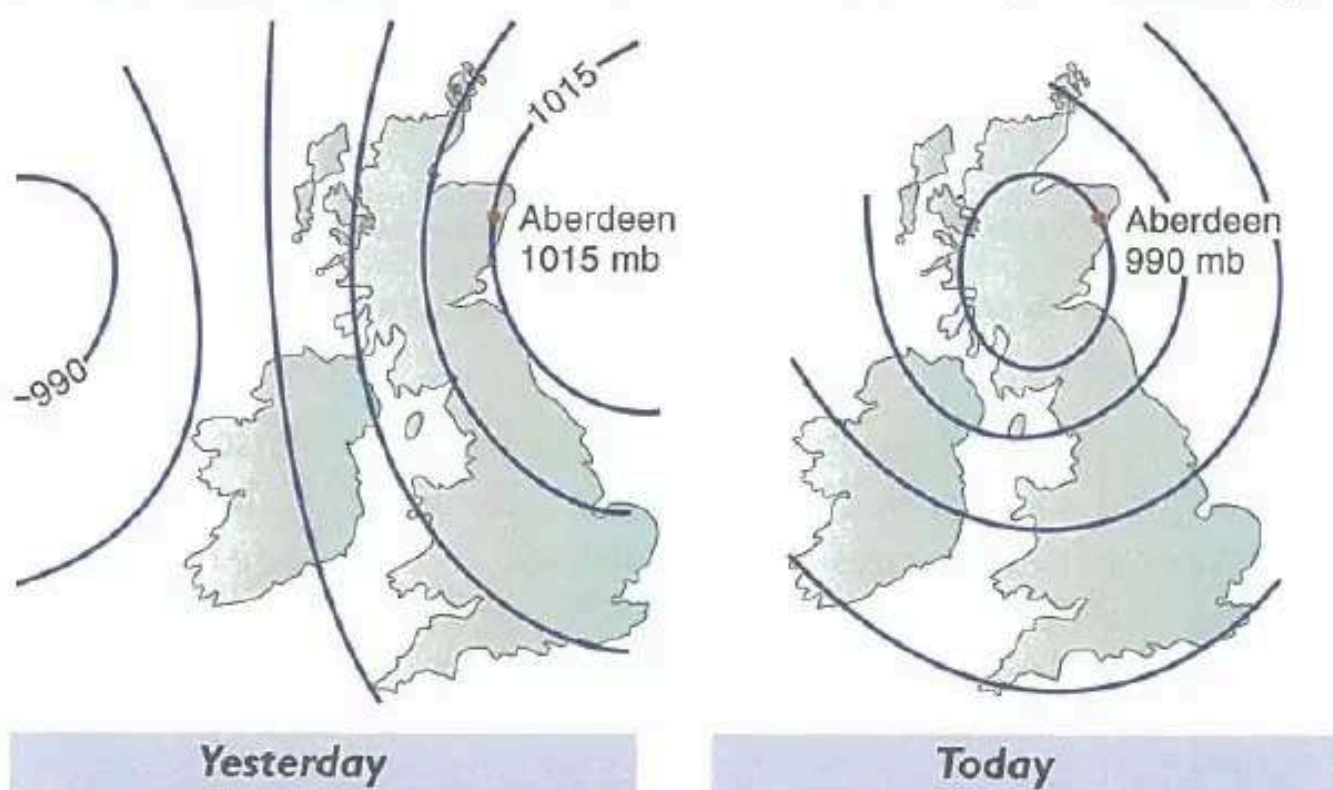
Now complete **Exercises 6 – Vertical Nav-2.**



Variations in Mean Sea Level Pressure – QNH

On maps and charts the height of terrain is given as height above mean sea level (amsl). It is therefore essential that a pilot knows the aircraft's height above mean sea level so that he can relate this to the height of terrain or obstructions and determine if there is sufficient vertical separation.

So far in our discussion the altimeter has only measured height above the ISA MSL datum of 1013 hPa. In reality, mean sea level pressure varies from day to day, and indeed from hour to hour, as the various high- and low-pressure systems move across the surface of the earth. You will notice this on daily weather maps.



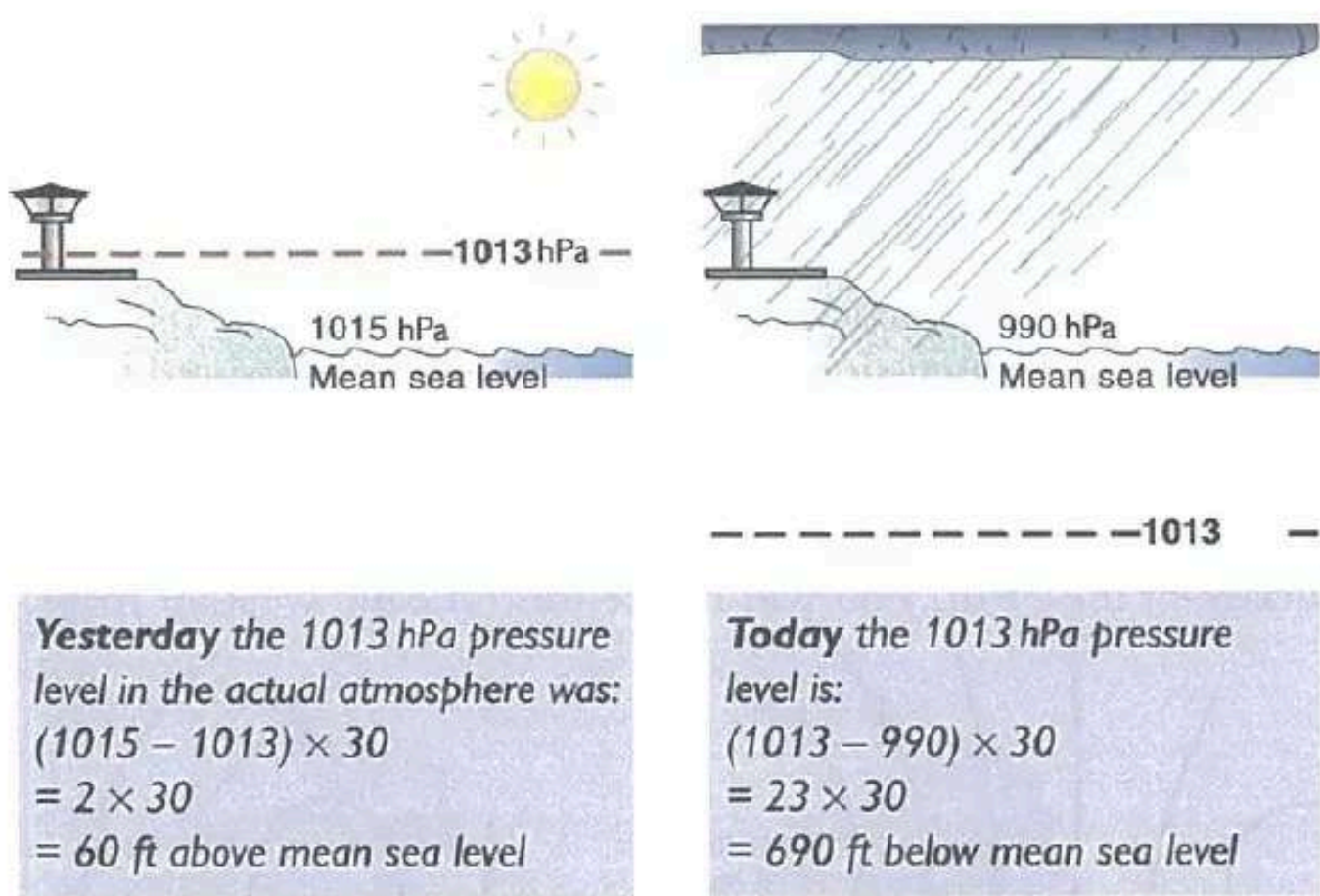
■ Figure 6-13 Two different synoptic situations

Consider the situations illustrated in Figure 6-13. Yesterday, a high-pressure system of 1015 hPa was sitting over Aberdeen. In 24 hours, the pressure system has moved on, and today Aberdeen is experiencing a lower pressure of 990 hPa. A profile of the atmosphere over Aberdeen on each of these days is shown in Figure 6-14.



■ Figure 6-14 Profile of MSL pressure situations – yesterday and today

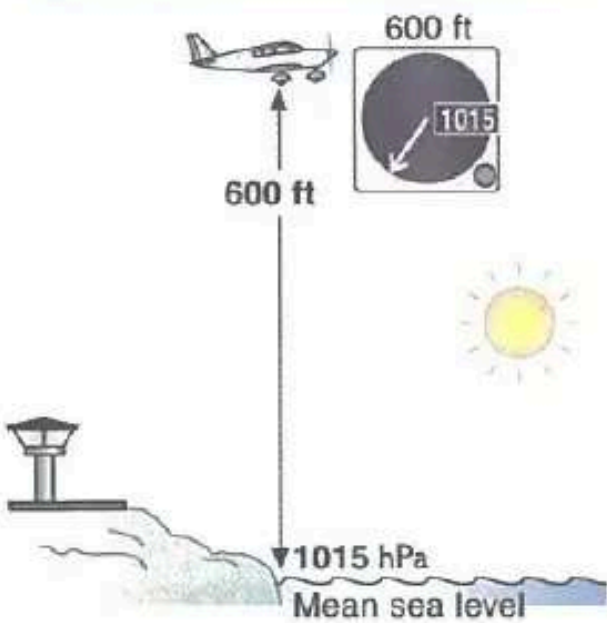
Figure 6-15 shows the position of the 1013 hPa pressure level in relation to each of these situations.



■ Figure 6-15 The standard pressure level in relation to the MSL pressure

Since height amsl is vital information to a pilot, instead of this unwieldy calculation being required in the cockpit, the design of the altimeter incorporates a small subscale and knob geared to the altimeter pointer. By rotating the knob, the desired pressure datum from which height will be measured is set in the subscale.

Yesterday we would have set 1015 hPa in the subscale to measure altitude (height amsl).



Altimeter now reads height above the 1015 hPa pressure level.

Today we set 990 hPa (today's MSL pressure level) for the altimeter to indicate altitude.



Altimeter now reads height above the 990 hPa pressure level.

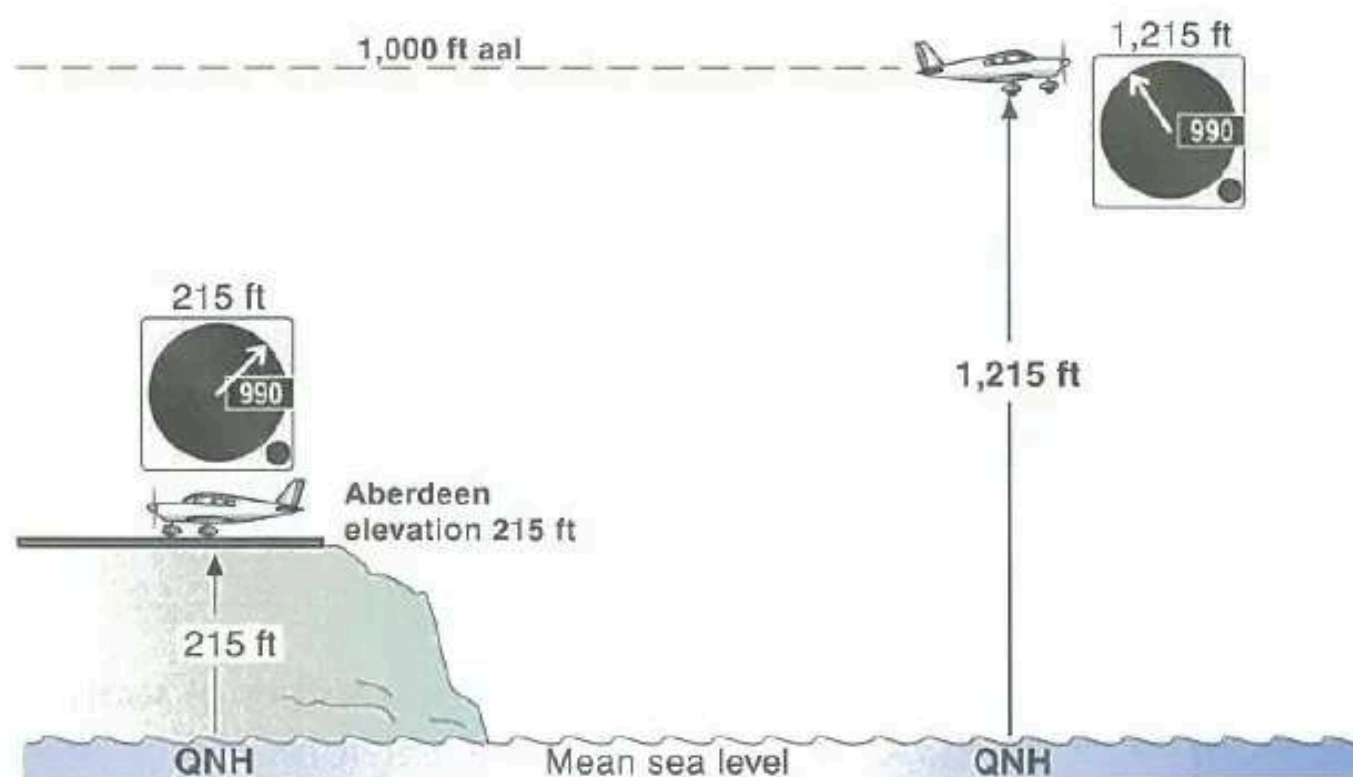
■ Figure 6-16 When MSL pressure is set in the altimeter subscale, the altimeter indicates height amsl for that pressure situation

QNH

The Q-code name for the altimeter subscale setting which gives us this altitude is QNH (i.e. the MSL pressure level in the *actual* atmosphere). The QNH is the atmospheric pressure corresponding to mean sea level pressure at that place and time.

*An altimeter with **QNH** set on its subscale will indicate **altitude** – height above mean sea level (amsl).*

From the previous example, we can see that yesterday in Aberdeen the QNH was 1015 hPa; today in Aberdeen the QNH is 990 hPa.



■ Figure 6-17 **Today in Aberdeen the QNH is 990 hPa**

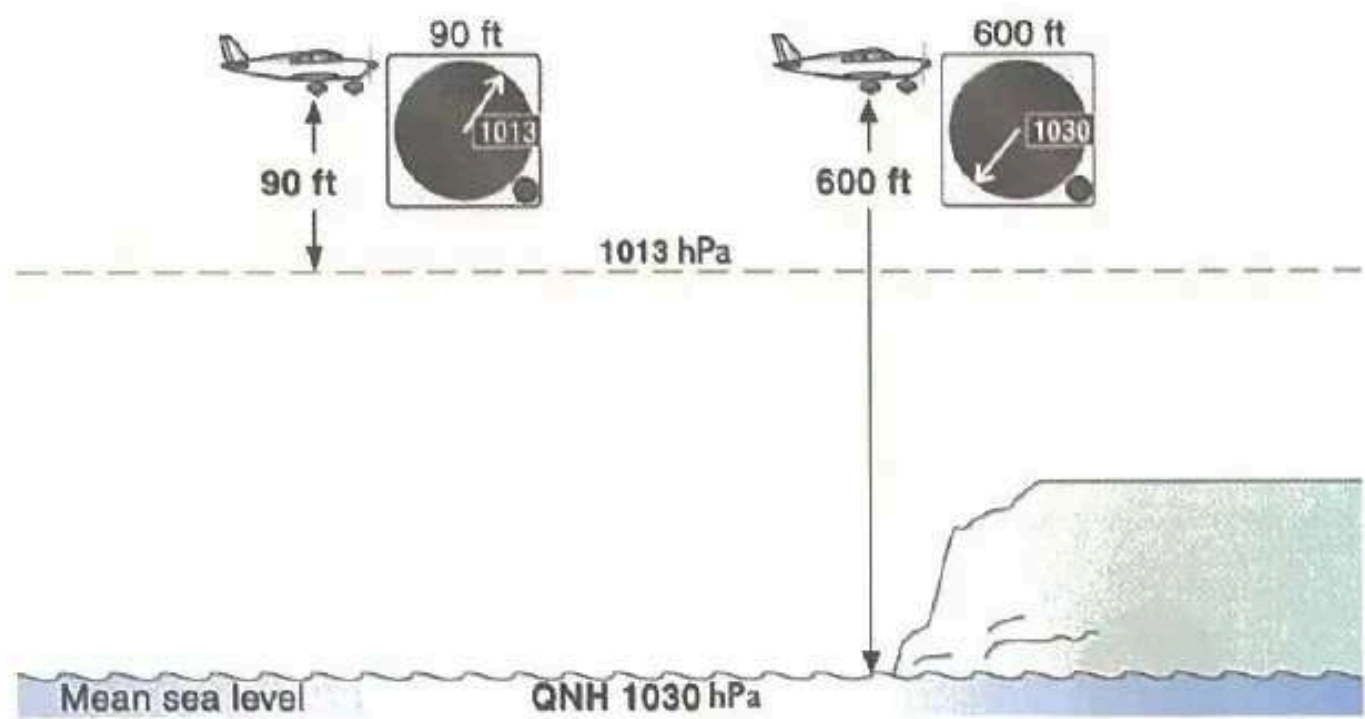
The official elevation of a particular aerodrome is that of the highest point on the landing area. Since most aerodromes are fairly level, it gives the pilot who is taxiing an opportunity to check his altimeter prior to take-off. With QNH set, the altimeter of an aeroplane on the ground should indicate close to aerodrome elevation.

The Altimeter Subscale

The altimeter subscale is controlled by a knob which the pilot can turn. It is connected to the pointer and mechanically geared in the ratio of approximately 1 hPa to 30 ft, i.e. altering the subscale setting by 1 hectopascal will alter the indicated height by about 30 ft. Altering the subscale setting by 10 hectopascals will alter the altimeter indication by about 300 ft.

An easy means of determining pressure altitude is simply to wind 1013 into the subscale. The altimeter will then indicate the height in the ISA above the 1013 hPa pressure level: pressure altitude.

EXAMPLE 12 Suppose that the MSL pressure is 1030 (i.e. QNH) and the aeroplane is 600 ft amsl. With QNH 1030 set in the subscale, the altimeter will indicate an altitude of 600 ft amsl. The pressure altitude can be found (without any calculation) by winding 1013 into the subscale. In this case the pressure altitude is 90 ft.



■ Figure 6-18 **Example 12**

Wind on hectopascals, wind on height (and vice versa).

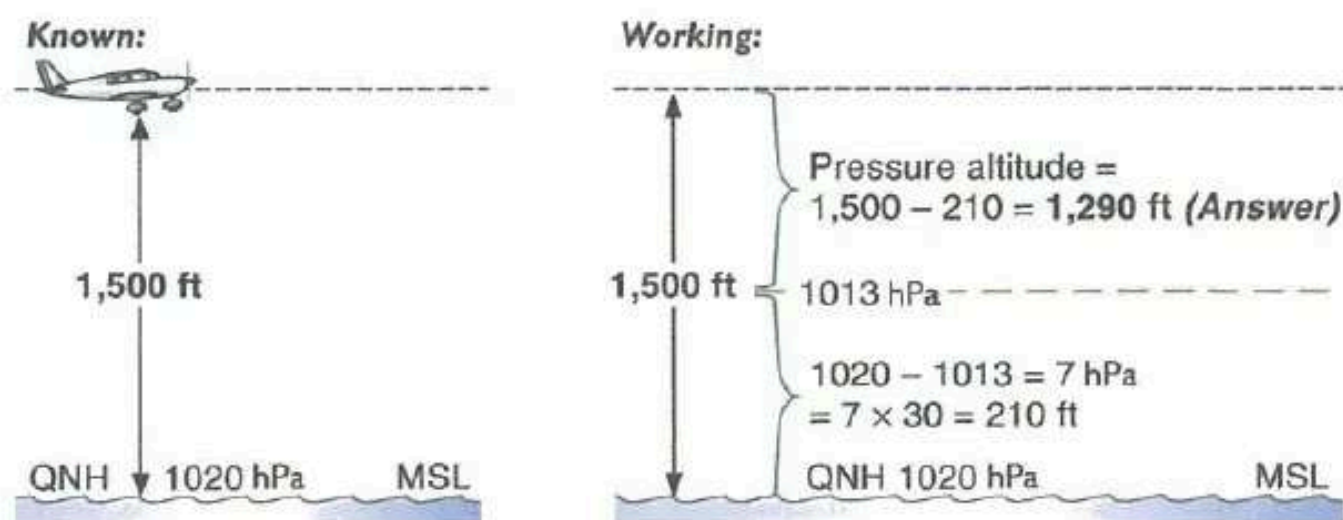
The correct QNH can be obtained from a number of sources, including Air Traffic Control, meteorological forecasts (at least approximately) and your own observations. By ‘your own observations’, we mean that if you are on the ground at an aerodrome whose elevation you know, then, after turning the knob until the altimeter indicates the aerodrome elevation, the reading in the subscale will be the current aerodrome QNH.

To illustrate this, suppose you arrive by car at a country aerodrome, elevation 1,290 ft, to go flying, and find that the altimeter indicates something quite different from 1,290. Simply by turning the knob until the pointers of the altimeter indicate 1,290 ft, which is the altitude (height amsl) of the aircraft, you will have wound the current QNH onto the subscale.

It is obvious that, for the altimeter indication to have any significance, the pilot must be aware of the setting on the subscale.

EXAMPLE 13 You are flying overhead Elstree aerodrome with the current QNH of 1020 set in the subscale, and the altimeter indicating 1,500 ft. What pressure altitude is that?

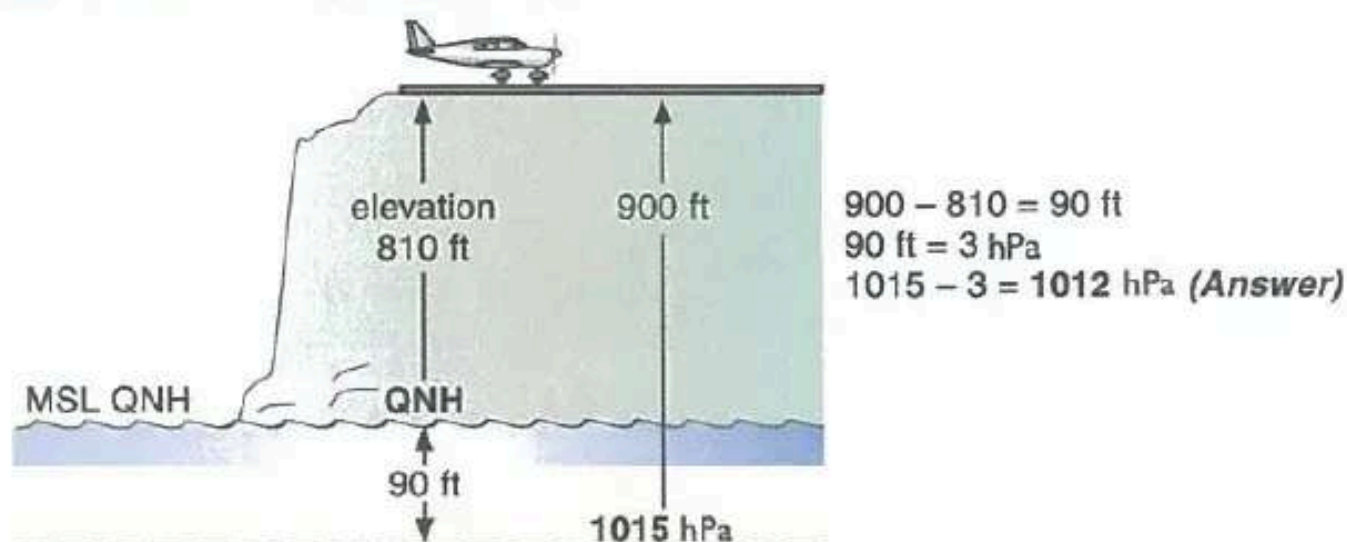
ANSWER 1,290 ft (Figure 6-19)



■ Figure 6-19 Example 13 answer: 1,290 ft

EXAMPLE 14 Your aircraft is sitting on the tarmac at Compton Abbas airfield, elevation 810 feet. The altimeter reads 900 ft (i.e. 90 ft too high) with 1015 set on the subscale. What is the airfield QNH (i.e. what is the mean sea level pressure at Compton Abbas at that time, assuming the atmosphere extends down to MSL)?

ANSWER With 1015 set in the subscale, the altimeter over-reads the known elevation by 90 ft. By winding off this 90 ft, we wind off 3 hPa, i.e. QNH is 1012.



■ Figure 6-20 Example 14 answer: QNH 1012 hPa

Now complete **Exercises 6 – Vertical Nav-3.**



Height Above Aerodrome Level – QFE

It is very convenient for circuit operations if the altimeter can be set to indicate height above aerodrome level (aal). This means that a 1,000 ft circuit at, for example, Leicester airfield (elevation 469 ft) can be achieved with the altimeter indicating 1,000 ft, rather than 1,469 ft if QNH was set in the subscale. It is a satisfactory procedure in the UK to set QFE in the subscale when flying in the circuit.

On the aerodrome, the altimeter should indicate within ± 50 ft of zero with QFE set in the subscale. QFE is the pressure at aerodrome level. In flight, the altimeter, with QFE set, will indicate height above the runway. (Having departed the circuit area, however, QFE is of little value, since the surrounding terrain will probably be at a different level from the aerodrome.)

QFE is the pressure at aerodrome level.

EXAMPLE 15 You plan to do some circuits at an aerodrome (elevation 749 ft amsl). On the ground you adjust the subscale of your altimeter until the altimeter indicates 0 feet, which it does with, say, 996 set in the subscale.

Calculate the current QFE, QNH and pressure altitude of this aerodrome, and the altimeter readings you would expect with these subscale settings when the aeroplane is on the ground.

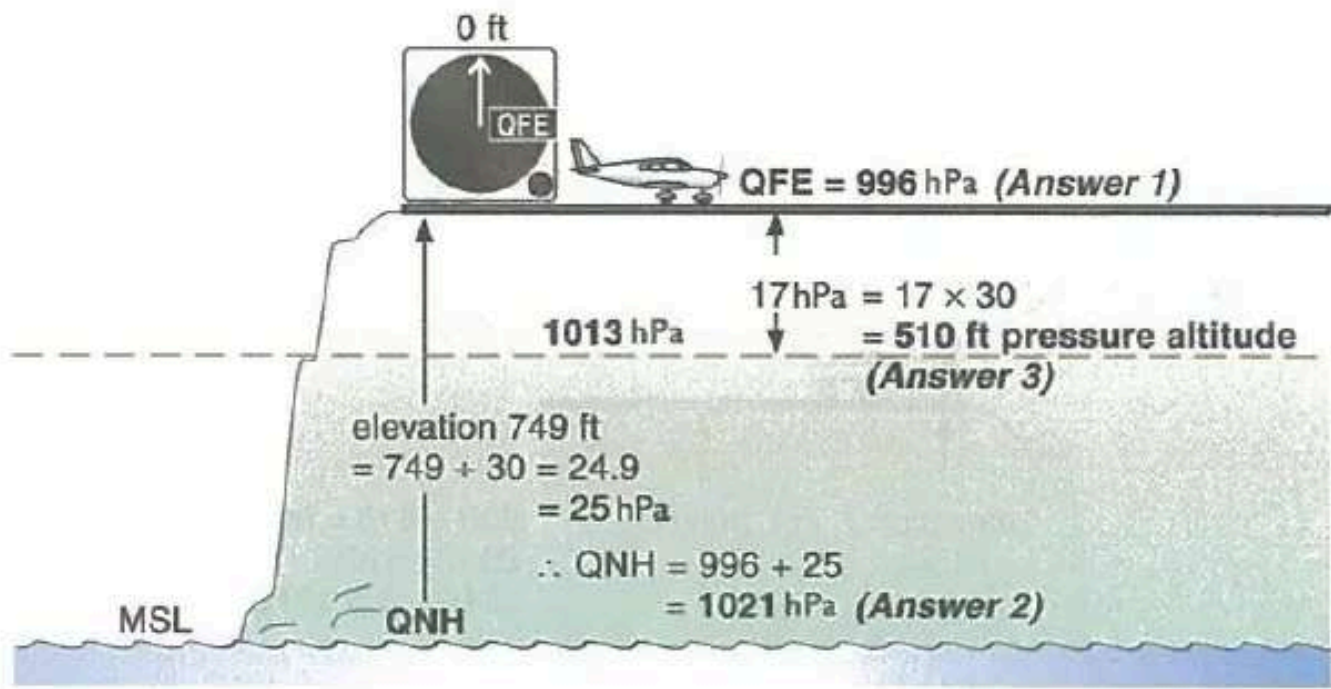


Figure 6-21 Example 15 answers: QFE 996 hPa, QNH 1021 hPa, pressure altitude 510 ft

Remember that the information your altimeter gives you depends on what you have set in the subscale.

For terrain clearance, use QNH.

Now complete Exercises 6 – Vertical Nav-4.

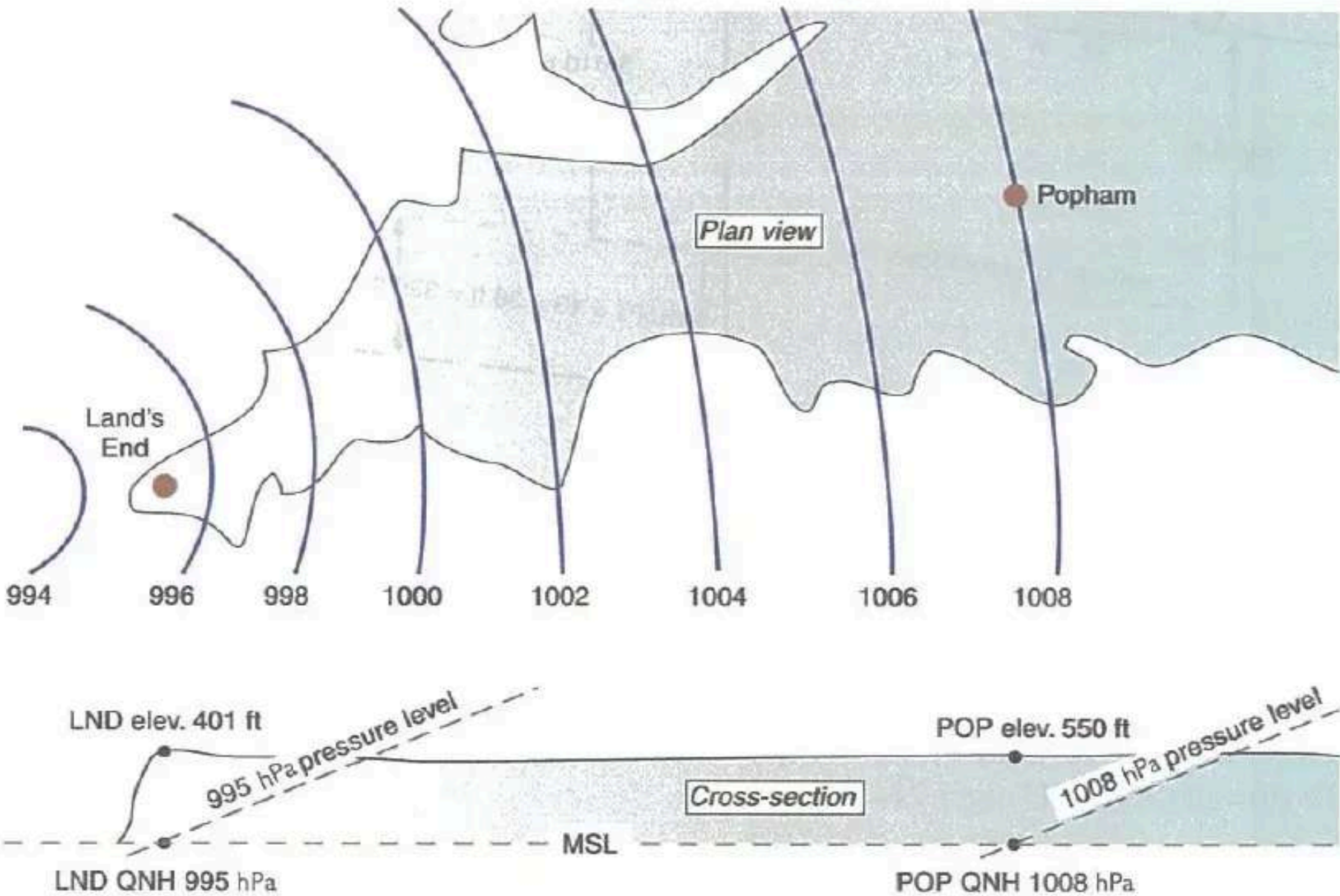


Flying With Changing Sea Level Pressures

Consider an aeroplane flying from Land's End to Popham, with the barometric pressure over the south-west of England as shown in Figure 6-22.

We depart Land's End with the Land's End QNH of 995 set on the subscale (i.e. MSL pressure at Land's End), and cruise 3,000 ft above this with our altimeter indicating 3,000 ft.

Tracking towards Popham, we are flying into an area of higher pressure, and so the 995 pressure level will be gradually rising. If we maintain 3,000 ft indicated on the altimeter, with 995 set, we will in fact be climbing with respect to sea level.

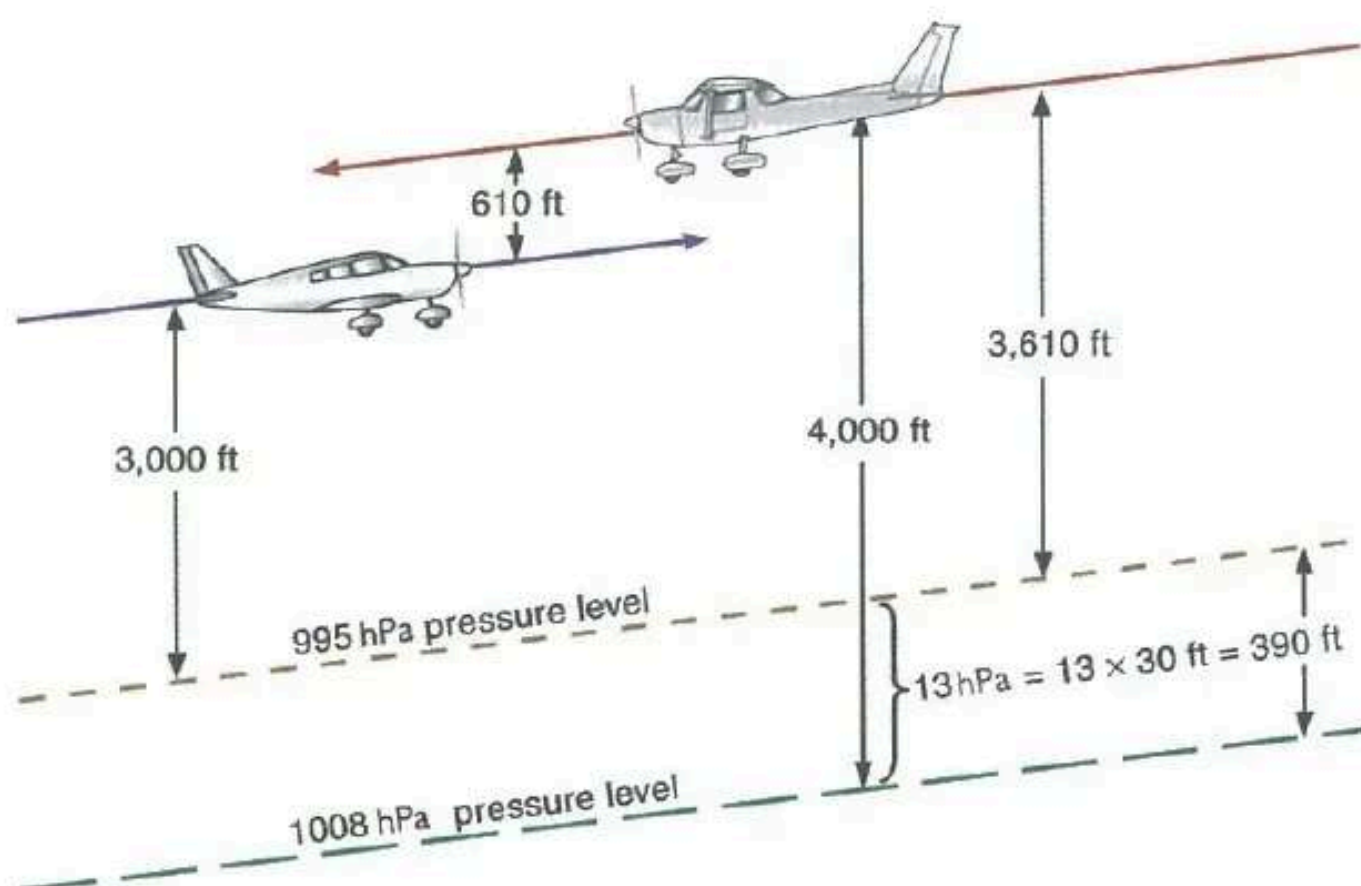


■ Figure 6-22 **Plan and cross-section views of this synoptic situation**

Conversely, another aircraft flying in the opposite direction with the Popham QNH of 1008 set on its subscale and cruising at 4,000 ft will in fact be gradually descending. Can you spot the two inherent dangers?

- Remember that two of the most important tasks for pilots are:
- to avoid hitting the ground unexpectedly; and
 - to avoid colliding with other aircraft.

The aircraft coming from Popham, flying from a higher-pressure area towards an area of lower pressure, will actually be gradually descending if the altimeter reading 4,000 ft continues to have 1008 hPa set on the subscale. The aircraft will in fact be lower than 4,000 ft amsl, so terrain clearance could be a problem. When flying 'From *high* to *low*, beware below!' (Danger No. 1)



■ Figure 6-23 **Flying from high to low, beware below!**

The two pilots may think that they have 1,000 ft vertical separation because the altimeter in one aeroplane indicates 3,000 ft, and the altimeter in the other indicates 4,000 ft. In fact, the vertical separation is only 610 ft, because 4,000 ft above the 1008 hPa pressure level is only $(4,000 - 390) = 3,610$ ft above the 995 hPa pressure level. (Danger No. 2)

Regional Pressure Setting (Regional QNH)

The practical solution to both of these problems (terrain clearance and traffic separation) because of changing QNHs is to have all aircraft which are cruising in the same area, below the transition altitude, use the same altimeter subscale setting. For this reason the UK is divided into a number of Altimeter Setting Regions (ASRs) – see Vol. 2 of *The Air Pilot's Manual*.

Regional Pressure Setting can be obtained from:

- all aerodromes with Air Traffic Services;
- any Air/Ground ATC channel;
- by land-line (for pre-flight planning);
- the London and Scottish ATCCs; or
- the Manchester Sub-Centre.

When cruising en route, the appropriate subscale setting is the **Regional Pressure Setting (RPS)**, also known colloquially as the **Regional QNH** or the **Area QNH**, which is the current QNH for that region at that time. Its value will be updated by ATC at least every hour.

The QNH will in fact vary slightly throughout the Altimeter Setting Region, depending on the pressure pattern. To be on the conservative side, the Regional Pressure Setting is the lowest

forecast QNH value for that hour, and so will be at sea level or slightly higher. This ensures that the aircraft will be at or slightly higher than the altitude indicated, and not lower.

For example, with Regional Pressure Setting set and the altimeter indicating 2,000 ft, the aeroplane should be 2,000 ft amsl or slightly higher. No Aerodrome QNH in that region will be lower than the value of the Regional Pressure Setting.

When cruising cross-country, set Regional Pressure Setting, and update it when you fly into another region, or whenever ATC communicates an amended value.

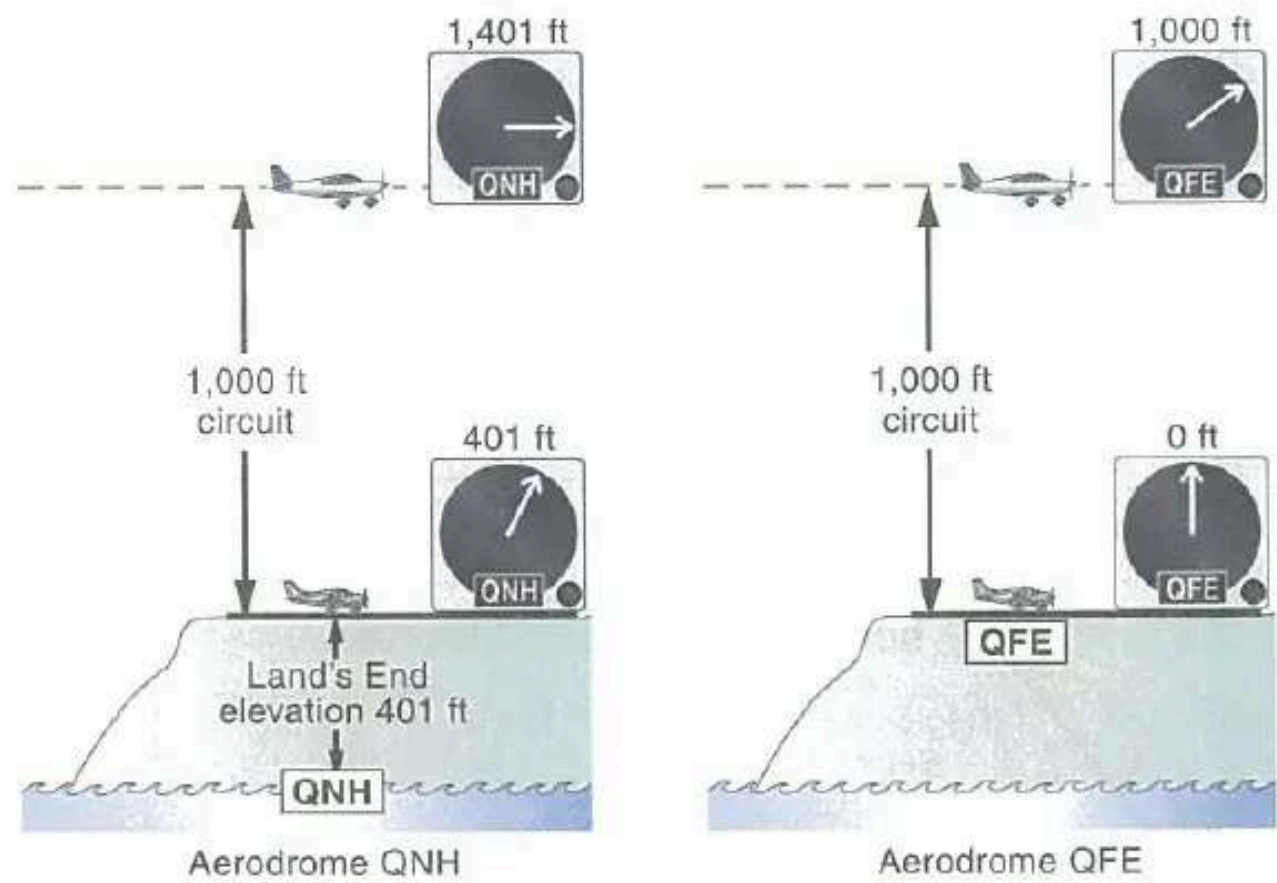
Altimetry Procedures for Cross-Country Flights

In the UK, a typical private cross-country flight in a light aircraft is usually conducted at or below 3,000 ft amsl. We will use a flight from Land’s End to Popham as an example.

1. Set Aerodrome QNH or QFE for Take-Off

With Aerodrome QNH set, the altimeter indicates height amsl, which is useful information in the circuit area and during the climb-out, especially over high terrain or if ATC requires you to report altitude. On the ground, the altimeter should indicate aerodrome elevation, which at Land’s End is 401 ft amsl.

If you choose to set QFE, the altimeter will indicate height aal, and this reading has little significance away from the vicinity of the aerodrome. On the ground, with QFE set, the altimeter should indicate close to zero (within ± 50 ft).

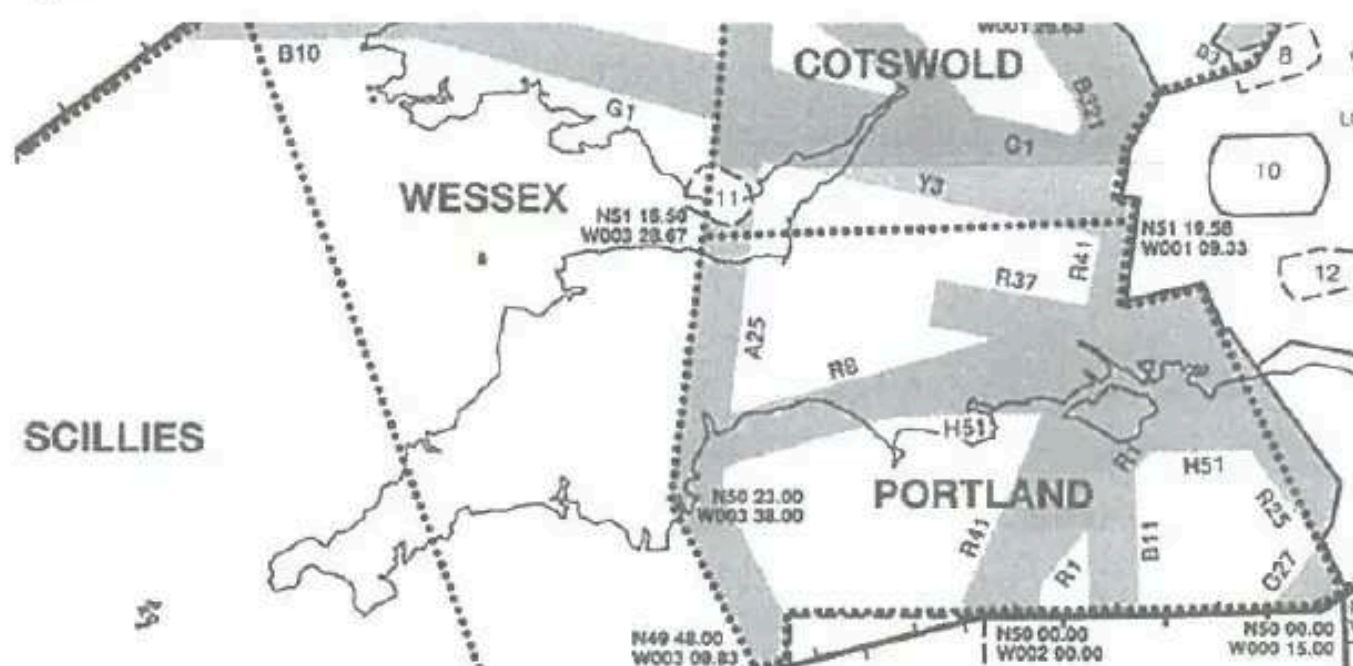


■ Figure 6-24 Set Aerodrome QNH, or Aerodrome QFE, for circuit operations

2. Set Regional Pressure Setting if Cruising Below the Transition Altitude

On reaching cruising altitude 3,000 ft, the Scillies Regional Pressure Setting should be set. Flying with the altimeter indicating 3,000 ft should ensure that the height above mean sea level is 3,000 ft or slightly more (since the Regional Pressure Setting is always the lowest QNH for that area).

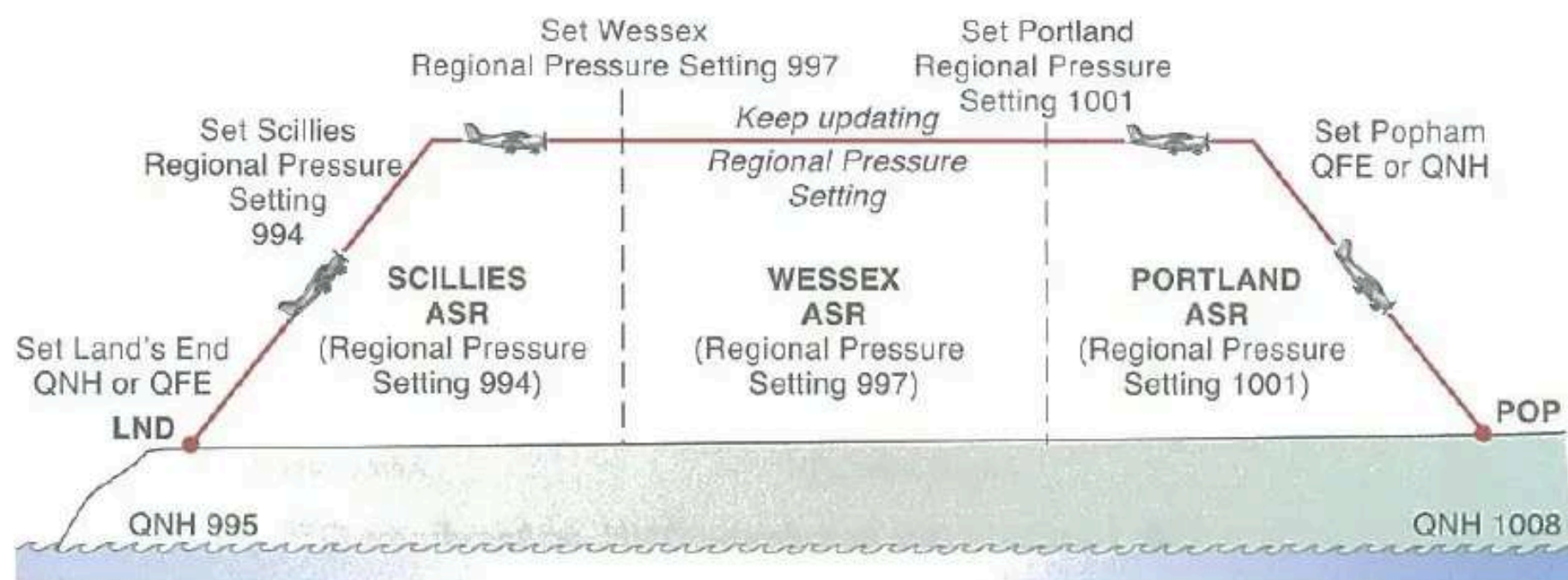
The Regional Pressure Setting should be updated as you cross each ASR boundary (shown on aeronautical charts) – in this case passing from Scillies ASR to the Wessex ASR to the Portland ASR, or whenever the appropriate Regional Pressure Setting is updated by ATC.



■ Figure 6-25 Excerpt from the ASR map

With Regional Pressure Setting set in the subscale, the pilot can evaluate:

- terrain clearance (the height of terrain is found from the aeronautical charts); and
- vertical separation from other aircraft.

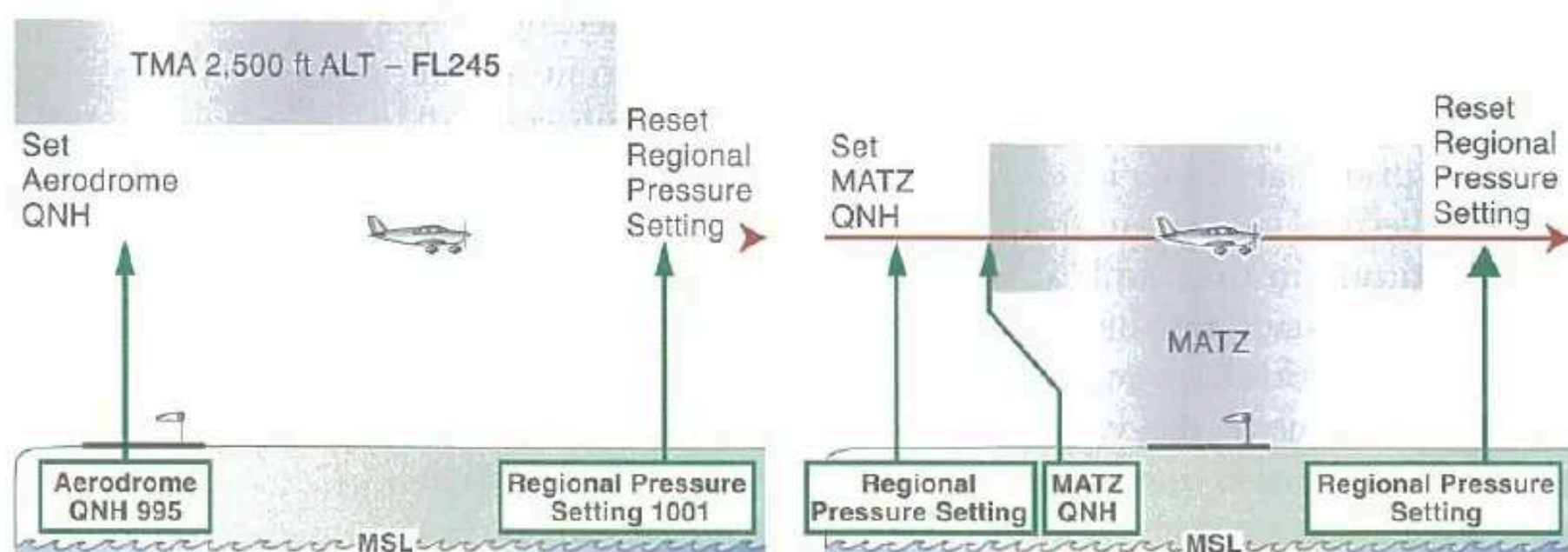


■ Figure 6-26 When cruising below the transition altitude, set Regional Pressure Setting

NOTE It is insufficient to assume that errors in horizontal tracking will safeguard you from other traffic in your vicinity operating at or near the same level. Modern tracking aids, such as VOR, GPS, and inertial navigation systems, have made extremely accurate tracking possible.

En route you may be required, when flying beneath a Terminal Control Area (TMA) or Control Area (CTA), to set the QNH of an aerodrome situated beneath that area, to assist in separation from other aeroplanes or to ensure that you do not inadvertently penetrate the controlled airspace above you. The Aerodrome QNHs will not differ greatly (if at all) from their Regional Pressure Setting.

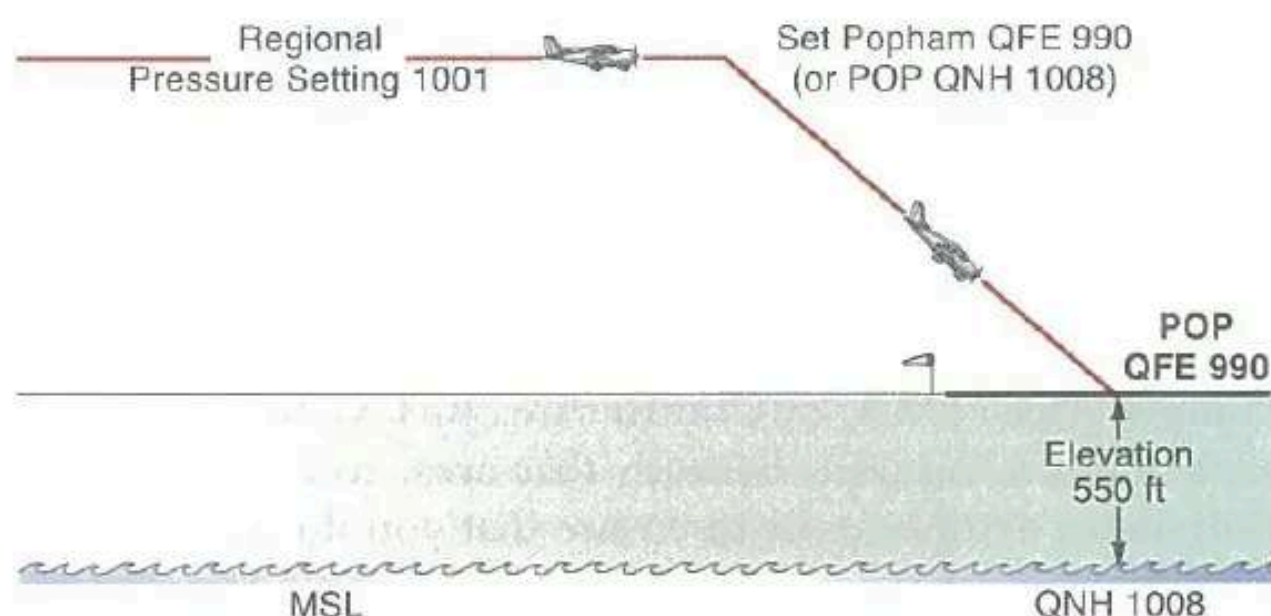
When transiting a Military Aerodrome Traffic Zone (MATZ) where military aircraft may be operating, you may be required to set the MATZ Aerodrome QNH for separation purposes. Once clear of the MATZ, Regional Pressure Setting should be reset on the subscale.



■ Figure 6-27 **Aerodrome QNH (rather than Regional Pressure Setting) may be required en route**

3. Set QFE or Aerodrome QNH when Approaching for Landing

In preparation for joining the circuit, set the aerodrome QFE (or QNH if preferred) for the destination aerodrome as the circuit area is approached. For instance, approaching Popham, you would set QFE 990, or QNH 1008, on the altimeter subscale. (Note: In some countries only QNH is used.)



■ **Figure 6-28** *Approaching the circuit, set aerodrome QFE (or QNH if preferred)*

Cruising Above the Transition Altitude

The highest terrain in the UK is 4,406 ft amsl at Ben Nevis in Scotland. Obviously most of the UK is much lower than this. To avoid the need for continually updating the altimeter setting, it is common practice when flying at levels where high terrain is not a problem to set standard pressure 1013 mb on the subscale. All aircraft cruising at these levels, with 1013 set, thereby obtain adequate vertical separation from each other.

The altitude in the climb at which the transition from QNH to 1013 mb is made is called the **transition altitude**. In the UK, the transition altitude is generally 3,000 ft amsl. There are some exceptions to this in the vicinity of some major airports, where the transition altitude may be higher because traffic density is high and where the high rates of climb and descent of jet aircraft is a consideration.

In other parts of the world where the terrain is much higher, the transition altitude is much higher. In the USA the transition altitude is 18,000 ft; in Papua New Guinea (a mountainous country just north of Australia) it is 20,000 ft; in Australia it is 10,000 ft.

Exceptions, at the time of printing, to the usual UK transition altitude of 3,000 ft are noted below. You need not remember them; they are included here for interest.

- London Terminal Control Area (TMA): 6,000 ft;
- Manchester TMA: 5,000 ft;
- Scottish TMA (covering Edinburgh, Glasgow and Prestwick): 6,000 ft;
- Aberdeen CTR/TMA: 6,000 ft;
- Belfast CTR/TMA: 6,000 ft;
- Teesside CTR/CTA: 6,000 ft;
- Edinburgh CTR: 6,000 ft;
- Glasgow CTR: 6,000 ft;

- Leeds Bradford CTR/CTA, 5,000 ft;
- Birmingham CTR/CTA: 4,000 ft;
- Solent CTR 4,000 ft.

Choice of Altimeter Setting for VFR Flights

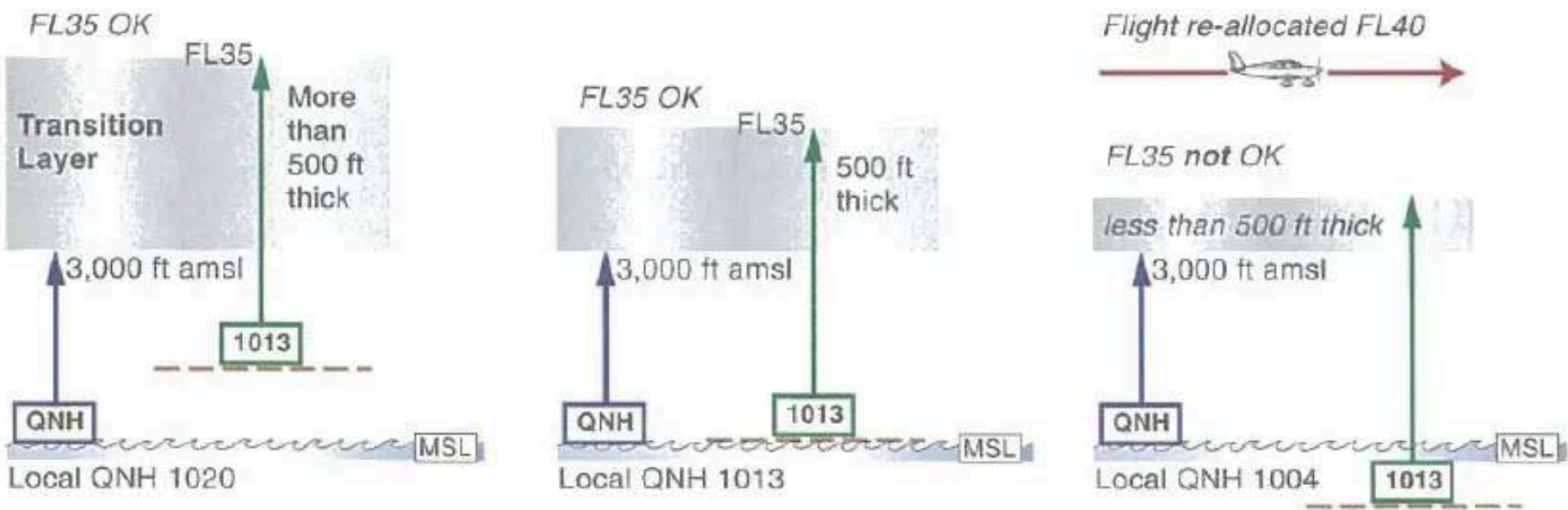
Outside controlled airspace and above the transition altitude, VFR flights may cruise with Regional Pressure Setting, but it is advisable to use 1013 and cruise on flight levels as aircraft operating on Instrument Flight Rules are required to do.

In controlled airspace and above the transition altitude, aircraft should cruise at flight levels (with 1013 hPa set), rather than at altitudes (with QNH set).

The Transition Layer

There would be a possibility of conflict if aircraft cruising above the transition altitude were on Regional Pressure Setting and aircraft only slightly above it were on the 1013 reference datum. For this reason there is a layer above the transition altitude in which cruising flight should not occur, to ensure satisfactory vertical separation of at least 500 ft.

In practice, flight levels are nominated in 500-foot steps, e.g. FL35, FL40, FL45, FL50, FL55, etc. Since the 1013 hPa reference datum may not be at mean sea level, it is possible (indeed most likely) that the actual vertical spacing between a transition altitude of 3,000 ft and a transition level of FL35 will not be 500 ft, as illustrated in Figure 6-29.

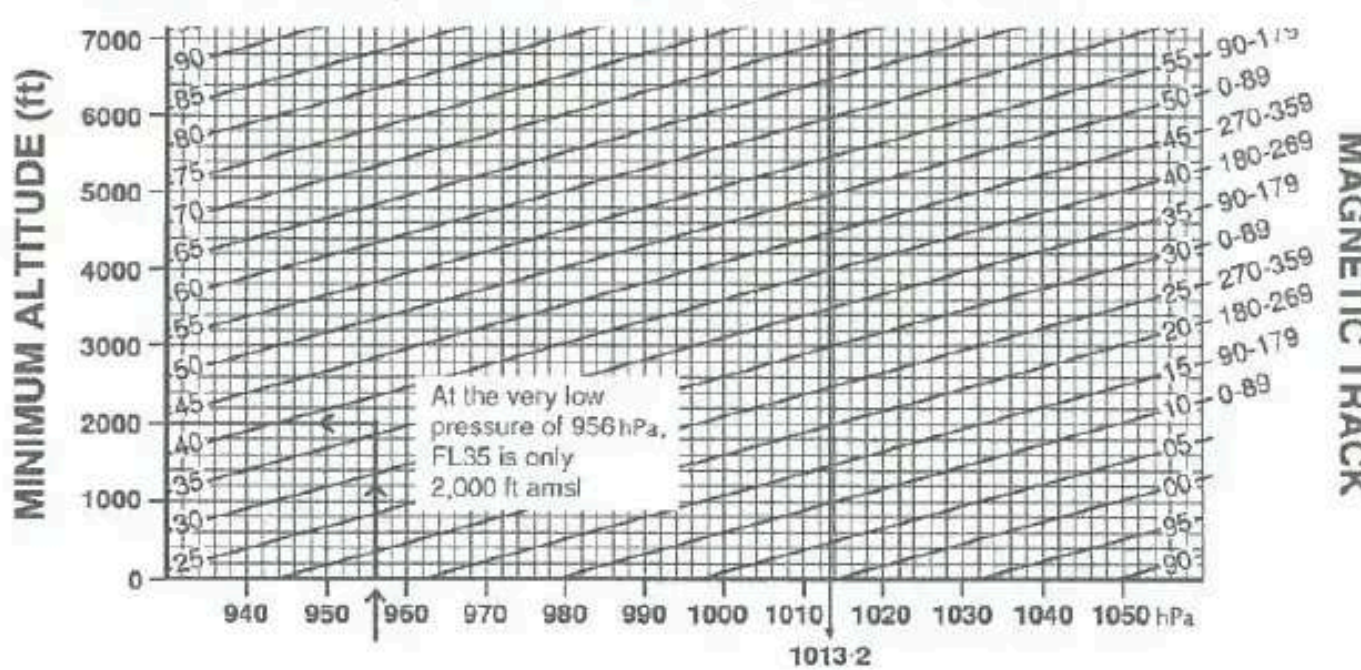


■ Figure 6-29 **The thickness of the transition layer varies with QNH**

As you can see, the transition altitude remains fixed, but the transition level moves up or down according to the QNH. If the Regional Pressure Setting is less than 1013 hPa, FL35 is lower than 3,500 ft amsl and there is less than 500 ft vertical separation between a transition altitude of 3,000 ft and FL35. So you would need to cruise at FL40 to retain a 500 ft or better vertical separation from an aircraft cruising at 3,000 ft on Regional Pressure Setting.

If needed, a table to convert altitudes to flight levels (and vice versa) is contained in the UK Aeronautical Information Publication (AIP ENR) and is reproduced in Vol. 2 of this series.

- To use the table (Figure 6-30):
- take a vertical line upwards from the QNH value along the bottom axis until it meets the appropriate (sloping) flight level line; then
 - read horizontally across to the equivalent altitude.



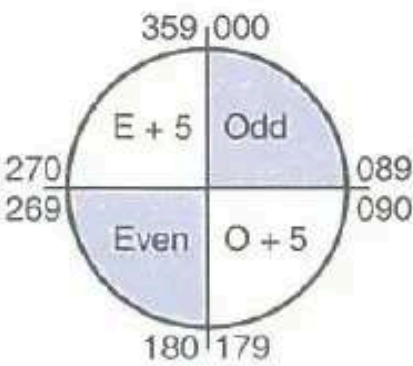
■ Figure 6-30 Using the altitude/flight level conversion table

Summary

THE TRANSITION ALTITUDE is the altitude at, or below, which the vertical position of aircraft is controlled by reference to altitudes, i.e. with regional QNH set. In the UK, it is usually 3,000 ft amsl.

THE TRANSITION LEVEL is the flight level at, or above, which the vertical position of aircraft is controlled with reference to flight levels, i.e. with 1013 set. In the UK, it is usually FL35 (or higher if QNH is less than 1013).

THE TRANSITION LAYER is the airspace between the transition altitude and the transition level. It varies in thickness, depending on the Regional Pressure Setting.



■ Figure 6-31
The quadrantal rule (advisable to follow, but not mandatory if VFR and outside controlled airspace)

Cruising Level Selection

As a student pilot, or a basic-PPL holder (no IMC or Instrument Rating), you will be restricted to flying under Visual Flight Rules (VFR) and as such should always be clear of cloud, satisfy minimum visibility requirements and be in sight of the ground if below 3,000 ft amsl.

It is not mandatory for VFR flights to cruise at any particular altitude or flight level, but it is recommended (in the UK AIP) that VFR flights adopt the IFR cruising level system, known as the quadrantal rule, if cruising above the transition altitude.

MAGNETIC TRACK	CRUISING FLIGHT LEVEL
000°M to 089°M	FLs in odd thousands of feet; FL30, 50, 70, etc.
090°M to 179°M	FLs in odd thousands plus 500 , FL35, 55, 75, etc.
180°M to 269°M	FLs in even thousands; FL40, 60, 80, etc.
270°M to 359°M	FLs in even thousands plus 500 , FL85, etc.

NOTE This rule will be changing when the UK adopts SERA (Standard European Rules of the Air) in late 2014.

Now complete **Exercises 6 – Vertical Nav-5**.



Safety Altitude (or Safety Height)

In selecting an appropriate altitude at which to fly on a cross country flight in VMC and under VFR, it is useful to bear in mind the following:

- The need to comply with the requirements of the low flying rules (rule 5) including:
 - separation from obstacles,
 - suitable clearance over congested areas,
 - the glide-clear requirements;
- Selecting a higher altitude at which to fly gives you more options in the event of an in-flight emergency;
- Selecting altitude other than the obvious 1,500, 2,000, 2,500 ft amsl is less likely to lead to a conflict with other traffic;
- It is useful to have calculated the minimum safe altitude (msa) for the route that you are flying in order to have an awareness of the terrain clearance that would be required should you inadvertently become IMC.

It is certainly not necessary on a VFR cross-country route to fly at a calculated MSA (Minimum Safe Altitude) since your separation from obstacles and terrain should be determined visually. The instrument flight rules specify a legal minimum of 1,000 ft clearance from the highest fixed obstacle within 5 nm of

the aircraft (or 2,000 ft in mountainous terrain) but there is no requirement for PPLs flying VFR in VMC to adhere to those rules.

In the PPL examinations you may come across a definition of Safety Altitude where the examiner has based his calculation on the addition of 1,000 ft to the highest fixed obstacle within 5 nm of the track or to the maximum elevation figure within a particular quadrangle of the chart. This is a definition for the examination and does not need to be adhered to in practical VFR planning.

The Effect of Temperature Variations

So far we have only considered pressure variations from the International Standard Atmosphere. Temperature variations from ISA play a role not only in vertical navigation and altimetry, but also in aeroplane performance (see Vol. 4 of this series). ISA MSL temperature is +15°C, and the temperature lapse rate is 2°C/1,000 ft (decrease with altitude).

TEMPERATURE STRUCTURE IN THE ISA	
Pressure Altitudes	ISA Temperatures
above 36,000 ft	constant at -57°C
36,000 ft	-57°C (ISA = +15 - [2 × 36] = +15 - 72 = -57°C)
20,000 ft	-25°C (ISA = +15 - [2 × 20] = +15 - 40 = -25°C)
3,000 ft	+9°C (ISA = +15 - [2 × 3] = +15 - 6 = +9°C)
2,000 ft	+11°C (ISA = +15 - [2 × 2] = +15 - 4 = +11°C)
1,000 ft	+13°C (ISA = +15 - 2 = +13°C)
MSL 1013 hPa	+15°C

Temperature Deviation from ISA

Suppose that in the ‘real’ atmosphere, the temperature at 2,000 ft is not +11°C as in the ISA, but +16°C, i.e. it is 5°C warmer than in the ISA. This can be expressed as +16°C, a simple temperature, or as a deviation from the ISA, which in this case is ISA+5. In aviation it is common practice to use this ISA deviation means of describing temperature.

EXAMPLE 16 Express -10°C at FL80 as a deviation from the ISA.

At FL80, i.e. pressure altitude 8,000 ft
ISA = +15 - (2 × 8)
= +15 - 16
= -1°C

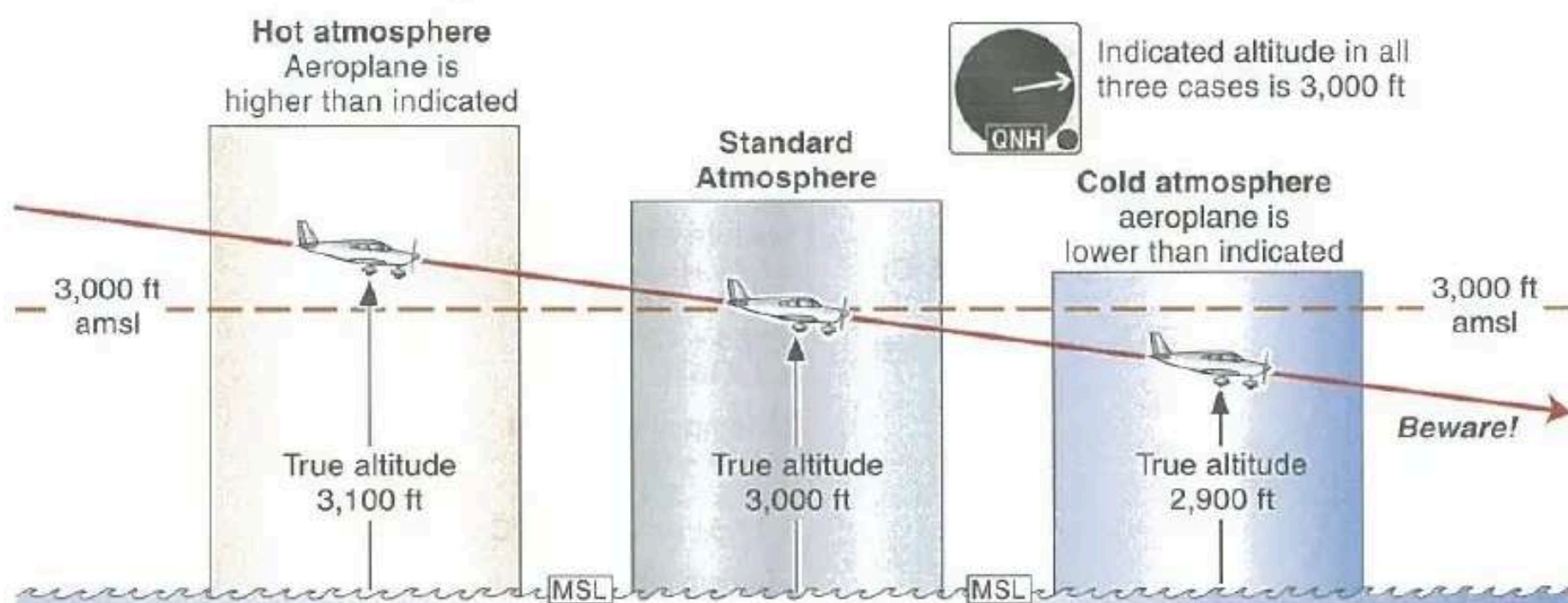
ANSWER Now -10°C is colder than -1°C by 9°C, and this is expressed as ISA-9.

In Cold Air, the Altimeter Over-Reads

The altimeter is essentially a barometer that converts pressure measurements to altitude. It is calibrated according to the International Standard Atmosphere. On a cold day when the air is more dense, various pressure levels at altitude will be lower than on a warm day.

The 913 hPa pressure level that equates with a pressure altitude of 3,000 ft in the International Standard Atmosphere may be at only 2,900 ft. The altimeter, however, because it is calibrated according to the ISA, will indicate this as 3,000 ft even though the aeroplane is at 2,900 ft. On a day that is warmer than ISA, when the altimeter indicates 3,000 ft, the aeroplane may in fact be slightly higher.

This is not significant for separation between aircraft, since all altimeters will be affected identically. It is important for terrain separation, however, so when flying from a high to low temperature, beware below because the altimeter will read too high.



■ Figure 6-32 ***When flying from high to low temperature, beware below!***
(Same rule as for pressure.)

Density Altitude (or Density Height)

Density is the mass per unit volume, or if you like, the number of molecules in each unit volume, and is affected directly by variations in temperature. Aeroplane performance (the speed at which it can fly, and the height to which it can climb) depends, amongst a number of factors, mainly on the ambient air density. Consequently it is important to be able to calculate density altitude.

Density altitude is the atmospheric density expressed in terms of altitude in the International Standard Atmosphere which corresponds to that density.

If, for example, at 1,000 ft amsl in the actual atmosphere the density is the same as the density at 2,400 ft in ISA, i.e. our altitude is 1,000 ft amsl but our density altitude is 2,400 ft, the aircraft and engines will perform as if the aeroplane were at 2,400 ft.

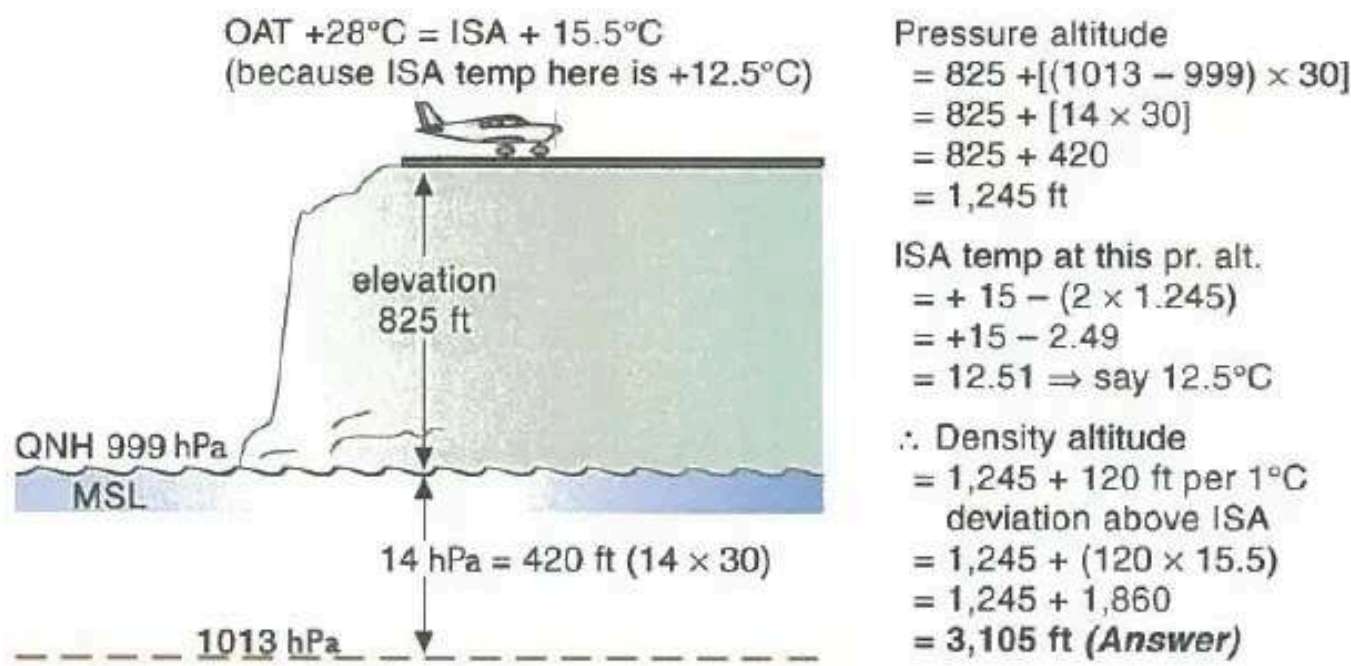
How to Calculate Density Altitude

It is impractical for a pilot to have the equipment necessary to measure air density, so we make use of two pieces of information already available and on which density depends: pressure altitude and temperature. Density altitude can be calculated in three ways:

- 1. By correcting pressure altitude for ISA temperature deviation by 120 ft per 1°C.
- 2. Graphically, as in most aeroplane performance charts, on which both pressure altitude and temperature are the criteria used in entering the graph. Though density altitude itself may not be specified, it is implied by these other two parameters.
- 3. By navigation computer.

EXAMPLE 17 Calculate the density altitude at Huddersfield (Crosland Moor), elevation 825 ft, if the current QNH is 999, and the OAT (outside air temperature) +28°C.

ANSWER PA 1,245 ft, DA 3,105 ft

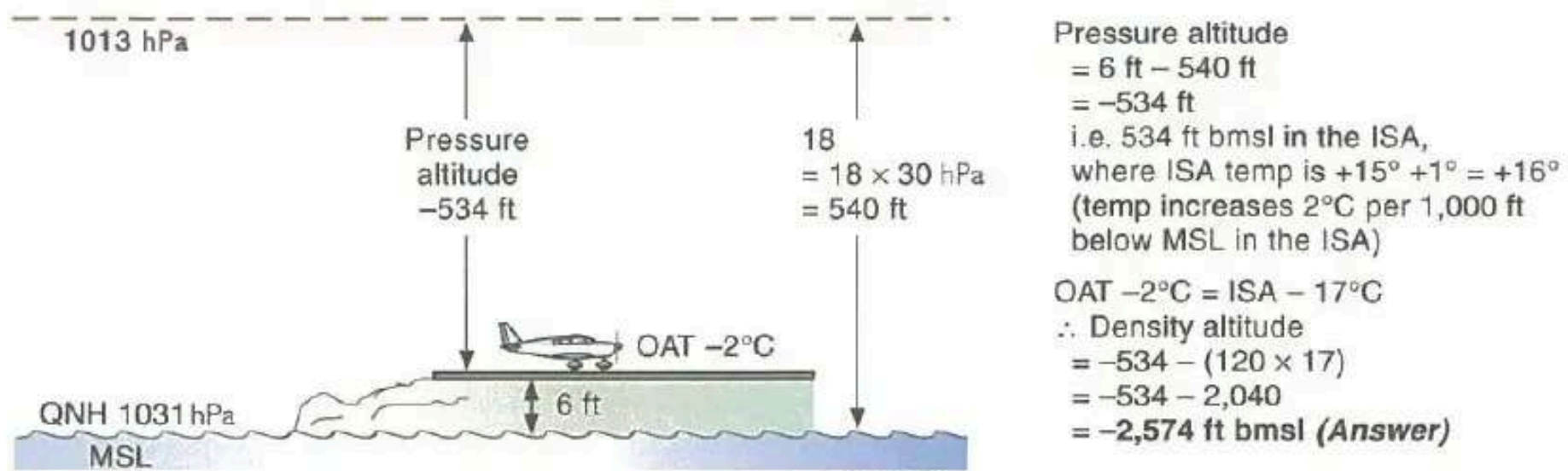


■ Figure 6-33 Example 17

NOTE Even though the aircraft is at less than 1,000 ft amsl, the engines will perform, and the aircraft will fly, as if it were at about 3,000 ft in the International Standard Atmosphere, i.e. it will have a much poorer performance. As a good pilot you should be aware of poor performance when hot and high.

EXAMPLE 18 Calculate the density altitude of Shoreham, elevation 6 ft, QNH 1031 hPa, OAT –2°C.

ANSWER PA – 534 ft, DA – 2,574, i.e. below msl



■ Figure 6-34 **Example 18**

NOTE A high QNH (i.e. high air pressure) increases density as does the low temperature, with the temperature usually being the critical factor to watch out for. In this particular case of a low aerodrome elevation and a low temperature at Shoreham, we could expect the engines and the aeroplane to perform very well.

Any graph or table in an aeroplane Performance or Flight Manual that has both pressure altitude and temperature on it means that density height is being allowed for, and will not have to be calculated directly as above or by computer.

Now complete **Exercises 6 – Vertical Nav-6.**



Time

Time is of great importance to the air navigator, and the clock is one of the basic instruments used in the cockpit. Time enables you to:

- **regulate** affairs on board your aeroplane;
- **measure** the progress of your flight;
- **anticipate** arrival time at certain positions;
- **calculate** a safe endurance for flight;
- **estimate** when weather conditions at the destination are likely to improve;
- **measure** rest periods between flights ... and so on.

Time is also used to measure the earth's rotation. We relate the rotation of our planet Earth to the position of celestial or heavenly bodies, such as the sun and other stars. By using time we can specify the beginning of day, sunrise, noon, sunset, commencement of night, midnight, moonrise, moonset, and so on.

To all navigators – land, sea and air – time is of vital importance, and a subject that must be mastered.

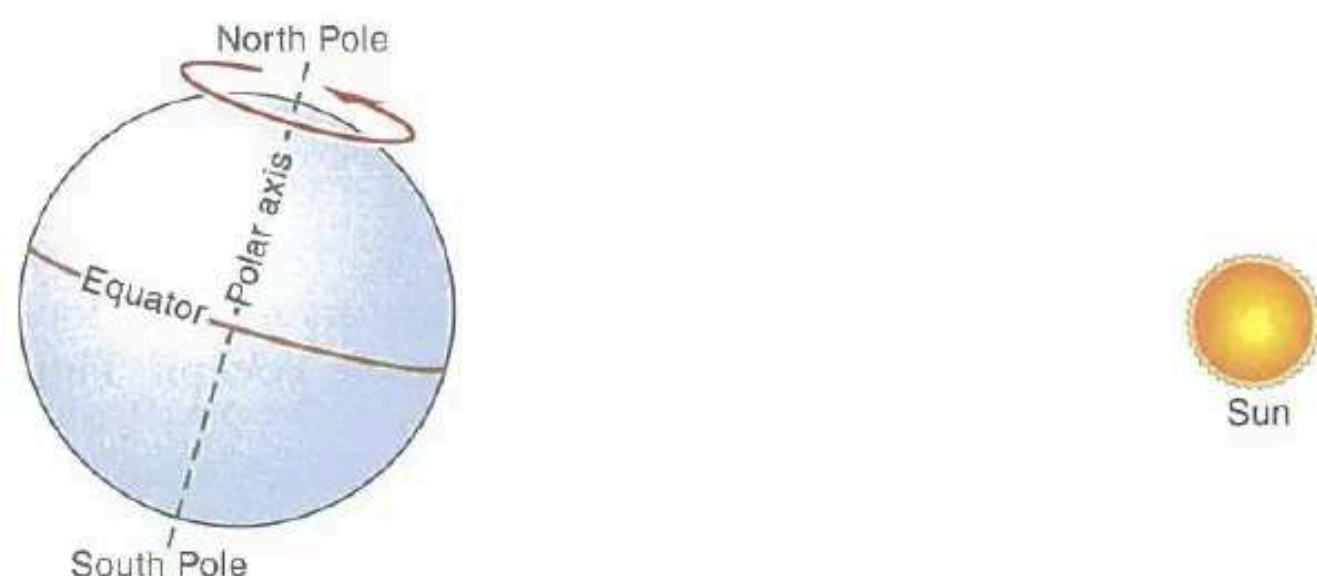
Motion of the Earth

To measure the passage of time, we need to relate it to some repetitive event. For our ancestors, and indeed for us, a suitable recurring event is the apparent passage of the sun across our skies – its highest point in the sky simply indicated when the shadows that it casts are shortest. The sun appears to cross our skies once in every day.

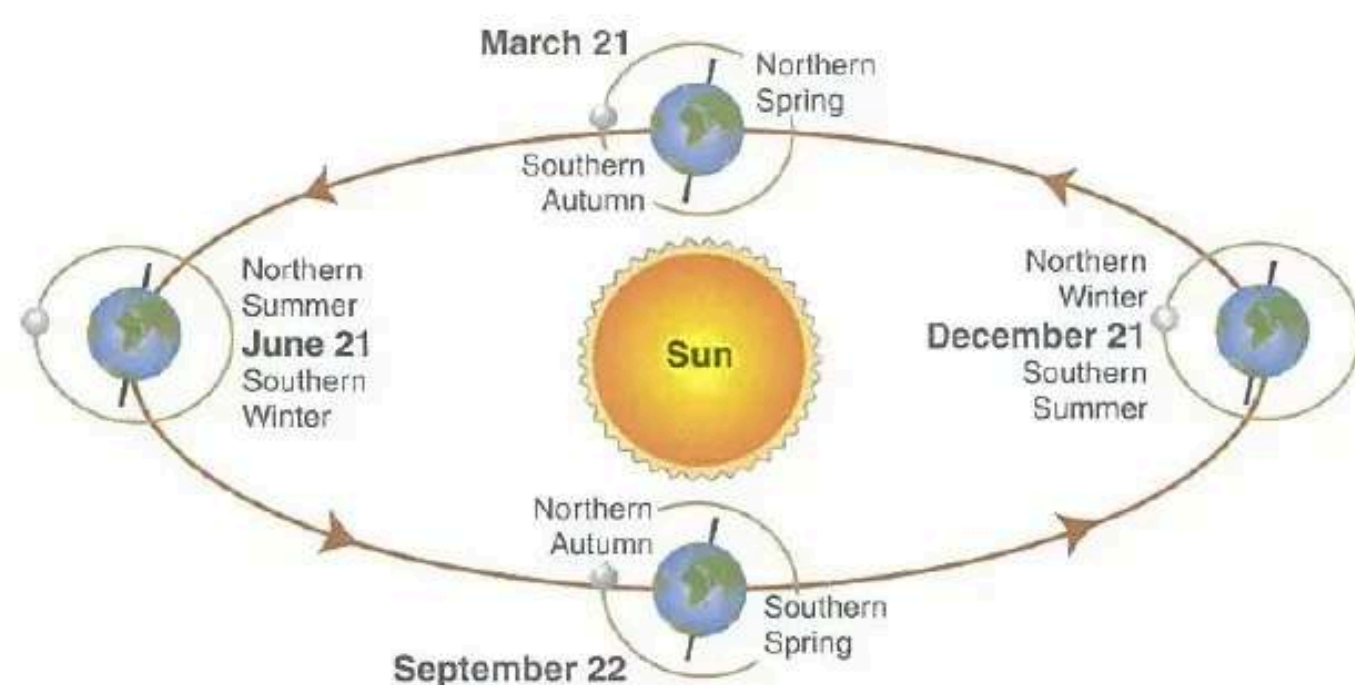
On a longer time scale, we notice the regular passage of the seasons – spring, summer, autumn and winter – a complete cycle of these being called one year.

The sun has been used as a simple clock for thousands of years. Whereas early man thought that it was the sun which moved around the earth, we now know that this is not the case. It is, in fact, the rotation of the earth on its axis that causes the appearance of the sun travelling across our skies each day, hence the term 'apparent passage of the sun'.

As man's knowledge increased it was realised that one day is the approximate time span of one revolution of the earth on its own axis. One year is the approximate time span of one complete orbit of the earth about the sun.



■ Figure 7-1 *The earth rotates about its own axis once every 24 hours*



■ Figure 7-2 *The earth rotates around the sun once a year*

There is a third fundamental type of time apart from the rotation of the earth about its own axis and the orbiting of the earth around the sun. It is international atomic time (TAI), where atomic vibrations, such as those occurring at very short time intervals in quartz crystals, are used to define the SI second and calibrate clocks extremely accurately. As this was endorsed internationally as recently as 1971 you can see that the subject of time is still not a closed book. In fact, only in 1985 was the international basis for standard time changed from Greenwich Mean Time (GMT) to Coordinated Universal Time (UTC).

Each day is divided into 24 hours, each hour further divided into 60 equal minutes, and each of these minutes further divided into 60 equal seconds.

To complete one orbit of the sun, the earth takes about 365 and $\frac{1}{4}$ solar days. It is convenient to have a whole number of days in a year, and so we define the civil year as 365 days. At the end of each 4 years, when the extra $\frac{1}{4}$ day each year adds up to one whole day, the extra day is added in to give a leap year of 366 days. This keeps the calendar reasonably in step with the seasons.

Measurement and Expression of Time

To measure time, use is again made of a repetitive event, such as the swinging of a pendulum, or the atomic vibrations within a quartz crystal, to design clocks that measure hours, minutes and seconds.

Each day is divided into 24 hours, which begins at midnight (00 hours 00 minutes), then proceeds through midday (1200) to midnight (2400), at which instant the next day begins (0000).

The hours are numbered from 00 hours to 24 hours (rather than 0 to 12 a.m. and 0 to 12 p.m.), and the 60 minutes of each hour are numbered from 00 min to 59 min. The term a.m. means *ante meridiem*, in the sense that the sun has yet to pass over your meridian of longitude, so the time is before noon; p.m. means *post meridiem*, because the sun has passed overhead and the time is afternoon.

For flight planning and navigation purposes we do not usually refer to the year or the month, but only the **day** of the month as the **date**, followed by the time in **hours** and **minutes**. As most air navigation occurs within a few hours, and only rarely in excess of 30 hours, we can be reasonably confident of which year and month we are talking about, so there is no need to specify them.

Seconds, which are $\frac{1}{60}$ of a minute, are too short a time interval for practical navigation, so date/time is usually expressed as a six figure date/time group.

Six-Figure Date/Time Group

In the six-figure date/time group:

- the **date** is a two-figure group for the day of the month from 00 to 31, and is followed by:
- the **time**, written as a four-figure group on a 24 hour clock – the first two figures representing the hours from 00 to 24, and the last two figures representing the minutes from 00 through to 59.

EXAMPLE 1 Express 13th of September, 2006, 10:35 a.m. as a six-figure date/time group.

<i>date</i>	<i>time</i>	
13	10	35
	<i>hr</i>	<i>min</i>

ANSWER 131035

EXAMPLE 2 Express 3:21 p.m. on March 17th, 2007, as a six-figure date/time group.

$$\begin{array}{r}
 3:21 \text{ p.m.} = 1200 \text{ noon} \\
 + \quad 321 \\
 \hline
 1521 \text{ on the 24 hr clock}
 \end{array}$$

ANSWER 171521